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Test Assets and Weak Factors*

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Abstract

Weak factors—those to which test assets have limited exposure—pose a significant challenge and have been extensively studied in empirical asset pricing. Meanwhile, despite its empirical relevance, the selection of test assets has received comparatively less systematic study. We introduce supervised-PCA (SPCA), a novel methodology addressing both of these problems and bridging these seemingly unrelated literature strands. SPCA iterates supervised selection, principal-component estimation, and factor projection. It enables risk premia estimation and factor model diagnosis even when weak factors are present and not all factors are observed. We establish SPCA's asymptotic properties and showcase its empirical applications.

Key words: Supervised PCA, risk premium, stochastic discount factor, APT, Ridge, Lasso

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1 Introduction

Estimation and inference on factor models are central elements of empirical work in asset pricing. Typically, a researcher starts with a given factor, for example an aggregate liquidity factor, motivated by economic theory. The objective of the researcher is to estimate and test its risk premium. To proceed, the researcher needs to decide which test assets to use in the estimation. While the literature has made a variety of choices for test assets, little work has been dedicated to investigating rigorously and systematically how they should be chosen.

Another issue the researcher has to face is the potential presence of weak factors in the data. Broadly speaking, the factor of interest to the researcher is one of many factors potentially driving returns. Among the various factors, some may be weak: these are factors to which the available test assets have little or no exposure. Naturally, this makes it difficult to learn about them (for example, about their risk premia) using the available assets. More importantly, however, this lack of exposure contaminates inference about the *entire* model, even when the other factors are strong. Specifically, as the literature has shown, the presence of a weak factor driving returns biases the estimation of the risk premia of *all* the factors, including the one of interest to the researcher (whether that factor itself is strong or weak) as well as the inference about the pricing ability of the model. To make things worse, a weak factor could be latent (i.e., not observable) – so that we may not even know it exists in the first place.

In this paper, we document an important connection between the selection of test assets and the long-standing problem of weak factors in asset pricing. Exploiting this connection, we propose a novel methodology, supervised principal component analysis (SPCA), that serves a dual purpose: first, it provides a well-founded basis for the selection of test assets, and second, it leverages the selection to mitigate the bias in risk premium estimation for the factor of interest to the researcher, irrespective of its strength and the strength of (known or unknown) factors in the panel of test asset returns.

The connection we emphasize between weak factors and test assets is that the strength or weakness of a factor (whether it is observable or latent), should not be viewed as a property of the factor itself, as typical in the asset pricing literature; rather, it should be viewed as a property of the set of test assets used in the estimation. As an example, a liquidity factor may be weak in a cross-section of portfolios sorted by, say, size and value, but may be strong in a cross-section of assets sorted by characteristics that capture well exposure to liquidity.

This perspective provides clear guidance on how to choose test assets: select them in a way that yields a consistent estimate of the risk premium of the factor chosen by the researcher, which is robust to the presence of observable or latent weak factors among those driving returns. This asset selection criterion is statistical in nature, and aims to bolster the strength of weak factors and alleviate their detrimental effects on estimation. Our approach therefore offers an agnostic selection and estimation technique. It serves as a complement to alternative selection strategies found in the

literature, where researchers often use strong economic priors or ad-hoc methodologies to determine which test assets to include and which to exclude.

Estimating and testing the risk premium of a factor of interest requires properly controlling for all the other factors relevant to investors (whether they are observed or latent), in order to avoid an omitted variable bias (see for example Giglio and Xiu (2021)).¹ Giglio and Xiu (2021) propose to do so by first estimating a latent factor model for the stochastic discount factor (SDF) using principal component analysis (PCA), and then using it to estimate the factor of interest's risk premium. This approach eliminates the need for explicit specification of all the control factors (or, intuitively, it can be viewed as using the latent factors estimated via PCA as control factors). However, their PCA-based approach relies on a crucial assumption that, while convenient, may be violated in many empirical applications – the assumption that all the latent factors driving the SDF are pervasive (i.e., strong). Our SPCA procedure also utilizes PCA for extracting latent factors while remaining agnostic about the identities of the control factors. However, it exploits correlations with the factor of interest as a guiding criterion for selecting a subset of test assets, to which PCA is then applied. This integration of supervised asset selection with PCA results in a versatile methodology that remains robust even in scenarios where certain factors are omitted, including cases where these omitted factors are weak.

In a nutshell, given a factor g_t specified ex-ante by the researcher, the procedure estimates its risk premium as follows. We start from a large universe of potential test assets. In a first step of the procedure (selection step), we compute the univariate correlation of each asset's return with g_t . We select a relatively small portion of assets, only keeping those with sufficiently high correlation (in absolute value): these are assets that are particularly informative about the factor of interest g_t . We then compute the first principal component of the returns of these assets (PCA step), which will be our first estimated latent factor. Next, we remove via linear projection from both g_t and all the returns of the test assets the part explained by this first latent factor (projection step). We then go back to the selection step, computing the univariate correlation of the *residuals* of the factor and the *residuals* of the assets from the projection step. Again, we select from the universe of test assets a subset for which this correlation is especially high, and compute the first principal component of these residuals. This will be our second estimated latent factor. We then further remove (from g_t and the test assets) the part explained by this second estimated factor as well, and iterate again on the residuals. We repeat this procedure $\hat{p}$ times, where $\hat{p}$ is a tuning parameter which can be determined by some validation step. In the most desirable scenario, $\hat{p}$ serves as a desirable estimate of the actual number of factors, p , in the data. This procedure recovers from the data latent factors that are informative about the factor of interest g_t . Importantly, the fact that at each iteration only test assets that are sufficiently correlated with the factor g_t are selected ensures that not only strong,

¹This is only necessary when the factor of interest is not itself a tradable portfolio (i.e., it is a non-tradable factor, like a macroeconomic risk). If the factor of interest is itself a portfolio (also referred to as tradable factor), like in the case of the CAPM, the computation of the risk premium just requires computing the average excess return of the portfolio. In practice, most economic models have predictions about the risk premia of non-tradable factors.

but also weak factors (relative to the entire cross-section) are captured by the procedure – contrary to standard PCA that uses *all* assets at all steps to extract latent factors. Finally, a time-series regression of g_t on the p latent factors allows us to estimate the risk premium for g_t by linking it to the risk premia for these latent factors, yielding a consistent estimator of the risk premium of g_t . The latent factors themselves can be thought of as the part of the SDF that is related to g_t and determines its risk premium.

While the supervision of g_t aids in the recovery of factors, including weak ones, it is important to note that this procedure may not retrieve *all* the factors driving the cross-section of returns (i.e., the entire SDF). It specifically ensures the recovery of factors correlated with g_t , while uncorrelated factors, particularly if they are weak, may remain unrecoverable (so it may be that $\hat{p} < p$). Fortunately, but crucially, the omission of these factors by SPCA does not affect the consistency of the risk premium estimation for g_t , since such factors do not contribute to the pricing of g_t . That said, complete recovery of all factors remains feasible, contingent on including multiple variables in the target g_t and ensuring that each latent factor has at least one variable in g_t with a non-vanishing exposure to it.

Beyond the estimation of factor risk premia, SPCA can also be used to diagnose omitted factors in a model based on a set of observable factors in g_t . Supervised by g_t , SPCA recovers all the latent factors that drive the SDF and correlate with g_t . We prove that SPCA consistently recovers the true SDF if and only if g_t is spanned by all factors that drive the SDF. We apply this result to diagnose whether g_t misses any factors. This diagnosis on g_t can be executed as a simple comparison between the maximal Sharpe ratio achieved by g_t and that achieved by the factors recovered by SPCA. When the latter is larger than the former, it indicates that g_t misses some factor, and that the researchers should seek a better model. On the other hand, if the latter is smaller, it implies that g_t contains factors to which the given cross-section of test assets have insufficient exposures. In such a scenario, a richer set of test assets is needed. This type of diagnostic test for factor models helps us go beyond standard statistical tests of the models, that may reject the models but that do not shed light on the reason why the model is rejected (for example, distinguishing whether the rejection is due to omitted factors or to the presence of alphas in the asset returns).

The choice of test assets in the literature has mainly followed one of three approaches. The vast majority of the literature has adopted a “standard” set of portfolios sorted by a few characteristics, such as size and value, following the seminal work by Fama and French (1993). A second approach, taken more recently, e.g., Kozak et al. (2020), has been to expand this cross-section to include portfolios sorted by a much larger set of characteristics discovered in the last decades, on the order of hundreds of portfolios. Finally, a third approach, see, e.g., Ang et al. (2006), has been more “targeted” around the specific factor of interest: sorting assets into portfolios by their estimated exposure to the factor, and then estimating risk premia using only these sorted portfolios, that is, using a small cross-section expected to be particularly informative about that factor.

It is useful to contrast the asset selection procedure of SPCA with the three standard approaches

to choose test assets summarized above. Using a standard, small cross-section (like the size- and value-sorted portfolios) to estimate risk premia has the problem that except for size and value, which are strong factors in this cross-section, many other factors are weak: these test assets do not contain sufficient information to identify their risk premia. Using a large cross-section of test assets (the second approach) may appear, on the surface, to address this issue: these assets contain returns that are exposed to a large number of underlying factors. However, and importantly, if only a *few* of those many assets are exposed to some factor, whereas most others are not, that factor will, again, be weak in this large cross-section, disrupting inference on the risk premium. Finally, the third approach – building targeted portfolios of assets sorted by the exposure to the factor of interest to the researcher – is affected by the omitted factor problem, since it considers univariate exposures only (exposures with the factor of interest may also capture correlated exposures to other risks in the economy); in general, it will fail in a multi-factor context.

The premise of SPCA is that in an ideal scenario (the population model), asset selection would be unnecessary. In this ideal context, all assets within the available universe should be utilized, devoid of any *a priori* economic restrictions dictating their inclusion or exclusion. This notion aligns with the second approach mentioned earlier, which involves employing all available test assets. However, this approach proves unviable in practice when weak factors are present. The intuition behind this limitation stems from the existence of a substantial number of potentially “uninformative” test assets. In finite samples, these uninformative assets dilute the signal-to-noise ratio, ultimately leading to distortions in the estimation of risk premia (through an error-in-variables mechanism). As we will elucidate in greater detail later, to mitigate this bias it is enough to have a large subset of assets that exhibit sufficient informativeness about the latent factors driving test asset returns. This subset can adequately serve as test assets, enabling the consistent estimation of risk premia. At the heart of SPCA lies the fundamental step of identifying the most informative assets. SPCA accomplishes this through correlation screening, reflecting the same underlying principle as the third approach discussed earlier. However, SPCA goes beyond this initial selection step and incorporates additional steps, which are needed to yield a general and formally rigorous approach to estimation in the presence of weak factors.

In the paper, we derive the asymptotic properties of SPCA, in a setting that allows for weak factors and test assets with highly correlated risk exposures. The latter scenario potentially involves the same (asymptotically) rank-deficiency issue as weak factors. We also analyze in this setting alternative estimators that have been proposed in the recent literature, which rely on PCA, Ridge, Lasso, and Partial Least Squares (PLS). We show that the PCA (and some other variations of it), Ridge, and PLS are inconsistent in the presence of weak factors, and that the Lasso approach is consistent for the estimation of the SDF, as well as risk premia estimation, but is not as efficient as SPCA in general. Additionally, we perform an extensive set of simulations to study the performance of SPCA in different scenarios. These simulations isolate issues with the standard two-pass regressions, so that we can easily compare SPCA with other estimators. The simulations confirm the robustness

of SPCA to both omitted factors and weak factors, as well as measurement error, which SPCA also tackles.

As expected, a trade-off exists between robustness and efficiency. In scenarios where all factors are strong, the PCA-based approach by Giglio and Xiu (2021) is consistent and likely to outperform SPCA in terms of efficiency. The potential efficiency loss associated with SPCA arises from its selective use of test assets when all of them are in fact informative, or the possibility that it may not recover all factors driving returns (since, as we explain later, it does not necessarily recover factors that are uncorrelated with the factor of interest). However, the PCA-based estimator is biased in the presence of weak factors, a major concern in empirical applications. We therefore advocate for using SPCA to estimate risk premia due to its robustness when weak factors are potentially present. This is because in cases where consistency is compromised, prioritizing efficiency becomes irrelevant, as estimators that are precisely centered around the incorrect value are not preferable. In practical applications, the scree plot serves as a reliable tool for preemptively assessing the presence of weak factors. This assessment relies only on test asset returns, regardless of the specific attributes of the factor of interest. Hence, the choice between SPCA and PCA should *not* be contingent upon the specific properties of the factor of interest. Unless the scree plot indicates that all factors driving the returns of the test assets are distinctly strong, there is no theoretical basis to favor PCA over SPCA.

Finally, we illustrate the use of SPCA in an empirical exercise, both for estimating risk premia of a variety of tradable and nontradable factors proposed in the asset pricing literature, and for diagnosing observable factor models. We use the large cross-section of test portfolios produced by Chen and Zimmermann (2020) and Hou et al. (2020), covering more than 900 and 1600 portfolios, respectively, for the period 1976-2020. We apply SPCA to estimate factor risk premia, and evaluate its out-of-sample performance. Almost none of the non-tradable factors are priced, except for the intermediary capital factor. We also explore the robustness of SPCA to the weakness of factors, by artificially changing the set of test assets used in the estimation: for example, we show that SPCA is able to recover the risk premium for momentum even when momentum assets are removed from the original set of test assets (and therefore the momentum factor is weak in the cross-section). Finally, we illustrate empirically how SPCA can be used to diagnose whether observable factor models are missing important priced factors.

1.1 Related literature

The problem of weak factors in latent factor models is closely connected to that of weak factors in observable factor models, which has been widely examined in the literature. The seminal contribution of Kan and Zhang (1999) shows that the inference on risk premia estimates from Fama-MacBeth regressions becomes invalid when a “useless” factor – a factor to which test assets have zero exposures – is included in the model. Kleibergen (2009) further points out the failure of the standard inference

even for strong factors, if betas are relatively small.² In our paper, we show that the same logic applies in the context of latent factor models: if some (latent) factors are weak in a cross-section, the PCA estimator will not be able to disentangle them from idiosyncratic error, leading to biases in the estimated factors and their risk premia.

The issue of weak factors is particularly important in empirical work in asset pricing, because most economically-motivated factors (e.g., most macroeconomic factors) do seem to be weak in practice. Moreover, a statistical problem analogous to weak factors arises when betas are collinear, that is, some factors are redundant in terms of explaining the variation of expected returns (as we describe in the paper, this case is equivalent to the issue of weak factors, and is also handled by SPCA). This is again a relevant issue in practice due to the existence of hundreds of factors discovered in the literature, see, e.g., [Harvey et al. \(2016\)](#), many of which are close cousins and do not add any explanatory power ([Feng et al. \(2020\)](#)).

The weak factor problem appears to be caused by having seemingly more factors than necessary, which is why some suggest eliminating such factors ([Bryzgalova \(2015\)](#)) or shrinking their risk premia estimates ([Bryzgalova et al. \(2019\)](#)), so as to improve the estimates for strong factors. We instead argue that the weak factor problem is fundamentally an issue of test asset selection. Since weaker factors may still be priced, our solution is to accommodate them using an adapted procedure with carefully selected test assets.³

Several recent papers have proposed different methodologies to deal with weak factors. [Lettau and Pelger \(2020\)](#) are among the first to study the issue of weak latent factors in a related problem, that of estimation of the SDF. They propose an estimator of the SDF in the presence of weak factors, which generalizes PCA with a penalty term that accounts for expected returns; they refer to the estimator as risk premium PCA (rpPCA). Their objective, i.e., the recovery of SDF, is different from ours, but the SDF estimated using this procedure can still be used to estimate risk premia, since risk premia are covariances with the SDF. Whereas this estimator features desirable properties as explored by [Lettau and Pelger \(2020\)](#), we show that it is inconsistent for estimating risk premia in the weak-factor setting we consider.⁴ [Anatolyev and Mikusheva \(2021\)](#) propose an complementary four-

²Related literature also include [Gospodinov et al. \(2013\)](#) and [Gospodinov et al. \(2014\)](#). On the other hand, [Pesaran and Smith \(2019\)](#) investigate the impact of factor strength and pricing error on risk premium estimation. They point out that the conventional two-pass risk premium estimator converges at a lower rate as the factors become weaker.

³It is worth noting that whereas some theories assume that only strong factors can be priced, this is not true in general for two reasons. First, many theoretical models – e.g., the consumption-CAPM – are silent on what assets are traded in equilibrium, and if markets are incomplete, it may very well be that some priced factors may not be reflected in many of the assets that are traded. Second, even if investors may have access to many assets exposed to a particular factor, the econometrician may not, making the factor weak for the set of test assets available to the econometrician.

⁴[Lettau and Pelger \(2020\)](#) focus their analysis on the case where factors are extremely weak – so much so that they are not statistically distinguishable from idiosyncratic noise. In that case, no estimator can be consistent for either risk premia or the SDF. They show that indeed, rpPCA does not recover consistently the SDF, but it correlates with the SDF more than the SDF estimator obtained from standard PCA. Rather than focusing on this extreme case of weak factors, our theory covers a range of factor weaknesses, which includes the cases from strong to very weak, and which still permits consistent estimation of factors and risk premia. Formally, we study the case where the minimum eigenvalues of the factor component in the covariance matrix of returns diverges whereas the largest eigenvalue due to the idiosyncratic errors is bounded.

split approach to dealing with weak factors, based on sample-splitting and instrumental variables. This alternative procedure works well to address the weak factor bias, though it does not deal with omitted priced factors or with measurement error in the factors.⁵

Our paper also relates to a literature that has explored different methods to form portfolios to test asset pricing models, like [Ahn et al. \(2009\)](#) or [Bryzgalova et al. \(2020\)](#). These methods are useful in helping to build or expand the starting cross-section for SPCA. In this paper, we use the simpler approach of working with an existing large cross-section of portfolios sorted by firm characteristics, as in [Chen and Zimmermann \(2020\)](#) and [Hou et al. \(2020\)](#).

The concept of supervised-PCA originated from a cancer diagnosis technique applied to DNA microarray data by [Bair and Tibshirani \(2004\)](#), and was later formalized by [Bair et al. \(2006\)](#) in a prediction framework, in which some predictors are not correlated with the latent factors that drive the outcome of interest. [Bair et al. \(2006\)](#) suggest a screening step using marginal correlations between predictors and the outcome variable to select the subset of useful predictors, before applying the standard PCA to this subset.⁶ They prove the consistency of this so-called SPCA procedure, but relying on a restrictive identification assumption that any important predictor must also have a substantial marginal correlation with the outcome. We provide several examples of multivariate factor models in which this assumption fails. While the screening step of our SPCA procedure shares the spirit with theirs (in the sense that their outcome variable is our factor of interest, and their predictors are our test assets), our projection step and the subsequent iteration procedure are new, and are introduced precisely to eliminate the strong identification assumption used in the existing statistics literature. Also, our focus is not on prediction per se, but instead on inference on parameters (i.e., risk premia), which involves additional steps and more intricate analysis for the asymptotic theory.

2 Methodology

To rigorously address the challenge of weak factors, our approach begins with the specification of a general Data Generating Process (DGP). It is crucial to underscore that within this population model, the concept of weak factors holds no relevance. In population, researchers aiming to identify the risk premium of a factor like g_t would ideally utilize all available assets for this purpose.

⁵Our paper also relates to a growing strand of econometrics literature on weak factor models. [Bai and Ng \(2021\)](#) show that PCA can recover moderately weak factors at the cost of efficiency. [Bai and Ng \(2008\)](#) and [Huang et al. \(2021\)](#) propose supervised learning methods in the context of factor-based forecasting. [Fan et al. \(2021\)](#) also exploit information from observed proxies to improve the estimation of factor models, and [Wan et al. \(2023\)](#) consider moderately weak factors as in [Bai and Ng \(2021\)](#) in this context. [Fan and Liao \(2022\)](#) propose to extract factors by diversifying away idiosyncratic noise directly. [Uematsu and Yamagata \(2022a\)](#) adopt a variant of the sparse PCA algorithm proposed in [Uematsu et al. \(2019\)](#) to estimate a sparsity-induced weak factor model. [Uematsu and Yamagata \(2022b\)](#) provide inference results in that sparse model. [Freyaldenhoven \(2019\)](#) and [Bailey et al. \(2020\)](#) adopt a similar framework for estimating factor count and strength.

⁶The screening approach has also been adopted in the contexts, such as classification and regression, see [Fan and Fan \(2008\)](#), [Fan and Lv \(2008\)](#).

However, the real-world (finite-sample) scenario diverges from this idealized population model. We encounter practical constraints, such as a large number of assets (large N), relatively short time spans (small T), and a significant proportion of assets that are only weakly correlated with the target variable g_t . Following the introduction of the DGP, we proceed to characterize this finite sample context using asymptotic concepts, formally defining the notion of weak factors. This particular asymptotic perspective is useful as it enables us to investigate the issues of weak factors arising in finite samples with existing estimators, and understand the properties of our proposed solution.

2.1 Model Setup

We study a standard linear factor model setup. Suppose that an $N \times 1$ vector of test asset excess returns, r_t , follows:

$$r_t = \beta\gamma + \beta v_t + u_t, \quad E(v_t) = E(u_t) = 0 \text{ and } \text{Cov}(v_t, u_t) = 0, \quad (1)$$

where β is an $N \times p$ matrix of factor exposures, v_t is a $p \times 1$ vector of innovations of p factors f_t (i.e., $v_t = f_t - \mu_f$, where $\mu_f = E(f_t)$), and u_t is an $N \times 1$ vector of idiosyncratic errors.

We assume that the vector of factor innovations v_t is not fully observable. Specifically, we allow the asset pricing factors f_t to be either latent or observable. In the former case, innovations v_t are naturally also latent. Even in the latter case, when a factor f_t is observable, its innovation v_t is not directly observable because μ_f is an unknown parameter.

Also, note that we model risk exposures (β) as constant: we implicitly assume that the test assets are portfolios sorted so that their factor exposures are modelled as constant, as in [Giglio and Xiu \(2021\)](#). Alternatively, one could work directly with individual stocks (which generally have time-varying risk exposure), combining our procedure with the methodologies of [Gagliardini et al. \(2016\)](#), [Kelly et al. \(2019\)](#), or [Kim et al. \(2020\)](#) to account for the time-variation in betas.

We situate our discussion within the framework of two standard asset pricing exercises: the estimation of risk premia and the recovery of the SDF.

In this model, an SDF can be defined in terms of asset pricing factors v_t as

$$m_t = 1 - \gamma^\top \Sigma_v^{-1} v_t, \quad (2)$$

where Σ_v is the covariance matrix of factor innovations, see, e.g., [Giglio and Xiu \(2021\)](#). It also makes sense to consider the SDF represented in terms of the set of tradable test asset returns:

$$\tilde{m}_t = 1 - b^\top (r_t - E(r_t)), \quad (3)$$

where b is an $N \times 1$ vector of SDF loadings which satisfies $E(r_t) = \Sigma b$, where Σ is the covariance matrix of r_t , see, e.g., [Kozak et al. \(2020\)](#). The relationship between the two SDFs depends on the degree of completeness of markets. As will be shown later, these two forms of the SDF are

asymptotically equivalent in the asymptotic scheme we consider, with the number of assets N going to infinity, so that there is no ambiguity with respect to which estimand we consider.

In addition to the SDF, we are also interested in estimating the risk premium of one or more observable factors, summarized in a $d \times 1$ vector, g_t . It is important to emphasize that g_t is a proxy for some risks, constructed or otherwise chosen by the researcher ex-ante, not necessarily tradable, and typically motivated from economic theory or narratives. Following Giglio and Xiu (2021), we do not impose that g_t is part of or identical to v_t ; instead, we assume g_t and v_t are just (potentially) correlated:

$$g_t = \xi + \eta v_t + z_t, \quad (4)$$

where $\xi = E(g_t)$, η is a $d \times p$ matrix, and z_t is measurement error orthogonal to v_t .⁷ This model clearly nests the classic linear asset pricing model with observable factors only, in which case we can set $\eta = \mathbb{I}_p$ and $z_t = 0$. To price g_t , we can simply use the SDF given by (4), as g_t 's risk premium is given by $\gamma_g = -\text{Cov}(g_t, m_t) = \eta\gamma$.

To characterize the strength of a factor, we need to set up an asymptotic environment in which weak factors may arise. First, let us introduce some useful notation. We use the notation $a \lesssim b$ to denote $a \leq Kb$ for some constant $K > 0$, and if $a \lesssim b$ and $b \lesssim a$, we write $a \asymp b$ for short. This notation therefore indicates the relative asymptotic growth rates of a and b : for example, if $a \lesssim 1$, it means that a is bounded above by some constant. We use similar notation $\lesssim_P$ and $\asymp_P$ for bounded in probability. Also, for any matrix A , we use $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ to denote its minimum and maximum eigenvalues, and $\lambda_i(A)$ is the i -th largest eigenvalue.

The environment in which we study weak factors is quite general, and is characterized by three assumptions. First, we assume that both N and T go to ∞ (at arbitrary rates, unless we specify otherwise), whereas the number of factors p is fixed. Letting N go to infinity in addition to T is rather natural in the asset pricing context, as motivated in theory by Ross' APT (Ross (1976)) and given the proliferation of "anomalies" generated by the empirical literature in the past decades. Second, we assume that the $p \times p$ factor covariance matrix of the factor innovations, Σ_v , is asymptotically non-singular: $1 \lesssim \lambda_{\min}(\Sigma_v) \leq \lambda_{\max}(\Sigma_v) \lesssim 1$. This assumption is rather weak, as it only rules out factors whose risks are (asymptotically) negligible or exploding. Finally, we also maintain the assumption that $\|\Sigma_u\| \lesssim 1$, where $\|\cdot\|$ indicates the spectral norm of a matrix, so that there exists no factor structure in the residuals u_t . This assumption is widely adopted in the so-called approximate factor models proposed by Chamberlain and Rothschild (1983).⁸

We are now ready to characterize the strength of factors, as an exclusive function of test assets'

⁷When g_t is nontradable, measurement error could arise as the econometrician is implementing an empirical counterpart of some theory-predicted factor; when g_t is tradable, it captures the non-diversified errors in the portfolio.

⁸We only need u_t to be stationary (so that Σ_u is well defined) when we discuss the SDF in Section 2.3. For risk premia estimation, we instead impose a weaker condition, namely, Assumption B.4, which plays a similar role as $\|\Sigma_u\| \lesssim 1$.

exposures to the factors, as opposed to a property of the factors themselves. We formalize here the idea that, for instance, a momentum factor could be a strong factor when the test assets are momentum-sorted portfolios, but this same factor may be weak when the test assets are portfolios sorted by size or value: the latter portfolios may diversify away the exposures to the momentum factor, and therefore may be uninformative about momentum risk.

In the econometrics literature on factor models (for example, Bai and Ng (2002)), the setup described in (1) is typically complemented by the assumption that $\lambda_i(\beta^\top \beta) \asymp N$ for $i = 1, 2, \dots, p$: all eigenvalues of the matrix $\beta^\top \beta$ grow at rate N , so that *all* factors are *pervasive*. Informally, even as the number of test assets N is large, there is a sufficiently large number of assets that are well exposed to each of the risk factors (their β with respect to all factors are non-vanishing for a large number of assets). Under this assumption, as we will review later, standard PCA works well to recover the latent factors v_t .

This is the point of departure of our paper: we study theoretically situations in which this pervasiveness assumption fails, with respect to some or even all factors. Formally, we define the presence of *weak factors* as the case in which some of those eigenvalues, $\lambda_i(\beta^\top \beta)$, grow at a slower rate than N (which will be made more precise later). Intuitively, in this case, while the number of test assets N is large, many test assets may have small or zero exposures to some or all of the factors, making those factors weak. The lack of test asset exposures to a factor makes it more difficult for standard PCA to recover this factor; and in more extreme cases, PCA completely fails to recover either the factors or their loadings. In our setting, the strength/weakness of a factor is actually not a binary distinction. Rather, we allow for a continuum of strength/weakness of factors, determined by how large the exposures to the risk factors are (formally, by the asymptotic behavior of the eigenvalues $\lambda_i(\beta^\top \beta)$).

How relevant do we expect these weak factors to be in practice? Consider Figure 4, the scree plot of the eigenvalues of returns from our empirical analysis, which uses a large cross-section of 950 assets. The figure shows that the first one or two eigenvalues are clearly much larger than the others. But the absence of clear gaps among the remaining eigenvalues suggests that several factors beyond the first two may be weak. Despite the large cross-section, their eigenvalues remain relatively small and difficult to distinguish from idiosyncratic error.

Our model naturally allows g_t to be weak, since the true factors in v_t are potentially weak and the observable factors in g_t inherit this weakness through their loading on v_t , η . However, as N and T increase, the risk premium associated with g_t , $\eta\gamma$, may not necessarily converge to zero. This is because neither the risk exposure of g_t to v_t , represented by η , nor the risk premiums of v_t , denoted as γ , are guaranteed to diminish asymptotically. In simpler terms, weak factors in this model can still have non-zero risk premia as the sample size and the cross-sectional dimension grow.

2.2 Estimating Risk Premia when Factors are Weak

We begin our analysis with risk premia estimation.

2.2.1 The Benchmark PCA-based Estimator

Giglio and Xiu (2021) study this problem in a similar setup as in this paper, except that all factors in v_t are assumed to be strong. They propose a three-pass procedure to estimate g_t 's risk premium $\eta\gamma$: 1) apply PCA to the sample covariance matrix of returns to obtain estimates of the latent factors, $\hat{v}_t$; 2) use Fama-MacBeth regressions to recover the risk premia of $\hat{v}_t$, $\hat{\gamma}$; 3) use time series regressions of g_t on $\hat{v}_t$ to estimate $\hat{\eta}$. The product of the estimates at steps 2 and 3 yields $\hat{\eta}\hat{\gamma}$, the estimate of risk premia. We summarize this procedure in the following algorithm:

Algorithm 1 (PCA-based Estimator of Risk Premia). *The estimator proceeds as follows:*

Inputs: $\bar{R}$ and $\bar{G}$, the matrices of demeaned returns and demeaned g_t , respectively.⁹

S1. Apply singular-value decomposition (SVD) on $\bar{R}$, and write the first p right singular vectors as ξ . The estimated factors are given by $\hat{V} = \sqrt{T}\xi^\top$.

S2. Estimate the risk premia of $\hat{V}$ by $\hat{\gamma} = (\hat{\beta}^\top \hat{\beta})^{-1} \hat{\beta}^\top \bar{r}$, where $\hat{\beta} = \bar{R} \hat{V}^\top (\hat{V} \hat{V}^\top)^{-1}$ and $\bar{r}$ is the vector of average excess returns.

S3. Estimate the factor loading of g_t on v_t by $\hat{\eta} = \bar{G} \hat{V}^\top (\hat{V} \hat{V}^\top)^{-1}$.

Outputs: $\hat{V}$, $\hat{\eta}$, $\hat{\gamma}$, and $\hat{\gamma}_g^{PCA} = \hat{\eta}\hat{\gamma}$.

As discussed in Giglio and Xiu (2021), one interpretation of this estimator is that it builds a mimicking portfolio for the factor g_t , by projecting it onto the first p principal components of the space of returns (which effectively are p portfolios that span the factor space). A mimicking portfolio would be ideally built directly using *all* possible assets. But when N is large, this can be inefficient or even infeasible (if $N > T$). The three-step estimator effectively regularizes the mimicking portfolio problem by using only p portfolios appropriately constructed as basis assets, i.e., the principal components of the returns. Giglio and Xiu (2021) establish the consistency of this estimator and derive its asymptotic inference, in the case that all latent factors are strong. This procedure also recovers the SDF, because it estimates all latent factors, $\hat{v}_t$ (columns of $\hat{V}$), that drive the SDF, consistently, along with their SDF loadings as in (2), $\hat{\Sigma}_v^{-1}\hat{\gamma}$.

This estimator is appealing for its simplicity, efficiency, and, importantly, robustness to missing factors (since the identity of any factors beyond g_t does not need to be specified). Unfortunately, it fails precisely when some latent factors are weak, which we will show next.

To explain why this is the case, it is sufficient to consider a one-factor model with $p = d = 1$ and $\Sigma_v = 1$, in which case the covariance matrix of returns satisfies: $\Sigma = \beta\beta^\top + \Sigma_u$. This matrix has

⁹For any time series of vectors $\{a_t\}_{t=1}^T$, we denote $\bar{a} = \frac{1}{T} \sum_{t=1}^T a_t$. In addition, we write $\bar{a}_t = a_t - \bar{a}$. We use the capital letter A to denote the matrix $(a_1, a_2, \dots, a_T)$, and write $\bar{A} = A - \bar{a}\iota_T^\top$ correspondingly.

a noisy low rank structure in that $\beta\beta^\top$ has rank 1, whereas Σ_u is a full-rank covariance matrix. To make the exposition simple, we also assume that the factor of interest g_t has no measurement error, i.e., $z_t = 0$ and $g_t = \eta v_t$.

As discussed above, the problem of weak factors stems from the fact that many assets may not have sufficiently strong exposure to the factor of interest, which hinders the construction of its mimicking portfolio, and in turn, the estimation of its risk premium. This intuition applies also when the weak factor is latent (v_t). In this case, the manifestation of the weak factor problem is that PCA will fail to recover this factor.

Estimation of the latent factors v_t via PCA involves recovering the matrix of risk exposures β from the covariance matrix of realized returns, $\hat{\Sigma}$. A successful recovery of β via PCA of realized returns therefore requires a favorable signal-to-noise ratio. If the “signal”, as measured by $\|\beta\|$, dominates the “noise”, which arises from the idiosyncratic component Σ_u and the estimation error in the sample covariance matrix $\hat{\Sigma} - \Sigma$, the first sample eigenvector of $\hat{\Sigma}$ would (approximately) span the same space spanned by the true β . Thus using $\hat{\beta}$, effectively the eigenvector of $\hat{\Sigma}$, in the cross-sectional regression step (step 2 of the estimator) would yield a consistent estimator of the risk premium of the estimated latent factor, which in turn leads to a consistent estimator of the risk premium of g_t . Otherwise, if the signal $\|\beta\|$ is so weak that the estimation error in $\hat{\beta}$ dominates, there would be a non-vanishing angle between the space spanned by $\hat{\beta}$ and that by β , which eventually results in an inconsistent estimate of the risk premium.

The bias generated by weak factors in a latent factor setting is analogous to the weak factor issue documented in the literature in the case of fully observable factors (Kan and Zhang (1999)). When the true betas are close to zero, the estimated betas will be dominated by measurement error. But estimating risk premia requires comparing the average returns of assets with different betas (e.g., computing the slope in a cross-sectional regression); measurement error in the betas thereby induces a bias in the risk premium estimate.

Proposition 1 below shows that the PCA-based risk premium estimator is consistent only if $N/(\|\beta\|^2 T) \rightarrow 0$. This condition formalizes our notion of whether a factor is strong or weak. In a one-factor model, the factor is weak if this condition fails. We generalize this definition for the case of multiple factors later.

Proposition 1. *Suppose that test asset returns follow a single-factor model in the form of (1) with $p = 1$, g_t satisfies (4) with $d = 1$, and u_t and v_t i.i.d. normally distributed and independent from each other and $z_t = 0$. In addition, suppose that β satisfies $N/(\|\beta\|^2 T) \rightarrow B \geq 0$ and $\|\beta\| \rightarrow \infty$. Then we have $\hat{\gamma}_g^{PCA} \xrightarrow{P} (1 + B)^{-1} \eta \gamma$.*

In the presence of strong factors, $\|\beta\| \asymp \sqrt{N}$, which leads to $B = 0$ as $T \rightarrow \infty$, so there is no bias. In general, the consistency depends on the relative magnitude of N , T , and $\|\beta\|$. When N and T are of the same order, $\|\beta\| \rightarrow \infty$ is sufficient for the consistency of risk premia estimation. This makes sense in that the eigenvalue of returns corresponding to this factor is proportional to $\|\beta\|^2$,

whereas the eigenvalues for the idiosyncratic errors are bounded, so that $\|\beta\| \rightarrow \infty$ guarantees the separation between factors and errors and hence the identification of factors.

This example also shows that the risk premium estimator could be biased even if we have consistent estimator of the factors. In fact, the estimated factors in $\hat{V}$ are consistent under the assumptions of Proposition 1 in the sense that $|\text{Corr}(\hat{V}, V)| \xrightarrow{P} 1$.¹⁰ However, estimating a large-dimensional vector β given $\hat{V}$ remains a challenging problem, which requires this additional condition, $B = 0$, to achieve consistency.

Section A of the online appendix studies how several other estimators perform in a weak-factor setting, including partial least squares (PLS), Ridge regression, and risk premium PCA (rp-PCA). The analysis there reveals that these estimators exhibit failures that mirror that of PCA, despite PLS leveraging information from g_t for supervision and rp-PCA being specifically designed for weak factors. None of these estimators, therefore, can address the bias originating from the presence of weak factors.

2.2.2 Our Solution: Supervised-PCA and Test Asset Selection

The results in the previous section shed light on the detrimental influence of weak factors on the PCA-based estimator (as well as other existing approaches).¹¹ Given that it is not possible to apply PCA to *all assets* without incurring a bias if there are weak factors in the cross-section, it becomes imperative to adopt a form of test asset selection to alleviate or eliminate the bias. Is it possible to enhance the signal-to-noise ratio and effectively resolve this weak factor issue by eliminating uninformative assets? The answer is affirmative.

As we mention in the introduction, an important difference with the prior literature is that we do not view the weakness of a factor as a property of the factor itself; rather, we see it as a property of the universe of test assets that are used in the estimation. This leads us to find a potential solution in modifying the set of test assets. The solution we propose in this paper is to *screen* test assets and only keep those that have nontrivial exposure to the factor of interest g_t . Then, if the factor is strong *within this smaller set of test assets*, it is possible to apply PCA (or other procedures discussed in the appendix) to recover its risk premium. The key idea behind the screening approach is to remove those uninformative assets, focusing the estimation on a set of assets whose exposures are large and dominate the estimation error in β .

To proceed with this idea, we formalize the problem by imposing an assumption that there exists a subset $I_0 \subset \langle N \rangle$,¹² within which test assets feature a strong factor structure. In other words, there

¹⁰We can further establish that a sufficient condition for consistent recovery of factors is $N/(\|\beta\|^4 T) \rightarrow 0$, which clearly holds in the setup of Proposition 1.

¹¹These results should not be regarded as evidence against the use of above estimators in all scenarios. Rather, we only establish that for data generating processes in the regime of weak factors we define, none of these estimators are consistent. It is however possible that, for some alternative sequences of data generating processes, or for purposes other than risk premia estimation or SDF recovery, these estimators may perform well.

¹²We use $\langle N \rangle$ to denote the set of integers: $\{1, 2, \dots, N\}$.

exists a subset of assets that are sufficiently *informative* about latent factors driving test asset returns. To be clear, we do not make any assumption about the remaining test assets in the complement set of I_0 — they may or may not be informative. Such a set is thus not uniquely defined. In this regard, this assumption is relatively mild.

To see how this assumption helps, note that in the population model of Proposition 1, the expected excess return of g_t 's mimicking portfolio *built only with test assets in I_0* is

$$\text{Cov}(g_t, r_{t,[I_0]})\text{Cov}(r_{t,[I_0]})^{-1}\text{E}(r_{t,[I_0]}) = \eta \Sigma_v \beta_{[I_0]}^\top (\beta_{[I_0]} \Sigma_v \beta_{[I_0]}^\top + \Sigma_{u,[I_0]})^{-1} \beta_{[I_0]} \gamma,$$

where $r_{t,[I_0]}$ denotes the vector of returns of test assets in I_0 , and $\beta_{[I_0]}$ is their corresponding beta.¹³ It can be shown that (see the proof of a more general setting in Proposition C.4 of the online appendix)

$$\text{Cov}(g_t, r_{t,[I_0]})\text{Cov}(r_{t,[I_0]})^{-1}\text{E}(r_{t,[I_0]}) = \eta \gamma + O(\|\beta_{[I_0]}\|^{-2}). \quad (5)$$

Since test assets in I_0 feature a strong factor structure, $\|\beta_{[I_0]}\|^2 \asymp |I_0| =: N_0$,¹⁴ the approximation error is thereby $O(N_0^{-1})$. This result establishes the fact that in population using a smaller number of sufficiently informative assets leads to an asymptotically vanishing error in approximating the risk premium. Moreover, it holds that $N_0/(\|\beta_{[I_0]}\|^2 T) = O(T^{-1})$, i.e., factors are pervasive within this subset. Therefore, as long as we locate a subset that satisfies the properties of I_0 , we can estimate risk premia consistently with PCA by only using test assets within this subset.

In practice, it is the researcher who decides which test assets to employ in an empirical study. Assuming that a strong factor structure exists at least within a subset of test assets seems practical and plausible. That said, this assumption does rule out the case in which exposures to a factor are uniformly small for all test assets. That is, if there are latent factors where exposures from all assets are zero or very small, this factor may be missed by SPCA (though it will be missed by all other estimators too).

Now the problem is that we do not know ex-ante such a set, i.e., which assets are informative about the latent factor v_t . Rather than selecting assets arbitrarily, the idea of SPCA revolves around selecting the most informative assets based on their covariances with g_t . In the DGP of Proposition 1, the group of assets exhibiting high covariances with g_t comprises those with large magnitudes of β s. Therefore screening via correlation selects a subset of assets satisfying the desirable properties of I_0 .

Our proposed screening strategy echoes some of the practice in the empirical asset pricing literature. Very often, test assets are formulated using the exact characteristics-sorted portfolios that the factor of interest is generated from. For instance, Fama and French (1993) use size and value double-sorted portfolios as test assets when estimating a factor model that includes size and value as factors. In other cases, for nontradable factors, portfolios are sorted based on individual stock betas

¹³We use $A_{[I]}$ to denote a submatrix of A whose rows are indexed in I .

¹⁴For an index set $I \subset \langle N \rangle$, we use $|I|$ to denote its cardinality.

with respect to the factor of interest.

These choices seldom are justified formally, and are often only valid in very special cases. For example, building portfolios by sorting stocks on beta with respect to g_t may inadvertently incorporate compensation for other correlated risks, introducing a bias when omitted factors exist in the asset pricing model that is used to calculate the betas, not to mention the issue of propagation of errors that arise in the estimation of the beta. Similarly, using Fama-French portfolios as test assets assumes implicitly that they span the investment universe. This assumption contradicts the recent asset pricing literature, from which numerous factors or anomalies emerge. While our methodology formalizes the insight behind these traditional procedures, the fundamental motivation behind our approach is precisely to circumvent the adoption of arbitrary priors when selecting assets.

We next formally present our SPCA procedure in the simple one factor setting as discussed in the previous proposition, which helps illustrate the intuition behind our proposal and facilitates the comparison with existing estimators (the next section is devoted to the general case).

Algorithm 2 (SPCA-based Estimator of Risk Premia for a Single Factor Model ($p = 1$)). *The procedure is as follows:*

Inputs: $\bar{R}$ and $\bar{G}$, a $1 \times T$ vector.¹⁵

S1. Select a subset $\hat{I} \subset \langle N \rangle$: $\hat{I} = \left\{ i \mid T^{-1} |\bar{R}_{[i]} \bar{G}^\top| \geq c_q \right\}$, where c_q is the $(1 - q)$ -quantile of $\{T^{-1} |\bar{R}_{[i]} \bar{G}^\top|\}_{i \in \langle N \rangle}$.

S2. Repeat S1. – S3. of Algorithm 1 with selected return matrix $\bar{R}_{[\hat{I}]}$ and $\bar{G}$, and $p = 1$.

Outputs: $\hat{\gamma}_g^{SPCA} := \hat{\eta} \hat{\gamma}$, $\hat{V}$, $\hat{\eta}$, and $\hat{\gamma}$.

SPCA (Algorithm 2) adds the screening step, S1, to the PCA-based risk-premium estimation method of Giglio and Xiu (2021) (Algorithm 1). In this step, out of the N assets available, only a subset $\hat{I}$ is selected, and the three steps of Algorithm 1 are applied to this subset only.

The selection is operated by computing the absolute value of the covariance between each of the N assets and the factor g_t : $(T^{-1} |\bar{R}_{[i]} \bar{G}^\top|$ for each asset i). Only those assets for which the magnitude of this covariance is large enough are selected: specifically, the top $q\%$ of them (for example if q is 0.2, only the top 20% of assets ranked by absolute covariance with g_t are selected). Therefore, SPCA involves a tuning parameter, q , which plays a crucial role in determining how many assets we use to extract the factor. Note that the fact that $\hat{I}$ incorporates information from the target, g_t , reflects the distinctive nature of a supervised procedure (from which the name *supervised-PCA*).

We next prove that SPCA is consistent in the presence of weak factors.

Proposition 2. *Suppose that $\log N/T \rightarrow 0$ and test asset returns follow a single-factor model in the form of (1) and that g_t satisfies (4), with u_t , v_t , and z_t i.i.d. normally distributed and independent from each other. The loading matrix β satisfies $\|\beta\|_{\text{MAX}} \lesssim 1$ and there exists a subset $I_0 \subset \langle N \rangle$*

¹⁵We discuss the case of a multivariate ($d \times T$) $\bar{G}$ in Section 2.2.4.

such that $\|\beta_{[I_0]}\| \asymp \sqrt{N_0}$ where $N_0 = |I_0| \rightarrow \infty$. Then, for any choice of q in Algorithm 2 such that $qN/N_0 \rightarrow 0$ ¹⁶ and $qN \rightarrow \infty$, and that $|\beta|_{\{qN+1\}} \leq (1+\delta)^{-1}|\beta|_{\{qN\}}$ ¹⁷ for some $\delta > 0$, where $|\beta|_{\{k\}}$ denotes the k th largest value in $\{|\beta_{[i]}|\}_{i \in [N]}$, we have $\hat{\gamma}_g^{SPCA} \xrightarrow{P} \eta\gamma$.

To gain a better understanding of the intuition, let us delve into some key steps of the proof, which is detailed in the online appendix. Given a specific choice of the tuning parameter q , we can identify the population counterpart of $\hat{I}$, denoted as I . This set I consists of the qN largest entries of β in terms of their magnitudes, as specified before Assumption B.7 in the appendix.¹⁸

The proof of Proposition 2 establishes the consistency of the selected set $\hat{I}$ (which contains the top qN test assets with the largest sample covariances with g_t) with respect to I in the following sense: $P(\hat{I} = I) \rightarrow 1$.

This result is valid for two reasons. Firstly, the estimation error for the (population) covariance with g_t for any test asset is of order $T^{-1/2}$. By applying the large deviation bound in high-dimensional statistics, we can establish that the estimation error for covariances between g_t and all test assets is uniformly bounded by $(\log N)^{1/2}T^{-1/2}$. Consequently, to ensure consistent estimation of all covariances, it is necessary that $\log N/T \rightarrow 0$.

Secondly, the condition that there exists I_0 such that $\|\beta_{[I_0]}\|^2 \asymp N_0$ and $qN/N_0 \rightarrow 0$ guarantee the existence of at least qN test assets with non-zero population covariances with g_t . Thus, according to the definition of I , the smallest population covariance with g_t among all test assets in I must be non-zero. This suggests that $\|\beta_{[I]}\|^2 \asymp |I| = qN$. Furthermore, since we assume a non-vanishing gap between the (qN) th and $(qN+1)$ th population covariances, it thereby follows that the set of test assets with largest population covariances must coincide with those having the largest sample covariances, because the vanishing estimation error is dominated by this non-vanishing gap in the asymptotic context.

Given that the identified set I can function as I_0 (since $\|\beta_{[I]}\|^2 \asymp |I|$), and as demonstrated in equation (5), we can directly approximate the risk premium of g_t using its mimicking portfolio built on this subset I of test assets. The consistency of risk premium estimate thereby follows from the consistency of $\hat{I}$ in the recovery of I .

Proposition 1 and Propositions A.1 - A.3 in the online appendix show that in the single factor case, the consistency of PCA, Ridge, PLS, and rpPCA requires $B = 0$. Suppose $\|\beta\|^2 = N^v$, for some $v > 0$, then $B = 0$ is equivalent to $N^{1-v}/T \rightarrow 0$. The consistency of SPCA, as shown by Proposition 2, nonetheless, only requires $\log N/T \rightarrow 0$.¹⁹

¹⁶It may be tempting to use $qN/N_0 \xrightarrow{P} \text{const} < 1$. However, this is not viable because N_0 and I_0 are not precisely defined in the assumption $\|\beta_{[I_0]}\| \asymp \sqrt{N_0}$. That is, if we replace N_0 by $N_0/2$, the previous assumption still holds but qN/N_0 might be greater than 1.

¹⁷This technical condition on $|\beta|_{\{qN+1\}}$ simply states that the test assets should have (asymptotically) distinct risk exposure. It is a rather mild assumption that simplifies the proof.

¹⁸It is crucial to distinguish between I and I_0 . I is uniquely defined for each q that satisfies the conditions of I_0 , whereas I_0 is a general mathematical abstraction not uniquely defined.

¹⁹Another idea that shares the spirit of SPCA is the scaled-PCA proposed by Huang et al. (2021), which uses regression coefficients of $\bar{G}$ on $\bar{R}$ to weight $\bar{R}$ before feeding it into the PCA procedure. An advantage of the scaled

2.2.3 SPCA in the General Case: Selection and Projection

Propositions 1 - 2 focus on an unrealistic single-factor model since they are meant to illustrate the failure of PCA due to the presence of a weak factor as well as the intuition behind our procedure. In general, the DGP of returns is likely driven by more than one factor; in addition, these factors will generally have different strength in any specific cross-section of test assets. Note also that g_t could have more than one dimension in the general setup (4). In this section, we show how to generalize SPCA to the case where multiple factors of distinct strength are present.

To begin with, in the same spirit of Proposition 1, we can show that a general necessary condition for the consistency of PCA in a multi-factor model is that

$$N/(\lambda_{\min}(\beta^\top \beta)T) \rightarrow 0. \quad (6)$$

If this holds, it means that even the weakest one among all p factors in (1) is sufficiently strong that it can be recovered by PCA. Then, the three-pass estimator of Giglio and Xiu (2021) would properly recover risk premium for any factor g_t . We thereby define weak factors as those for which test asset exposures fail condition (6). This is a compact formal description of the non-ideal finite-sample environment encountered in practice.²⁰

Just like in the single-factor case, in the multi-factor case condition (6) can fail if one of the factors is not pervasive. But in the multi-factor case, it can also happen that all factors are individually strong, and condition (6) still fails because the factors' exposures are highly correlated. Consider, for example, a two-factor model where the beta matrix has the following form:

$$\beta = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}, \quad (7)$$

where β_{11} and β_{12} are $N_0 \times 1$ vectors, β_{21} and β_{22} are $(N - N_0) \times 1$ vectors, and N_0 is small relative to N . Suppose that $\beta_{21} = \beta_{22}$. In this setup, we can identify two groups of test assets. The first one is a small group of N_0 test assets, with exposures β_{11} to the first factor and β_{12} to the second factor. The second is a large group of $(N - N_0)$ assets, that have the *same* exposure to both factors (since $\beta_{21} = \beta_{22}$). In this case, we can show that condition (6) can fail: even if each factor is strong individually, there is a "rank deficiency" issue in the betas. The reason is that most of the asset

PCA approach is that it does not involve any tuning parameter. Nonetheless, the scaled PCA still assigns weights of $1/\sqrt{T}$ magnitude to assets that have zero-correlations with the target variable, whereas our approach assigns zero weights to such assets. As a result, our procedure only requires $\log N$ to be small relative to T , whereas both the scaled PCA and PCA require N to grow no faster than a certain polynomial rate relative to T .

²⁰Note that r_t is related to g_t through v_t . The loading of g_t on v_t is a low dimensional parameter η specific to each g_t , whereas the loading of r_t on v_t is a high-dimensional vector β independent of g_t . The advantage of formulating the condition in terms of $\lambda_{\min}(\beta^\top \beta)$ without η guarantees the applicability of our conclusion across all factors of interest.

(group 2) do not contain information that can separate the risk premia of the two factors, because they are equally exposed to the two. This loss of information turns out to have exactly the same effect on estimation and inference as the weak factor issue.²¹ We need a procedure that consistently estimates risk premia in this case as well.

It is also important to note that in the general case with multiple factors of potentially different strength, a simple extension of Algorithm 2, operating an initial screening (S1) and then extracting *multiple* factors via PCA (S2) would *not* actually work to recover all factors. To see this, take (7) again as an example. Suppose now that $\beta_{21} \neq \beta_{22}$, but $\beta_{22} = 0$: that is, most of the assets have zero exposure to the second factor. Consequently, the first factor is strong, while the second factor is weak.²² Now suppose that $\eta = (1, 1)$, implying that the observed factor g is correlated with both factors and, by extension, with all the test assets. In this scenario, the determination of which assets to exclude via screening hinges on the betas of these test assets. Should a majority of the selected assets pertain to the second group, the subsequent application of PCA in step S2 would only recover the first factor. This would occur if condition (6) fails for the selected assets. On the other hand, if many of the selected assets belong to the first group, PCA applied to them has the potential to recover both factors. In this scenario, the first principal component may capture a linear combination of both the strong and weak factors.

This example demonstrates that even though screening assets ensures that the *first* principal component after screening recovers one factor (which could be the strong factor, the weak factor or their mixture on the basis of the original cross-section), there is no guarantee that this procedure can solve the weak factor issue in one shot.

Next we provide another example, that shows that in some situations screening can sometimes eliminate *too many* assets, making a strong factor model become weak or even rank-deficient. Suppose that β has the following form:

$$\beta = \left[\begin{array}{c|c} \beta_{11} & \beta_{11} \\ \hline 0 & \beta_{22} \end{array} \right], \quad (8)$$

where β_{11} and β_{22} are $N/2 \times 1$ non-zero vectors satisfying $\|\beta_{11}\| \asymp \|\beta_{22}\| \asymp \sqrt{N}$. Clearly, β is full-rank and both factors are strong. Therefore, a standard PCA procedure should work smoothly. Suppose in addition that $\eta = (1, 0)$ (i.e., $g_t = v_{1t}$) and that v_{1t} and v_{2t} are uncorrelated. Then it implies that g_t is uncorrelated with the second half of test assets in r_t , so only those test assets within the first half would remain, should screening be applied with g_t before extracting the principal components. In this example, however, the *remaining* test assets have perfectly correlated exposures to both factors,

²¹Formally, we can show that $\lambda_{\min}(\beta^\top \beta) \leq \|\beta_{11} - \beta_{12}\|^2 / 2 \lesssim N_0$. As a result, $N/(\lambda_{\min}(\beta^\top \beta)T) \gtrsim N/(N_0 T)$, which does not necessarily converge to 0 if N_0 and T are small, so that the condition (6) could fail.

²²It is easy to show that in this case $\lambda_{\min}(\beta^\top \beta) \leq \|\beta_{12}\|^2 \lesssim N_0$.

so that effectively only *one* factor, $v_{1t} + v_{2t}$, is left. This example shows once again that the one-step supervised procedure (screening once and then applying PCA) may fail at extracting all factors in a multi-factor setting.²³

To address the aforementioned issues, we propose a multi-step version of SPCA, that iteratively conducts selection and projection. Step S1 of Algorithm 2 described above – valid when there is only one factor – can help identify one strong factor from a selected subset of test assets. In a nutshell, the multi-step SPCA, described below in Algorithm 3 iteratively applies Algorithm 2 to extract a new factor, with a projection step designed to ensure that each new factor is orthogonal to the estimated factors in the previous steps, similar to the factors extracted by the standard PCA.

Formally, the algorithm is given by:

Algorithm 3 (Selection and Projection). *The iterative SPCA procedure for risk premia estimation is as follows:*

Inputs: $\bar{R}_{(1)} := \bar{R}$, $\bar{r}_{(1)} := \bar{r}$, and $\bar{G}_{(1)} := \bar{G}$, a $d \times T$ vector.

S1. For $k = 1, 2, \dots$ iterate the following steps using $\bar{R}_{(k)}$, $\bar{r}_{(k)}$, and $\bar{G}_{(k)}$:

- a. Select an appropriate subset $\hat{I}_k \subset \langle N \rangle$.*
- b. Repeat S1. – S3. of Algorithm 1 with selected return matrix $(\bar{R}_{(k)})_{[\hat{I}_k]}$ and $\bar{G}_{(k)}$ to extract only the first principle component. Denote the estimates as $\hat{V}_{(k)}$, $\hat{\eta}_{(k)}$, $\hat{\gamma}_{(k)}$.*
- c. Estimate the exposure of $\bar{R}_{(k)}$ to $\hat{V}_{(k)}$ by $\hat{\beta}_{(k)} = T^{-1} \bar{R}_{(k)} \hat{V}_{(k)}^\top$.*
- d. Obtain $\bar{R}_{(k+1)} = \bar{R}_{(k)} - \hat{\beta}_{(k)} \hat{V}_{(k)}$, $\bar{r}_{(k+1)} = \bar{r}_{(k)} - \hat{\beta}_{(k)} \hat{\gamma}_{(k)}$, and $\bar{G}_{(k+1)} = \bar{G}_{(k)} - \hat{\eta}_{(k)} \hat{V}_{(k)}$.*

Stop at $k = \hat{p}$, where $\hat{p}$ is chosen based on some proper stopping rule.

S2. Estimate risk premia by $\hat{\gamma}_g^{SPCA} = \sum_{k=1}^{\hat{p}} \hat{\eta}_{(k)} \hat{\gamma}_{(k)}$.

Outputs: $\hat{\gamma}_g^{SPCA}$, $\hat{\eta} = (\hat{\eta}_{(1)}^\top, \dots, \hat{\eta}_{(\hat{p})}^\top)^\top$, $\hat{\gamma} = (\hat{\gamma}_{(1)}, \dots, \hat{\gamma}_{(\hat{p})})^\top$, $\hat{V} = (\hat{V}_{(1)}^\top, \dots, \hat{V}_{(\hat{p})}^\top)^\top$ and $\hat{\beta} = (\hat{\beta}_{(1)}, \dots, \hat{\beta}_{(\hat{p})})$.

Each iteration k of the procedure recovers one latent factor $\hat{V}_{(k)}$, estimates its risk premium $\hat{\gamma}_{(k)}$, and the exposure of g_t to that factor, $\hat{\eta}_{(k)}$. In step S1, there is first asset selection (S1.a). Next, the three-step estimator of risk premia of Giglio and Xiu (2021) is applied using the selected assets (S1.b) to recover the k th factor $\hat{V}_{(k)}$ in addition to $\hat{\gamma}_{(k)}$ and $\hat{\eta}_{(k)}$, which are specific to that factor. Then, in S1.c, we project the returns of all assets (not just those selected) on the estimated factor $\hat{V}_{(k)}$, and in step S1.d we compute the residuals of this projection for returns and the factor g_t itself. Therefore, at the end of step S1, we have completely eliminated the effect of the k th factor on returns and the target factor g_t . We then repeat S1 again, this time using the residuals of returns and g_t ,

²³This one-step procedure was originally called Supervised PCA, as proposed by Bair et al. (2006) in the context of prediction. We propose below an iterative version that can cope with a general multi-factor model. We still use the term Supervised PCA for this iterative procedure.

looking for the next factor. Iteration continues for $\hat{p}$ steps. At the end, step S2 combines the $\hat{\gamma}_{(k)}$ and the $\hat{\eta}_{(k)}$ obtained at each step into an estimator $\hat{\gamma}_g^{SPCA}$ for the risk premia of g_t .

Algorithm 3 requires an appropriate choice of $\hat{I}_k$ and a stopping rule. One possible choice for $\hat{I}_k$ is:²⁴

$$\hat{I}_k = \left\{ i \mid T^{-1} \left\| (\bar{R}_{(k)})_{[i]} \bar{G}_{(k)}^\top \right\|_{\text{MAX}} \geq c_q^{(k)} \right\},$$

where $c_q^{(k)}$ is the $(1 - q)$ th-quantile of $\left\{ T^{-1} \left\| (\bar{R}_{(k)})_{[i]} \bar{G}_{(k)}^\top \right\|_{\text{MAX}} \right\}_{i \in \langle N \rangle}$.

(9)

Correspondingly, we set the stopping criterion as:

$$c_q^{(k)} < c, \quad \text{for some threshold } c.$$
(10)

In other words, we select test assets that have predictive power for at least one variable in g_t and stop when most test assets are uncorrelated with all variables in g_t . With a good choice of tuning parameters, q and c , the iteration stops as soon as most projected residuals of returns appear uncorrelated with the projected residuals of g_t , which implies that all factors that are correlated with g_t are successfully recovered.

It is helpful to revisit the aforementioned examples and understand how the new procedure fixes issues with the one-step SPCA. Recall that in example (7), $\beta_{22} = 0$ and $g_t = v_{1t} + v_{2t}$. As discussed previously, screening will select a subset of q assets that are spread across both groups of assets since they are all correlated with g_t . Consequently, applying PCA to them will identify a factor that is in general spanned by v_{1t} and v_{2t} . Even if this first step only recovers the strong factor v_{1t} , once we project r_t and g_t onto this factor following Algorithm 2, both residuals should only depend on v_{2t} . Subsequently, applying screening again to these residuals will leave us with only the test assets within the first group of assets, to which applying PCA can recover v_{2t} . In cases where a linear combination of v_{1t} and v_{2t} are recovered in the first step, after projection the residuals feature a strong factor (again a linear combination of v_{1t} and v_{2t} but orthogonal to the first linear combination), since the second group of $N - N_0$ assets have exposure to it. Therefore, a subsequent screening and PCA suffice to recover this factor.

Similarly in example (8), the second half of the assets will be eliminated in the first step when using $g_t = v_{1t}$ to screen test assets. The returns for the remaining (first half) assets load on $v_{1t} + v_{2t}$ with a common loading matrix β_{11} . Applying PCA to these assets thereby finds $(v_{1t} + v_{2t})/\sqrt{2}$ as the first factor (up to a sign, assuming v_{1t} and v_{2t} share the same variance). Following Algorithm 2, we then obtain residuals from projections of r_t and g_t onto this factor. It is easy to see that the residuals

²⁴Using covariance for screening allows us to replace all $\bar{G}_{(k)}$ in the definition of $\hat{I}_k$ and Algorithm 3 by $\bar{G}$, that is, only the projections of $\bar{R}_{(k)}$ and $\bar{r}_{(k)}$ are needed, because this replacement would not affect the covariance between $\bar{G}_{(k)}$ and $\bar{R}_{(k)}$, and in turn, the test assets after screening and the estimates of $\hat{\eta}_{(k)}$. We use this fact in the proofs, which simplifies the notation. We can also use correlation instead of covariance in constructing $\hat{I}_k$. While this does not affect the asymptotic analysis, we find correlation screening performs slightly better in finite samples.

of the second half of r_t and the residuals of g_t both load on a single strong factor $(v_{1t} - v_{2t})/\sqrt{2}$ yet the first half of the residuals are purely idiosyncratic. Applying screening plus PCA will successfully recover this factor, and hence the span of the factor space.

To formally establish the consistency of this estimator, we introduce an assumption akin to the single factor case. Specifically, we require that a subset of assets, indexed by I_0 , satisfies that all factors are strong within this subset. In other words, $\lambda_{\min}(\beta_{[I_0]}^\top \beta_{[I_0]}) \asymp N_0$, where $N_0 = |I_0| \rightarrow \infty$. Because the number of factors, p , is finite, such a subset I_0 always exists as long as for each factor we can locate a sufficiently large subset, respectively, within which this factor can be extracted consistently.²⁵ Proposition C.4 of the online appendix establishes that test assets in such a subset suffice to serve as basis assets, building on which a mimicking portfolio can approximate the risk premia of any observable factor. With this identification assumption, along with moment conditions given in the appendix, the following theorem establishes the consistency of the SPCA estimator:

Theorem 1. *Suppose that test asset returns in r_t follow (1), the factor proxies in g_t satisfy (4), and that Assumptions B.1-B.8 hold. If $\log(NT)(N_0^{-1} + T^{-1}) \rightarrow 0$ then for any tuning parameters c and q that satisfy*

$$c \rightarrow 0, \quad c^{-1}(\log NT)^{1/2}(q^{-1/2}N^{-1/2} + T^{-1/2}) \rightarrow 0, \quad qN/N_0 \rightarrow 0, \quad (11)$$

we have $\hat{\gamma}_g^{SPCA} \xrightarrow{P} \eta\gamma$.

The screening step in Algorithm 3 ensures that the selected test assets or their residuals must encompass one strong factor, as they have high correlations with g_t . As the SPCA procedure unfolds, each iteration selects a distinct subset of test assets. By amalgamating all such subsets, we obtain a subset of assets within which all factors are potentially strong, given that the number of factors is finite. However, this procedure may not recover all factors that drive returns. The number of factors that SPCA can recover depends on the interplay between η and β as well as the tuning parameters in a complex manner.²⁶ Some of the factors that SPCA omits might even be strong! Intuitively, only factors correlated with g_t are guaranteed to be recovered. This is the trade-off that arises for using g_t as a supervisory signal.²⁷ Nonetheless, missing any factors in the SDF that are uncorrelated with g_t does not affect the consistency of the estimate of the risk premia of g_t . This holds true because such factors do not help price g_t . Of course, this result will need to be strengthened if the objective is to recover the entire SDF, a problem we tackle in Section 2.3.

It is also worth mentioning that this consistency result does not rely on Gaussian error assumptions nor on an assumption that all factors have the same strength with respect to all test assets. The assumption on the relative size of N and T is also quite flexible, in contrast with existing results

²⁵This assumption is weak in that it does not imply all factors should have identical strength with respect to the entire cross-section of assets in r_t . In addition, different groups of assets could be exposed to different factors.

²⁶We explicitly characterize this number, denoted by $\tilde{p}$, given in the online appendix following Assumption B.7.

²⁷In the context of forecasting, Giglio et al. (2023) provide convergence rate of the estimated factor space, spanned by the factors that are correlated with the variables used for supervision in a similar SPCA procedure.

on factor models in the literature, where N cannot grow at a rate exceeding a certain polynomial function of T .

2.2.4 Asymptotic Inference on Risk Premia

In this section we develop the asymptotic distribution of the risk premia estimator from Algorithm 3. Naturally, deriving asymptotic inference requires stronger assumptions than those required for consistency discussed above. To consistently estimate the risk premia of g_t , one only needs recover factors that are correlated with g_t . Nonetheless, if SPCA misses factors that are in the SDF but are not correlated with g_t , consistency is maintained, but inference is undermined, because the omitted factors may contribute a higher-order error that invalidates the central limit result.

More specifically, the conditions in Theorem 1 do not guarantee that $\hat{\gamma}_g^{SPCA}$ converges to $\eta\gamma$ at the desirable rate $T^{-1/2}$. The major obstacle lies in the recovery of factors not strongly correlated with g_t , which we can explain with the previous single-factor example.

Recall that we use the sample correlation/covariance between r_t and g_t to screen test assets. Condition (11) necessitates two key considerations: First, it requires that $c \rightarrow 0$, allowing the iteration procedure to continue until the selected r_t exhibit asymptotically diminishing correlations with g_t . Simultaneously, it demands that $c\sqrt{T} \rightarrow \infty$ and $c\sqrt{qN} \rightarrow \infty$. In other words, c must be sufficiently large to supersede the estimation error in covariance estimates during the screening step, which is of order $T^{-1/2}$,²⁸ and to dominate error in the construction of residuals in the projection step when multiple steps are involved, an error of order $T^{-1/2} + (qN)^{-1/2}$.

However, for any given threshold, say, $c = T^{-1/4}$, if it happens that $\eta \asymp T^{-1/3} < T^{-1/4}$, then screening based on g_t 's correlation with r_t will likely not select any assets, which in turn leads to the termination of Algorithm 3 and no discovery of factors. Our procedure thereby gives a risk premium estimate of 0, which is certainly consistent, but the estimation error is of an order $T^{-1/3} > T^{-1/2}$, so that the usual central limit theorem (CLT) fails.

In general, this problem arises due to the possibility of not identifying all factors in the DGP. Once all factors are recovered, the central limit theorem holds regardless of the magnitude of η . Therefore to make correct inference, we need a stronger assumption that eliminates scenarios like this.

It appears that if $\eta \in \mathbb{R}^{d \times p}$ meets the condition $\lambda_{\min}(\eta^\top \eta) \gtrsim 1$, we can rule out the possibility of missing factors. This condition necessitates that each latent factor maintains a correlation with at least one of the observable variables within g_t . Consequently, this implies that d must be greater than or equal to p , meaning we require g_t to possess at least the same number of variables as the true number of factors. Meanwhile, our algorithm will not select more factors than needed, as we stop the iteration as soon as $c_q^{(k)}$ is sufficiently small (below c), at which points no common factors are left in the residuals of g_t and r_t . We thus obtain the consistency result on the number of factors,

²⁸Even if g_t is uncorrelated with the test assets, their sample covariances can be as large as $T^{-1/2}$.

which in turn leads to the CLT result on risk premia. Formally, we have

Theorem 2. *Under the same assumptions as Theorem 1, if we further have $T^{-1/2}N_0 \rightarrow \infty$, Assumption B.9 and $\lambda_{\min}(\eta^\top \eta) \gtrsim 1$, then for any tuning parameters c and q in (9) and (10) satisfying*

$$c \rightarrow 0, \quad c^{-1}(\log NT)^{1/2}(q^{-1/2}N^{-1/2} + T^{-1/2}) \rightarrow 0, \quad qN/N_0 \rightarrow 0, \quad q^{-1}N^{-1}T^{1/2} \rightarrow 0,$$

we have that $\hat{p}$ defined in Algorithm 3 satisfies $P(\hat{p} = p) \rightarrow 1$, and that the estimator constructed via Algorithm 3 satisfies $\hat{\gamma}_g^{SPCA} - \eta\gamma = O_P(T^{-1/2}) + O_P(q^{-1}N^{-1})$. Furthermore, we obtain a central limit result:

$$\sqrt{T}(\hat{\gamma}_g^{SPCA} - \eta\gamma) \xrightarrow{d} \mathcal{N}(0, \Phi),$$

where Φ is given by

$$\Phi = (\gamma^\top \Sigma_v^{-1} \otimes \mathbb{I}_d) \Pi_{11} (\Sigma_v^{-1} \gamma \otimes \mathbb{I}_d) + (\gamma^\top \Sigma_v^{-1} \otimes \mathbb{I}_d) \Pi_{12} \eta^\top + \eta \Pi_{12}^\top (\Sigma_v^{-1} \gamma \otimes \mathbb{I}_d) + \eta \Pi_{22} \eta^\top,$$

and Π_{11} , Π_{12} , and Π_{22} are specified by Assumption B.9.

In regard to our theoretical findings, several key points merit attention. Firstly, Theorem 2 hinges on the existence of a tuning parameter, q , which must satisfy two conditions: $q^{-1}N^{-1}T^{1/2} \rightarrow 0$ and $qN/N_0 \rightarrow 0$. A necessary condition for the existence of such a q is thus $q^{-1}N^{-1}T^{1/2} \times qN/N_0 = T^{1/2}/N_0 \rightarrow 0$.

Secondly, the estimation error of $\hat{\gamma}_g^{SPCA} - \eta\gamma$ consists of two components. A portion of this error stems from the error accumulation at each step of the iteration in Algorithm 3. This accumulated error is compounded in each step k at most by a factor of $\sqrt{|\hat{I}_k|/\hat{\lambda}_{(k)}}$, where $\hat{\lambda}_{(k)} = \left\| (\bar{R}_{(k)})_{[\hat{I}_k]} \right\|^2 / T$. Importantly, the assumption that there exists a subset within which factors are pervasive ensures that $\hat{\lambda}_{(k)} \asymp_P qN = |\hat{I}_k|$, implying that the accumulated error is only magnified by a constant factor with each iteration of SPCA. Ultimately, our proof establishes that this iterative process results in an overall estimation error in risk premia estimates, that is of the order $O_P(T^{-1/2} + q^{-1}N^{-1})$. The condition $q^{-1}N^{-1}T^{1/2} \rightarrow 0$ thus guarantees that the $O_P(q^{-1}N^{-1})$ term does not influence the asymptotic distribution. The derivation of the error rate for an iterative procedure is non-trivial, constituting our primary contribution to the econometric literature on factor models.

Thirdly, the estimation error of the factor loading has no impact on the asymptotic variance of risk premia, as the expression of Φ demonstrates. This stands in contrast to the classical Fama-MacBeth regression setting, where the Shanken's adjustment term (Shanken (1992)) is a crucial factor. This difference is due to the fact that when dealing with a large cross-sectional dimension ($N \rightarrow \infty$), it can be shown that the Shanken's adjustment term vanishes asymptotically.²⁹ To make inference

²⁹For a more detailed discussion on this point, please refer to equation 45 of Giglio et al. (2022), and the discussion that follows it.

feasible, we implement the same Newey-West-type estimator for Φ as in Section 4.5 of Giglio and Xiu (2021), since each component of Φ can be estimated from the outputs of the SPCA algorithm. These estimates are consistent up to some rotation matrices which will cancel each other and yield a consistent estimate of Φ .

Fourthly, Theorem 2 suggests that, with a high probability, we can expect a perfect recovery of the number of factors p . It is important to note that in any finite sample, achieving a perfect recovery can be challenging. However, the assumptions made here are considerably less stringent compared to the prevalent factor assumptions found in the literature, as seen in works like Bai (2003) and Bai and Ng (2002). In these previous studies, inference theory for factor models also relies on the perfect recovery of the count of (strong) factors. We delve into an exploration of the finite sample behavior of SPCA in simulations in Section 3.

Lastly, in the special scenario when the returns of test assets are *exclusively* driven by strong factors, SPCA is asymptotically equivalent to PCA, contingent upon the appropriate selection of the tuning parameters c and q . Otherwise, SPCA is less efficient – either due to an excessively small tuning parameter q to the extent that the $O_P(q^{-1}N^{-1})$ term plays a dominant role in the estimation error in finite sample (note that PCA corresponds to the case of $q = 1$), or to the fact that some factors (specifically those uncorrelated with g_t) may not be recovered by SPCA. The former loss of efficiency can be mitigated through careful tuning parameter selection; the latter typically hinges on the unknown values of β and η , which can be resolved with a multivariate target satisfying $\lambda_{\min}(\eta^\top \eta) \gtrsim 1$.

2.2.5 The Case of Observable Factors

The previous discussion does not assume any knowledge of the identities of the factors v_t in (1). If v_t corresponds to innovations of known and observable factors, denoted by f_t , say, the Fama-French five factors, our procedure can be greatly simplified. It is meaningful to study this case, because it is common in the empirical literature, albeit having perfect knowledge of the factor model is a rather strong assumption.

Suppose first that factors in f_t are tradable. If the factor of interest g_t is one of the factors in f_t (therefore also tradable), then we can estimate the risk premium of g_t by simply taking its time-series average. If g_t is either spanned by f_t or not tradable, then a simple time series regression of g_t onto the factors f_t can recover its loading, η , which along with the risk premia estimates of f_t by their time-series averages yields the risk premium estimate of g_t . These scenarios are simple, and do not require cross-sectional regressions.

If some of the observed factors in f_t are not tradable, say, GDP growth, then a cross-sectional regression is necessary, which effectively constructs the mimicking portfolios for the non-tradable factors. In this setting, a weak factor problem potentially arises as documented in the literature, see, e.g., Kan and Zhang (1999), Kleibergen (2009). To tackle this issue, one could adopt a simplified

version of Algorithm 3, to supervise the construction of mimicking portfolios for each of the observed non-tradable factors (in this case GDP growth), while using residuals from the projection of test asset returns onto tradable factors as new test assets.

2.2.6 The Case of Unknown Zero-beta Rate

We have so far considered the case where the zero-beta rate is known. When it is not known, we need modify our SPCA procedure slightly. Suppose that the DGP of returns follows

$$r_t = \gamma_0 \iota + \beta \gamma + \beta v_t + u_t, \quad (12)$$

where γ_0 is the zero-beta rate, and ι is a vector of 1s.

To proceed, we multiply $\mathbb{M}_t = \mathbb{I}_N - N^{-1} \iota \iota^\top$, from the left on both sides of equation (12). This results in a similar form of (1):

$$\tilde{r}_t = \tilde{\beta} \gamma + \tilde{\beta} v_t + \tilde{u}_t,$$

where $\tilde{a}_t = \mathbb{M}_t a_t$, for $a = r, \beta$, and u . Subsequently, we can readily apply Algorithm 3 to the transformed returns, $\tilde{r}_t$. To better grasp the reasoning behind this adjustment, let us consider a one-factor scenario. Choosing assets with strong absolute correlations with $\tilde{r}_t$ amounts to selecting assets characterized by large magnitudes of $\tilde{\beta}$. This choice, in turn, leads to the selection of assets that exhibit high cross-sectional dispersion in their β values.

2.2.7 Tuning Parameter Selection

While the enhanced robustness to weak factors is an advantage, it comes at the expense of introducing an additional tuning parameter. To employ the SPCA estimator, we need make choices regarding two tuning parameters: q and c . The parameter q governs the subset size employed in PCA construction, while c determines the stopping rule and consequently the number of factors, p . In contrast, PCA (and other estimators like PLS) essentially require the selection of only p . We have established in Theorem 2 that we can consistently recover p , as long as certain conditions are met by q and c .

In theory, the textbook approach to choosing a tuning parameter for parameter estimation revolves around the analytical minimization of the root-mean-squared error (RMSE) of the estimator.³⁰ This approach effectively balances the trade-off between bias and variance inherent in the estimation. Regrettably, this method necessitates intricate finite sample analytical calculations of the RMSE, often relying on strong assumptions regarding the DGP. In our context, assumptions of normal distribution for returns and certain distributional properties and sparsity conditions for betas are likely necessary. Complicating matters further, our iterative SPCA procedure compounds the

³⁰Note that in the realm of machine learning, the prevailing approach involves leaning on the prediction RMSE derived from a validation sample, where the actual values of the prediction target are available. This stands in contrast to the estimation problem, where the true values are never known.

difficulty of this analysis, rendering it practically infeasible. Additionally, this RMSE-based criterion primarily hinges on statistical considerations, lacking economic relevance.

In lieu of this, we instead opt for the utilization of the R^2 of the hedging portfolio for g_t built by SPCA as a criterion, which is both simpler to apply and justified from an economic perspective. Recall that any estimator of risk premia for a nontradable factor explicitly or implicitly builds a hedging portfolio, or a factor-mimicking portfolio, for g_t , and computes the risk premium as the average excess return of that portfolio. The empirical R^2 obtained by different estimators then has an economic meaning: it reveals the hedging efficacy of the factor mimicking portfolios constructed (explicitly or implicitly) by any risk premia estimators.³¹

Beyond the economic motivation, the R^2 is a useful criterion from a statistical perspective, because attaining an optimal R^2 in a validation sample stands as a sufficient condition for valid selection of tuning parameters, which in turn guarantees consistency of risk premia estimates, see Proposition C.5 in the online appendix for a rigorous statement. To provide an intuitive illustration of this result, consider the following: When an estimator attains the maximal R^2 , it must have recovered all the driving factors behind g_t . This holds true regardless of whether g_t loads onto all or just a subset of the factors within v_t . Conversely, an estimator which delivers an R^2 lower than this must miss some of the variation in g_t explainable by v_t . Whether that yields an inconsistent estimate of the risk premium depends on which part of the variation is missed by the hedging portfolio.

Furthermore, in practice we can consider directly tuning the parameter p instead of c , as it offers greater interpretability, restricts itself to integer values, and is well-informed by the scree plot, providing insights into reasonable ranges for p . Regarding the parameter q , opting for larger values makes SPCA's performance resemble that of PCA, thus reducing its robustness against weak factors. Conversely, smaller values of q raise the risk of overfitting, resulting in a high in-sample R^2 but a low out-of-sample one. We suggest tuning $\lfloor qN \rfloor$ instead of q , because the former can only take integer values, and that multiple choices of the latter may lead to the same integer values of the former.

In our applications, we select tuning parameters based on 3-fold cross-validation in a training sample, that proceeds as follows. We split the sample into 3 folds. We then use each of the three folds, in turn, for validation while the other two are used for training. We select the optimal tuning parameters according to the average time series R^2 in the validation folds.

2.3 Recovery of the Stochastic Discount Factor

The estimation of risk premia for observable factors g_t , studied in Section 2.2, is a natural application of the supervised PCA approach, since g_t can be used to supervise the latent factor extraction. In this section we explore another application in which observable factors help extract latent factors: a *diagnostic procedure* for observable factor models.

³¹To be clear, while comparing R^2 s provides an insightful depiction of the empirical performance of the hedging portfolios, this cannot be interpreted as proof of the superiority of one estimator over another (which is instead established based on the theoretical properties, like consistency and efficiency, discussed in the previous sections).

The asset pricing literature has proposed a variety of models composed of a small number of tradable factors g_t : the CAPM, the Fama-French 3 or 5 factor models, etc. These models are typically evaluated by computing the alphas of a universe of test assets, and testing whether these alphas are different from zero. This is clearly a valid test for a model, but it gives only limited insights about the reason why the model is (as is often the case) rejected statistically. In particular, these tests do not distinguish whether the model fails because of the presence of true alphas, or because the model is missing some factors that are priced in the selected cross-section. Our SPCA procedure helps shed light on this by recovering strong and weak latent factors that drive the cross-section of returns, and evaluating whether those factors are indeed spanned by the observable factor model g_t (and therefore the model is missing some factors).

A last point relates to the universe of test assets. The asset pricing literature (e.g. [Lewellen et al. \(2010\)](#)) has emphasized that using a large cross-section of test assets is important for evaluating asset pricing model, as it can improve the power of the tests. There is, however, a downside in expanding the set of test assets: the possibility that many of the added assets may have little exposure to some factors, introducing a weak factor problem. The ability of SPCA to handle weak factors also frees the researcher from worrying about adding assets to the universe, not only in risk premia estimation, but also in performing diagnostic tests like the one we explore in this section.

2.3.1 Consistency of the SDF Estimator

We first prove that, under certain conditions, SPCA does consistently recover the SDF even in the presence of weak factors. Using the outputs of Algorithm 3, we can estimate the SDF as:

$$\hat{m}_t^{SPCA} = 1 - \hat{\gamma}^\top \hat{v}_t, \quad \text{where } \hat{v}_1, \dots, \hat{v}_T \text{ are the columns of } \hat{V}. \quad (13)$$

In the Appendix, we prove the following theorem, which not only shows the consistency of the recovery of the SDF, but also derives the rate at which the recovery occurs.

Theorem 3. *Suppose the same assumptions as in Theorem 2 hold. In addition, we have Assumption B.10. Then the estimator (13) satisfies*

$$\frac{1}{T} \sum_{t=1}^T |\hat{m}_t^{SPCA} - m_t|^2 \lesssim_P \frac{1}{T} + \frac{\log N_0}{N_0}. \quad (14)$$

As the theorem illustrates, it is possible to consistently estimate the entire time-series of the SDF in terms of the average ℓ_2 -distance. However, to achieve this, several conditions must be met. Firstly, for every weak latent factor in v_t , there must be a sufficiently large subset of assets with exposure to that factor. This requirement, denoted by N_0 being large, is also necessary for the consistent estimation of risk premia.

In addition, for each latent factor in v_t , there must be at least one observable factor in g_t that

is correlated with that latent factor. This second assumption is not only needed for asymptotic inference on risk premia but also for SDF recovery. In cases where g_t does not correlate with a latent factor, that latent factor can potentially be missed by SPCA, thereby hindering SDF recovery.

2.3.2 Comparison with Alternative Procedures of SDF Estimation

There are a number of alternative approaches for SDF estimation with latent factors proposed in the literature, e.g., the selection/shrinkage approach by [Kozak et al. \(2020\)](#) and the risk premia PCA by [Lettau and Pelger \(2020\)](#). In what follows, we provide a theoretical comparison of Lasso- and Ridge-based estimators in our general framework where factors can potentially be weak. The ridge estimator shares the same spirit of PCA-based estimators as shown by [Giglio and Xiu \(2021\)](#) and propositions in previous sections. Examining the asymptotic behavior of these two approaches will provide useful insights that may guide their applications in practice. Developing the asymptotic guarantee of these estimators is yet another contribution we make to the existing literature on SDF recovery.

[Kozak et al. \(2020\)](#) consider an SDF in the form of (3), whereas we represent it as in (2). Prior to the asymptotic analysis of their estimators, we first establish the asymptotic equivalence of these two definitions in our large- N setting:

Proposition 3. *Suppose that test asset returns in r_t follow (1), and Assumption B.10 holds. Then as $N \rightarrow \infty$, we have*

$$\frac{1}{T} \sum_{t=1}^T |m_t - \tilde{m}_t|^2 \lesssim_P \frac{1}{\lambda_{\min}(\beta^\top \beta)}.$$

Effectively, Proposition 3 proves that there is no ambiguity with respect to the definition of the estimand, since the two estimands are asymptotically equivalent as long as $\lambda_{\min}(\beta^\top \beta) \rightarrow \infty$. Given that this exact assumption is necessary for Theorem 3, and that $\lambda_{\min}(\beta^\top \beta) \gtrsim N_0$, we can replace m_t in the left-hand side of (14) by $\tilde{m}_t$.

[Kozak et al. \(2020\)](#) suggest estimating the SDF by solving an optimization problem:

$$\hat{b} = \arg \min_b \left\{ (\bar{r} - \hat{\Sigma}b)^\top \hat{\Sigma}^{-1} (\bar{r} - \hat{\Sigma}b) + p_\mu(b) \right\}, \quad (15)$$

with which the estimated SDF is given by

$$\hat{m}_t = 1 - \hat{b}^\top (r_t - \bar{r}). \quad (16)$$

In the above, $\hat{\Sigma}$ is the sample covariance matrix of r_t and $p_\mu(b)$ is a penalty term through which economic priors are imposed. Depending on the penalty function, we will denote the resulting estimator of m by $\hat{m}_t^{Ridge}$ or $\hat{m}_t^{Lasso}$.

The objective function in (15) appears to require the inverse of the sample covariance matrix $\widehat{\Sigma}^{-1}$, which is not well-defined when $N > T$. Instead, we suggest optimizing an equivalent but different form of (15):

$$\widehat{b} = \arg \min_b \left\{ b^\top \widehat{\Sigma} b - 2b^\top \bar{r} + b^\top \widehat{\Sigma} b + p_\mu(b) \right\}, \quad (17)$$

which avoids the calculation of $\widehat{\Sigma}^{-1}$.

The following result sheds light on the asymptotic properties of this estimator in the cases of $p_\mu(b) = \mu \|b\|_1$ and $p_\mu(b) = \mu \|b\|^2$, respectively.³²

Theorem 4. *We investigate two distinct scenarios.*

- (a) *Suppose that r_t is driven by p latent factors as in (1). With $p_\mu(b) = \mu \|b\|^2$, if $(N+T)/(\lambda_p T) \rightarrow 0$ and Assumptions B.4-B.7, B.10-B.12 hold, we have*

$$\frac{1}{T} \sum_{t=1}^T |\widehat{m}_t^{\text{Ridge}} - m_t|^2 \lesssim_P \frac{1}{T} + \frac{N+T}{\lambda_p T},$$

where λ_p is the p -th largest eigenvalue of $\beta \Sigma_v \beta^\top$. Since $\lambda_p \asymp \lambda_{\min}(\beta^\top \beta)$, we can replace m_t in the above equation by $\tilde{m}_t$.

- (b) *Suppose that the true SDF satisfies $E(\tilde{m}_t^2) \lesssim 1$. With $p_\mu(b) = \mu \|b\|_1$, if Assumptions B.10, B.11 hold, we have*

$$\frac{1}{T} \sum_{t=1}^T |\widehat{m}_t^{\text{Lasso}} - \tilde{m}_t|^2 \lesssim_P \|b\|_1 \sqrt{\frac{\log N}{T}}. \quad (18)$$

If, in addition, we have that $\lambda_{\min}(\Sigma) \gtrsim 1$, and $\|b\|_0^2 \log N/T \rightarrow 0$, then we have a stronger result

$$\frac{1}{T} \sum_{t=1}^T |\widehat{m}_t^{\text{Lasso}} - \tilde{m}_t|^2 \lesssim_P \|b\|_0 \frac{\log N}{T}. \quad (19)$$

Interestingly, both the Ridge and Lasso approaches deliver consistent estimates of the SDF, albeit under distinct sets of assumptions.

In the case of the Ridge approach, its convergence rate hinges significantly on the strength of the weakest factor. If condition (6) is not met, the SDF consistency is compromised. The failure of this condition is a clear symptom of weak factors, precisely the type of scenario for which our SPCA estimator is designed.

Furthermore, when considering the estimator utilizing the Lasso penalty, it replaces the explicit factor model assumption on r_t with a sparsity assumption on the vector b . This sparsity assumption

³²We use $\|\cdot\|_0$, $\|\cdot\|_1$, and $\|\cdot\|$ to denote the ℓ_0 -, ℓ_1 -, and ℓ_2 -norms of a vector, respectively.

dictates that the SDF should be represented as a sparse linear combination of the test assets but imposes no explicit assumptions on the DGP of these test assets. This implies that the Lasso estimator remains consistent regardless of the strength of the factors but converges at a rather slow rate, as indicated in (18), which is $\|b\|_1 \sqrt{\log N/T}$. Consequently, it is not as efficient as our SPCA estimator, which leverages the factor structure to achieve faster convergence. Nevertheless, under a much stronger sparsity assumption where $\|b\|_0^2 \log N/T \rightarrow 0$, the Lasso estimator can attain a comparable convergence rate to that of the SPCA. This more stringent notion of sparsity essentially asserts that the set of true factors must be part of the test assets. In contrast, our SPCA estimator allows for the presence of idiosyncratic components in any of the test assets, making it a more practical assumption in real-world applications.

We can adapt any SDF estimator to obtain an estimator of risk premia, because of the relationship $-\text{Cov}(m_t, g_t) = \eta\gamma$. In light of this, we have a Lasso-based risk premia estimator:³³

$$\hat{\gamma}_g^{Lasso} = -\frac{1}{T} \sum_{t=1}^T \hat{m}_t^{Lasso} \times (g_t - \bar{g}).$$

Furthermore, the consistency of the SDF estimator translates to the consistency of the resulting risk premia estimator.³⁴ Deriving a valid inference procedure is possible for Lasso-based risk premia estimator, if we employ an additional de-biasing step, see, Feng et al. (2020), which is beyond the scope of the current paper.

2.3.3 Diagnosis of SDF Models using Sharpe Ratios

We now discuss the diagnosis of SDF models that consist of tradable factors exclusively. Recall that the projection of the SDF on the space of returns achieves the highest possible Sharpe ratio. Given that the factors recovered by SPCA are themselves portfolios, as long as SPCA recovers the entire SDF these factors should achieve the maximal Sharpe ratio. We can then diagnose a model g_t by comparing its Sharpe ratio with that achieved by the estimated SDF supervised by g_t . If g_t contains all the factors that drive the SDF, then the maximal Sharpe ratio achieved by factors in g_t should be on par with the Sharpe ratio of the SDF. Otherwise, if g_t achieves a lower Sharpe ratio, it is a sign that g_t is missing some factors; if g_t 's Sharpe ratio is higher than that achieved by SPCA, it indicates that g_t has alpha relative to the entire cross-section of test asset returns.

For this purpose, it is more convenient to rewrite our SPCA estimator of the SDF given by equation (13) in the form of portfolio returns as in (16), so that we can directly evaluate its Sharpe

³³The SDF-induced Ridge estimator is numerically equivalent to (A.1), so we do not introduce it again.

³⁴By Assumption B.11(1), Cauchy-Schwartz and triangle inequalities, we have

$$\|\hat{\gamma}_g^{Lasso} - \gamma_g\|_{\text{MAX}} \lesssim_P \sqrt{\frac{1}{T} \sum_{t=1}^T |\hat{m}_t^{Lasso} - \tilde{m}_t|^2} + \sqrt{\frac{\log N}{T}}.$$

ratio. In other words, we need an SPCA based estimate of b in the definition of SDF given by equation (3). Formally, we provide the following algorithm:³⁵

Algorithm 4. *The SPCA based procedure for estimating SDF loadings is as follows:*

Inputs: $\bar{R}_{(1)} := \bar{R}$, $\bar{r}_{(1)} := \bar{r}$, and $\bar{G}_{(1)} := \bar{G}$, a $d \times T$ vector.

S1. For $k = 1, 2, \dots$ iterate the following steps using $\bar{R}_{(k)}$, $\bar{r}_{(k)}$, and $\bar{G}_{(k)}$ and construct an $N \times p$ matrix B :

- a. Run S1.a of Algorithm 3 to obtain $\hat{I}_k$*
- b. Run S1. - S3. of Algorithm 1 with selected return matrix $(\bar{R}_{(k)})_{[\hat{I}_k]}$ and $\bar{G}_{(k)}$. Construct the k th column of B as: $B_{[\hat{I}_k],k} = \varsigma_{(k)}$ and $B_{[\hat{I}_k^c],k} = 0$, where $\varsigma_{(k)}$ is the left singular vector of $(\bar{R}_{(k)})_{[\hat{I}_k]}$. Also, obtain $\hat{V}_{(k)}$ and $\hat{\eta}_{(k)}$.*
- c. Run S1.c of Algorithm 3 to obtain $\hat{\beta}_{(k)}$.*
- d. Run S1.d of Algorithm 3 to obtain $\bar{R}_{(k+1)}$ and $\bar{G}_{(k+1)}$.*

Stop at $k = \hat{p}$, where $\hat{p}$ is chosen based on some proper stopping rule.

S2. Estimate the SDF loading b as:

$$\hat{b}^{SPCA} = TB (B^\top \bar{R} \bar{R}^\top B)^{-1} B^\top \bar{r}. \quad (20)$$

Outputs: $\hat{b}^{SPCA}$

Similarly, we can construct estimates of b using PCA and PLS.³⁶ With $\hat{b}$ it is convenient to build SDFs (optimal portfolios) and evaluate their Sharpe ratio.

Theorem 5. *Under the same assumptions as Theorem 1, if Assumption B.10 holds, the Sharpe ratio of the optimal portfolio constructed by $\hat{b}^{SPCA}$ in (20) satisfies*

$$\sqrt{\gamma^\top \Sigma_v^{-1} \gamma} \geq \lim_{N, T \rightarrow \infty} \frac{\hat{b}^{SPCA \top} \mathbf{E}(r_t)}{\sqrt{\hat{b}^{SPCA \top} \Sigma \hat{b}^{SPCA}}} \geq \sqrt{\gamma^\top \eta^\top (\eta \Sigma_v \eta^\top)^\dagger \eta \gamma}, \quad (21)$$

where $\dagger$ denotes the Moore–Penrose inverse of a matrix.

³⁵The effectiveness of this procedure stems from the fact that the SPCA estimates of $\hat{V}$ can be written as a rotation of $B^\top \bar{R}$. Given that b is invariant to rotations of factors, we can exploit this invariance property to construct a convenient estimator $\hat{b}$. To elaborate, if we use $B^\top \bar{R}$ as the factors, denoted by, $\tilde{V}$, with their risk premia and covariance denoted as $\tilde{\gamma}$ and $\tilde{\Sigma}$ respectively, we can express the SDF as $m_t = 1 - \tilde{\gamma}^\top (\tilde{\Sigma}_v)^{-1} \tilde{v}_t = 1 - \tilde{\gamma}^\top (\tilde{\Sigma}_v)^{-1} \tilde{v}_t = 1 - \tilde{\gamma}^\top (\tilde{\Sigma}_v)^{-1} B^\top (r_t - \bar{r})$. Consequently, we can deduce that:

$$\hat{b} = B(\tilde{\Sigma}_v)^{-1} \tilde{\gamma} = B\left(\frac{1}{T} B^\top \bar{R} \bar{R}^\top B\right)^{-1} B^\top \bar{r}.$$

³⁶For PCA, the k th column of B can be chosen as the left singular vectors of $\bar{R}$, then (20) yields the standard PCA based SDF loadings. For PLS, B is a similar weight matrix given by the iterative procedure. We compare these SDF estimators in simulations.

In the inequality (21), the upper bound corresponds to the optimal Sharpe ratio of the SDF, while the middle term represents the optimal Sharpe ratio achieved by the SPCA estimator. Meanwhile, the lower bound corresponds to the optimal Sharpe ratio achieved by $\eta(v_t + \gamma)$. This lower bound also matches the bound attained by g_t , except for any undiversified idiosyncratic errors that may persist in g_t . These errors would further reduce the Sharpe ratio, but for the sake of our discussion exclusively on observable factor models in the literature, we follow the convention and assume that g_t comprises well-diversified portfolios, so we can ignore this aspect in this section. A sufficient condition for the upper and lower bounds to be equal is that $\lambda_{\min}(\eta^\top \eta) \gtrsim 1$. In this case, the SPCA-based SDF estimator also achieves the optimal Sharpe ratio. This result is not surprising, especially considering the consistency result outlined in Theorem 3.

Theorem 5 serves as the basis for diagnosing SDF models. We do not observe the left side of the equation (the true maximal Sharpe ratio), but can estimate and compare the middle term (Sharpe ratio obtained by the SPCA-recovered SDF) and the right term (Sharpe ratio of g_t). If we find in the data that the Sharpe ratio from SPCA is higher, then we learn that g_t must be missing a factor. If we instead find that the Sharpe ratio from g_t is higher, it means that there are factors in g_t that are insufficiently represented in r_t (for example, if none of the assets in r_t has exposure to those factors): this points to an insufficiently rich set of test assets r_t .³⁷

3 Simulations

In this section, we study the finite sample performance of our SPCA procedure using Monte Carlo simulations.

3.1 Results on Risk Premia

We implement a number of risk premia estimators for comparison, some of which are robust to omitted or weak factors, including PCA and its related estimators (Ridge, PLS, and rpPCA), Lasso, as well as the four-split estimator by Anatolyev and Mikusheva (2021).³⁸ Both the standard two-pass and four-split methods directly use g_t as if they were the true factors in their regressions. The PCA, rpPCA, Ridge, and Lasso effectively construct the SDF first without knowledge of g_t , then estimate the risk premia of g_t factor by factor, using the covariance between each factor and the resulting SDF. PLS and SPCA use all variables in g_t to supervise the estimation procedure.

To implement the SPCA estimator, we select the tuning parameters p and $\lfloor qN \rfloor$ by cross-validation using the procedure detailed in Section 2.2.7. To ensure a conservative basis for comparison, all methods, except for SPCA, use optimal (albeit infeasible) tuning parameters. Specifically,

³⁷Of course, it can also be that the two Sharpe ratios are the same. In that case, g_t and the latent-factor model recovered by SPCA are equivalent in terms of their pricing ability.

³⁸The four-split estimator, which does not rely on dimension reduction, selection, or shrinkage techniques, is valid in the presence of weak observable factors and strong omitted factors that are *not* priced. However, it does not have asymptotic guarantees against omitted and priced strong/weak factors, or measurement error in the observed factors.

for PCA, PLS and rpPCA, we make use of the true number of factors, $p = 4$, even though it is difficult to obtain a consistent estimator of p in the regime of weak factors. The tuning parameter μ of the Ridge estimator is determined via maximum likelihood estimation, with perfect knowledge of $\Sigma = \text{Cov}(r_t)$ and $E(r)$. The second tuning parameter of rpPCA is selected by maximizing the theoretical Sharpe ratio of the estimated SDF, using, again, perfect knowledge of Σ and $E(r)$. Due to limited sample size, estimating the sample mean and sample covariances in a separate validation sample is rather challenging, which would further deteriorate their performance.

To demonstrate and compare the performance of different estimators, we consider various DGPs of returns and/or the observed variables in g_t .

We start with the benchmark case (a), in which all factors are strong and observed. Specifically, we consider a 4-factor DGP as given by equation (1), where the first three factors are calibrated to match the three Fama-French factors (RmRf, SMB, HML) as in Giglio and Xiu (2021), and the last one is a potentially weak factor, denoted by V . We calibrate the parameters such that the monthly Sharpe ratio for the optimal portfolio out of these factors is about 0.25. The realizations of u_t are generated independently from a Gaussian distribution with mean 0 and a vector of standard deviations, σ_u .³⁹ We consider an inhomogenous setting where the cross-section of σ_u is generated independently from an inverse-gamma distribution so that the standard deviation of u_t varies from 0.01 to 0.09. For comparison, the standard deviations of the four factors are calibrated as 0.04, 0.03, 0.03, and 0.02. The loadings of RmRf are generated independently from $\mathcal{N}(1, 1)$ and the loadings of SMB and HML are generated independently from $\mathcal{N}(0, 1)$. We generate the exposure to the fourth factor V , $\beta_{i,V}$, independently from a Gaussian mixture distribution, with probability a from $\mathcal{N}(0, 1)$ and $1 - a$ from $\mathcal{N}(0, 0.1^2)$. Based on our calibration, we choose $a = 0.5$, so that the factor V is sufficiently strong with respect to the cross-section of assets in simulations. g_t includes exactly these four factors in the DGP (RmRF, SMB, HML, and V), and we set $\eta = \mathbb{I}_4$, and measurement error is absent.

In scenario b), we choose $a = 0.05$ so that V is weak in that for almost all test assets their factor loadings to V are tiny: only 5% of the assets have nontrivial exposure to this factor. In scenario c), the DGP is the same as that of the benchmark case, except that we add Gaussian measurement error, z_t , to each of the factors in g_t . In scenario d), we simulate β for V according to $\beta_{i,V} = -\beta_{i,HML} + e_i$ instead, where e_i s are generated independently from the same mixture Gaussian distribution as above with $a = 0.05$. In this case, the loading matrices of V and HML are very similar, which (almost) leads to a rank deficient factor loading matrix due to highly correlated exposures. The variable g_t contains all four factors with no measurement error. In scenario e), we consider the same DGP of returns as in scenario d), but in g_t we omit the HML factor. Finally, in scenario f), we further add measurement error to scenario d).

For each of these six scenarios (including the benchmark), we plot in Figure 1 the histograms of

³⁹Although it is conceivable to employ a more complex covariance matrix for u_t , calibrating such a model can be a challenging endeavor. We thereby simulate u_t s that are cross-sectionally uncorrelated for simplicity.

the estimated risk premium of V (one entry in g_t) for all estimators. If an estimator is consistent, then the histogram is expected to be centered around the true risk premium of V , whose value is represented by a vertical dashed line. This is indeed the case for SPCA in *all* scenarios. It is also the case for almost all estimators in the benchmark scenario, a), when factors are strong (except for Lasso and Ridge, which have a large shrinkage bias). This suggests that the latter two estimators are not suitable for *inference* on risk premia. Furthermore, in scenario b), when weak factors are present, only SPCA and four-split are consistent. The same is true for scenario d) in which a similar rank-deficiency issue arises. In scenario c) the four-split estimator becomes inconsistent due to measurement error, and it is also ill-behaved in scenario e) because the omitted variable, HML, is priced. The PCA and PLS estimators are consistent in scenario c) but also fail in e), because they are robust to measurement error but not to omitted weak factors. The standard two-pass estimator is only consistent in the benchmark scenario. Overall, the simulation evidence is in agreement with our theoretical predictions.

Next, we focus on the last scenario f), which includes the case of weak factors as well as measurement error. For this case, we report in Table 1 the bias and the RMSE (root-mean-square error) of all estimators for various sample size T . The four rows in each panel provide the results of risk premia estimation for RmRf, SMB, HML, and the weak factor V , respectively. We find that our SPCA approach has smaller biases for the weak factors, whereas the remaining estimators have larger biases and RMSEs, which agrees with our theoretical analysis and Figure 1. Notably, PLS ranks the second. Except for two-pass, all estimators perform better in terms of RMSE as T increases.

In the online appendix, we also report a scenario similar to c) except that the last factor is a pure noise. In other words, the DGP is driven by the first three factors, but econometricians, lacking knowledge of the true model, include these three factors alongside this pure noise variable in their attempt to estimate risk premia. This scenario closely resembles the one extensively discussed by Kan and Zhang (1999) and Kleibergen (2009). For the sake of comparison, PLS and SPCA incorporate this pure noise variable along with the aforementioned three factors into g_t . The histograms corresponding to the risk premium estimates associated with the noise factor suggest that SPCA, PCA, PLS, rpPCA, Lasso, and Ridge remain consistent and cluster around zero. The consistency stems from the fact that none of these methods involve a cross-sectional regression on the estimated beta for the noise factor. In contrast, the four-split and two-pass methods seem to exhibit considerable variances.

We then investigate the finite sample performance of the inference result developed in Theorem 2. Figure 2 plots histograms of the standardized risk premia estimators using the estimated asymptotic standard errors for SPCA and PCA, respectively, using the DGP in scenario f) as an example. The histograms of PCA deviate from the standard Gaussian distribution for the two highly correlated factors, V and HML. In contrast, the histograms corresponding to SPCA closely align with the standard Gaussian distribution, showcasing significantly reduced bias for these two factors. A portion of this small bias stems from the population-level approximation as demonstrated in (5) (see also Proposition C.4). This phenomenon thereby likely persists irrespective of the value of T . Finally,

we also investigate the statistical power of SPCA in strong and weak cases, respectively, and draw a comparative analysis with PCA. We report these results in the online appendix.

3.2 Results on SDF recovery

Next, we study the finite sample behavior of the SDF estimators. We compare the performance of SPCA, PCA, rpPCA, Lasso and Ridge estimators in scenario f). We report in Table 2 the MSE of the SDF estimators where the true SDF is defined by equation (3). We also include the tuned number of factors determined through our SPCA approach. Additionally, we report in Table 3 the out-of-sample Sharpe ratios of different methods, given by $\hat{b}^\top E(r) / \sqrt{\hat{b}^\top \Sigma \hat{b}}$, where $E(r)$ and Σ are the true mean and covariance of all test assets and $\hat{b}$ is the estimated SDF loading using each method. We find that in terms of the RMSE, SPCA outperforms all other methods, and that rpPCA performs the worst. That said, rpPCA performs the best in terms of the out-of-sample Sharpe ratio, closely followed by the SPCA. The advantage of rpPCA in terms of the Sharpe ratio largely comes from our (infeasible) use of $E(r_t)$ and $\text{Cov}(r_t)$ in a Sharpe ratio criterion when selecting its tuning parameters. Last but not least, the tuning parameter $\hat{p}$ is found to be in close proximity to the truth value of 4.

Finally, we investigate the pattern of out-of-sample Sharpe ratios for various models g_t in Figure 3. The setting resembles scenario (f), except that we consider different models g_t to examine the role of g_t in supervising the procedure. We report Sharpe ratios as a function of number of factors $\hat{p}$ used in the PCA and SPCA procedure. For SPCA, we select $\lfloor qN \rfloor$ via cross-validation using the time series R^2 for each given $\hat{p}$. The sample size T is fixed at 240. The theoretical value of the optimal Sharpe ratio is 0.263, as shown in Table 3, though in finite sample the maximum Sharpe ratio achieved by SPCA is around 0.233.

We consider four cases of $g_t = \eta v_t + z_t$ here. In case (a), we set $\eta = \mathbb{I}_4$, so all factors are included in g_t to supervise the procedure. In case (b), only the factor V and HML are included in g_t . In case (c), we fix $\eta = (1, 0, 0, 0)$, that is, g_t only includes the (strong) market factor. Finally, in case (d), we let $\eta = \gamma^\top \Sigma_v^{-1}$, so that g_t is a noisy measure of the SDF. In light of Theorem 5, SPCA should achieve the maximal out-of-sample Sharpe ratio in cases (a) and (d), provided appropriate tuning parameters. Figure 3 confirms this result. In case (a), SPCA reaches its highest Sharpe ratio out-of-sample precisely at $\hat{p} = 4$, and the Sharpe ratio declines slightly as $\hat{p}$ increases beyond 4, since these additional factors only add noise. Case (d) exhibits a similar pattern. In contrast, the PCA approach cannot achieve the maximal Sharpe ratio, even as $\hat{p}$ increases to 10, because PCA cannot recover the weak factor, which contributes to the SDF. In case (b), SPCA is supervised by two factors with highly correlated loadings, so it can recover the part of the SDF spanned by the weak factors. With a large enough p , we force the procedure of SPCA to continue, then it will also extract the strong factors and achieve the maximal Sharpe ratio. In case (c), however, SPCA and PCA provide similar results — neither achieves the optimum — because g_t only includes the market factor, which does not help SPCA recover the missing weak factor.

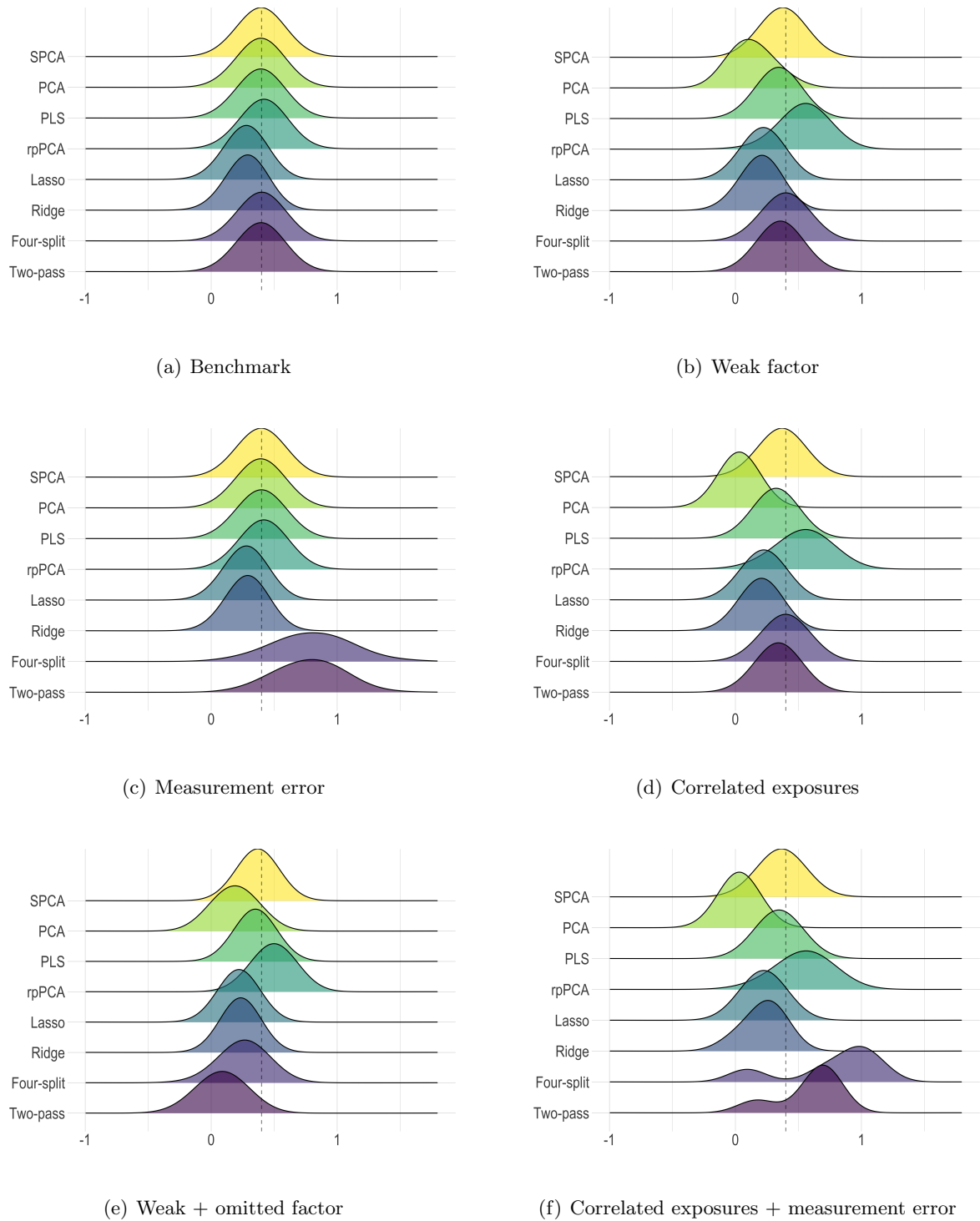


Figure 1: Histogram of Risk Premium Estimates of V

Note: The figure provides histograms of the risk premium estimates in six scenarios for eight estimators we compare, including SPCA, PCA, PLS, rpPCA, Lasso, Ridge, four-split, and the standard two-pass estimator. We simulate the models with $N = 1,000$ and $T = 240$. The number of Monte Carlo repetitions is 1,000.

T	Param	True	SPCA		PCA		rpPCA		PLS	
			Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE
120	RmRf	53.7	0.3	39.0	0.3	38.8	0.3	40.9	0.4	38.9
	SMB	21.7	0.3	29.0	0.2	28.5	0.1	31.7	0.3	28.8
	HML	25.4	-4.5	27.0	-36.1	42.6	15.4	38.0	-9.6	27.8
	V	40.0	-5.1	18.8	-35.9	37.9	13.8	30.5	-10.2	19.9
240	RmRf	53.7	0.8	29.7	0.4	29.6	0.9	30.5	0.8	29.7
	SMB	21.7	0.3	19.9	0.0	19.5	0.6	20.7	0.3	19.8
	HML	25.4	-3.2	18.4	-37.0	39.9	14.7	29.1	-5.2	18.6
	V	40.0	-3.5	13.2	-36.9	37.8	14.1	24.5	-5.6	13.7
480	RmRf	53.7	0.0	20.3	-0.1	20.2	0.2	20.4	0.0	20.2
	SMB	21.7	-0.2	14.1	-0.4	14.0	0.0	14.4	-0.2	14.1
	HML	25.4	-2.6	13.5	-20.7	25.9	10.5	17.5	-2.8	13.6
	V	40.0	-3.2	9.5	-21.1	24.5	9.6	14.9	-3.4	9.5
T	Param	True	Lasso		Ridge		Four-split		Two-pass	
			Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE
120	RmRf	53.7	-10.8	30.9	-40.2	41.1	38.3	57.3	8.1	23.3
	SMB	21.7	-5.1	19.2	-8.2	12.0	70.3	82.2	40.1	45.7
	HML	25.4	-24.2	30.2	-11.9	14.8	66.6	79.1	36.4	42.4
	V	40.0	-25.1	28.6	-26.5	27.9	52.0	67.2	21.8	30.8
240	RmRf	53.7	-1.8	27.5	-31.5	33.5	23.2	43.0	5.0	22.0
	SMB	21.7	-1.7	17.3	0.5	11.6	55.2	66.0	37.0	42.7
	HML	25.4	-18.1	23.9	-3.2	12.0	51.5	62.9	33.3	39.5
	V	40.0	-17.6	21.1	-17.8	21.2	36.8	51.7	18.6	28.4
480	RmRf	53.7	-2.0	19.7	-29.1	29.5	29.1	37.8	20.2	25.8
	SMB	21.7	-2.0	12.6	2.9	5.3	61.1	65.7	52.2	54.6
	HML	25.4	-21.1	24.0	-0.8	4.6	57.4	62.3	48.5	51.1
	V	40.0	-18.0	19.8	-15.5	16.1	42.7	49.1	33.9	37.5

Table 1: Simulation Results for Risk Premia Estimators

Note: In this table, we report the bias (Column “Bias”) and the root-mean-square error (Column “RMSE”) of the risk premia estimates using SPCA, PCA, rpPCA, Lasso, PLS, Ridge, four-split, and the standard two-pass regression approaches, respectively. The true data-generating process, given by scenario f), has four factors, driven by RmRf, SMB, HML, and V, whereas we estimate the risk premia for noisy versions of these four factors. Their true risk premia are provided in Column “True.” We fix $N = 1,000$ while varying $T = 120, 240$, and 480 in this experiment. All values reported are in basis points.

T	SPCA		PCA	rpPCA	PLS	Lasso	Ridge
	$\hat{p}$	MSE	MSE	MSE	MSE	MSE	MSE
120	4.540	0.041	0.082	0.380	0.045	0.051	0.068
	(0.769)	(0.028)	(0.026)	(0.120)	(0.027)	(0.022)	(0.016)
240	4.315	0.020	0.069	0.150	0.022	0.040	0.052
	(0.598)	(0.013)	(0.013)	(0.048)	(0.013)	(0.014)	(0.012)
480	4.430	0.011	0.034	0.027	0.011	0.025	0.031
	(0.669)	(0.007)	(0.016)	(0.015)	(0.007)	(0.009)	(0.008)

Table 2: Simulation Results for SDF estimators

Note: In this table, we report the mean-squared errors (Column “MSE”) defined by $\frac{1}{T} \sum_{t=1}^T |\hat{m}_t - \tilde{m}_t|^2$ for various SDF estimates using SPCA, PCA, rpPCA, PLS, Lasso, and Ridge approaches, respectively. The reported MSEs are the sample average over 1,000 Monte Carlo repetitions and their standard deviations are reported in the brackets. We also report the mean and standard deviation of the estimated number of factors $\hat{p}$ using the SPCA approach. The true data-generating process, given by scenario f), has four factors, driven by RmRf, SMB, HML, and a weak factor V, whereas we estimate the SDF using a vector of factor proxies, g_t , that includes noisy versions of the four factors. We compare three scenarios with $T = 120, 240$, and 480 , where $N = 1,000$ is fixed.

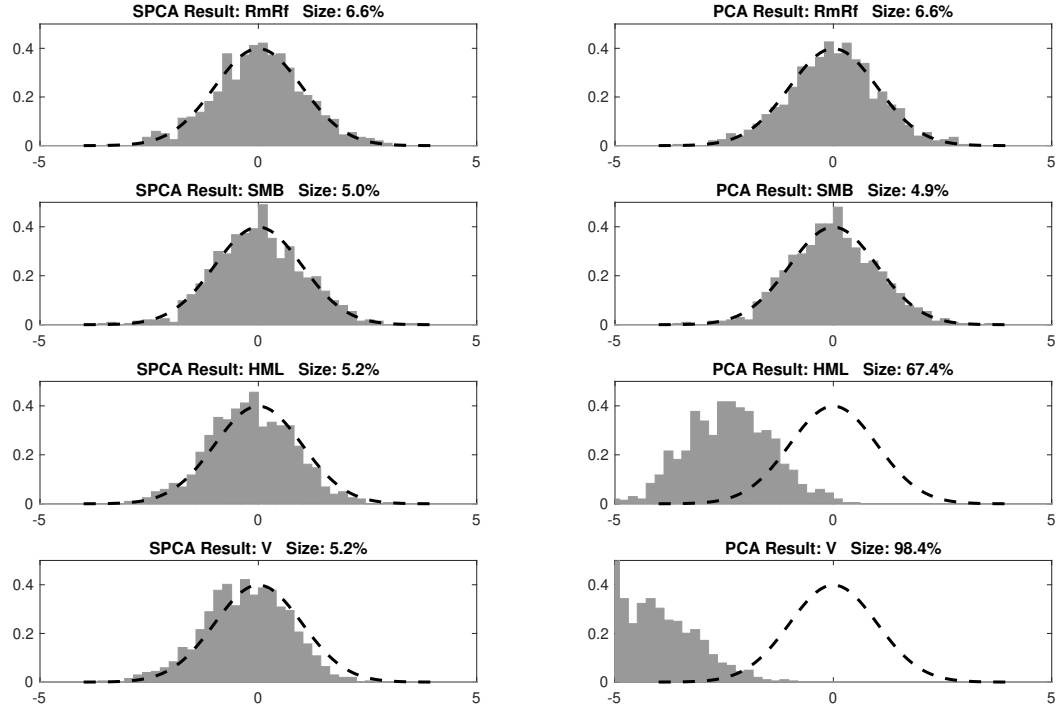


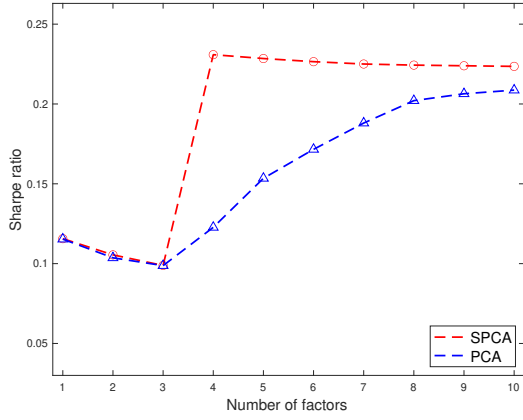
Figure 2: Histogram of the Standardized Estimates in Simulations

Note: The left panels provide the histograms of the standardized SPCA estimates as in Algorithm 3 with asymptotic standard errors given by Theorem 2, whereas the right panels provide those of the standardized PCA-based risk premia estimates as in Algorithm 1. We simulate the model in scenario f) with $N = 1,000$ and $T = 240$. The number of Monte Carlo repetitions is 1,000. These standardized statistics serve as the basis for testing the null hypotheses that the risk premia are equal to their true values. The sizes of these t-tests at 5% level are reported in the figure subtitles, allowing us to assess the tail behavior of our asymptotic approximations.

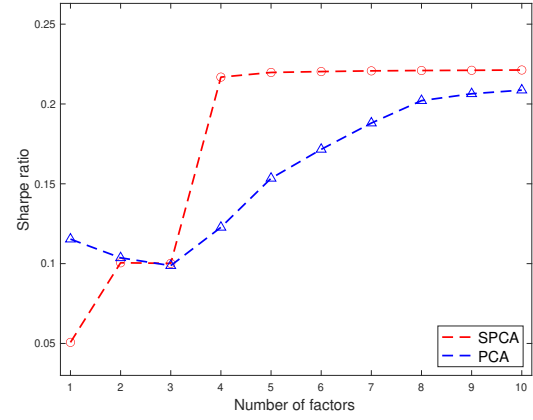
T	SPCA	PCA	rpPCA	PLS	Lasso	Ridge	Theoretical Value
120	0.206 (0.043)	0.094 (0.038)	0.212 (0.040)	0.181 (0.049)	0.173 (0.049)	0.160 (0.057)	0.263
240	0.233 (0.025)	0.104 (0.030)	0.238 (0.026)	0.224 (0.029)	0.210 (0.036)	0.206 (0.039)	0.263
480	0.248 (0.013)	0.201 (0.048)	0.254 (0.009)	0.246 (0.014)	0.226 (0.023)	0.232 (0.018)	0.263

Table 3: Simulation Results for Out-of-Sample Sharpe Ratios of Optimal Portfolios

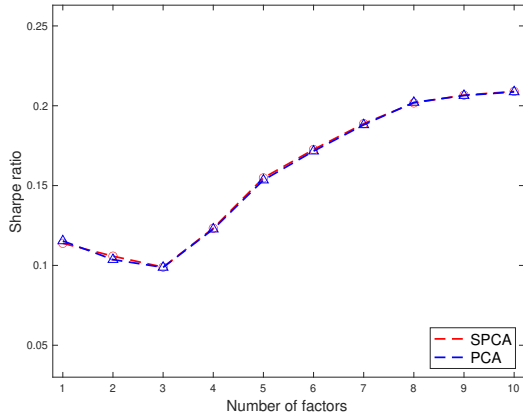
Note: In this table, we report the mean and standard deviation of the out-of-sample Sharpe ratios for various optimal portfolios constructed by SPCA, PCA, rpPCA, PLS, Lasso, and Ridge approaches, respectively. The true data-generating process, given by scenario f), has four factors, driven by RmRf, SMB, HML, and a weak factor V , whereas we estimate the SDF using a vector of factor proxies, g_t , that includes noisy versions of the four factors. The reported Sharpe ratios are the sample average over 1,000 Monte Carlo repetitions and their standard errors are reported in the brackets. Column “Theoretical Value” provides the benchmark Sharpe ratio calculated by $b^T E(r) / \sqrt{b^T \Sigma b}$ using true parameter values. We compare three scenarios with $T = 120, 240$, and 480 , where $N = 1,000$ is fixed.



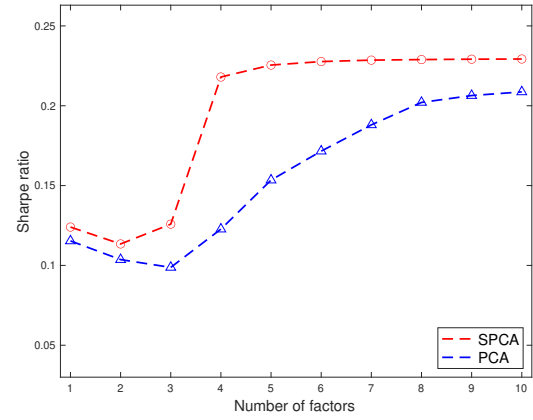
(a) $\eta = \mathbb{I}_4$



(b) $\eta = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$



(c) $\eta = (1 \ 0 \ 0 \ 0)$



(d) $\eta = \gamma^\top \Sigma_v^{-1}$

Figure 3: Out-of-sample Sharpe Ratio Patterns with Different Models of g_t

Note: Each panel reports the out-of-sample Sharpe ratios for PCA (blue) and SPCA (red) as a function of number of factors, p , for a specific model of $g_t = \eta v_t + z_t$.

4 Empirical Analysis

In this section we apply SPCA to estimate the risk premia of a variety of observable factors, and to diagnose observable factor models.

4.1 Data

Our main dataset is the [Chen and Zimmermann \(2020\)](#) data, which includes a large number of equity portfolios sorted by characteristics. Specifically, we employ the April 2021 release of the data. For each characteristic considered, [Chen and Zimmermann \(2020\)](#) construct a variable number of portfolios (as many as are used in the original papers that introduced the anomaly in the literature: typically 2, 5, or 10). Not all test assets are available for the entire time period; for our analysis, we study the time period 1976m3 to 2020m12, for which 901 test portfolios are available without missing values. To these sorted portfolios, we add 49 industry portfolios from Ken French's website. All of our results are at the monthly frequency.⁴⁰

We also consider an alternative dataset, proposed by [Hou et al. \(2020\)](#), that includes for the same period 1672 portfolios sorted by characteristics without missing values. [Hou et al. \(2020\)](#) classify their portfolios in six groups: momentum, value, investment, profitability, intangibles, frictions. The two datasets of [Hou et al. \(2020\)](#) and [Chen and Zimmermann \(2020\)](#) are similar and yield comparable results. Rather than producing two versions of each result using the two datasets, we choose [Chen and Zimmermann \(2020\)](#) to be our main dataset and report the robustness of the main results using the [Hou et al. \(2020\)](#) data (e.g., see section 4.2.6). What both datasets have in common is that they capture a wide universe of anomaly equity portfolios discovered in the last four decades of asset pricing research.

We consider both tradable and nontradable factors in our analysis, focusing on the best-known ones from the literature. The tradable factors are: the market (in excess of the risk-free rate); size (SMB); value (HML); profitability (RMW); investment (CMA); momentum (MOM); betting-against-beta (BAB, from [Frazzini and Pedersen \(2014\)](#)); and quality-minus-junk (QMJ, from [Asness et al. \(2013\)](#)). The nontradable factors are: the liquidity factor from [Pástor and Stambaugh \(2003\)](#); the intermediary capital factor from [He et al. \(2017\)](#); AR(1) innovations in industrial production growth (IP); VAR(1) innovations in the first three principal components of 279 macro-finance variables from [Ludvigson and Ng \(2010\)](#); AR(1) innovations in the three uncertainty indexes of [Jurado et al. \(2015\)](#), representing financial uncertainty, macroeconomic uncertainty, and real uncertainty; AR(1) innovations in the term spread, the credit spread, and the unemployment rate; AR(1) innovations in two sentiment indexes, one from [Huang et al. \(2015\)](#) and one from [Baker and Wurgler \(2006\)](#); oil price growth AR(1) innovations; and consumption growth AR(1) innovations.⁴¹

⁴⁰The theory is silent on what the correct frequency of the data to study is. Here we follow the literature and focus on monthly frequency; we leave for future work a more comprehensive study and comparison across frequencies.

⁴¹The market factor, SMB, HML, RMW, CMA and MOM are from Ken French's website. BAB and QMJ are from AQR's website. The liquidity factor is from Lubos Pastor's website. The intermediary capital factor is from Asaf

4.2 Estimation of Risk Premia using SPCA

In this section we estimate the risk premia of a variety of tradable and nontradable factors. We begin by discussing some details of the implementation of the estimator.

4.2.1 Choice of Tuning Parameters and Implementation Details

To apply SPCA to the estimation of the risk premia and to evaluate its out-of-sample performance, we split the sample period into two equal-sized subsamples. The first half of the sample (training period) is used to choose the tuning parameters and produce the risk premium estimate. The second half of the sample (evaluation period) is used to evaluate the out-of-sample performance of the estimator and the choice of the tuning parameter.

For ease of presentation, we choose to select only one tuning parameter, q (or, equivalently, the number of assets selected $\lfloor qN \rfloor$), for each plausible choice of p (the number of factors) in our analysis. This approach reduces the number of tuning parameters to only one, and also conveniently serves as a robustness check with respect to the number of factors.

To determine reasonable candidates for p , we examine the factor structure of the panel of test asset returns. Figure 4 provides the scree plot of the log of the first 25 eigenvalues. There appear to be at least three strong factors. In addition, it appears that factors 4-11 might also be relevant, but weak.

The scree plot therefore directly confirms a result documented in different forms in the literature: that in the panel of returns, there appear to be weak factors, in which case standard estimators of risk premia would be biased: it is precisely the scenario where the theory and the simulations indicate SPCA would be preferred. Motivated by the scree plot, in the empirical study below we highlight results for p equal to 3, 5, 7, and 11, therefore showing the robustness of our results to a wide range of model dimensions.

To choose the tuning parameter q , we adopt the same R^2 criterion as in simulations to evaluate the estimator's out-of-sample performance, namely, the hedging ability of the portfolio built by SPCA for g_t . Guided by this *statistical* justification, in our empirical work we choose q by 3-fold cross-validation (100 runs) within the training sample, maximizing the hedging R^2 for g_t . Appendix E describes in detail the steps for the cross-validation. Once we have tuned q , we use it to compute the SPCA risk premium estimate for g_t .

In addition to its (statistical) role for choosing the tuning parameter, this R^2 can also be used to evaluate the out of sample performance of the estimator, precisely because of its *economic* interpretation as the performance of the hedging portfolio associated with the estimator. Achieving a higher

Manela's website. The macro principal components and the uncertainty indexes are from Sydney Ludvigson's website. Industrial production, the credit spread, unemployment rate, the term spread, and oil price are from Fred-MD. The Huang et al. (2015) sentiment index is from Huang's webpage. The Baker and Wurgler (2006) sentiment index is from Wurgler's website. The consumption factor was built from NIPA data using the methodology of Schorfheide et al. (2018).

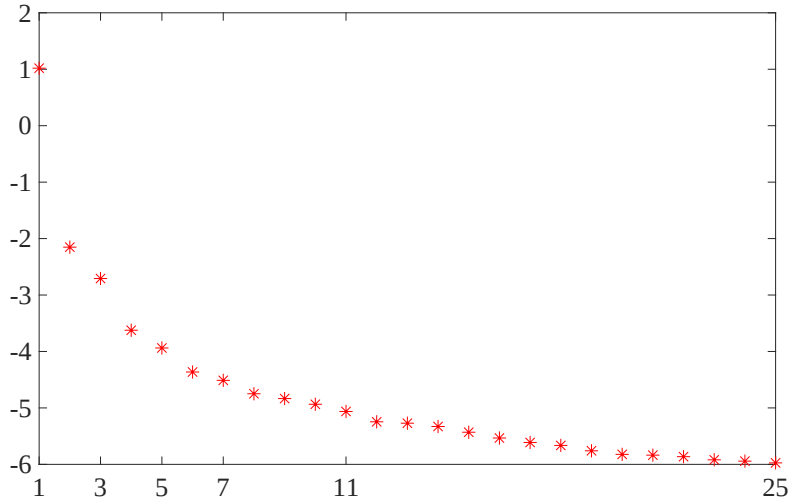


Figure 4: Logarithm of the First 25 Eigenvalues in the Chen-Zimmerman data

Note: The figure plots the logarithm of the first 25 eigenvalues of the data, obtained from [Chen and Zimmermann \(2020\)](#) plus 49 industry portfolios, covering the period 1976-2020.

R^2 out of sample means that the estimator is better able to capture the variation in the risk that the portfolio aims to hedge. While achieving a higher R^2 does not imply that an estimator is “better” in a statistical sense (which is instead what the theory and simulations are devoted to), it does provide a useful diagnostic on the estimator and on the choice of the tuning parameter: if a tuning parameter is chosen properly in the training sample, yielding a portfolio that hedges g_t well, we expect this hedging performance to persist in the evaluation sample. Following this motivation, using the test asset returns from the evaluation period, together with the hedging portfolio weights estimated in the training period, we compute the return of the hedging portfolio in the evaluation sample, and we calculate the hedging R^2 in the evaluation sample. This calculation does *not* re-estimate any parameter or coefficient in the evaluation data, and is thus a fully out-of-sample R^2 . Appendix E also describes the evaluation steps in algorithmic form.

4.2.2 Results: Estimation of Risk Premia and Out-of-sample Evaluation

We report the main empirical results in Table 4 and Figures 5 and 6. Each row of Table 4 corresponds to one factor; the first 8 are tradable, the rest are nontradable. For tradable factors, the first two columns show the average excess return of the factor, in the training sample and in the evaluation sample, respectively; these numbers correspond to model-free estimates of the risk premia of tradable factors, and can be directly compared with the model-based estimate obtained from SPCA.

The next columns of the table show the SPCA results in 4 groups of columns, corresponding to the number of latent factors $p = 3, 5, 7$, and 11, respectively. For each choice of p , we report the

risk-premium estimate (obtained in the training sample, in bp per month), the number of assets selected by SPCA (determined by q), and the out-of-sample R^2 obtained in the evaluation period. These estimates are obtained factor by factor: that is, in each case, g_t contains one factor, and the asset selection is driven by that factor only. In the last two columns of the table, we repeat the exercise (with $p = 11$) but estimate all risk premia simultaneously: g_t contains all the factors and the selection of the assets is based on all of them simultaneously (so that $d \geq p$ as opposed to $d = 1$). As discussed above, in this case, assets are sorted by the maximum of the correlation with any of the factors in g_t for the purpose of the selection step. In theory, both approaches are consistent. In practice, estimating risk premia factor by factor has the advantage that the latent factors zoom in immediately on the assets relevant for each factor. On the other hand, the joint estimation is required to satisfy the more stringent assumptions for the CLT of Section 2.2.4.

Consider first the market portfolio (first row of the table), a strong factor in this dataset. The average return of the market in the training sample is 74bp per month, and 62bp in the evaluation period. The SPCA estimates of the market risk premium, for the four chosen values of p , are 68, 70, 72, and 74bp per month, respectively, all close to the average excess return. To obtain these estimates, SPCA estimates the latent factors picking, in each iteration, 100 assets out of the total of 950. Finally, the portfolio that SPCA builds to hedge the market achieves, not surprisingly, a very high out-of-sample R^2 , above 0.98 for all p .

To better understand the performance of the estimator and the tuning parameter choice, we can examine the heatmap in Figure 5, panel (a), which focuses on the market factor. In the heatmap, the x axis reports the number of factors p ; the y axis reports the number of test assets selected by SPCA (in turn determined by q); for each combination of p and q , the heatmap reports the out-of-sample R^2 of the hedging portfolio built by SPCA. As discussed above, looking at this outcome has both an economic interpretation (since the hedging performance is of direct interest to investors) and a statistical interpretation (to evaluate whether the choice of q operated in the training sample appears to be valid out of sample). The heatmap therefore gives a complete description of the out-of-sample properties of the SPCA-based hedging portfolio as a function of the two parameters p and q .

Panel (a) shows that for all combinations of p and q (that is, throughout the heatmap), out-of-sample R^2 s are overall very high for the market portfolio, above 85%. However, there appears to be a subset of the parameter space where hedging performance is especially good: combinations with high p and low q . The red marks in the heatmap correspond to the values of q chosen by cross-validation (CV) in the training sample (one for each value of p considered in the table: 3, 5, 7, 11). Ideally, the values of q chosen by CV in the training sample would yield a hedging portfolio that performs well out of sample: that is, the marks should lie in areas in the heatmap with high out-of-sample R^2 s. This is indeed the case, as the figure shows, indicating good out-of-sample performance of the tuning parameter selection procedure.

Consider now another tradable factor, CMA, in the 5th row of Table 4. Like for the market, the estimated risk premium for CMA is not significantly different from the average excess return of the

Table 4: Risk premia estimates

	Avg. ret.		Avg. ret.		3 Latent Factors		5 Latent Factors		7 Latent Factors		11 Latent Factors		Joint estim, 11 factors	
	(train.)	(eval.)	RP	# Assets	R^2	RP	# Assets	R^2	RP	# Assets	R^2	RP	# Assets	R^2
Market	74	62	68	100	0.98	70	100	0.98	72	100	0.99	74	100	0.99
HML	39	-7	50	100	0.70	37	100	0.79	39	150	0.78	44	250	0.79
SMB	12	25	15	100	0.82	5	100	0.85	10	100	0.85	10	100	0.85
RMW	37	28	-8	100	-0.18	40	100	0.56	33	100	0.61	27	150	0.66
CMA	26	19	36	250	0.41	40	100	0.55	27	200	0.55	31	350	0.53
Momentum	91	30	67	100	0.79	86	100	0.87	102	100	0.87	101	100	0.88
BAB	126	56	112	100	0.43	120	100	0.38	112	150	0.35	128	150	0.45
QMJ	41	39	-9	100	0.43	28	100	0.81	31	100	0.80	36	150	0.78
Liquidity			70	550	0.01	85	650	0.02	83	700	0.04	95	900	0.03
Intermed. Cap.			112	100	0.59	101	100	0.56	121	150	0.55	116	350	0.52
IP growth			-4	950	-0.01	-4	950	-0.02	-5	950	-0.03	-2	950	0.00
LN 1			225	550	-0.28	202	650	-0.19	150	700	-0.11	54	950	-0.12
LN 2			-70	950	-0.05	-79	950	-0.12	-24	950	-0.16	-29	950	-0.17
LN 3			96	400	0.03	86	650	0.06	16	700	0.06	-21	850	0.05
Consumption			2	950	-0.01	3	950	0.00	3	950	-0.01	2	950	-0.01
Fin. Unc.			-61	350	-0.08	-48	750	0.00	-40	850	0.09	-41	950	0.10
Real Unc.			-6	950	0.05	-7	950	0.04	-9	950	0.04	-11	950	0.06
Macro Unc.			-7	950	0.08	-10	950	0.08	-10	950	0.08	-16	950	0.09
Term			229	950	-0.11	81	950	-0.36	-57	950	-0.54	262	950	-0.59
Credit			41	950	-0.03	62	950	-0.03	41	950	-0.02	-43	950	-0.03
Unempl.			65	950	0.00	109	950	-0.01	112	950	-0.01	110	950	0.00
Sentiment HJTZ			-24	950	0.01	-27	950	-0.03	-18	950	-0.06	-40	950	-0.07
Sentiment BW			57	950	0.00	64	950	0.00	50	950	0.01	16	950	-0.02
Oil			-37	950	-0.05	-62	950	-0.02	-42	950	-0.03	-20	950	-0.02

Note: In this table, we report the estimation results for tradable and nontradable factors using SPCA. The first two columns report the average excess returns for tradable factors, in the training sample (first half of the sample period) and in the evaluation sample (second half of the sample period). The remaining columns report, for different values of the number of factors p , the risk premia estimates (in basis points per month, computed in the training period), the number of assets selected by SPCA (governed by the parameter q), and the out-of-sample R^2 of the implied hedging portfolio. The last two columns report risk premia estimates and standard errors including all factors in g_t simultaneously, with $p = 11$. Sample is the [Chen and Zimmermann \(2020\)](#) test portfolios plus 49 industry portfolios, over the period 1976-2020.

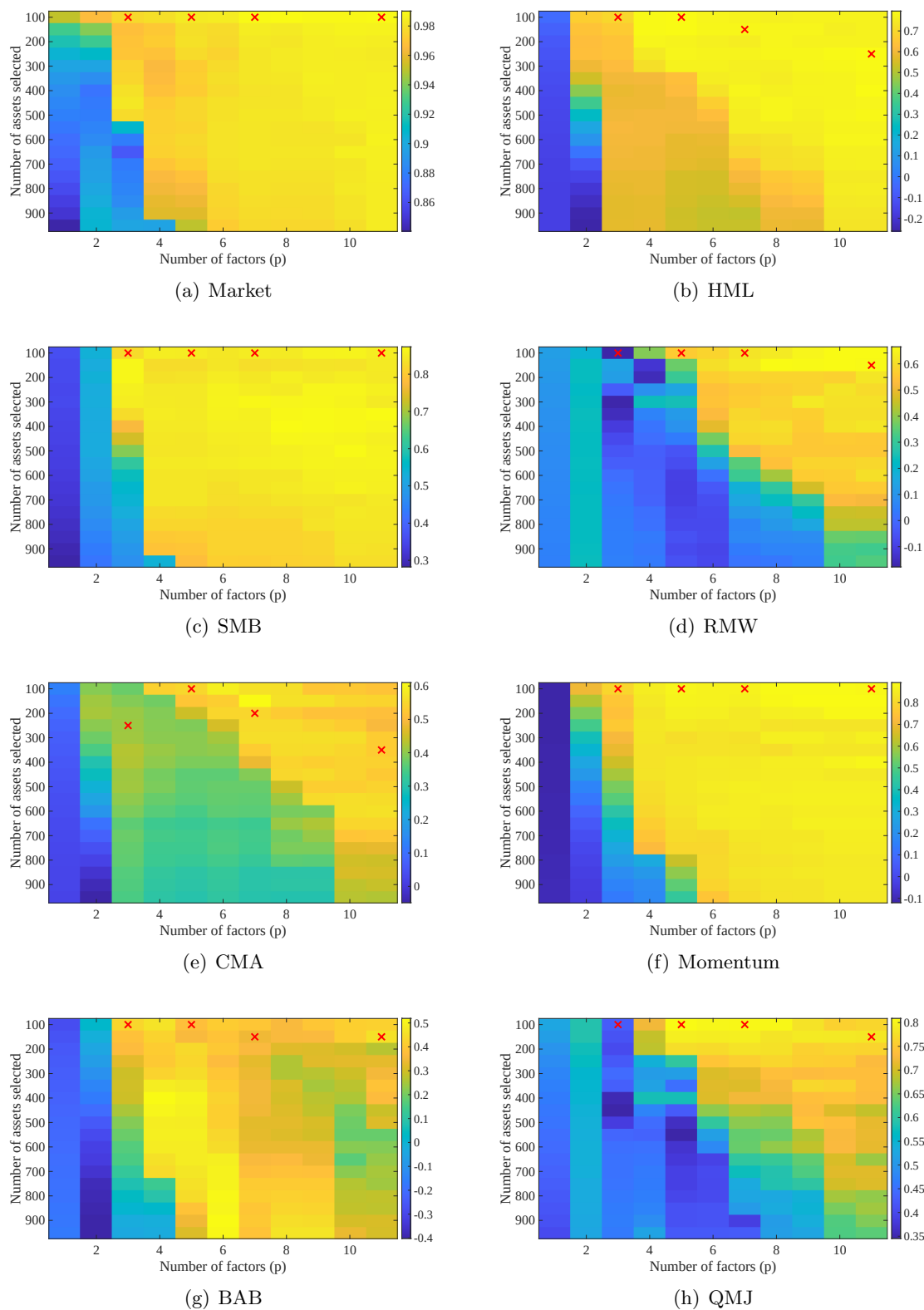


Figure 5: Out-of-sample R^2 Heatmaps, Tradable Factors

Note: Each panel reports the out-of-sample R^2 heatmap for a different factor. X-axis reports p . Y-axis reports the number of assets selected, governed by q . The colors in the heatmap correspond to the out-of-sample R^2 of the SPCA-implied hedging portfolio for the factor g_t ; this R^2 is computed entirely in the evaluation period. The red marks are the points chosen by CV within the training sample.

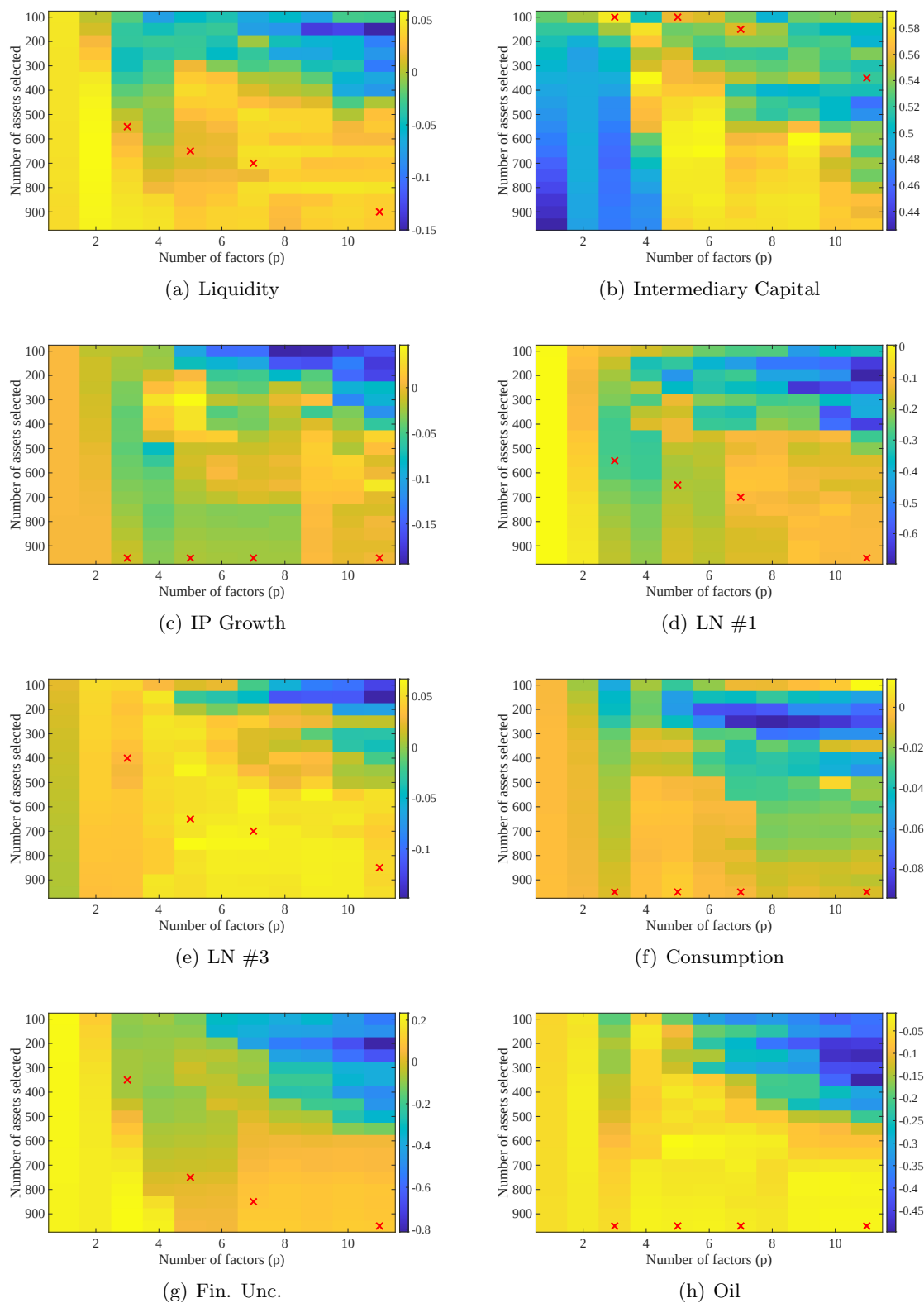


Figure 6: Out-of-sample R^2 Heatmaps, Nontradable Factors

Note: Same as Figure 5, but for a subset of nontradable factors.

factor. The number of assets selected by SPCA ranges between 100 and 350, and the out-of-sample R^2 is above 50%, indicating that the hedge portfolio built by our latent factor model is able to capture the majority of the variation on CMA out of sample.⁴²

The heatmap of the out-of-sample R^2 for the hedging portfolio of this factor is panel (e) of Figure 5. The figure shows that for the case of CMA, different combinations of p and q yield very different out-of-sample hedging performance, with R^2 s ranging from above 50% to below 0. Ideally, if the tuning parameter were chosen properly, we would hope to see the hedging portfolio also does well out of sample. The red marks in the figure show that this is indeed the case, especially for $p = 5$ and above.

These heatmaps also allow us to compare the results with the PCA-based estimator of Giglio and Xiu (2021). This is because the last row of the heatmap corresponds to the case $q = 1$, that is, all assets are used to estimate the factors; so PCA corresponds to a particular choice for the tuning parameter. Looking across the various panels of Figure 5, it is clear that while for some factors (like the market) similar R^2 can be obtained by PCA and SPCA, for other factors (like CMA and RMW) the out-of-sample R^2 s obtained by SPCA are substantially higher than those obtainable by PCA, for any choice of p (graphically: the area with the highest R^2 s is concentrated in the upper part of the heatmap, where $q < 1$). These results indicate that the optimal tuning parameter (both in and out of sample) is below 1: the strengthening of the signal in the cross-section of test assets that SPCA operates is deemed desirable by the training exercise, and is validated to be useful in the validation sample. This is not surprising given that the scree plot has shown the presence of several weak factors in the data.

One additional advantage of SPCA that is clearly visible in the heatmaps is that SPCA often manages to achieve the same (or better) R^2 than PCA, while estimating a much smaller number of factors. For example, consider the momentum factor in panel (f). The last row of the heatmap shows that extracting factors via PCA achieves an R^2 above 70% only once at least 6 factors are included; SPCA gets there even with 3 factors. The reason is intuitive: SPCA focuses on the test assets that are most informative about g_t , and therefore can zoom in quickly on the most relevant latent factors.

For nontradable factors, we cannot compare the risk premium estimate from SPCA with the average excess return; beyond relying on the theory and simulations, we can look at the out-of-sample R^2 for suggestive evidence about the empirical performance of the estimator. Note that it is well known in the literature that it is difficult to hedge nontradable factors, like consumption or IP growth, in equity markets. We will however show that SPCA gives a hedging portfolio that

⁴²Given that the universe of test assets includes portfolios sorted by the same characteristics used to construct the tradable factors like CMA, one may wonder why an out-of-sample R^2 of 100% is not always obtained for tradable factors. The reason is that SPCA attempts to build a hedging portfolio for the target g_t with factors that must also explain covariation among the universe of test assets. An advantage of our approach is that the hedging portfolio is able to avoid fitting the “measurement error” component in g_t , which, as discussed above, can be thought of as non-diversified idiosyncratic error for tradable factors, or more literally measurement error for nontradables.

successfully hedges at least a part of the variation in many nontradable factors.

Consider first the liquidity factor of Pástor and Stambaugh (2003), in row 9 of Table 4 and panel (a) of Figure 6. The out-of-sample R^2 achieved by SPCA is above 0 (up to 4%), and the estimated risk premium appears to be high (between 70 and 95bp per month). Panel (a) of Figure 6 shows how strongly this R^2 depends on p and q . Among all combinations of parameters, a large fraction actually delivers a negative out-of-sample R^2 . This simply stresses how difficult it is to hedge this factor (like most macro factors) using equity markets, and indicates again the relatively good performance of SPCA as tuned in the training sample.

The remainder of the table and of the two figures shows the results for all the other factors (for reasons of space, the heatmaps only report a subset of the factors, while the table reports them all). A few interesting patterns emerge. First, for tradable factors, SPCA gives risk premia estimates that are always close to the model-free estimates obtained from average excess returns: the two are never statistically different at the 5% level (with the only exception of QMJ with $p = 3$). Second, confirming the previous literature, nontradable factors are much harder to hedge than tradable factors; in fact, for several factors – like the first two JLN macro factors – we do not get positive R^2 at all. For those factors, there is so little exposure in equity returns that SPCA cannot build a proper hedging portfolio. However, SPCA is able to hedge out of sample at least a part of the variation of many factors, like the third LN factor, the three uncertainty measures, the liquidity factor and the intermediary capital factor (for which it achieves an R^2 above 50%). Third, the risk premia estimated by SPCA – for those factors where SPCA can actually hedge some of the variation – make economic sense: for example, the liquidity and intermediary factors command significantly positive risk premia, whereas the three uncertainty measures command negative risk premia.

4.2.3 Asset Selection

To better understand how SPCA estimates risk premia, we can study which assets are selected when extracting the latent factors. Table 5 shows, for four representative factors (two tradables, Momentum and RMW, and two nontradables, liquidity and intermediary capital), the top 10 test assets (by absolute value of correlation) selected at each step. The names of the portfolios follow Chen and Zimmermann (2020), with the numbers indicating the quintile or decile of the characteristic.

Consider Momentum, in the first set of rows of the table. To extract the first latent factor, SPCA selects the assets with the highest correlation with the momentum factor. As the table shows, the highest correlation is achieved by IntMom09 (an intermediate momentum portfolio). The correlation is 0.44. The other assets with high correlation are all momentum-related, not surprisingly. In the next columns, the table shows the assets selected at the second iteration of SPCA, after orthogonalizing g_t and the test assets to the first factor. Interestingly, the correlations among these residuals are even higher, up to 0.79 for a different momentum sort (Mom12mOffSeason, momentum without the seasonal component). This suggests that the first factor captures some of the asset variation that is

Table 5: Assets Selected by SPCA

Factor #1			Factor #2			Factor #3		
	Asset	Corr	Asset	Corr		Asset	Corr	
Mom	IntMom09	0.44	Mom12mOffSeason02	0.79		Mom12m08	0.64	
	IntMom10	0.4	Mom12mOffSeason03	0.76		BMdec05	0.63	
	MomVol10	0.37	Size01	0.74		IntMom03	0.63	
	MomVol09	0.36	ResidualMomentum01	0.73		SP05	0.62	
	IntMom08	0.36	ResidualMomentum02	0.73		ShareIss5Y05	0.62	
	Mom12m10	0.36	NumEarnIncrease01	0.72		BookLeverage02	0.62	
	FirmAgeMom05	0.35	ShareIss5Y01	0.7		cfp05	0.61	
	Mom12mOffSeason10	0.34	MomVol03	0.69		BMdec04	0.61	
	Mom12mOffSeason09	0.33	CompEquIss01	0.68		ShareIss1Y05	0.6	
	Mom12m09	0.33	Mom12m03	0.68		LReversal04	0.6	
RMW	Industry:Gold	0.27	OperProf05	0.54		OperProfRD01	0.53	
	MomOffSeason10	0.27	OperProfRD09	0.53		RoE01	0.47	
	AccrualsBM02	0.27	CBOperProf09	0.5		GP01	0.45	
	DelEqu05	0.27	RoE05	0.49		CBOperProf02	0.45	
	LReversal05	0.27	CBOperProf10	0.49		DolVol01	0.44	
	roaq01	0.26	Leverage02	0.49		OperProfRD02	0.44	
	AssetGrowth10	0.26	OperProfRD08	0.49		CBOperProf01	0.43	
	DolVol05	0.25	realestate03	0.49		OperProf01	0.41	
	ChEQ05	0.25	GP05	0.49		RoE02	0.4	
	Price05	0.25	GP04	0.48		VolMkt02	0.4	
Liq.	InvGrowth06	0.47	InvGrowth06	0.28		InvGrowth06	0.3	
	NetPayoutYield07	0.47	BetaFP09	0.26		DolVol01	0.27	
	PayoutYield05	0.46	EntMult06	0.25		XFIN08	0.26	
	PayoutYield07	0.46	NetPayoutYield07	0.24		MeanRankRevGrowth01	0.26	
	BetaFP03	0.46	PayoutYield07	0.24		BetaFP03	0.25	
	DeLTIO2	0.46	PayoutYield05	0.24		ShortInterest01	0.25	
	IntanBM03	0.46	cfp04	0.23		BetaFP09	0.24	
	EntMult06	0.46	BetaFP10	0.23		EntMult06	0.24	
	VolMkt04	0.46	XFIN08	0.23		PayoutYield07	0.24	
	PayoutYield06	0.46	ShortInterest01	0.22		ChEQ04	0.23	
Interm.	Industry:Banks	0.9	Industry:banks	0.76		Industry:banks	0.7	
	Industry:Fin	0.84	Industry:Fin	0.56		Industry:Fin	0.47	
	IntMom05	0.8	DelEqu02	0.46		DebtIssuance02	0.38	
	EquityDuration04	0.8	grcapx3y02	0.44		NOA10	0.36	
	IdioVolAHT05	0.8	OScore02	0.43		ChAssetTurnover04	0.35	
	IdioVol3F05	0.79	GrLTNOA10	0.43		HerfAsset05	0.35	
	MaxRet08	0.79	ChAssetTurnover04	0.43		ShareRepurchase01	0.35	
	Illiquidity01	0.79	IntMom05	0.43		HerfBE05	0.35	
	IdioRisk05	0.79	IdioVolAHT05	0.42		DelEqu05	0.32	
	CBOperProf03	0.78	Tax01	0.42		Beta05	0.32	

Note: For each factor (one per panel) the table shows the top-10 assets selected by SPCA in extracting the latent factors. Assets are sorted by absolute value of the correlation. For each factor from 1 to 3, the table reports the names of the portfolios selected, and the absolute value of the correlation with g_t . Naming convention for the portfolios follows [Chen and Zimmermann \(2020\)](#).

not exclusively specific to momentum (for example, part of the market factor), which the projection step of SPCA removes. The residuals of the factors and the portfolios are then *more* correlated

than the original factors and portfolios with the influence of the first factor eliminated. In any case, momentum portfolios appear again at the second iteration, and, in part, at the third iteration.

The remainder of the table shows which assets are selected at the different iterations for RMW, Liquidity, and Intermediary Capital. For RMW (a profitability factor), the assets selected are often based on accounting measures, like asset growth, accruals, leverage, and operating profits. For liquidity, portfolios sorted by payout yield and beta seem to play an important role in hedging the risk. Finally, for intermediary capital, the portfolios selected by SPCA relate to idiosyncratic volatility, liquidity, as well as two industry portfolios (not surprisingly, banking and financials).

The selection of particularly informative assets is the central mechanism through which SPCA addresses the issue of weak factors. It is also responsible for the parsimony of SPCA to the number of factors used, as we noted above: given that SPCA zooms in on the most informative assets, it can build a good hedging portfolio even with a small number of factors.

4.2.4 Strength of the Factors

We next report the strength of the factors extracted by SPCA at each step. To make the results comparable across iterations of SPCA, and between SPCA and PCA, we compute the strength of a latent factor as the eigenvalue of the factor normalized by the number of assets used to extract it. Figure 7 reports, in each panel, the log normalized eigenvalues for the factors extracted from PCA (dashed line) and for the factors extracted by SPCA, grouped across panels for the various targets (since the factors extracted by SPCA are different for different targets g_t): panels (a) and (b) show the factors extracted when the targets are the tradable factors, panels (c) use a subset of nontradables, and (d) the remaining nontradables. The figure shows eigenvalues corresponding to the first 5 factors.

As expected, the log eigenvalues for PCA decrease as lower and lower-variance factors are extracted. This is also mostly (but not always) the case for SPCA, where however we see a large difference across factors. For some factors (like most nontradables, which, as discussed above, are mostly noise factors), SPCA chooses a large number of assets, so the results look very similar to PCA (e.g. see panel (d)). For factors where SPCA chooses a small number of assets (e.g., intermediary capital and many tradables) we see that the strength of the factor extracted is higher than with PCA. This effect is strongest for the first eigenvalue (the log scale hides it somewhat), but is there for subsequent factors as well. In general, it appears that SPCA indeed strengthens the factor extracted from the cross-section, compared to PCA, and especially so when fewer assets are selected.⁴³

⁴³One caveat is that once the main factors are extracted, and mostly noise is left in the cross-section, noise itself could lead to higher normalized eigenvalues. This is why the criterion for tuning the parameter q of SPCA is the out-of-sample R^2 of the hedging portfolio, and not this measure of factor strength.

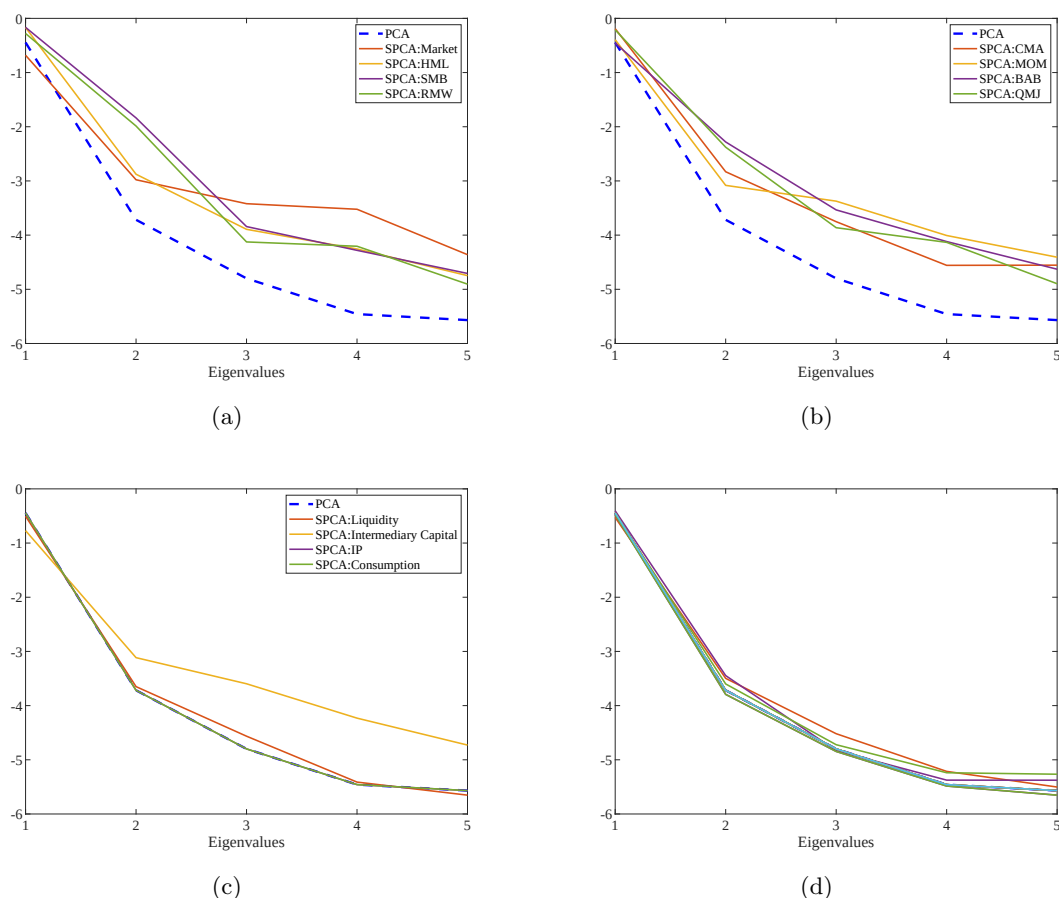


Figure 7: Strength of the Latent Factors

Note: Each panel of the figure shows the log eigenvalues extracted by PCA from the universe of all assets in the training sample, as well as the log eigenvalues extracted by SPCA at each iteration (for the first 5 factors), for the tuning parameter selected by CV. All eigenvalues are normalized by the number of assets used, which is a measure of strength of the factor that is directly comparable. Panels (a) and (b) study two groups of tradable factors, panel (c) a selection of the nontradables, and panel (d) the remaining nontradables.

4.2.5 SPCA and the Universe of Test Assets

The fact that SPCA estimates the latent factors using the most informative assets also makes it particularly robust to the universe of test assets used in the estimation. We explore this here in detail by considering three factors, value, momentum, and profitability, for which we can easily identify test assets that are informative about them. Specifically, we consider (for this section only) the dataset from Hou et al. (2020), which, as discussed in Section 4.1, collects test portfolios by characteristics in six groups, among which one is labeled “value vs. growth”, one “momentum”, and one “profitability.” We can then ask: how does SPCA perform in estimating the value risk premium if we exclude the value and growth sorts from the universe? Similarly, how does it perform in estimating the momentum and profitability risk premia if momentum and profitability test assets, respectively,

are removed? When the corresponding sorted portfolios are removed, the factors naturally become weaker. However, we should expect SPCA to still perform well, at least as long as sufficient exposure to the factor is present in the remaining test assets. On the contrary, we expect PCA's performance to deteriorate more sharply.

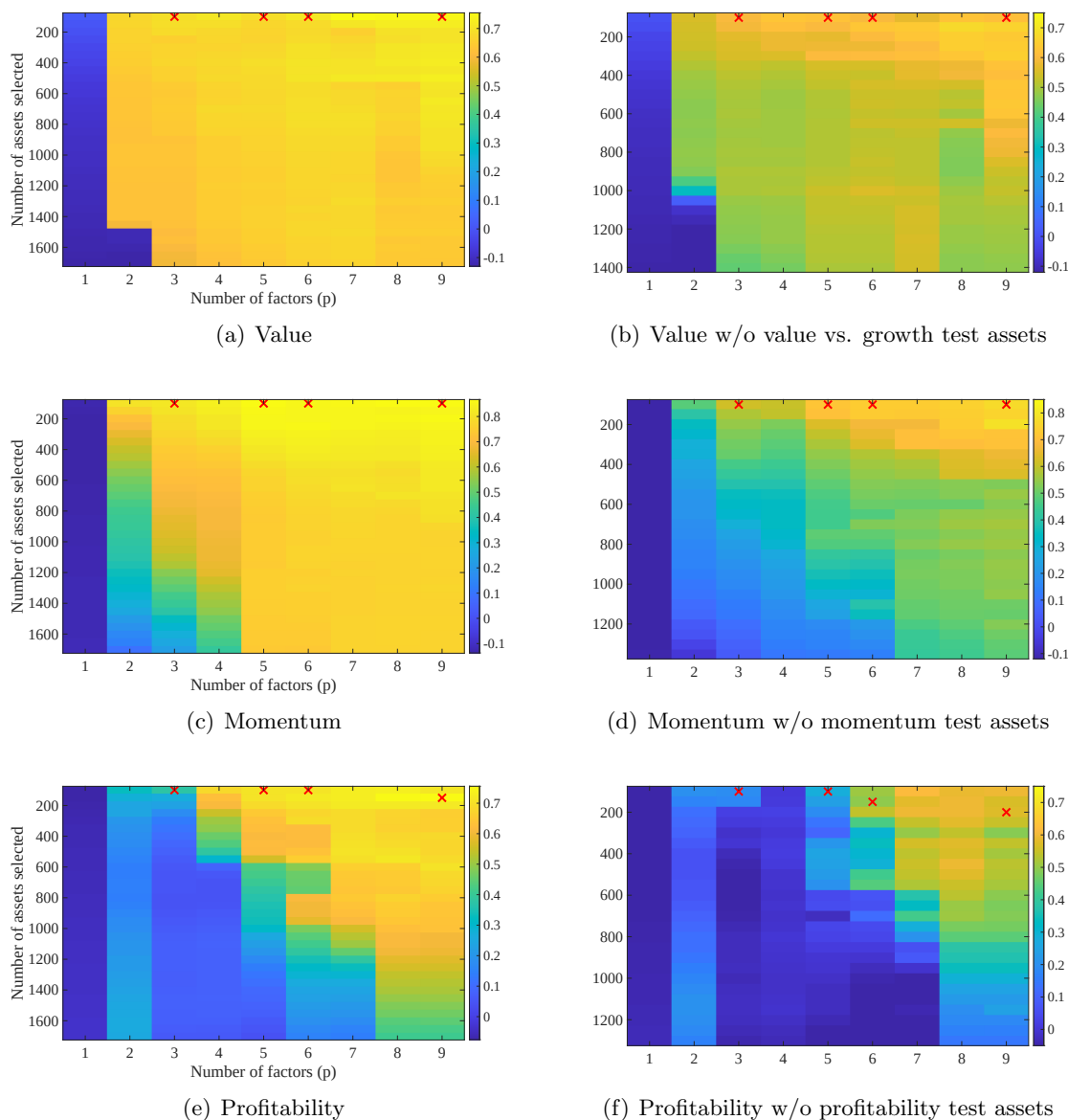


Figure 8: Varying the Universe of Test Assets

Note: For value, momentum and RMW (profitability), the figure shows the out-of-sample R^2 heatmaps when all the test assets from Hou et al. (2020) are used in the estimation (left), and when value portfolios, momentum portfolios, or profitability portfolios, respectively, are excluded (right).

We again look at the performance of SPCA through the lens of the hedging portfolio R^2 , based on the following economic motivation: if SPCA is successful in extracting weak latent factors that g_t

loads on, we can hope to see this reflected in a higher hedging ability of the corresponding hedging portfolio. Figure 8 reports the out-of-sample time-series R^2 heatmap for the three factors: value, momentum and profitability. On the left of each row we can see the R^2 obtained using all assets from the Hou et al. (2020) dataset; on the right we can see the results excluding the test assets corresponding to each factor. By looking at the last row of each heatmap, which corresponds to the PCA estimate with no selection, it is clear that the hedging performance of a portfolio built via PCA deteriorates significantly when the most informative assets are removed. Consider for example the case $p = 9$. For value, the PCA hedging portfolio's out-of-sample R^2 decreases from 64% to 47%, as value and growth assets are removed; SPCA's R^2 decreases by substantially less, from 74% to 62%. In the case of momentum, the R^2 decreases from 76% to 48% for PCA, but only from 86% to 77% for SPCA. Finally, for profitability, the R^2 decreases from 41% to 14% for PCA, but only from 71% to 60% for SPCA. In all cases, the SPCA portfolio hedging ability deteriorates little when the relative sorts are removed and the factor is made weaker, whereas the deterioration is much larger for PCA.

To sum up, these empirical results mirror the simulations in Section 3, which show that SPCA performs well (as measured by the hedging ability of the corresponding portfolio) even when the factor of interest is weak in the universe of test assets considered. This is important in practical applications: given a certain factor g_t , we do not know ex ante if many or just a few assets are exposed to that factor, so a method like SPCA that is robust to that possibility yields an important advantage.

4.2.6 Robustness

We conclude by reporting in Table 6 a version of Table 4 obtained using the Hou et al. (2020) dataset instead of the Chen and Zimmermann (2020) data. The results are qualitatively similar to the ones obtained using the Chen and Zimmermann (2020) data, and, with a few exceptions, not statistically different. This confirms that, broadly, the results do not depend on using one particular universe of test assets. That said, the results also suggest some differences between these two universes of test assets, which our analysis in the next section sheds some light on.

4.3 Diagnosing Factor Models via SPCA

In the previous section we applied SPCA to the estimation of risk premia. In this section, we illustrate the use of SPCA to diagnose missing factors in observable-factor models, applying the theory developed in Section 2.3. Recall that given an observable-factor model g_t , and a set of test assets r_t , we can use SPCA to recover the latent-factor SDF (using g_t to supervise the extraction of weak factors). If we find that the Sharpe ratio achieved by the latent factors recovered by SPCA is higher than that achieved by g_t , we can conclude that the factor model using g_t to span the SDF, is missing some factor. This is not just a test of whether g_t explains r_t , as it instead focuses on shedding light on *why* a model may be rejected in the data.

Table 6: Risk Premia Estimates, [Hou et al. \(2020\)](#) Data

	Avg. ret.		3 Latent Factors			5 Latent Factors			6 Latent Factors			9 Latent Factors			Joint estim, 9 factors	
	(train.)	(eval.)	RP	# Assets	R^2	RP	# Assets	R^2	RP	# Assets	R^2	RP	# Assets	R^2	RP	Stderr
Market	74	62	72	100	0.98	74	100	0.99	74	100	0.99	74	100	0.99	71	26
HML	39	-7	22	100	0.69	20	100	0.69	16	100	0.71	18	100	0.74	24	16
SMB	12	25	-12	100	0.74	-13	100	0.72	-16	100	0.74	-15	100	0.79	-9	18
RMW	37	28	12	100	0.38	26	100	0.71	25	100	0.71	36	150	0.76	18	9
CMA	26	19	8	100	0.70	11	100	0.65	12	100	0.66	4	300	0.65	8	11
Momentum	91	30	68	100	0.81	60	100	0.86	57	100	0.85	55	100	0.86	58	20
BAB	126	56	47	100	0.08	37	100	0.07	31	100	0.08	27	200	0.04	37	12
QMJ	41	39	-3	150	0.68	15	100	0.82	15	100	0.83	17	150	0.80	6	10
Liquidity			28	1700	0.05	35	1700	0.06	42	1700	0.06	35	1700	0.06	37	18
Intermed. Cap.			107	100	0.49	98	100	0.46	91	100	0.51	63	100	0.43	90	37
IP growth			-2	1700	0.01	-4	1700	-0.02	-3	1700	-0.01	-3	1700	-0.01	-1	2
LN 1			171	1200	-0.11	215	1650	-0.15	151	1700	-0.11	169	1700	-0.08	110	93
LN 2			-19	1700	-0.08	-17	1700	-0.08	-13	1700	-0.08	-4	1700	-0.12	-83	55
LN 3			16	1000	0.03	69	1550	0.04	26	1700	0.02	15	1700	0.03	11	62
Consumption			0	1700	0.00	0	1700	0.00	1	1700	0.00	0	1700	0.00	1	1
Fin. Unc.			-5	1600	0.18	-15	1700	0.16	-15	1700	0.16	-9	1700	0.14	-18	11
Real Unc.			-4	1700	0.02	-5	1700	0.02	-8	1700	0.02	-6	1700	-0.03	-11	6
Macro Unc.			-2	1700	0.05	-4	1700	0.05	-6	1700	0.05	-4	1700	0.04	-9	5
Term			-11	1700	-0.11	24	1700	-0.10	77	1700	-0.08	24	1700	-0.14	261	240
Credit			24	1700	-0.02	29	1700	-0.03	0	1700	-0.06	8	1700	-0.09	16	40
Unempl.			42	1700	0.00	116	1700	-0.01	112	1700	-0.01	101	1700	-0.02	89	61
Sentiment HJTZ			-44	1700	0.01	-39	1700	0.01	-22	1700	0.02	-20	1700	0.02	-39	44
Sentiment BW			-29	1700	0.03	-31	1700	0.02	-21	1700	0.02	-25	1700	-0.01	9	43
Oil			-8	1600	-0.03	-39	1500	0.00	-35	1600	0.00	-26	1550	-0.01	-47	28

Note: Same as Figure 4, but using the characteristic-sorted portfolios from [Hou et al. \(2020\)](#) instead of those from [Chen and Zimmermann \(2020\)](#).

We consider five observable factor models: the CAPM, the Fama-French 3-factor model (FF3), the Fama-French 5-factor model (FF5), and finally two richer models: one with the FF5 factors plus momentum, and one with FF5 plus momentum, BAB, and QMJ. We diagnose these models using both the CZ and the HXZ datasets.

We divide the sample into two parts as in Section 2.2, and use the first half for training (and selection of the tuning parameter) and the second half for out-of-sample evaluation. Maximal Sharpe ratios achieved using the factors in g_t and using the factors from SPCA are calculated out of sample.

Figure 9 reports the results. Each panel corresponds to a different model. The x axis in each figure corresponds to the number of factors extracted via SPCA. The y axis is the out-of-sample Sharpe ratio. The Sharpe ratio achieved by g_t is represented by a dashed solid line, which naturally does not depend on the number of latent factors. In each graph, we overlay the SPCA results with the HXZ and CZ data, respectively, using different markers (blue triangles for HXZ and red circles for CZ). Not surprisingly, the out-of-sample Sharpe ratios are somewhat noisy; we also plot fitted lines using raw estimates to help visualize the trend.

Consider panel (a), in which g_t is just the market. The market in our out-of-sample period achieves a Sharpe ratio of 0.45 (dashed line). SPCA factors extracted using g_t achieve significantly higher Sharpe ratios, both in the HXZ and CZ data. The Sharpe ratio increases with the number of factors, indicating that the CAPM misses several sources of risk. Results for the FF3 and FF5 models (panels (b) and (c)) are similar: for both, once the number of factors is sufficiently large, SPCA produces a Sharpe ratio that is superior to either model. Interestingly, however, the FF5 model does perform as well as SPCA in the HXZ data. This suggests that relative to the the universe of test assets in the HXZ dataset, the FF5 model appears to be spanned by almost all sources of risk driving this dataset (but not so in the CZ dataset).

As more observable factors are added to these models, we should expect the the Sharpe ratio of the model to increase, as long as more latent factors adds risk factors and not noise. We indeed find that this is the case: when adding momentum (panel (d)) and momentum, BAB and QMJ (panel (e)) to g_t , the improvement of SPCA becomes lower (though it still remains nonzero in the CZ data). Overall, this suggests that these richer models do a better job in capturing the fundamental sources of risk in these dataset, although some amount of misspecification remains visible in the CZ dataset.

The differences between the results using the HXZ and CZ datasets also emphasize the importance of the choice of test assets. Ideally, to have as powerful tests as possible, we would want to have a large and varied universe of test assets. The number of assets in a datasets is, however, not a perfect proxy for the richness of the universe in terms of risk exposures. In fact, as we have remarked in this paper, a universe with large N but low exposures to some factors can introduce a weak factor problem. Here we see another case in which the size of the dataset does not necessarily translate into richer risk exposure: HXZ contains more assets than CZ; yet, the results in this section show that using the test assets, r_t , from CZ, SPCA diagnoses additional factors compared to the ones diagnosed using HXZ (this could reflect, for example, a different construction of the portfolios in the

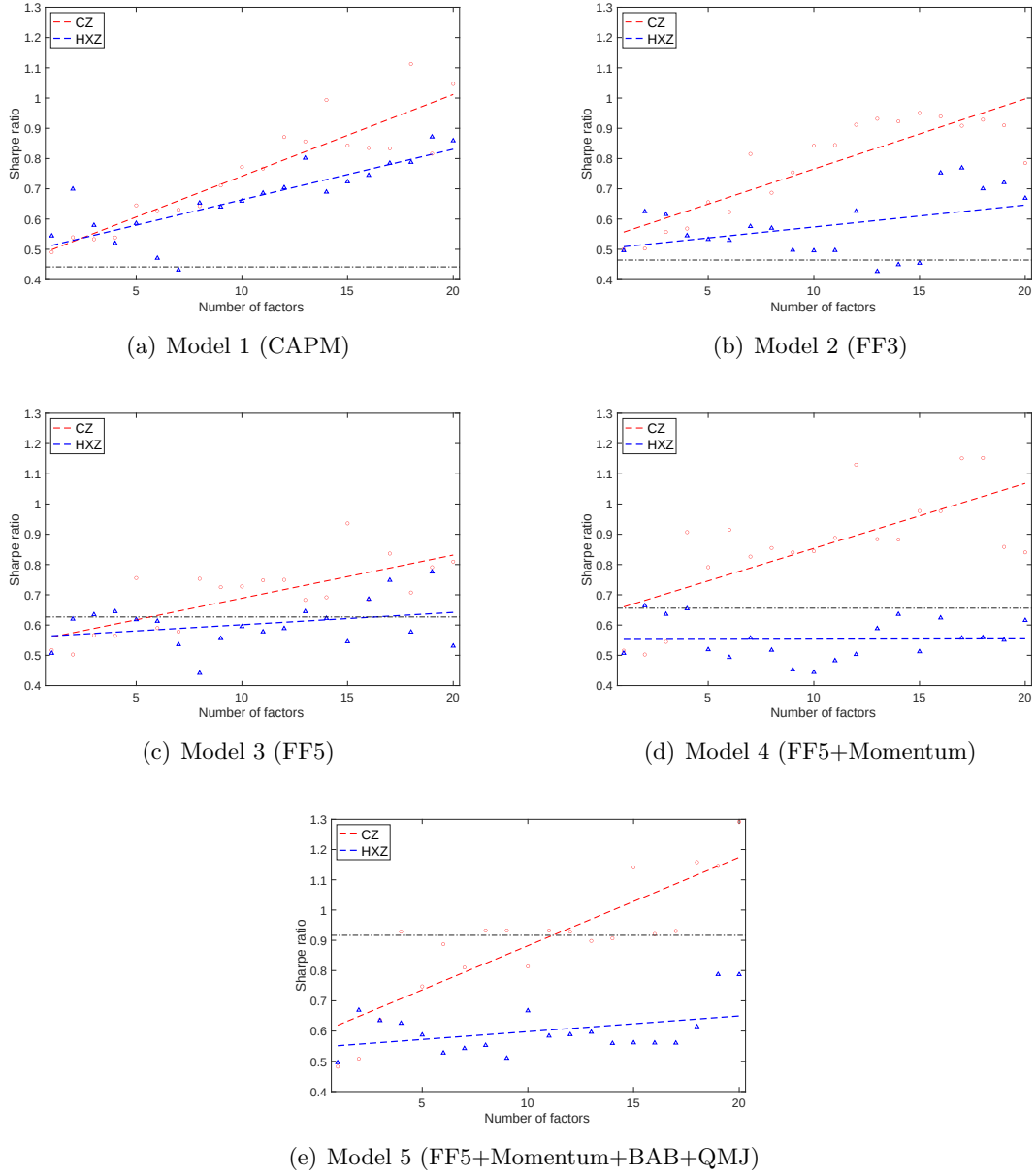


Figure 9: Out-of-sample Sharpe Ratios of Different Factor Models

Note: Each panel reports the out-of-sample Sharpe ratio of an observable-factor model g_t (dashed line), together with the out-of-sample Sharpe ratio obtained from the factors recovered using SPCA, in the HXZ data (triangles) and in CZ (circles). The x axis corresponds to the number of factors used in SPCA (p).

different datasets, or a different selection of characteristics).

Overall, these results illustrate that the ability of SPCA to recover weak latent factors can prove useful as a diagnostic tool for observable factor models, and once again highlight the importance of the choice of test assets in performing asset pricing tests.

5 Conclusions

The choice of test assets plays a fundamental role in empirical asset pricing tests. The recent explosion of anomaly discoveries and related characteristics in the empirical literature has provided researchers with a large universe of potential test assets to choose from. On the one hand, the availability of so many different characteristics gives us hope that the returns of these portfolios can help us uncover and identify the pricing of various dimensions of risk, including those that are not well captured by standard cross-sections. On the other hand, the large dimensionality goes hand in hand with the weak factor issue: a factor may well be captured by *some* assets within the large cross-section, but if most assets do not have exposure to that factor, it will be weak and inference will be incorrect.

Traditional methodologies take the cross-section of assets as given. In this paper, we present a new methodology, SPCA, that instead actively selects assets in order to estimate risk premia of factors of interest, whether they are strong or weak, and at the same time addresses the issue of potentially omitted factors, again regardless of whether they are strong or weak. In addition, SPCA can exploit its ability to recover weak latent factors to help diagnose omitted factors in observable-factor models. The paper confirms the good performance of SPCA for both of these tasks in a variety of simulations, and illustrates the application of the methodology in various empirical contexts.

While the road to a full understanding of risk and risk premia in financial markets is still long, we believe that addressing systematically the issue of weak factors in empirical asset pricing is an important step forward, that opens the door to the study of factors that, while important to investors, may be not pervasive either in the standard cross-sections or in the recently developed large universes of test assets.

Two pressing issues on the debates related to the factor zoo are the economic interpretability and the overwhelming amount of degrees of freedom in empirical asset pricing research. The central issue we address in this paper is to evaluate factors motivated by economic theories. Our proposal eliminates two critical degrees of freedom altogether from this exercise: the choice of control factors when estimating risk premia of economically motivated factors, and the choice of test assets used for estimation and testing. Our study thereby contributes to a promising agenda developing a fusion of asset pricing theory and machine learning. It does so by using the factor structure as a main theoretical foundation, and applying to it tools and results from machine learning, in order to exploit these statistical advances while maintaining economic interpretability.

References

- Ahn, D.-H., J. Conrad, and R. F. Dittmar (2009). Basis assets. *The Review of Financial Studies* 22(12), 5133–5174.
- Anatolyev, S. and A. Mikusheva (2021). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, forthcoming.

- Ang, A., R. Hodrick, Y. Xing, and X. Zhang (2006). The cross-section of volatility and expected returns. *Journal of Finance* 61, 259–299.
- Asness, C. S., A. Frazzini, and L. H. Pedersen (2013). Quality Minus Junk. Technical report, AQR.
- Bai, J. (2003). Inferential Theory for Factor Models of Large Dimensions. *Econometrica* 71(1), 135–171.
- Bai, J. and S. Ng (2002). Determining the number of factors in approximate factor models. *Econometrica* 70, 191–221.
- Bai, J. and S. Ng (2008). Forecasting economic time series using targeted predictors. *Journal of Econometrics* 146(2), 304–317.
- Bai, J. and S. Ng (2021). Approximate factor models with weaker loading. Technical report, Columbia University.
- Bailey, N., G. Kapetanios, and M. H. Pesaran (2020). Measurement of factor strength: Theory and practice.
- Bair, E., T. Hastie, D. Paul, and R. Tibshirani (2006). Prediction by supervised principal components. *Journal of the American Statistical Association* 101(473), 119–137.
- Bair, E. and R. Tibshirani (2004). Semi-supervised methods to predict patient survival from gene expression data. *PLoS Biology* 2(4), 511–522.
- Baker, M. and J. Wurgler (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance* 61(4), 1645–1680.
- Bryzgalova, S. (2015). Spurious Factors in Linear Asset Pricing Models. Technical report, Stanford University.
- Bryzgalova, S., J. Huang, and C. Julliard (2019). Bayesian solutions for the factor zoo: We just ran two quadrillion models. *Available at SSRN 3481736*.
- Bryzgalova, S., M. Pelger, and J. Zhu (2020). Forest through the trees: Building cross-sections of asset returns. Technical report, London School of Business and Stanford University.
- Chamberlain, G. and M. Rothschild (1983). Arbitrage, factor structure, and mean-variance analysis on large asset markets. *Econometrica* 51, 1281–1304.
- Chen, A. Y. and T. Zimmermann (2020). Open source cross-sectional asset pricing. *Available at SSRN*.
- Fama, E. F. and K. R. French (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33(1), 3–56.
- Fan, J. and Y. Fan (2008). High dimensional classification using features annealed independence rules. *The Annals of Statistics* 36, 2605–2637.
- Fan, J., Y. Ke, and Y. Liao (2021). Augmented factor models with applications to validating market risk factors and forecasting bond risk premia. *Journal of Econometrics* 222(1), 269–294.

- Fan, J. and Y. Liao (2022). Learning latent factors from diversified projections and its applications to over-estimated and weak factors. *Journal of the American Statistical Association* 117(538), 909–924.
- Fan, J. and J. Lv (2008). Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society, B* 70(5), 849–911.
- Feng, G., S. Giglio, and D. Xiu (2020). Taming the factor zoo: A test of new factors. *Journal of Finance* 75(3), 1327–1370.
- Frazzini, A. and L. H. Pedersen (2014). Betting against beta. *Journal of Financial Economics* 111(1), 1–25.
- Freyaldenhoven, S. (2019). A generalized factor model with local factors.
- Gagliardini, P., E. Ossola, and O. Scaillet (2016). Time-varying risk premium in large cross-sectional equity datasets. *Econometrica* 84(3), 985–1046.
- Giglio, S., B. Kelly, and D. Xiu (2022). Factor models, machine learning, and asset pricing. *Annual Review of Financial Economics* 14, 337–368.
- Giglio, S., D. Xiu, and D. Zhang (2023). Prediction when factors are weak. Technical report, University of Chicago.
- Giglio, S. W. and D. Xiu (2021). Asset pricing with omitted factors. *Journal of Political Economy* 129(7), 1947–1990.
- Gospodinov, N., R. Kan, and C. Robotti (2013). Chi-squared tests for evaluation and comparison of asset pricing models. *Journal of Econometrics* 173(1), 108–125.
- Gospodinov, N., R. Kan, and C. Robotti (2014). Misspecification-Robust Inference in Linear Asset-Pricing Models with Irrelevant Risk Factors. *The Review of Financial Studies* 27(7), 2139–2170.
- Harvey, C. R., Y. Liu, and H. Zhu (2016). ...and the Cross-Section of Expected Returns. *The Review of Financial Studies* 29(1), 5–68.
- He, Z., B. Kelly, and A. Manela (2017). Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics* 126(1), 1–35.
- Hou, K., C. Xue, and L. Zhang (2020). Replicating anomalies. *Review of Financial Studies* 33(5), 2019–2133.
- Huang, D., F. Jiang, K. Li, G. Tong, and G. Zhou (2021). Scaled pca: A new approach to dimension reduction. *Management Science*, forthcoming.
- Huang, D., F. Jiang, J. Tu, and G. Zhou (2015). Investor sentiment aligned: A powerful predictor of stock returns. *The Review of Financial Studies* 28(3), 791–837.
- Jurado, K., S. C. Ludvigson, and S. Ng (2015). Measuring uncertainty. *The American Economic Review* 105(3), 1177–1216.
- Kan, R. and C. Zhang (1999). Two-Pass Tests of Asset Pricing Models with Useless Factors. *The*

- Journal of Finance* 54(1), 203–235.
- Kelly, B., S. Pruitt, and Y. Su (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics* 134(3), 501–524.
- Kim, S., R. A. Korajczyk, and A. Neuhierl (2020). Arbitrage portfolios. *Review of Financial Studies*, *Forthcoming*.
- Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics* 149(2), 149–173.
- Kozak, S., S. Nagel, and S. Santosh (2020). Shrinking the cross-section. *Journal of Financial Economics* 135(2), 271–292.
- Lettau, M. and M. Pelger (2020). Estimating latent asset-pricing factors. *Journal of Econometrics* 218, 1–31.
- Lewellen, J., S. Nagel, and J. Shanken (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics* 96(2), 175–194.
- Ludvigson, S. C. and S. Ng (2010). A factor analysis of bond risk premia. In A. Ulah and D. E. A. Giles (Eds.), *Handbook of empirical economics and finance*, Volume 1, Chapter 12, pp. 313–372. Chapman and Hall, Boca Raton, FL.
- Pástor, L. and R. F. Stambaugh (2003). Liquidity risk and expected stock returns. *Journal of Political Economy* 111(3), 642–685.
- Pesaran, M. H. and R. Smith (2019). The role of factor strength and pricing errors for estimation and inference in asset pricing models.
- Ross, S. A. (1976). The Arbitrage Theory of Capital Asset Pricing. *Journal of Economics Theory* 13, 341–360.
- Schorfheide, F., D. Song, and A. Yaron (2018). Identifying long-run risks: A bayesian mixed-frequency approach. *Econometrica* 86(2), 617–654.
- Shanken, J. (1992). On the Estimation of Beta Pricing Models. *The Review of Financial Studies* 5(1), 1–33.
- Uematsu, Y., Y. Fan, K. Chen, J. Lv, and W. Lin (2019). Sofar: Large-scale association network learning. *IEEE transactions on information theory* 65(8), 4924–4939.
- Uematsu, Y. and T. Yamagata (2022a). Estimation of sparsity-induced weak factor models. *Journal of Business & Economic Statistics* 41(1), 213–227.
- Uematsu, Y. and T. Yamagata (2022b). Inference in sparsity-induced weak factor models. *Journal of Business & Economic Statistics* 41(1), 126–139.
- Wan, R., Y. Li, W. Lu, and R. Song (2023). Mining the factor zoo: Estimation of latent factor models with sufficient proxies. *Journal of Econometrics*.

Internet Appendix for Test Assets and Weak Factors

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Abstract

This internet appendix provides alternative estimators, model assumptions, supplementary theoretical and simulation findings, implementation details in the empirical analysis, and complete mathematical proofs.

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A Alternative Estimators and Their Asymptotic Behavior

While the literature has proposed several different estimators of the SDF and risk premia, their properties in a general weak factor setting like ours have not been investigated. In this section we revisit a number of estimation procedures, and show that they are inconsistent in the presence of weak factors, using a simple model with a single weak factor.

We focus our discussion of alternative estimators on those that can be used when factors are latent. In this setting, the researcher does not need to know the identities of all true factors, which yields a risk premium estimator that is robust to potentially omitted factors.

PLS

As reviewed in the main text, [Giglio and Xiu \(2021\)](#) show that the PCA-based estimation procedure effectively constructs a mimicking portfolio for g_t via a principal component regression (PCR) on r_t , which amounts to a projection of g_t onto the first few PCs of the sample covariance matrix of r_t . This is an unsupervised approach, in that the PCs are obtained without any information from g_t . Therefore, PCA might be misled by large idiosyncratic errors in r_t when the signal is not sufficiently strong. In contrast with PCA, partial least squares (PLS) is a supervised procedure, which has been shown to work better than PCA in other settings, see, e.g., [Kelly and Pruitt \(2013\)](#). In the same spirit, we propose a PLS-based approach for risk premia estimation, exploiting variation of returns that is relevant to the target factor of interest.

The key difference between the two approaches is that PCA seeks linear combinations of r_t that maximize variation, ignoring information from the target g_t , whereas PLS seeks linear combinations that have the largest covariance with g_t . The PLS-based risk premia estimator effectively uses PLS instead of PCA in the first step of Algorithm 1 described in the main text.

We formulate a general PLS-based algorithm for a $d \times 1$ vector of g_t below:

Algorithm 1 (PLS-based Estimator of Risk Premia). *The estimator proceeds as follows:*

Inputs: $\bar{R}_{(1)} := \bar{R}$, $\bar{r}_{(1)} := \bar{r}$ and $\bar{G}$, a $d \times T$ matrix.

S1. For $k = 1, 2, \dots, p$, repeat the following steps using $\bar{R}_{(k)}$, $\bar{r}_{(k)}$ and $\bar{G}$.

- a. Obtain the weight vector $\hat{w}$ from the largest left singular vector of $\bar{R}_{(k)}\bar{G}^\top$.*
- b. Estimate the k th factor as $\hat{V}_{(k)} = \sqrt{T}\hat{w}^\top \bar{R}_{(k)} / \|\hat{w}^\top \bar{R}_{(k)}\|$.*
- c. Estimate the risk premium of $\hat{V}_{(k)}$ by $\hat{\gamma}_{(k)} = \sqrt{T}\hat{w}^\top \bar{r}_{(k)} / \|\hat{w}^\top \bar{R}_{(k)}\|$.*
- d. Estimate the k th factor loading of r_t by $\hat{\beta}_{(k)} = T^{-1}\bar{R}_{(k)}\hat{V}_{(k)}^\top$.*
- e. Remove $\hat{V}_{(k)}$ to obtain residuals for the next step: $\bar{R}_{(k+1)} = \bar{R}_{(k)} - \hat{\beta}_{(k)}\hat{V}_{(k)}$ and $\bar{r}_{(k+1)} = \bar{r}_{(k)} - \hat{\beta}_{(k)}\hat{\gamma}_{(k)}$.*

S2. Estimate the factor loading of g_t on v_t by $\hat{\eta} = T^{-1}\bar{G}\hat{V}^\top$, where $\hat{V} = (\hat{V}_{(1)}^\top, \dots, \hat{V}_{(p)}^\top)^\top$, and denote their risk premia estimated above as $\hat{\gamma} = (\hat{\gamma}_{(1)}, \dots, \hat{\gamma}_{(p)})^\top$.

Output: $\hat{\gamma}_g^{PLS} = \hat{\eta}\hat{\gamma}$.

The PLS estimator has a closed-form formula if $\bar{G}$ is a $1 \times T$ vector and a single-factor is extracted ($p = 1$):

$$\hat{\gamma}_g^{PLS} = \|\bar{G}\bar{R}^\top \bar{R}\|^{-2} \bar{G}\bar{R}^\top \bar{R}\bar{G}^\top \bar{G}\bar{R}^\top \bar{r}.$$

While the PLS procedure seems appealing, the next proposition shows that this approach is asymptotically equivalent to the PCA-based procedure, hence it fails in exactly the same weak factor setting as PCA.

Proposition A.1. *Suppose that test asset returns follow a single-factor model in the form of (1) with $p = 1$, g_t satisfies (4) with $d = 1$, u_t and v_t i.i.d. normally distributed and independent from each other, and $z_t = 0$. In addition, suppose that β satisfies $N/(\|\beta\|^2 T) \rightarrow B \geq 0$ and $\|\beta\| \rightarrow \infty$. Then we have $\hat{\gamma}_g^{PLS} \xrightarrow{P} (1 + B)^{-1} \eta\gamma$.*

Intuitively, the covariance information embedded in the objective function of PLS is dominated by its variance component, hence PLS yields the same asymptotic behavior as PCA with respect to estimating β , and therefore risk premia.

Ridge

Next, we consider an alternative, ridge-regression-based approach to the construction of mimicking portfolios, which instead directly regularizes the projection of g_t on the vector of returns. The Ridge-based estimator can be written as:

$$\hat{\gamma}_g^{Ridge} = \bar{G}\bar{R}^\top (\bar{R}\bar{R}^\top + \mu \mathbb{I}_N)^{-1} \bar{r}, \quad (\text{A.1})$$

where $\mu > 0$ is some tuning parameter. In the case of pervasive factors, Giglio and Xiu (2021) show that the ridge estimator yields a consistent estimate of $\eta\gamma$. However, we show that the ridge estimator also fails in the presence of weak factors:

Proposition A.2. *Suppose that test asset returns follow a single-factor model in the form of (1) with $p = 1$, g_t satisfies (4) with $d = 1$, u_t and v_t i.i.d. normally distributed and independent from each other, and $z_t = 0$. In addition, suppose that β satisfies $N/(\|\beta\|^2 T) \rightarrow B \geq 0$ and $\|\beta\| \rightarrow \infty$, and the tuning parameter μ satisfies $\mu/(\|\beta\|^2 T) \rightarrow D$ for some constant $D \geq 0$ such that $B + D > 0$. Then we have $\hat{\gamma}_g^{Ridge} \xrightarrow{P} (1 + B + D)^{-1} \eta\gamma$.*

Intuitively, the Ridge-based estimator fails because the tuning parameter μ in the ridge procedure serves as a threshold that suppresses the influence of eigenvectors corresponding to small eigenvalues, just like in PCA and PLS (which explains the appearance of B in the limit). The presence of μ also leads to a shrinkage bias to the first few eigenvectors (i.e., factors), which is why an extra term D appears in the limit as well.

Risk Premium PCA

Finally, we consider an estimator based on the approach of Lettau and Pelger (2020). This approach was designed to estimate a latent-factor SDF, but can also be used to estimate the risk premium of a factor g_t , by replacing the PCA step of Algorithm 1 with the risk premia PCA procedure of Lettau and Pelger (2020):

Algorithm 2 (rpPCA-based Estimator of Risk Premia). *The estimator proceeds as follows:*

Inputs: $\bar{R}$ and $\bar{G}$.

- S1. Apply PCA on $T^{-1}RR^\top + \mu\bar{r}\bar{r}^\top$, where μ is a tuning parameter, and write the first p eigenvectors as $\hat{\varsigma}$. The estimated factors are given by $\hat{V} = \hat{\varsigma}^\top \bar{R}$.
- S2. Estimate the risk premia of $\hat{V}$ by $\hat{\gamma} = \hat{\varsigma}^\top \bar{r}$.
- S3. Estimate the factor loading of g_t on v_t by $\hat{\eta} = \bar{G}\hat{V}^\top(\hat{V}\hat{V}^\top)^{-1}$.

Outputs: $\hat{\gamma}_g^{rpPCA} = \hat{\eta}\hat{\gamma}$.

Standard PCA is applied to the covariance matrix of returns, that is $T^{-1}RR^\top - \bar{r}\bar{r}^\top$. Lettau and Pelger (2020) show that assigning a larger weight $\mu > -1$ to the term related to average returns (the second term) improves the Sharpe ratio of the estimated SDF. Lettau and Pelger (2020) derive asymptotic properties of rpPCA in a setting where all factors are weak and N and T increase to infinity at the same rate. The setting they analyze is one where all factors are extremely weak, so that they cannot be recovered – specifically, the strength of weak factors remains indistinguishable from that of idiosyncratic errors as N and T increase. Under this assumption, consistent estimation of the SDF is impossible, including by rpPCA, which, despite being more correlated with the SDF than PCA, is also inconsistent. In contrast, we preclude this extreme case from our discussion because no estimators under consideration could achieve consistency and a harmless modeling choice would be to treat these extremely weak factors as noise: their risk premia cannot be distinguished from alpha. The weak-factor setting we investigate permits consistency, and allows for asymptotic comparison of different estimators. The following proposition shows that, like PCA, rpPCA is also inconsistent for estimating risk premia.

Proposition A.3. *Suppose that test asset returns follow a single-factor model in the form of (1) with $p = 1$, g_t satisfies (4) with $d = 1$, u_t and v_t i.i.d. normally distributed and independent from each other, and $z_t = 0$. In addition, suppose that β satisfies $N/(\|\beta\|^2 T) \rightarrow B \geq 0$ and $\|\beta\| \rightarrow \infty$, that the factor has a non-zero risk premia, i.e., $\gamma \neq 0$. Then for some tuning parameter $\mu > -1$, we have*

$$\hat{\gamma}_g^{rpPCA} \xrightarrow{P} w(1+B)^{-1}\eta\gamma + (1-w)\eta(\gamma + \gamma^{-1}B),$$

where w is a constant that depends on B, μ, γ , explicitly given by equation (F.39) in the proof.

Proposition A.3 shows that this rpPCA estimator is inconsistent in the presence of a weak factor, with a more involved bias term compared to the above estimators. Like PCA and PLS, this estimator is consistent when all factors are strong ($B = 0$). When $B > 0$, the estimator is inconsistent.

Different asymptotic settings can affect the asymptotic behavior of rpPCA. For example, if one assumes that the tuning parameter $\mu \rightarrow \infty$, the rpPCA estimator converges to $\eta(\gamma + \gamma^{-1}B)$ (so it still displays a bias). If we further assume $\gamma \rightarrow \infty$ (while keeping $\eta\gamma$ constant), this estimator can be consistent as long as $\eta\gamma^{-1}B \xrightarrow{P} 0$. This suggests that rpPCA can be robust to weak factors if the information about β from the expected return $\beta\gamma$ dominates the information from return covariances (when $\gamma \rightarrow \infty$). But this is only the case if factors have diverging Sharpe ratios, i.e., $\Sigma_v^{-1/2}\gamma \rightarrow \infty$.

B Model Assumptions

To derive the asymptotic properties of the SPCA and alternative estimators, we need the following high-level assumptions, which can be easily verified by standard and more primitive assumptions. We start with assumptions that characterize the data generating process (DGP) of returns and factor proxies.

Assumption B.1. *The factor innovation V satisfies:*

$$\|\bar{v}\| \lesssim_P T^{-1/2}, \quad \|T^{-1}VV^\top - \Sigma_v\| \lesssim_P T^{-1/2}, \quad \|V\|_{\text{MAX}} \lesssim_P (\log T)^{1/2},$$

where $\Sigma_v \in \mathbb{R}^{p \times p}$ is a positive-definite matrix with $\lambda_p(\Sigma_v) \gtrsim 1$ and $\lambda_1(\Sigma_v) \lesssim 1$.

Assumption B.2. *The residual innovation Z satisfies:*

$$\|\bar{z}\| \lesssim_P T^{-1/2}, \quad \|T^{-1}ZZ^\top - \Sigma_z\| \lesssim_P T^{-1/2}, \quad \|Z\|_{\text{MAX}} \lesssim_P (\log T)^{1/2},$$

where $\Sigma_z \in \mathbb{R}^{d \times d}$ is a positive-definite matrix with $\lambda_d(\Sigma_z) \gtrsim 1$ and $\lambda_1(\Sigma_z) \lesssim 1$. In addition,

$$\|ZV^\top\| \lesssim_P T^{1/2}.$$

Assumptions B.1 and B.2 impose rather weak conditions on the time series behavior of the factors and measurement error. Since v_t and z_t have a finite cross-sectional dimension, both assumptions hold if these processes are stationary, strong mixing, and satisfy some moment conditions.

Assumption B.3. *The factor loading matrix β satisfies*

$$\|\beta\|_{\text{MAX}} \lesssim 1, \quad \lambda_p(\beta_{[I_0]}^\top \beta_{[I_0]}) \gtrsim N_0,$$

for some index set I_0 , where $N_0 = |I_0|$.

Assumption B.3 implies that there exists a subset of test assets, within which all latent factors are strong. It does not imply all factors should have identical strength with respect to the entire cross-section of assets in r_t .

Next, we need the following moment conditions.

Assumption B.4. *The idiosyncratic component U satisfies:*

$$\|U\|_{\text{MAX}} \lesssim_P (\log T)^{1/2} + (\log N)^{1/2}, \quad \|\bar{u}\|_{\text{MAX}} \lesssim_P T^{-1/2}(\log N)^{1/2}.$$

In addition, for any non-random subset $I \subset \langle N \rangle$,

$$\|U_{[I]}\| \lesssim_P |I|^{1/2} + T^{1/2}, \quad \|\bar{u}_{[I]}\| \lesssim_P |I|^{1/2}T^{-1/2}.$$

Assumption B.4 imposes restrictions on the time-series dependence and heteroskedasticity of u_t . We do not necessarily need stationarity on u_t . That said, the first two inequalities can be shown by some large deviation theorem, see, e.g., Fan et al. (2011); the last two inequalities can be shown by random matrix theory, see Bai and Silverstein (2009), if u_t is i.i.d. both in time and in the cross-section.

Assumption B.5. *For any non-random subset $I \subset \langle N \rangle$, the factor loading $\beta_{[I]}$ and the idiosyncratic error $U_{[I]}$ satisfy the following conditions:*

$$\begin{aligned} (i) \quad & \left\| (\beta_{[I]}^\top \beta_{[I]})^{-1/2} \beta_{[I]}^\top U_{[I]} \right\| \lesssim_P T^{1/2}. \\ (ii) \quad & \left\| (\beta_{[I]}^\top \beta_{[I]})^{-1/2} \beta_{[I]}^\top U_{[I]} \iota_T \right\| \lesssim_P T^{1/2}. \end{aligned}$$

If $\beta_{[I]}^\top \beta_{[I]}$ is singular, we need replace the matrix inverse above by the Moore-Penrose inverse.

Assumption B.6. *The following conditions hold for U , V , β , and any non-random subset $I \subset \langle N \rangle$:*

$$\begin{aligned} (i) \quad & \|U_{[I]} V^\top\| \lesssim_P |I|^{1/2} T^{1/2}, \quad \|U_{[I]} V^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{1/2}. \\ (ii) \quad & \left\| (\beta_{[I]}^\top \beta_{[I]})^{-1/2} \beta_{[I]}^\top U_{[I]} V^\top \right\| \lesssim_P T^{1/2}. \end{aligned}$$

Assumption B.7. *The following conditions hold for U , Z , β , and any non-random subset $I \subset \langle N \rangle$:*

$$\begin{aligned} (i) \quad & \|U_{[I]} Z^\top\| \lesssim_P |I|^{1/2} T^{1/2}, \quad \|U_{[I]} Z^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{1/2}. \\ (ii) \quad & \left\| (\beta_{[I]}^\top \beta_{[I]})^{-1/2} \beta_{[I]}^\top U_{[I]} Z^\top \right\| \lesssim_P T^{1/2}. \end{aligned}$$

Assumptions B.5 - B.7 resemble Assumptions A.7, A.9, and A.10 of Giglio and Xiu (2021), except that here we impose their stronger versions which hold for any non-random subset $I \subset \langle N \rangle$. Of course, these two sets of assumptions are equivalent if u_t is identically distributed along the cross-sectional dimension.

In the main text, we denote the selected subsets in the SPCA procedure as $\widehat{I}_k$, $k = 1, 2, \dots$. We now define their population counterparts. Because SPCA is an iterative procedure, we need these quantities to characterize the limiting behavior of the procedure.

Without loss of generality, we consider the case $\Sigma_v = \mathbb{I}_p$ here. In the general case, we can simply replace β and η by $\beta' = \beta \Sigma_v^{1/2}$ and $\eta' = \eta \Sigma_v^{1/2}$ in the following definitions. In detail, we start with $a_i^{(1)} := \|\beta_{[i]} \eta^\top\|_{\text{MAX}}$ and define $I_1 := \{a_i^{(1)} \geq c_{qN}^{(1)}\}$, where $c_{qN}^{(1)}$ is the (qN) th largest value in $\{a_i^{(1)}\}_{i=1, \dots, N}$. Then, we denote the largest right singular vector of $\beta_{(1)} := \beta_{[I_1]}$ by b_1 . For $k > 1$, we obtain $a_i^{(k)} := \|\beta_{[i]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top\|_{\text{MAX}}$, $I_k := \{a_i^{(k)} \geq c_{qN}^{(k)}\}$ and b_k is the largest right singular vector of $\beta_{(k)} := \beta_{[I_k]} \prod_{j < k} \mathbb{M}_{b_j}$. This procedure is stopped at step $\tilde{p}$ (for some $\tilde{p}$ that is not necessarily equal to p) if $c_{qN}^{(\tilde{p}+1)} < c$. In a nutshell, I_k 's are what we will select if we do SPCA directly on $\beta \in \mathbb{R}^{N \times p}$ and $\eta \in \mathbb{R}^{d \times p}$, while $\widehat{I}_k$'s are obtained by SPCA on $\bar{R} \in \mathbb{R}^{N \times T}$ and $\bar{G} \in \mathbb{R}^{d \times T}$. We need the following assumption to guarantee the selection consistency, that is, $P(\widehat{I}_k = I_k) \rightarrow 1$ for any $1 \leq k \leq \tilde{p}$.

Assumption B.8. We assume that $\beta_{(k)}$, $a_i^{(k)}$ and c in the above procedure satisfy:

(i) $\sigma_1(\beta_{(k)})$ and $\sigma_2(\beta_{(k)})$ are distinct in the sense that there exists a constant $\delta > 0$ such that

$$\sigma_2(\beta_{(k)}) \leq (1 + \delta)^{-1} \sigma_1(\beta_{(k)}).$$

(ii) $c_{qN}^{(k)}$ and $c_{qN+1}^{(k)}$ are distinct in the sense that there exists a constant $\delta > 0$ such that

$$c_{qN+1}^{(k)} \leq (1 + \delta)^{-1} c_{qN}^{(k)},$$

where $c_{qN}^{(k)}$ and $c_{qN+1}^{(k)}$ are the (qN) th and $(qN+1)$ th largest value in $\{a_i^{(k)}\}_{i=1, \dots, N}$, respectively.

(iii) $c_{qN}^{(\tilde{p}+1)}$ and c are distinct in the sense that there exists a constant $\delta > 0$ such that

$$c_{qN}^{(\tilde{p}+1)} \leq (1 + \delta)^{-1} c.$$

Assumption B.8 requires that these singular values are distinguishable, so that their (relative) differences will not vanish asymptotically. This assumption is rather mild, despite not being very explicit.

Assumption B.9. As $T \rightarrow \infty$, the following joint central limit theorem holds:

$$T^{1/2} \begin{pmatrix} T^{-1} \text{vec}(ZV^\top) \\ \bar{v} \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{12}^\top & \Pi_{22} \end{pmatrix} \right),$$

where Π_{11} , Π_{12} , Π_{22} are $dp \times dp$, $dp \times p$, and $p \times p$ matrices, respectively, defined as:

$$\Pi_{11} = \lim_{T \rightarrow \infty} \frac{1}{T} \text{E}(\text{vec}(ZV^\top) \text{vec}(ZV^\top)^\top),$$

$$\begin{aligned}\Pi_{12} &= \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left(\text{vec}(ZV^\top) \iota_T^\top V^\top \right), \\ \Pi_{22} &= \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left(V \iota_T \iota_T^\top V^\top \right).\end{aligned}$$

Assumption B.9 characterizes the joint asymptotic distribution of $ZV^\top$ and $V \iota_T$. Since the dimensions of these random processes are finite, this CLT is a standard result of a central limit theory for mixing processes.

Now we introduce assumptions needed for the SDF estimation. Assumption B.10 ensures that the SDF concept is well defined. Assumption B.11 again can be shown by some large deviation result and certain central limit theorem.

Assumption B.10. Suppose that v_t and u_t are stationary time series independent of β , and that $\Sigma_v = \text{Cov}(v_t)$ and $\Sigma_u = \text{Cov}(u_t)$ satisfy $\lambda_{\min}(\Sigma_v) \gtrsim 1$ and $\lambda_{\max}(\Sigma_u) \lesssim 1$. Consequently, $\Sigma = \text{Cov}(r_t) = \beta \Sigma_v \beta^\top + \Sigma_u$.

Assumption B.11. The time series r_t and the SDF defined by $m_t = 1 - b^\top(r_t - \mathbb{E}(r))$ with $b = \Sigma^{-1} \mathbb{E}(r_t)$ satisfy:

$$\begin{aligned}(1) \quad & \left\| T^{-1} \sum_{t=1}^T (r_t - \bar{r}_t)(m_t - \bar{m}_t) - \text{Cov}(r_t, m_t) \right\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{-1/2}. \\ (2) \quad & \left\| T^{-1} \sum_{t=1}^T (r_t - \bar{r}_t)(r_t - \bar{r}_t)^\top - \text{Cov}(r_t) \right\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{-1/2}. \\ (3) \quad & \left| T^{-1} \sum_{t=1}^T m_t - \mathbb{E}(m_t) \right| \lesssim_P T^{-1/2}. \\ (4) \quad & \left\| T^{-1} \sum_{t=1}^T r_t - \mathbb{E}(r_t) \right\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{-1/2}.\end{aligned}$$

Finally, we need the following assumption for establishing the convergence of the ridge-based SDF estimator. It ensures that all eigenvalues of $\beta \Sigma_v \beta^\top$ are well separated. This assumption shares the spirit with Assumption B.8. A similar assumption has been adopted by, e.g., Wang and Fan (2017).

Assumption B.12. The eigenvalues of $\beta \Sigma_v \beta^\top$ are separated in the sense that

$$(\lambda_j - \lambda_{j+1})/\lambda_j \geq \delta$$

for some constant $\delta > 0$, where $\lambda_j := \lambda_j(\beta \Sigma_v \beta^\top)$ is the j th eigenvalue of $\beta \Sigma_v \beta^\top$.

C Additional Propositions

In this section, we present additional theoretical results. Proposition C.4 establishes that test assets in a subset I_0 are adequate to serve as basis assets, building on which a mimicking portfolio can

approximate the risk premium of any observable factor in g_t .

Proposition C.4. *Suppose that r_t and g_t follow (1) and (4), respectively, and that Assumption B.10 holds. Then for any subset $I_0 \subset \langle N \rangle$, we have*

$$\text{Cov}(g_t, r_{t,[I_0]}) \text{Cov}(r_{t,[I_0]})^{-1} \mathbb{E}(r_{t,[I_0]}) = \eta\gamma + O\left(1/\lambda_{\min}(\beta_{[I_0]}^\top \beta_{[I_0]})\right).$$

Next, we provide a result that sheds light on the effectiveness of out-of-sample R^2 as a criterion for tuning parameter selection. In the main text, we partition the complete dataset into two segments, one for training and the other for evaluation (testing). Within the training sample, we employ cross-validation to determine the optimal tuning parameters. A more detailed procedure is outlined in Appendix E. For the sake of simplicity, our theoretical analysis is based on a “validation” procedure instead of “cross-validation.” In this context, the phrase “out of sample” specifically refers to the validation sample, used to select the tuning parameters.

For each combination of tuning parameter values $\check{q}$ and $\check{p}$, the application of SPCA to the in-sample data produces factor estimates, with each estimate representing a portfolio. Consequently, this process gives rise to a mimicking portfolio for g_t , characterized by weights denoted as $w(\check{p}, \check{q}) \in \mathbb{R}^{d \times N}$. The expected return of this portfolio is thus estimated as $w(\check{p}, \check{q})\bar{r}$. We also write the matrix forms of de-meaned r_t and g_t out of sample as $\bar{R}_{\text{oos}} \in \mathbb{R}^{N \times T_{\text{oos}}}$ and $\bar{G}_{\text{oos}} \in \mathbb{R}^{d \times T_{\text{oos}}}$, where T_{oos} represents the sample size of out of sample data. The time series R^2 of the i th factor’s hedging portfolio out of sample is thus given by

$$R_i^2(\check{p}, \check{q}) = 1 - \frac{\|(\bar{G}_{\text{oos}})_{[i]} - (w(\check{p}, \check{q}))_{[i]} \bar{R}_{\text{oos}}\|^2}{\|(\bar{G}_{\text{oos}})_{[i]}\|^2}.$$

To derive theoretical results for parameter tuning using these R^2 values, we require additional assumptions about the underlying DGP out of sample. In essence, the following assumption asserts that the DGP remains unchanged between the in-sample and out-of-sample contexts.

Assumption C.13. *Assumptions B.1, B.2, and B.4 hold when V , U , Z , and T are replaced by V_{oos} , U_{oos} , Z_{oos} , and T_{oos} , respectively.*

Moreover, we need the following assumption regarding the relationship between in-sample estimates and out-of-sample DGP.

Assumption C.14. *U_{oos} , V_{oos} , Z_{oos} , and $\check{w} = w(\check{p}, \check{q}) \in \mathbb{R}^{d \times N}$ constructed by in-sample data satisfy:*

$$\|\check{w}_{[i]} U_{\text{oos}}\| \asymp_P \|\check{w}_{[i]}\| T_{\text{oos}}^{1/2}, \quad \|\check{w}_{[i]} U_{\text{oos}} A^\top\| \lesssim_P \|\check{w}_{[i]}\| T_{\text{oos}}^{1/2},$$

for $A = V_{\text{oos}}, Z_{\text{oos}}, \iota_T^\top$ and $i \leq d$.

Assumption C.14 shares the same spirit of Assumptions B.5 - B.7. However, the key distinction lies in the direct imposition of constraints on the relationship between $\check{w}$ and out-of-sample data.

Given that $\check{w}$ is constructed solely from in-sample data, these conditions can be interpreted as restrictions on the dependence between in-sample and out-of-sample data.

It is important to note that the first equality effectively imposes both an upper bound and a lower bound on $\|\check{w}_{[i]}U_{\text{OOS}}\|$. In the special case that $\text{Cov}(u_{\text{OOS},t}) = \Sigma_u$ and in-sample data is independent of the out-of-sample data, each element in $\check{w}_{[i]}U_{\text{OOS}}$ has variance $\check{w}_{[i]}\Sigma_u\check{w}_{[i]}^\top$, which is bounded within $\|\check{w}_{[i]}\|^2 \lambda_{\min}(\Sigma_u)$ and $\|\check{w}_{[i]}\|^2 \lambda_{\max}(\Sigma_u)$. Therefore, the first equality becomes a standard concentration result under the restriction $1 \lesssim \lambda_{\min}(\Sigma_u) \leq \lambda_{\max}(\Sigma_u) \lesssim 1$.

The next proposition shows that selecting tuning parameters using out of sample R^2 leads to consistent estimates of risk premia, both in sample and out of sample:

Proposition C.5. *Suppose that the out-of-sample DGP also follows (1) and (4), and satisfies Assumptions C.13-C.14. In addition, let p^* and q^* be defined by*

$$(p^*, q^*) = \operatorname{argmax}_{\check{p} \leq p_{\max}, \check{q} \in \mathcal{Q}} \sum_{i=1}^d R_i^2(\check{p}, \check{q}),$$

where $p_{\max}$ is some finite upper bound on the number of factors, and $\mathcal{Q} = \{N^{-\alpha_j} | j = 1, \dots, n_q\}$, $0 \leq \alpha_1 \leq \alpha_2 \dots < \alpha_{n_q} < 1$ is a finite grid of tuning parameter values. If $p_{\max} \geq p$, $N^{1-\alpha_{n_q}}/N_0 \rightarrow 0$ and $\log T/N^{1-\alpha_{n_q}} \rightarrow 0$, under assumptions of Theorem 1, as $T_{\text{OOS}} \rightarrow \infty$, we have $w^* = w(p^*, q^*)$ satisfies $\|w^*\bar{r}_{\text{OOS}} - \eta\gamma\| \xrightarrow{P} 0$. In addition, if $q^*N \log N = O(T)$, we have $\|\hat{\gamma}_g^{\text{SPCA}} - \eta\gamma\| = \|w^*\bar{r} - \eta\gamma\| \xrightarrow{P} 0$.

D Additional Simulation Results

In this section, we present a scenario akin to situation c) in the simulation setting of the main text, with the key distinction being that the final factor in this scenario is purely random noise. To elaborate, the DGP of r_t is driven by the first three factors. However, econometricians, who lack knowledge of the true model, include these three factors along with a random noise variable in their attempt to estimate risk premia. This scenario resembles a setting extensively discussed by Kan and Zhang (1999) and Kleibergen (2009).

For the sake of comparison, both PLS and SPCA incorporate this random noise variable alongside the aforementioned three factors, considering them collectively as g_t . The histograms provided in Figure D1 depicting the estimated risk premiums associated with this noise factor reveal that SPCA, PCA, PLS, rpPCA, Lasso, and Ridge methods produce estimates around zero – the true value. The consistency arises because none of these methods entail a cross-sectional regression on the estimated beta of the noise factor. In contrast, the four-split and two-pass methods seem to display substantial variances in this context.

Finally, we investigate the statistical power of SPCA in strong and weak cases, respectively, and draw a comparative analysis with PCA. We adopt the setting in scenario c) of the main text, as the case of strong factors. To simulate a weak factor scenario, we simply replace $a = 0.5$ in c) by

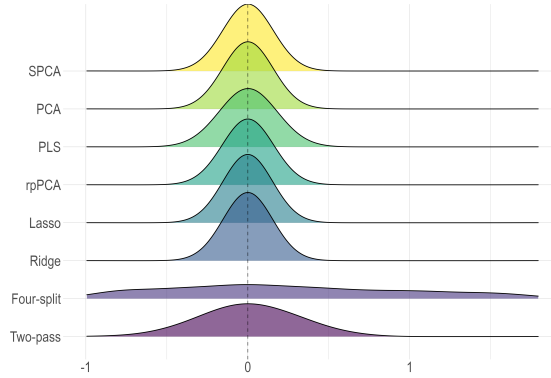


Figure D1: Histogram of Risk Premium Estimates of the noise factor

Note: The figure provides histograms of the risk premium estimates of the noise variable for eight estimators we compare, including SPCA, PCA, PLS, rpPCA, Lasso, Ridge, four-split, and the standard two-pass estimator. We simulate a model of returns driven by three strong factors, whereas g_t includes a pure noise variable, in addition to these three factors. All estimators attempt to estimate risk premia for the three factors and the noise variable altogether. We set $N = 1,000$ and $T = 240$. The number of Monte Carlo repetitions is 1,000.

$\alpha = 0.05$. We consider a null hypothesis that the risk premium of V is zero, whereas the true risk premium of V ranges from -0.01 to 0.01. In Figure D2, we present the rejection rates for both SPCA and PCA. The left panel demonstrates that when all factors are strong, SPCA and PCA yield almost identical results. However, the right panel indicates that SPCA exhibits greater power than PCA across most ranges of risk premium values. The rejection rate for SPCA is around 5% when the null hypothesis is true, and it escalates to 100% as the actual risk premium value diverges from zero.

E Implementation Details

In this section we detail the steps to compute the out-of-sample R^2 used in tuning parameter selection in the empirical analysis.

1. Inputs

- (a) Training sample data: returns for the N assets and target factor g_t , for the first half of the sample
- (b) Evaluation sample data: returns for the N assets and target factor g_t , for the second half of the sample

2. For each value of the number of factors p , execute the following steps:

- (a) Run 100 times the following cross-validation steps:

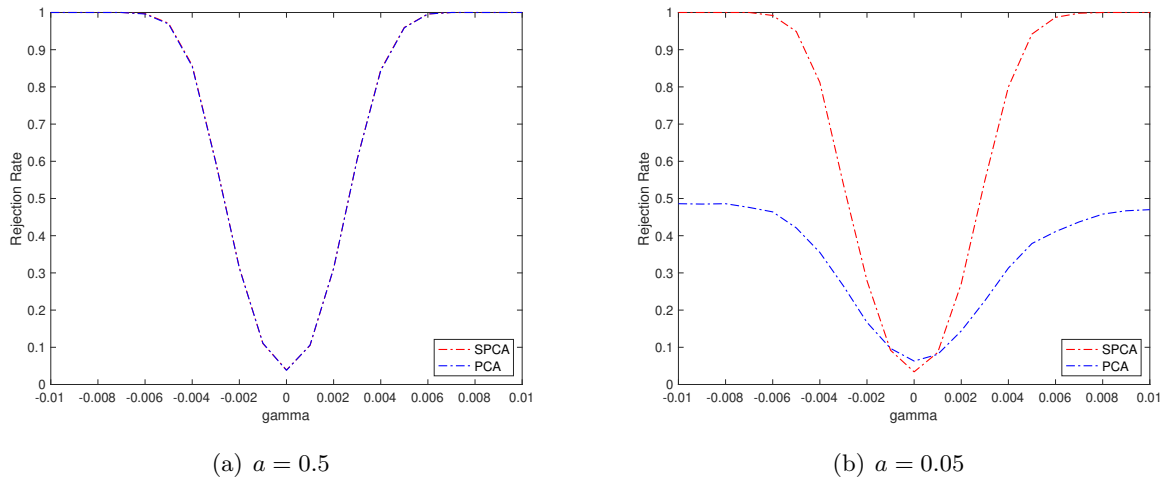


Figure D2: Rejection Rate

Note: We simulate the model in scenario c) of the main text, using two different values for the parameter a : 0.5 for the strong factor case, and 0.05 for the weak factor case. We fix $N = 1,000$ and $T = 240$. The null hypothesis we test is that the risk premium of V is equal to zero, whereas its actual value varies between -0.01 to 0.01. We conduct a total of 1,000 Monte Carlo repetitions.

- i. Divide the training sample data into three folds (subsamples), chosen randomly without replacement
- ii. Choose the first of the three folds as validation (and the other two folds as training)
- iii. For each value of $\lfloor qN \rfloor$ between 100 and the maximum number of assets in the universe N in increments of 50:
 - A. Estimate SPCA in the two training folds using p and $\lfloor qN \rfloor$
 - B. Compute the R^2 of the mimicking portfolio in the validation fold
- iv. Repeat steps ii and iii using folds 2 and then 3 as validation samples (with the remaining two folds as training in each case)
- v. Find the tuning parameter $\lfloor qN \rfloor$ (and therefore the corresponding q) that maximizes the average R^2 in the validation samples across the three folds
- (b) Choose $\lfloor qN \rfloor$ as the median of the 100 tuning choices obtained across the cross-validation runs in (a)
- (c) Estimate SPCA in the training sample using the tuning parameters p and the choice of $\lfloor qN \rfloor$ from (b).
- (d) Compute the out-of-sample R^2 achieved by the SPCA mimicking portfolio estimated in (c) in the evaluation sample.
- (e) Repeat (a)-(d) for every value of p .

F Mathematical Proofs

F.1 Proof of Proposition 1

Proof. Note that for any orthogonal matrix $\Gamma \in \mathbb{R}^{N \times N}$, the estimators based on PCA, PLS and Ridge on $R' = \Gamma R$ are the same as those based on R . Thus, without loss of generality, we can assume $\beta = (\lambda^{1/2}, 0, \dots, 0)^\top$, where $\lambda = \|\beta\|^2$. The same simplifying assumption is adopted in the proofs of Propositions 1, A.1, and A.2. Also, since $z_t = 0$, $\bar{G} = \eta \bar{V}$.

We start with $\hat{\gamma}_g^{PCA}$. We write $\bar{R}$ in the following form:

$$\bar{R} = \beta \bar{V} + \bar{U} = \begin{pmatrix} \sqrt{\lambda} \bar{V} + \bar{U}_1 \\ \bar{U}_2 \end{pmatrix}, \quad (\text{F.2})$$

where $\bar{U}_1$ is the first row of $\bar{U}$ and $\bar{U}_2$ contains the remaining rows. Correspondingly, we write the largest left singular vector of $\bar{R}$ as $\hat{\varsigma} = (\hat{\varsigma}_1, \hat{\varsigma}_2^\top)^\top$, where $\hat{\varsigma}_1$ is the first element of $\hat{\varsigma}$ and $\hat{\varsigma}_2$ is a vector of the remaining $N - 1$ entries of $\hat{\varsigma}$. Recall that in Algorithm 1, we denote $\hat{\xi}$ and $\hat{\varsigma}$ as the largest right and left singular vectors of $\bar{R}$ with the singular value $\sqrt{T\lambda}$, so that by simple algebra we have

$$\hat{\varsigma}_1 = \frac{(\sqrt{\lambda} \bar{V} + \bar{U}_1) \hat{\xi}}{\sqrt{T\lambda}}, \quad \hat{\varsigma}_2 = \frac{\bar{U}_2 \hat{\xi}}{\sqrt{T\lambda}}. \quad (\text{F.3})$$

Since the entries of U and V are i.i.d $\mathcal{N}(0, 1)$, we have

$$|T^{-1} \bar{V} \bar{V}^\top - 1| = |T^{-1} V (\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top) V^\top - 1| \leq |T^{-1} V V^\top - 1| + |\bar{v}|^2 \lesssim_P T^{-1/2},$$

where we use large deviation results $|T^{-1} V V^\top - 1| \lesssim_P T^{-1/2}$ and $|\bar{v}| \lesssim_P T^{-1/2}$ in the last equation. This equation also implies that $\|\bar{V}\| - \sqrt{T} \lesssim_P 1$.

Similarly, we can get $|T^{-1} \bar{U}_1 \bar{U}_1^\top - 1| \lesssim_P T^{-1/2}$ and $\|\bar{U}_1\| - \sqrt{T} \lesssim_P 1$.

In addition, by Lemma A.1 in Wang and Fan (2017), we have $\|N^{-1} U^\top U - \mathbb{I}_T\| \lesssim_P \sqrt{T/N}$, which leads to

$$\|N^{-1} \bar{U}^\top \bar{U} - (\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)\| = \|(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)(N^{-1} U^\top U - \mathbb{I}_T)(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)\| \lesssim_P \sqrt{T/N}.$$

Next, by direct calculation using the above inequalities we obtain

$$\left\| \frac{\bar{V}^\top \bar{U}_1 + \bar{U}_1^\top \bar{V}}{T\sqrt{\lambda}} + \frac{\bar{U}^\top \bar{U} - N(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)}{T\lambda} \right\| \lesssim_P \frac{1}{\sqrt{\lambda}} + \frac{\sqrt{NT}}{T\lambda} \lesssim_P \frac{1}{\sqrt{\lambda}}.$$

Together with (F.2), we have

$$\left\| \frac{\bar{R}^\top \bar{R}}{T\lambda} - \frac{\bar{V}^\top \bar{V}}{T} - \frac{N(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)}{T\lambda} \right\| \lesssim_P \frac{1}{\sqrt{\lambda}}. \quad (\text{F.4})$$

Because of this result, to study the eigenstructure of $\bar{R}^\top \bar{R}/(T\lambda)$, we need analyze the eigenstructure of

$$M := \frac{\bar{V}^\top \bar{V}}{T} + \frac{N(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)}{T\lambda} = \frac{\bar{V}^\top \bar{V}}{T} + \tilde{B}(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top),$$

where $\tilde{B} = N/(T\lambda)$ and the assumption of the proposition implies that $\tilde{B} \rightarrow B$ for a constant B .

Note that $\bar{V} \iota_T = 0$, the eigenvalues of M can be explicitly given by:

$$\lambda_i = \begin{cases} T^{-1} \bar{V} \bar{V}^\top + \tilde{B} & i = 1; \\ \tilde{B} & 2 \leq i \leq T-1; \\ 0 & i = T. \end{cases} \quad (\text{F.5})$$

and the first eigenvector is $\bar{V}^\top / \|\bar{V}^\top\|$. Since the largest eigenvalue of $\bar{R}^\top \bar{R}/(T\lambda)$ is $\hat{\lambda}/\lambda$ with its corresponding eigenvector $\hat{\xi}$, Weyl's theorem yields that

$$\frac{\hat{\lambda}}{\lambda} = \frac{\bar{V} \bar{V}^\top}{T} + \tilde{B} + O_P\left(\frac{1}{\sqrt{\lambda}}\right) = 1 + \tilde{B} + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right), \quad (\text{F.6})$$

and the sin-theta theorem in [Davis and Kahan \(1970\)](#) implies that

$$\left\| \mathbb{P}_{\bar{V}^\top} - \mathbb{P}_{\hat{\xi}} \right\| = \left\| \bar{V}^\top (\bar{V} \bar{V}^\top)^{-1} \bar{V} - \hat{\xi} \hat{\xi}^\top \right\| \lesssim_P \frac{1}{\sqrt{\lambda}}, \quad (\text{F.7})$$

which implies that $(\bar{V} \bar{V}^\top)^{-1} (\bar{V} \hat{\xi})^2 = \hat{\xi}^\top \bar{V}^\top (\bar{V} \bar{V}^\top)^{-1} \bar{V} \hat{\xi} = 1 + O_P(\lambda^{-1/2} + T^{-1/2})$. Together with $|T^{-1} \bar{V} \bar{V}^\top - 1| \lesssim T^{-1/2}$, we have

$$\frac{|\bar{V} \hat{\xi}|}{\sqrt{T}} = 1 + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right). \quad (\text{F.8})$$

It is easy to observe that the sign of $\hat{\xi}$ plays no role in the estimator $\hat{\gamma}_g^{PCA}$, we can choose $\hat{\xi}$ such that

$$\frac{\bar{V} \hat{\xi}}{\sqrt{T}} = 1 + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right). \quad (\text{F.9})$$

Recall that the risk premium estimator is $\hat{\gamma}_g^{PCA} = \hat{\eta} \hat{\gamma}$, where

$$\hat{\eta} = \frac{\bar{G} \hat{\xi}}{\sqrt{T}} \quad \text{and} \quad \hat{\gamma} = \frac{\hat{\varsigma}^\top \bar{r}}{\sqrt{\lambda}}. \quad (\text{F.10})$$

Using $\bar{G} = \eta \bar{V}$ and (F.9), we have

$$\hat{\eta} = \eta + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right). \quad (\text{F.11})$$

Write

$$\hat{\gamma} = \frac{\hat{\zeta}^\top \bar{r}}{\sqrt{\hat{\lambda}}} = \frac{\hat{\zeta}^\top \beta(\gamma + \bar{v})}{\sqrt{\hat{\lambda}}} + \frac{\hat{\zeta}^\top \bar{u}}{\sqrt{\hat{\lambda}}} = \frac{\sqrt{\lambda} \hat{\zeta}_1}{\sqrt{\hat{\lambda}}}(\gamma + \bar{v}) + \frac{\hat{\zeta}^\top \bar{u}}{\sqrt{\hat{\lambda}}}, \quad (\text{F.12})$$

where we use $\beta = (\sqrt{\lambda}, 0, \dots, 0)^\top$ in the last step. Now we analyze the two terms on the right hand side of (F.12) one by one. For the first term, using (F.3), we have

$$\frac{\sqrt{\lambda} \hat{\zeta}_1}{\sqrt{\hat{\lambda}}} = \frac{\lambda}{\hat{\lambda}} \frac{(\bar{V} + \lambda^{-1/2} \bar{U}_1) \hat{\xi}}{\sqrt{T}} = \frac{\lambda}{\hat{\lambda}} \left(\frac{\bar{V} \hat{\xi}}{\sqrt{T}} + \frac{\bar{U}_1 \hat{\xi}}{\sqrt{T\lambda}} \right).$$

Using (F.6) and (F.9) and $\|\bar{U}_1\| \lesssim_P \sqrt{T}$, it follows that

$$\frac{\sqrt{\lambda} \hat{\zeta}_1}{\sqrt{\hat{\lambda}}} = \frac{1}{1 + \tilde{B}} + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right). \quad (\text{F.13})$$

For the second term in (F.12), using (F.3) again, we can write

$$\frac{\hat{\zeta}^\top \bar{u}}{\sqrt{\hat{\lambda}}} = \frac{\hat{\zeta}_1 U_1 \iota_T}{T \sqrt{\hat{\lambda}}} + \frac{\hat{\zeta}_2^\top U_2 \iota_T}{T \sqrt{\hat{\lambda}}} = \frac{\hat{\zeta}_1 U_1 \iota_T}{T \sqrt{\hat{\lambda}}} + \frac{\hat{\xi}^\top (\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top) U_2^\top U_2 \iota_T}{T^{3/2} \hat{\lambda}}. \quad (\text{F.14})$$

The condition that entries of U are independent $\mathcal{N}(0, 1)$ implies that $\|U_1 \iota_T\| \lesssim_P \sqrt{T}$, with $\hat{\lambda}/\lambda \xrightarrow{P} 1 + B$ as shown in (F.6), the first term in (F.14) is of order $O_P(T^{-1/2} \lambda^{-1/2})$. For the second term in (F.14), using $\|(N-1)^{-1} U_2^\top U_2 - \mathbb{I}_T\| \lesssim_P \sqrt{T/N}$, we have

$$\begin{aligned} \left| \frac{\hat{\xi}^\top (\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top) U_2^\top U_2 \iota_T}{T^{3/2} \hat{\lambda}} \right| &\leq \left| \frac{(N-1) \hat{\xi}^\top (\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top) \iota_T}{T^{3/2} \hat{\lambda}} \right| + \frac{N-1}{T \hat{\lambda}} \|(N-1)^{-1} U_2^\top U_2 - \mathbb{I}_T\| \\ &= \frac{N-1}{T \hat{\lambda}} \|(N-1)^{-1} U_2^\top U_2 - \mathbb{I}_T\| \lesssim_P \frac{1}{\sqrt{\lambda}}, \end{aligned}$$

which leads to $|\hat{\lambda}^{-1/2} \hat{\zeta}^\top \bar{u}| \lesssim_P \lambda^{-1/2}$. Plugging this and (F.13) into (F.12), we obtain

$$\hat{\gamma} = \frac{\hat{\zeta}^\top \bar{r}}{\sqrt{\hat{\lambda}}} = \frac{\gamma}{1 + \tilde{B}} + O_P\left(\frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}\right), \quad (\text{F.15})$$

and thus $\hat{\gamma}_g^{PCA} \xrightarrow{P} (1+B)^{-1} \eta \gamma$ by (F.11), (F.15) and $\tilde{B} \rightarrow B$. $\square$

F.2 Proof of Proposition A.1

Proof. Recall that in Section A, we have

$$\hat{\gamma}_g^{PLS} = \|\bar{G}\bar{R}^\top\bar{R}\|^{-2}\bar{G}\bar{R}^\top\bar{R}\bar{G}^\top\bar{G}\bar{R}^\top\bar{r}. \quad (\text{F.16})$$

We analyze $\|\bar{G}\bar{R}^\top\bar{R}\|$, $\bar{G}\bar{R}^\top\bar{R}\bar{G}^\top$ and $\bar{G}\bar{R}^\top\bar{r}$ separately. Recall that from (F.4), we have

$$\left\| \frac{\bar{R}^\top\bar{R}}{T\lambda} - \frac{\bar{V}^\top\bar{V}}{T} - \tilde{B}(\mathbb{I}_T - T^{-1}\iota_T\iota_T^\top) \right\| \lesssim_P \frac{1}{\sqrt{\lambda}},$$

where $\tilde{B} = N/(T\lambda)$ satisfies $\tilde{B} \rightarrow B$. Together with $\bar{G} = \eta\bar{V}$ and $\|\bar{G}\| \lesssim_P \sqrt{T}$, we have

$$\begin{aligned} \frac{1}{T\lambda\sqrt{T}} \|\bar{G}\bar{R}^\top\bar{R}\| &= \frac{1}{\sqrt{T}} \left\| \bar{G} \left(\frac{\bar{V}^\top\bar{V}}{T} + \tilde{B}(\mathbb{I}_T - T^{-1}\iota_T\iota_T^\top) \right) \right\| + O_P\left(\frac{1}{\sqrt{\lambda}}\right) \\ &= \frac{\eta}{\sqrt{T}} \left\| \frac{\bar{V}^\top\bar{V}}{T} + \tilde{B}\bar{V} \right\| + O_P\left(\frac{1}{\sqrt{\lambda}}\right) \xrightarrow{P} \eta(1+B), \end{aligned} \quad (\text{F.17})$$

where we use $|T^{-1}\bar{V}\bar{V}^\top - 1| \lesssim_P T^{-1/2}$ and $\|\bar{V}\| - \sqrt{T} \lesssim_P 1$ in the last equation. For the same reason, by direct calculation we have

$$\begin{aligned} \frac{1}{T^2\lambda} \bar{G}\bar{R}^\top\bar{R}\bar{G}^\top &= \frac{1}{T} \bar{G} \left(\frac{\bar{V}^\top\bar{V}}{T} + \tilde{B}(\mathbb{I}_T - T^{-1}\iota_T\iota_T^\top) \right) \bar{G}^\top + O_P\left(\frac{1}{\sqrt{\lambda}}\right) \\ &= \eta^2 \frac{\bar{V}\bar{V}^\top\bar{V}\bar{V}^\top}{T^2} + \eta^2 \tilde{B} \frac{\bar{V}\bar{V}^\top}{T} + O_P\left(\frac{1}{\sqrt{\lambda}}\right) \xrightarrow{P} \eta^2(1+B). \end{aligned} \quad (\text{F.18})$$

Next, we write

$$\frac{1}{T\lambda} \bar{G}\bar{R}^\top\bar{r} = \frac{1}{T\lambda} \bar{G}\bar{R}^\top\beta(\gamma + \bar{v}) + \frac{1}{T\lambda} \bar{G}\bar{R}^\top\bar{u}. \quad (\text{F.19})$$

We analyze these two terms in (F.19) separately. For the first term, we can write $\bar{R}$ in the form of (F.2) as in the proof of Proposition 1. Then, using $\|\bar{U}_1\| \lesssim_P \sqrt{T}$ we have

$$\frac{1}{T\lambda} \bar{G}\bar{R}^\top\beta = \eta \frac{\bar{V}\bar{V}^\top}{T} + \eta \frac{\bar{V}\bar{U}_1^\top}{T\sqrt{\lambda}} = \eta \frac{\bar{V}\bar{V}^\top}{T} + O_P\left(\frac{1}{\sqrt{\lambda}}\right). \quad (\text{F.20})$$

For the second term in (F.19), we have

$$\begin{aligned} \frac{1}{T\lambda} \bar{G}\bar{R}^\top\bar{u} &= \eta \frac{1}{T^2\sqrt{\lambda}} \bar{V}\bar{V}^\top\bar{U}_1\iota_T + \eta \frac{1}{T^2\lambda} \bar{V}\bar{U}_1^\top U\iota_T = \eta \frac{1}{\sqrt{\lambda}} \frac{\bar{V}\bar{V}^\top}{T} \frac{\bar{U}_1\iota_T}{T} + \eta \frac{1}{T^2\lambda} \bar{V}U^\top U\iota_T \\ &= O_P\left(\frac{1}{\sqrt{T\lambda}}\right) + \eta \frac{N}{T^2\lambda} \bar{V} (N^{-1}U^\top U - \mathbb{I}_T) \iota_T + \eta \frac{N}{T^2\lambda} \bar{V}\iota_T = O_P\left(\frac{1}{\sqrt{T\lambda}}\right) + O_P\left(\frac{1}{\sqrt{\lambda}}\right), \end{aligned} \quad (\text{F.21})$$

where we use $\|N^{-1}U^\top U - \mathbb{I}_T\| \lesssim_P \sqrt{T/N}$ and $\bar{V}\iota_T = 0$ in the last equation. Plugging (F.20) and (F.21) into (F.19), we have

$$\frac{1}{T\lambda} \bar{G} \bar{R}^\top \bar{r} = \eta \frac{\bar{V} \bar{V}^\top}{T} (\gamma + \bar{v}) + O_P\left(\frac{1}{\sqrt{\lambda}}\right) \xrightarrow{P} \eta\gamma. \quad (\text{F.22})$$

Plug (F.17), (F.18), (F.22) into (F.16), we have

$$\hat{\gamma}_g^{PLS} \xrightarrow{P} \frac{1}{\eta^2(1+B)^2} \eta^2(1+B) \eta\gamma = \frac{1}{1+B} \eta\gamma.$$

□

F.3 Proof of Proposition A.2

Proof. Since $\text{Rank}(\bar{R}) \leq \min\{N, T-1\}$, and the assumptions of the proposition imply that $N/T \rightarrow \infty$, we thereby have a condensed SVD of $\bar{R}$ as

$$\bar{R} = \sqrt{T}(\hat{\varsigma}, \hat{\varsigma}_*) \hat{\Lambda}^{1/2} (\hat{\xi}, \hat{\xi}_*)^\top = \sqrt{T} \hat{\varsigma} \hat{\lambda}^{1/2} \hat{\xi}^\top + \sqrt{T} \hat{\varsigma}_* \hat{\Lambda}_*^{1/2} \hat{\xi}_*^\top,$$

where $\hat{\Lambda}^{1/2}$ is the diagonal matrix of $T-1$ singular values, $\hat{\varsigma}, \hat{\xi}$ are the left and right singular vectors corresponding to the largest singular value of $T^{-1/2} \bar{R}$, which is denoted by $\hat{\lambda}^{1/2}$. In addition, $\hat{\varsigma}_* \in \mathbb{R}^{N \times (T-2)}$ and $\hat{\xi}_* \in \mathbb{R}^{T \times (T-2)}$ are the singular vectors corresponding to the rest $T-2$ nonzero singular values, $\hat{\Lambda}_*^{1/2} \in \mathbb{R}^{(T-2) \times (T-2)}$. By direct calculation, we have

$$\sqrt{T} \bar{R}^\top (\bar{R} \bar{R}^\top + \mu I)^{-1} = (\hat{\xi}, \hat{\xi}_*) \hat{\Lambda}^{1/2} (\hat{\Lambda} + T^{-1} \mu I)^{-1} (\hat{\varsigma}, \hat{\varsigma}_*)^\top = \frac{\hat{\lambda}^{1/2}}{\hat{\lambda} + T^{-1} \mu} \hat{\xi} \hat{\varsigma}^\top + \hat{\xi}_* \hat{\Lambda}_*^{1/2} (\hat{\Lambda}_* + T^{-1} \mu I)^{-1} \hat{\varsigma}_*^\top,$$

and thus, with $\bar{G} = \eta \bar{V}$, the Ridge estimator can be written as

$$\begin{aligned} \hat{\gamma}_g^{\text{Ridge}} &= \bar{G} \bar{R}^\top (\bar{R} \bar{R}^\top + \mu I)^{-1} \bar{r} = \frac{\hat{\lambda}}{\hat{\lambda} + T^{-1} \mu} \frac{\eta \bar{V} \hat{\xi}}{\sqrt{T}} \frac{\hat{\varsigma}^\top \bar{r}}{\sqrt{\hat{\lambda}}} + \frac{\eta \bar{V} \hat{\xi}_*}{\sqrt{T}} \hat{\Lambda}_*^{1/2} (\hat{\Lambda}_* + T^{-1} \mu)^{-1} \hat{\varsigma}_*^\top \bar{r} \\ &= \frac{\hat{\lambda}}{\hat{\lambda} + T^{-1} \mu} \hat{\gamma}_g^{\text{PCA}} + \frac{\eta \bar{V} \hat{\xi}_*}{\sqrt{T}} \hat{\Lambda}_*^{1/2} (\hat{\Lambda}_* + T^{-1} \mu)^{-1} \hat{\varsigma}_*^\top \bar{r}. \end{aligned} \quad (\text{F.23})$$

Using (F.6) and the fact that $T^{-1} \lambda^{-1} \mu \rightarrow D$ and Proposition 1, we can show that the first term in (F.23) converges to $(1+B+D)^{-1} \eta\gamma$. With respect to the second term, as shown in (F.4), we have

$$\left\| \frac{\bar{R}^\top \bar{R}}{T\lambda} - \frac{\bar{V}^\top \bar{V}}{T} - \frac{N(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)}{T\lambda} \right\| \lesssim_P \frac{1}{\sqrt{\lambda}},$$

and the eigenvalues of

$$M = \frac{\bar{V}^\top \bar{V}}{T} + \frac{N(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top)}{T\lambda}$$

are given by (F.5), it then follows from Weyl's theorem that $\lambda_i(T^{-1}\lambda^{-1}\bar{R}^\top\bar{R}) = \tilde{B} + O_P(\lambda^{-1/2})$ for $2 \leq i \leq T-1$. Note that $\hat{\Lambda}_*^{1/2} \left(\hat{\Lambda}_* + T^{-1}\mu \right)^{-1}$ is a $(T-2) \times (T-2)$ diagonal matrix and the i th element on the diagonal is

$$\frac{\lambda_{i+1}(T^{-1}\bar{R}^\top\bar{R})^{1/2}}{\lambda_{i+1}(T^{-1}\bar{R}^\top\bar{R}) + T^{-1}\mu} = \frac{1}{\sqrt{\lambda}} \frac{\lambda_{i+1}(T^{-1}\lambda^{-1}\bar{R}^\top\bar{R})^{1/2}}{\lambda_{i+1}(T^{-1}\lambda^{-1}\bar{R}^\top\bar{R}) + T^{-1}\lambda^{-1}\mu}.$$

Together with $T^{-1}\lambda^{-1}\mu \rightarrow D$, we have

$$\left\| \hat{\Lambda}_*^{1/2} \left(\hat{\Lambda}_* + T^{-1}\mu \right)^{-1} \right\| = \max_{1 \leq i \leq T-2} \frac{\lambda_{i+1}(T^{-1}\bar{R}^\top\bar{R})^{1/2}}{\lambda_{i+1}(T^{-1}\bar{R}^\top\bar{R}) + T^{-1}\mu} \lesssim_P \frac{1}{\sqrt{\lambda}}. \quad (\text{F.24})$$

Also, with $\|\bar{u}\| \lesssim_P \sqrt{N/T}$, we have

$$\|\hat{\varsigma}_*^\top \bar{r}\| \leq \|\hat{\varsigma}_*^\top \beta(\gamma + \bar{v})\| + \|\hat{\varsigma}_*^\top \bar{u}\| \leq \|\beta(\gamma + \bar{v})\| + \|\bar{u}\| \lesssim_P \sqrt{\lambda} + \sqrt{N/T} \lesssim_P \sqrt{\lambda} \quad (\text{F.25})$$

and

$$\left\| \frac{\bar{V}\hat{\xi}_*}{\sqrt{T}} \right\|^2 = \left\| \frac{\bar{V}(\hat{\xi}, \hat{\xi}_*)}{\sqrt{T}} \right\|^2 - \left\| \frac{\bar{V}\hat{\xi}}{\sqrt{T}} \right\|^2 \leq \left\| \frac{\bar{V}}{\sqrt{T}} \right\|^2 - \left\| \frac{\bar{V}\hat{\xi}}{\sqrt{T}} \right\|^2 = 1 + O_P\left(\frac{1}{\sqrt{T}}\right) - \left\| \frac{\bar{V}\hat{\xi}}{\sqrt{T}} \right\|^2 \lesssim_P \frac{1}{\sqrt{\lambda}} + \frac{1}{\sqrt{T}}, \quad (\text{F.26})$$

where we use (F.9) in the last inequality. Consequently, using (F.24), (F.25) and (F.26), we have

$$\left| \frac{\eta \bar{V}\hat{\xi}_*}{\sqrt{T}} \hat{\Lambda}_*^{1/2} \left(\hat{\Lambda}_* + T^{-1}\mu \right)^{-1} \hat{\varsigma}_*^\top \bar{r} \right| \leq \left\| \frac{\eta \bar{V}\hat{\xi}_*}{\sqrt{T}} \right\| \left\| \hat{\Lambda}_*^{1/2} \left(\hat{\Lambda}_* + T^{-1}\mu \right)^{-1} \right\| \|\hat{\varsigma}_*^\top \bar{r}\| \lesssim T^{-1/4} + \lambda^{-1/4}.$$

By comparing this with the limit of the first term in (F.23), we obtain

$$\hat{\gamma}_g^{\text{Ridge}} \xrightarrow{P} \frac{1}{1+B+D} \eta \gamma.$$

□

F.4 Proof of Proposition A.3

Proof. By direct calculation, we can write

$$RR^\top + T\mu\bar{r}\bar{r}^\top = R \left(\mathbb{I}_T + \frac{\mu}{T} \iota_T \iota_T^\top \right) R^\top = R \left(\mathbb{I}_T + \frac{\tilde{\mu}}{T} \iota_T \iota_T^\top \right)^2 R^\top, \quad (\text{F.27})$$

where $\tilde{\mu} = \sqrt{\mu+1} - 1$. Hence, the eigenvectors of $RR^\top + T\mu\bar{r}\bar{r}^\top$ are equivalent to the left singular vectors of $R \left(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top \right)$. Let $\hat{\varsigma}$ and $\hat{\xi}$ denote the largest left and right singular vectors of

$R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)$. Note that $\hat{\xi}$ can be viewed as the largest eigenvector of

$$(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)R^\top R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top),$$

we analyze the eigenspace of this matrix first. Similar to (F.4) in the PCA case, we have the following approximation of $R^\top R$

$$\left\| \frac{R^\top R}{T\lambda} - \frac{\bar{V}^\top \bar{V}}{T} - \gamma \frac{\iota_T \bar{V} + \bar{V}^\top \iota_T^\top}{T} - \gamma^2 \frac{\iota_T \iota_T^\top}{T} - \frac{N}{T\lambda} \mathbb{I}_T \right\| \lesssim_P \frac{1}{\sqrt{T}} + \frac{1}{\sqrt{\lambda}}, \quad (\text{F.28})$$

by $|T^{-1}\bar{V}\bar{V}^\top - 1| \lesssim_P T^{-1/2}$, $\|\bar{U}_1\| \lesssim_P T^{1/2}$ and $\|N^{-1}\bar{U}^\top \bar{U} - (\mathbb{I}_T - T^{-1}\iota_T\iota_T^\top)\| \lesssim_P \sqrt{T/N}$.

Then, with (F.28) and $N/(T\lambda) \rightarrow B$, we have

$$\|T^{-1}\lambda^{-1}(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)R^\top R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top) - M^*\| = o_P(1) \quad (\text{F.29})$$

where the matrix M^* here is defined by

$$M^* := B\mathbb{I}_T + T^{-1}\bar{V}^\top \bar{V} + T^{-1}(1 + \tilde{\mu})\gamma(\iota_T \bar{V} + \bar{V}^\top \iota_T^\top) + T^{-1}\left((1 + \tilde{\mu})^2\gamma^2 + \tilde{\mu}^2 B + 2\tilde{\mu}B\right)\iota_T\iota_T^\top.$$

Recall that $\hat{\xi}$ is the eigenvector of $T^{-1}\lambda^{-1}(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)R^\top R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)$, we can analyze the eigenspace of M^* first and then use sin-theta theorem to characterize $\hat{\xi}$.

Firstly, the rank of $M^* - B\mathbb{I}_T$ is at most 2. Using the fact that $\bar{V}\iota_T = 0$, by direct calculation, we have the two nonzero eigenvalues of $M^* - B\mathbb{I}_T$ are the solutions of the equation

$$(x - a_1)(x - a_3) - a_2^2 = 0, \quad (\text{F.30})$$

where $a_1 = T^{-1}\|\bar{V}\|^2$, $a_2 = T^{-1/2}(1 + \tilde{\mu})\gamma\|\bar{V}\|$ and $a_3 = (1 + \tilde{\mu})^2\gamma^2 + \tilde{\mu}^2 B + 2\tilde{\mu}B$. Since the larger solution of (F.30) is

$$\frac{a_1 + a_3 + \sqrt{(a_1 - a_3)^2 + 4a_2^2}}{2} \geq a_1 > 0 \quad (\text{F.31})$$

with probability 1, it is also the largest eigenvalue of $M^* - B\mathbb{I}_T$. In addition, the second largest eigenvalue of $M^* - B\mathbb{I}_T$ should be distinct with $\lambda_1(M^* - B\mathbb{I}_T)$. To see this, if the second eigenvalue is the other solution of (F.30), we have $\lambda_1(M^* - B\mathbb{I}_T) - \lambda_2(M^* - B\mathbb{I}_T) = \sqrt{(a_1 - a_3)^2 + 4a_2^2} \geq \max\{2a_2, |a_1 - a_3|\} > 0$. If the second eigenvalue is 0 (in which case the second solution of the above equation must be negative), we have shown in (F.31) that $\lambda_1(M^* - B\mathbb{I}_T) - \lambda_2(M^* - B\mathbb{I}_T) = \lambda_1(M^* - B\mathbb{I}_T) \geq a_1 > 0$. In both cases, we have $\lambda_1(M^* - B\mathbb{I}_T) - \lambda_2(M^* - B\mathbb{I}_T) \geq \delta$ for some constant $\delta > 0$. Consequently,

$$\lambda_1(M^*) - \lambda_2(M^*) = \lambda_1(M^* - B\mathbb{I}_T) - \lambda_2(M^* - B\mathbb{I}_T) \geq \delta, \quad (\text{F.32})$$

for some constant $\delta > 0$. Now we calculate the first eigenvector of M^* , which should also be the first eigenvector of $M^* - B\mathbb{I}_T$. We use $\tilde{\xi}$ to denote this eigenvector. Since we already know that the largest eigenvalue of $\lambda_1(M^* - B\mathbb{I}_T)$ is a solution of (F.30), which means that $\tilde{\xi}$ should be in the space spanned by $\bar{V}^\top$ and ι_T . Writing $\tilde{\xi} = K_1 \|\bar{V}\|^{-1} \bar{V}^\top + K_2 T^{-1/2} \iota_T$ and plugging the largest eigenvalue of $\lambda_1(M^* - B\mathbb{I}_T)$ of (F.31) into $\lambda_1(M - B\mathbb{I}_T)\tilde{\xi} = (M - B\mathbb{I}_T)\tilde{\xi}$, we directly get

$$\frac{K_2}{K_1} = \frac{\sqrt{(a_1 - a_3)^2 + 4a_2^2} + a_3 - a_1}{2a_2}, \quad (\text{F.33})$$

which will pin down K_1 and K_2 because we also have $\|\tilde{\xi}\| = 1$.

Using $\|T^{-1}\lambda^{-1}(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)R^\top R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top) - M\| = o_P(1)$, (F.32) and sin-theta theorem, we have

$$\|\mathbb{P}_{\hat{\xi}} - \mathbb{P}_{\tilde{\xi}}\| \leq \frac{o_P(1)}{\delta - o_P(1)} = o_P(1),$$

which implies that $|\hat{\xi}^\top \tilde{\xi}| \xrightarrow{P} 1$ and consequently,

$$\|\hat{\xi} - K_1 \|\bar{V}\|^{-1} \bar{V}^\top - K_2 T^{-1/2} \iota_T\| = o_P(1) \quad \text{or} \quad \|\hat{\xi} + K_1 \|\bar{V}\|^{-1} \bar{V}^\top + K_2 T^{-1/2} \iota_T\| = o_P(1).$$

Since the sign of $\hat{\xi}$ plays no role in the estimator $\hat{\gamma}_g^{rpPCA}$, we can simply assume the former one.

Also, the relationship between singular vectors implies that

$$\hat{F} = \hat{\xi}^\top R = \|R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)\|^{-1} \hat{\xi}^\top (\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top) R^\top R. \quad (\text{F.34})$$

With the approximation of $R^\top R$ in (F.28), $\bar{V}\iota_T = 0$, $T^{-1}\bar{V}\bar{V}^\top = 1 + O_P(T^{-1/2})$ and $N/(T\lambda) \rightarrow B$, by direct calculation, we have

$$\left\| \|\bar{V}\|^{-1} \bar{V}(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top) R^\top R - \lambda T^{1/2} \left((1+B)\bar{V} + \gamma\iota_T^\top \right) \right\| = o_P(\lambda T), \quad (\text{F.35})$$

and

$$\left\| T^{-1/2} \iota_T^\top (\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top) R^\top R - \lambda T^{1/2} (1 + \tilde{\mu}) \left(\gamma\bar{V} + (\gamma^2 + B)\iota_T^\top \right) \right\| = o_P(\lambda T). \quad (\text{F.36})$$

Plugging (F.35), (F.36) and $\|\hat{\xi} - K_1 \|\bar{V}\|^{-1} \bar{V}^\top + K_2 T^{-1/2} \iota_T\| = o_P(1)$ into (F.34) we have

$$\left\| \|R(\mathbb{I}_T + T^{-1}\tilde{\mu}\iota_T\iota_T^\top)\| \hat{F} - \lambda T^{1/2} (L_1 \bar{V} + L_2 \iota_T^\top) \right\| = o_P(\lambda T), \quad (\text{F.37})$$

where

$$L_1 = K_1(1+B) + K_2(1+\tilde{\mu})\gamma, \quad L_2 = K_1\gamma + K_2(1+\tilde{\mu})(\gamma^2 + B). \quad (\text{F.38})$$

It is easy to observe that scalar plays no role in the estimator $\hat{\gamma}_g^{rpPCA}$, we can redefine

$$\hat{F}^* = \lambda^{-1} T^{-1/2} L_1^{-1} \|R(\mathbb{I}_T + T^{-1} \tilde{\mu} \iota_T \iota_T^\top)\| \hat{F}$$

and use $\hat{F}^*$ to create $\hat{\gamma}_g^{rpPCA}$. Then, (F.37) becomes $\|\hat{F}^* - \bar{V} - L_1^{-1} L_2 \iota_T^\top\| = o_P(T^{1/2})$. Consequently, $\|\hat{V} - \bar{V}\| = \|\hat{F}^*(\mathbb{I}_T - T^{-1} \iota_T \iota_T^\top) - \bar{V}\| = o_P(T^{1/2})$, $\hat{\gamma} = T^{-1} \hat{F}^* \iota_T = L_1^{-1} L_2 + o_P(1)$, and

$$\hat{\eta} = \bar{G} \hat{V}^\top (\hat{V} \hat{V}^\top)^{-1} = \eta \bar{V} \hat{V}^\top (\hat{V} \hat{V}^\top)^{-1} = \eta (\bar{V} \bar{V}^\top + o_P(T)) (\bar{V} \bar{V}^\top + o_P(T))^{-1} = \eta + o_P(1),$$

and the estimator $\hat{\gamma}_g^{rpPCA} = \hat{\eta} \hat{\gamma} \xrightarrow{P} \eta L_1^{-1} L_2$, where L_1 and L_2 are defined in (F.38).

In light of that $a_1 \xrightarrow{P} 1$, $a_2 \xrightarrow{P} (1 + \tilde{\mu})\gamma$, $\tilde{\mu} = \sqrt{1 + \mu} - 1$, $\hat{\gamma}_g^{rpPCA} \xrightarrow{P} \eta L_2 / L_1$, (F.33) and the definitions of L_1 and L_2 in (F.38), we have

$$\hat{\gamma}_g^{rpPCA} \xrightarrow{P} w(1 + B)^{-1} \eta \gamma + (1 - w) \eta (\gamma + \gamma^{-1} B),$$

where

$$w = \frac{2 + 2B}{1 + 2B + \sqrt{(1 - a)^2 + 4(1 + \mu)\gamma + a}}, \quad a = (1 + \mu)(\gamma^2 + B) - B. \quad (\text{F.39})$$

□

F.5 Proof of Proposition 2

Proof. Consider the set $I = \{|\beta_{[i]}| \geq \beta_{\{qN\}}\}$, where $|\beta|_{\{qN\}}$ is the (qN) th largest value in $\{|\beta_{[i]}|\}_{i \in \langle N \rangle}$. Since

$$T^{-1} \bar{R} \bar{G}^\top - \beta \eta^\top = \beta (T^{-1} \bar{V} \bar{V}^\top - 1) \eta^\top + T^{-1} \bar{U} \bar{V}^\top \eta^\top + T^{-1} \beta \bar{V} \bar{Z}^\top + T^{-1} \bar{U} \bar{Z}^\top,$$

we have

$$\begin{aligned} \|T^{-1} \bar{R} \bar{G}^\top - \beta \eta^\top\|_{\text{MAX}} &\lesssim \|\beta\|_{\text{MAX}} \|T^{-1} \bar{V} \bar{V}^\top - 1\| \|\eta\| + T^{-1} \|\bar{U} \bar{V}^\top\|_{\text{MAX}} \|\eta\| \\ &\quad + T^{-1} \|\beta\|_{\text{MAX}} \|\bar{V} \bar{Z}^\top\| + T^{-1} \|\bar{U} \bar{Z}^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{-1/2}. \end{aligned}$$

In other words, the difference between $T^{-1} \bar{R} \bar{G}^\top$ and $\beta \eta^\top$ for all test assets is bounded by $O_P((\log N)^{1/2} T^{-1/2})$, which is $o_P(1)$ under our assumption.

On the other hand, with the assumption that $\|\beta\|_{\text{MAX}} \lesssim 1$ and the definition of $|\beta|_{\{qN\}}$, we have $\|\beta_{[I_0]}\|^2 \lesssim qN + (N_0 - qN) |\beta|_{\{qN\}}^2$. Together with the assumption that $qN/N_0 \rightarrow 0$ and $\|\beta_{[I_0]}\| \asymp \sqrt{N_0}$, it leads to $|\beta|_{\{qN\}}^2 \gtrsim \|\beta_{[I_0]}\|^2 / N_0 \asymp 1$. Then, with the assumption that $|\beta|_{\{qN+1\}} \leq (1 + \delta)^{-1} |\beta|_{\{qN\}}$, we have that the difference between $|\beta|_{\{qN+1\}}$ and $|\beta|_{\{qN\}}$ should be at the same rate as $|\beta|_{\{qN\}} \gtrsim 1$, which is larger than the difference between $T^{-1} \bar{R} \bar{G}^\top$ and $\beta \eta^\top$. Therefore, with

probability approaching one, we have $\hat{I} = I$. In what follows, we only need consider the case of $\hat{I} = I$.

Since $qN/N_0 \rightarrow 0$, by the definition of I , we have $\|\beta_{[I]}\|/\sqrt{|I|} \geq \|\beta_{[I_0]}\|/\sqrt{|I_0|}$. Together with the assumption that $\|\beta_{[I_0]}\| \asymp \sqrt{N_0}$, $\|\beta_{[I_0]}\| \rightarrow \infty$ and $|I| = qN \rightarrow \infty$, we have $|I|/(T\|\beta_{[I]}\|^2) \rightarrow 0$ and $\|\beta_{[I]}\| \rightarrow \infty$. Now compared to the case with PCA, the expansion on $\hat{\gamma}_g^{SPCA}$ resembles that of (F.12), except for an extra term that depends on $\bar{Z}$ and the restriction of $\bar{r}$ on I :

$$\hat{\gamma}_g^{SPCA} = \frac{\eta \bar{V} \hat{\xi} \hat{\xi}^\top \bar{r}_{[I]}}{\sqrt{T} \sqrt{\hat{\lambda}}} + \frac{\bar{Z} \hat{\xi} \hat{\xi}^\top \bar{r}_{[I]}}{\sqrt{T} \sqrt{\hat{\lambda}}}. \quad (\text{F.40})$$

In restriction to the smaller set I , the first term matches exactly the case of $|I|/(T\|\beta_{[I]}\|^2) \rightarrow 0 = B$ in Proposition 1, which yields

$$\frac{\eta \bar{V} \hat{\xi} \hat{\xi}^\top \bar{r}_{[I]}}{\sqrt{T} \sqrt{\hat{\lambda}}} = \eta \gamma + o_P(1).$$

We now analyze the second term in (F.40). As shown in (F.15), we have

$$\left\| \frac{\hat{\xi}^\top \bar{r}_{[I]}}{\sqrt{\hat{\lambda}}} \right\| \lesssim_P 1,$$

so to prove that SPCA is consistent in this case, it is sufficient to show that $T^{-1/2} \|\bar{Z} \hat{\xi}\| \xrightarrow{P} 0$, where $\hat{\xi}$ is the largest right singular vector of $\bar{R}_{[I]}$. Similar to the proof of (F.7) in Proposition 1, we can show that the difference between projection matrices, $\mathbb{P}_{\hat{\xi}}$ and $\mathbb{P}_{\bar{V}^\top}$ is small by sin-theta theorem. That is to say, we have $\|\hat{\xi} \hat{\xi}^\top - \bar{V}^\top (\bar{V} \bar{V}^\top)^{-1} \bar{V}\| \xrightarrow{P} 0$. Then, with the fact that

$$\|\bar{Z} \bar{V}^\top (\bar{V} \bar{V}^\top)^{-1} \bar{V}\| \leq \|\bar{Z} \bar{V}^\top\| \|(\bar{V} \bar{V}^\top)^{-1}\| \|\bar{V}\| \lesssim_P T^{1/2} \times T^{-1} \times T^{1/2} \lesssim_P 1,$$

we have $T^{-1/2} \|\bar{Z} \hat{\xi} \hat{\xi}^\top\| \xrightarrow{P} 0$. Consequently,

$$T^{-1/2} \|\bar{Z} \hat{\xi}\| = T^{-1/2} \|\bar{Z} \hat{\xi} \hat{\xi}^\top \hat{\xi}\| \leq T^{-1/2} \|\bar{Z} \hat{\xi} \hat{\xi}^\top\| \|\hat{\xi}\| \xrightarrow{P} 0.$$

Hence, z_t does not affect the consistency of the SPCA estimator. This completes the proof. $\square$

F.6 Proof of Theorem 1

Proof. It is sufficient to consider the case $\Sigma_v = \mathbb{I}_p$. Otherwise, we can do transformation $V' = \Sigma_v^{-\frac{1}{2}} V$, $\beta'_{[I]} = \beta_{[I]} \Sigma_v^{\frac{1}{2}}$, $\eta' = \eta \Sigma_v^{\frac{1}{2}}$ and $\gamma' = \Sigma_v^{-\frac{1}{2}} \gamma$. All the Assumptions B.1-B.8 still hold for the new V' , $\beta'_{[I]}$. Therefore, we only need analyze the case of $\Sigma_v = \mathbb{I}_p$.

For notation simplicity, throughout the proofs of Theorems 1-3, we use $\tilde{R}_{(k)} := (\bar{R}_{(k)})_{[\hat{I}_k]}$ to denote the matrix on which we perform SVD in each step of Algorithm 3. Similarly, we use $\tilde{r}_{(k)} := (\bar{r}_{(k)})_{[\hat{I}_k]}$.

The first left and right singular vectors of $\tilde{R}_{(k)}$ are denoted by $\hat{\varsigma}_{(k)}$ and $\hat{\xi}_{(k)}$, while the largest singular value of $\tilde{R}_{(k)}$ is denoted by $\sqrt{T\hat{\lambda}_{(k)}}$. As a result, $\hat{\lambda}_{(k)} = T^{-1} \left\| \tilde{R}_{(k)} \right\|^2$ and

$$\hat{\varsigma}_{(k)} = \frac{\tilde{R}_{(k)} \hat{\xi}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}}, \quad \hat{\xi}_{(k)} = \frac{\tilde{R}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}}. \quad (\text{F.41})$$

Using the above notation, our estimated factor at k -th step is $\hat{V}_{(k)} = \sqrt{T} \hat{\xi}_{(k)}^\top \in \mathbb{R}^{1 \times T}$, the risk premium of this factor is given by $\hat{\gamma}_{(k)} = \hat{\lambda}_{(k)}^{-1/2} \hat{\varsigma}_{(k)}^\top \tilde{r}_{(k)}$, the loading matrix of R on this factor is $\hat{\beta}_{(k)} = T^{-1/2} \tilde{R} \hat{\xi}_{(k)}$, and the loading of G on this factor is $\hat{\eta}_{(k)} = T^{-1/2} \tilde{G} \hat{\xi}_{(k)}$. By footnote 24, we can use $\tilde{G}$ instead of $\tilde{G}_{(k)}$ in Algorithm 3 and throughout the proof. We denote $\hat{\eta} = (\hat{\eta}_{(1)}, \dots, \hat{\eta}_{(\bar{p})})$ and $\hat{\gamma} = (\hat{\gamma}_{(1)}, \dots, \hat{\gamma}_{(\bar{p})})^\top$, so the risk premium estimator is $\hat{\gamma}_g^{SCA} = \hat{\eta} \hat{\gamma}$.

To see more clear the relationship between $\tilde{R}_{(k)}$ and $\tilde{R}$, we define matrix $D_{(k)} \in \mathbb{R}^{|\hat{I}_k| \times N}$ iteratively:

$$D_{(k)} = \mathbb{I}_{[\hat{I}_k]} - \sum_{i=1}^{k-1} \tilde{R}_{[\hat{I}_k]} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top D_{(i)}}{\sqrt{T\hat{\lambda}_{(i)}}}$$

with $D_{(1)} = \mathbb{I}_{[\hat{I}_1]}$. We can show by induction that $\tilde{R}_{(k)} = D_{(k)} \tilde{R}$. In fact, by Lemma 2, we have $\hat{\xi}_{(i)}^\top \hat{\xi}_{(j)} = 0$. With $\hat{V}_{(i)} = \sqrt{T} \hat{\xi}_{(i)}^\top$ and the definition of $\tilde{R}_{(k)}$, we have

$$\tilde{R}_{(k)} := (\tilde{R}_{(k)})_{[\hat{I}_k]} = \tilde{R}_{[\hat{I}_k]} \prod_{i=1}^{k-1} \mathbb{M}_{\hat{V}_{(i)}} = \tilde{R}_{[\hat{I}_k]} \left(\mathbb{I}_T - \sum_{i=1}^{k-1} \hat{\xi}_{(i)} \hat{\xi}_{(i)}^\top \right), \quad (\text{F.42})$$

for $k > 1$ and when $k = 1$,

$$\tilde{R}_{(1)} = \tilde{R}_{[\hat{I}_1]} = \beta_{[\hat{I}_1]} \bar{V} + \bar{U}_{[\hat{I}_1]}.$$

Using (F.41), if $\tilde{R}_{(i)} = D_{(i)} \tilde{R}$ for $i < k$, we can write (F.42) as

$$\tilde{R}_{(k)} = \tilde{R}_{[\hat{I}_k]} \left(\mathbb{I}_T - \sum_{i=1}^{k-1} \hat{\xi}_{(i)} \hat{\xi}_{(i)}^\top \right) = \tilde{R}_{[\hat{I}_k]} - \sum_{i=1}^{k-1} \tilde{R}_{[\hat{I}_k]} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{R}_{(i)}}{\sqrt{T\hat{\lambda}_{(i)}}} = D_{(k)} \tilde{R} \quad (\text{F.43})$$

As $\tilde{R}_{(1)} = \tilde{R}_{[\hat{I}_1]} = D_{(1)} \tilde{R}$ holds immediately by the definition, we have $\tilde{R}_{(k)} = D_{(k)} \tilde{R}$ by induction. If we further define $\tilde{\beta}_{(k)} = D_{(k)} \beta$ and $\tilde{U}_{(k)} = D_{(k)} \bar{U}$, then $\tilde{R}_{(k)}$ can be written in the form $\tilde{R}_{(k)} = \tilde{\beta}_{(k)} \bar{V} + \tilde{U}_{(k)}$. Similarly, we can write

$$\tilde{r}_{(k)} = \tilde{\beta}_{(k)} (\gamma + \bar{v}) + \tilde{u}_{(k)}, \quad (\text{F.44})$$

where $\tilde{r}_{(k)} = D_{(k)} \bar{r}$ and $\tilde{u}_{(k)} = D_{(k)} \bar{u}$.

We also create similar representations for $\tilde{G}_{(k)} := \bar{G} \prod_{i=1}^{k-1} \mathbb{M}_{\tilde{V}_{(i)}^\top}$. Specifically, we have

$$\begin{aligned} \tilde{G}_{(k)} &:= \bar{G} \left(\mathbb{I}_T - \sum_{i=1}^{k-1} \hat{\xi}_{(i)} \hat{\xi}_{(i)}^\top \right) = \bar{G} - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{R}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} = \eta \bar{V} + \bar{Z} - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{\beta}_{(i)} \bar{V}}{\sqrt{T \hat{\lambda}_{(i)}}} - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \\ &= \left(\eta - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right) \bar{V} + \left(\bar{Z} - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right). \end{aligned}$$

In light of this equation, if we define

$$\tilde{\eta}_{(k)} := \eta - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}}, \quad \text{and} \quad \tilde{Z}_{(k)} := \bar{Z} - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\xi}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}}, \quad (\text{F.45})$$

$\tilde{G}_{(k)}$ can be written as $\tilde{G}_{(k)} = \tilde{\eta}_{(k)} \bar{V} + \tilde{Z}_{(k)}$.

To sum up, we have defined $\tilde{R}_{(k)}, \tilde{r}_{(k)}, \tilde{\beta}_{(k)}, \tilde{U}_{(k)}, \tilde{u}_{(k)}, \tilde{\eta}_{(k)}$ and $\tilde{Z}_{(k)}$ at the k th step of the algorithm. Note that $\tilde{\beta}_{(k)} \in \mathbb{R}^{|I_k| \times p}$ and $\tilde{\eta}_{(k)} \in \mathbb{R}^{d \times p}$ can be viewed as the loading of $\tilde{R}_{(k)}$ and $\tilde{G}_{(k)}$ on $\bar{V}$, but they are not the same as the estimators defined in Algorithm 3, $\hat{\beta}_{(k)} \in \mathbb{R}^{N \times 1}$ and $\hat{\eta}_{(k)} \in \mathbb{R}^{d \times 1}$, which are the estimated loadings of R and G on the k th factor.

By Lemma 4, we have $P(\hat{I}_k = I_k) \rightarrow 1$ for $k \leq \tilde{p}$ and $P(\hat{p} = \tilde{p}) \rightarrow 1$. Thus, we can impose that $\hat{I}_k = I_k$ for any k and $\hat{p} = \tilde{p}$ in what follows. In addition, Lemma 3(ii) and Lemma 4(iii) imply that $\hat{\lambda}_{(k)} \asymp qN$ and that $|I_k| = qN$. Therefore, the assumptions of Lemmas 6-9 hold.

Since our algorithm stops at $\tilde{p}$, it implies that at most $qN - 1$ test assets satisfy $T^{-1} \left\| \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_{\text{MAX}} = T^{-1} \left\| (\bar{R}_{(\tilde{p}+1)})_{[i]} \bar{G}^\top \right\|_{\text{MAX}} \geq c$. Let $\mathcal{S}$ denote the set of these assets. For asset $i \in \mathcal{S}$, we have

$$\left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_F^2 \lesssim \left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_{\text{MAX}}^2 \lesssim 1.$$

Consider the test assets in I_0 , we have

$$\begin{aligned} \sum_{i \in I_0} \left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_F^2 &= \sum_{i \in I_0 \cap \mathcal{S}} \left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_F^2 + \sum_{i \in I_0 \cap \mathcal{S}^c} \left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[i]}^\top \right\|_F^2 \\ &\lesssim_P qN + c^2 N_0 = o(N_0), \end{aligned} \quad (\text{F.46})$$

where we use the assumptions $c \rightarrow 0$ and $qN/N_0 \rightarrow 0$ in the last equation. Consequently, (F.46) leads to $\left\| T^{-1} \tilde{G}_{(\tilde{p}+1)} \bar{R}_{[I_0]}^\top \right\| = o_P(N_0^{1/2})$. Write

$$\tilde{G}_{(\tilde{p}+1)} \bar{R}_{[I_0]}^\top = \tilde{\eta}_{(\tilde{p}+1)} \bar{V} \bar{V}^\top \beta_{[I_0]} + \tilde{\eta}_{(\tilde{p}+1)} \bar{V} \bar{U}_{[I_0]}^\top + \bar{Z}_{(\tilde{p}+1)} \bar{V}^\top \beta_{[I_0]} + \bar{Z}_{(\tilde{p}+1)} \bar{U}_{[I_0]}^\top. \quad (\text{F.47})$$

Using (F.46), (F.47) and Lemma 8(i)(ii), we have

$$\left\| \tilde{\eta}_{(\tilde{p}+1)} \left(\bar{V} \bar{V}^\top \beta_{[I_0]} + \bar{V} \bar{U}_{[I_0]}^\top \right) \right\| = o_P \left(N_0^{1/2} T \right). \quad (\text{F.48})$$

Also, using Assumption B.6, Lemma 1(i) and Weyl's theorem, we have

$$|\sigma_p(\bar{V} \bar{V}^\top \beta_{[I_0]} + \bar{V} \bar{U}_{[I_0]}^\top) - \sigma_p(T \beta_{[I_0]})| \leq \left\| \bar{V} \bar{U}_{[I_0]}^\top \right\| + \|T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p\| \|T \beta_{[I_0]}\| \lesssim_P N_0^{1/2} T^{1/2}. \quad (\text{F.49})$$

Since Assumption B.3 implies that $\sigma_p(\beta_{[I_0]}) \asymp N_0^{1/2}$, we have $\sigma_p(\bar{V} \bar{V}^\top \beta_{[I_0]} + \bar{V} \bar{U}_{[I_0]}^\top) \asymp_P N_0^{1/2} T$. Using this result, (F.48) and the inequality $\left\| \tilde{\eta}_{(\tilde{p}+1)} \left(\bar{V} \bar{V}^\top \beta_{[I_0]} + \bar{V} \bar{U}_{[I_0]}^\top \right) \right\| \geq \sigma_p(\bar{V} \bar{V}^\top \beta_{[I_0]} + \bar{V} \bar{U}_{[I_0]}^\top) \|\tilde{\eta}_{(\tilde{p}+1)}\|$, we have $\|\tilde{\eta}_{(\tilde{p}+1)}\| \xrightarrow{P} 0$. That is, by definition of $\tilde{\eta}_{(\tilde{p}+1)}$ in (F.45),

$$\left\| \eta - \sum_{i=1}^{\tilde{p}} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right\| = o_P(1). \quad (\text{F.50})$$

Multiplying (F.50) by γ from the right-hand side, we have

$$\left\| \eta \gamma - \sum_{i=1}^{\tilde{p}} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \gamma \right\| = o_P(1). \quad (\text{F.51})$$

Recall that our final estimator of γ_g is

$$\hat{\gamma}_g^{SPCA} = \hat{\eta} \hat{\gamma} = \sum_{i=1}^{\tilde{p}} \hat{\eta}_{(i)} \hat{\gamma}_{(i)} = \sum_{i=1}^{\tilde{p}} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{r}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} = \sum_{i=1}^{\tilde{p}} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} (\gamma + \bar{v}) + \sum_{i=1}^{\tilde{p}} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{u}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}}. \quad (\text{F.52})$$

Combining (F.51) and (F.52), we have

$$\|\eta \gamma - \hat{\eta} \hat{\gamma}\| \leq \sum_{i=1}^{\tilde{p}} \left\| \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \bar{v} \right\| + \sum_{i=1}^{\tilde{p}} \left\| \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{u}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right\| + o_P(1). \quad (\text{F.53})$$

Using $\|\bar{G}\| \lesssim_P T^{1/2}$, Lemma 7(ii), Lemma 9(i) and the assumptions that $qN \rightarrow \infty$, we have

$$\left\| \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \bar{v} \right\| \leq \left\| \bar{G} \hat{\xi}_{(i)} \right\| \left\| \frac{\hat{\varsigma}_{(i)}^\top \tilde{\beta}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right\| \|\bar{v}\| = o_P(1),$$

and

$$\left\| \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{u}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right\| \leq \left\| \bar{G} \hat{\xi}_{(i)} \right\| \left\| \frac{\hat{\varsigma}_{(i)}^\top \tilde{u}_{(i)}}{\sqrt{T \hat{\lambda}_{(i)}}} \right\| = o_P(1).$$

Plugging them into (F.53) completes the proof.

F.7 Proof of Theorem 2

To derive the asymptotic distribution, we need a more intricate analysis. As in the proof of Theorem 1, we impose that $\hat{p} = \tilde{p}$ and $\hat{I}_k = I_k$, since Lemma 4 shows that both events occur with probability approaching 1.

Recall that in Algorithm 3 the SPCA estimator is written as $\hat{\gamma}_g^{SPCA} = \hat{\eta} \hat{\gamma} = \sum_{k=1}^{\hat{p}} \hat{\eta}_{(k)} \hat{\gamma}_{(k)}$, where $\hat{p}$ is the number of factors selected and, with the notation defined in the proof of Theorem 1,

$$\hat{\eta}_{(k)} = \frac{\bar{G} \hat{\xi}_{(k)}}{\sqrt{T}} = \frac{\eta \bar{V} \hat{\xi}_{(k)}}{\sqrt{T}} + \frac{\bar{Z} \hat{\xi}_{(k)}}{\sqrt{T}}, \quad \hat{\gamma}_{(k)} = \frac{\hat{\varsigma}_{(k)}^\top \tilde{r}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}} = \frac{\hat{\varsigma}_{(k)}^\top \tilde{\beta}_{(k)} (\gamma + \bar{v})}{\sqrt{\hat{\lambda}_{(k)}}} + \frac{\hat{\varsigma}_{(k)}^\top \tilde{u}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}}. \quad (\text{F.54})$$

Denote $H_1 = (h_{11}, \dots, h_{\hat{p}1})$, $H_2 = (h_{12}, \dots, h_{\hat{p}2})$, where

$$h_{k1} = T^{-1/2} \bar{V} \hat{\xi}_{(k)}, \quad h_{k2} = \hat{\lambda}_{(k)}^{-1/2} \tilde{\beta}_{(k)}^\top \hat{\varsigma}_{(k)}. \quad (\text{F.55})$$

Therefore, we can write (F.54) as

$$\hat{\eta}_{(k)} - \eta h_{k1} = \frac{\bar{Z} \hat{\xi}_{(k)}}{\sqrt{T}}, \quad \hat{\gamma}_{(k)} - h_{k2}^\top (\gamma + \bar{v}) = \frac{\hat{\varsigma}_{(k)}^\top \tilde{u}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}}. \quad (\text{F.56})$$

Since $\hat{\xi}_{(k)}$ and $\hat{\varsigma}_{(k)}$ are the largest singular vectors of $\tilde{R}_{(k)}$ with the singular value $\sqrt{T \hat{\lambda}_{(k)}}$, we have

$$\hat{\varsigma}_{(k)} = \frac{\tilde{R}_{(k)} \hat{\xi}_{(k)}}{\sqrt{T \hat{\lambda}_{(k)}}}, \quad \hat{\xi}_{(k)} = \frac{\tilde{R}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{T \hat{\lambda}_{(k)}}}. \quad (\text{F.57})$$

From (F.57), we have

$$\frac{\bar{Z} \hat{\xi}_{(k)}}{\sqrt{T}} = \frac{\bar{Z}}{\sqrt{T}} \frac{\tilde{R}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{T \hat{\lambda}_{(k)}}} = \frac{\bar{Z} \bar{V}^\top}{T} \frac{\tilde{\beta}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}} + \frac{\bar{Z} \tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{T \sqrt{\hat{\lambda}_{(k)}}} = \frac{\bar{Z} \bar{V}^\top}{T} h_{k2} + \frac{\bar{Z} \tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{T \sqrt{\hat{\lambda}_{(k)}}}.$$

Using Lemma 7(ii), we have

$$\left\| \frac{\bar{Z}\tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P \frac{1}{T} + \frac{1}{qN}, \quad \left\| \frac{\hat{\varsigma}_{(k)}^\top \tilde{u}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P \frac{1}{T} + \frac{1}{qN}.$$

Then, along with (F.56) and Lemma 1(vi), the above equations lead to

$$\left\| \hat{\eta} - \eta H_1 - \frac{ZV^\top}{T} H_2 \right\| \lesssim_P \frac{1}{T} + \frac{1}{qN}, \quad (\text{F.58})$$

and

$$\|\hat{\gamma} - H_2^\top \gamma - H_2^\top \bar{v}\| \lesssim_P \frac{1}{T} + \frac{1}{qN}. \quad (\text{F.59})$$

Combining (F.58) and (F.59), with $\|H_1\| \lesssim_P 1$, $\|H_2\| \lesssim_P 1$ from Lemma 9 and Assumptions B.1, B.2, we have

$$\left\| \hat{\eta}\hat{\gamma} - \eta H_1 H_2^\top (\gamma + \bar{v}) - \frac{ZV^\top}{T} H_2 H_2^\top \gamma \right\| \lesssim_P \frac{1}{T} + \frac{1}{qN}. \quad (\text{F.60})$$

As shown in Lemma 3(iv), under the assumption that $\lambda_p(\eta^\top \eta) \gtrsim 1$, we have $\tilde{p} = p$. Together with $P(\hat{p} = \tilde{p}) \rightarrow 1$, we can impose that $\hat{p} = p$ for derivations below. To analyze $H_1 H_2^\top$ and $H_2 H_2^\top$ in (F.60), using Lemma 9 and the assumptions on q , we have

$$\|H_2^\top H_2 - \mathbb{I}_p\| \leq \|H_1^\top H_2 - \mathbb{I}_p\| + \|H_1 - H_2\| \|H_2\| \lesssim_P T^{-1/2}. \quad (\text{F.61})$$

Then, for the term $H_2 H_2^\top$, we have

$$\|H_2 H_2^\top - \mathbb{I}_p\| = \max_{1 \leq i \leq p} |\lambda_i(H_2 H_2^\top) - 1| = \max_{1 \leq i \leq p} |\lambda_i(H_2^\top H_2) - 1| = \|H_2^\top H_2 - \mathbb{I}_p\| \lesssim_P T^{-1/2} \quad (\text{F.62})$$

since H_2 is a $p \times p$ matrix.

For the term $H_1 H_2^\top$, by Lemma 9, we have

$$\|H_1^\top H_2 - \mathbb{I}_p\| \lesssim_P \frac{1}{T} + \frac{1}{qN}. \quad (\text{F.63})$$

In addition, we have

$$\sigma_p(H_2) \|H_2 H_1^\top - \mathbb{I}_p\| \leq \|(H_2 H_1^\top - \mathbb{I}_p) H_2\| = \|H_2 (H_1^\top H_2 - \mathbb{I}_p)\| \leq \|H_2\| \|H_1^\top H_2 - \mathbb{I}_p\|. \quad (\text{F.64})$$

Since (F.61) implies that $\sigma_1(H_2)/\sigma_p(H_2) = \lambda_1(H_2H_2^\top)^{1/2}/\lambda_p(H_2H_2^\top)^{1/2} \lesssim_P 1$, (F.63) and (F.64) give

$$\|H_1H_2^\top - \mathbb{I}_p\| = \|H_2H_1^\top - \mathbb{I}_p\| \leq \frac{\sigma_1(H_2)}{\sigma_p(H_2)} \|H_1^\top H_2 - \mathbb{I}_p\| \lesssim_P \frac{1}{T} + \frac{1}{qN}. \quad (\text{F.65})$$

Combining (F.60), (F.62), (F.65) and the assumption $q^{-1}N^{-1}T^{1/2} \rightarrow 0$, we obtain

$$\|\widehat{\eta}\widehat{\gamma} - \eta(\gamma + \bar{v}) - T^{-1}ZV^\top\gamma\| \lesssim_P \frac{1}{T} + \frac{1}{qN}. \quad (\text{F.66})$$

(F.66) implies that $\|\widehat{\eta}\widehat{\gamma} - \eta\gamma\| \lesssim_P T^{-1/2} + (qN)^{-1}$. In addition, with the assumption $q^{-1}N^{-1}T^{1/2} \rightarrow 0$, (F.66) becomes $\|\widehat{\eta}\widehat{\gamma} - \eta(\gamma + \bar{v}) - T^{-1}ZV^\top\gamma\| = o_P(T^{-1/2})$. Using Delta method and Assumption B.9, it is straightforward to obtain: $\sqrt{T}(\widehat{\eta}\widehat{\gamma} - \eta\gamma) \xrightarrow{d} \mathcal{N}(0, \Phi)$, where Φ is as defined in Theorem 2. $\square$

F.8 Proof of Theorem 3

Proof. As shown in the proof of Theorem 2, we have $P(\widehat{p} = p) \rightarrow 1$ and $\mathbb{P}(\widehat{I}_k = I_k) \rightarrow 1$ for $k \leq p$. Thus, we impose $\widehat{p} = \tilde{p} = p$ and $\widehat{I}_k = I_k$ below. Using the same notation as in the proof of Theorem 2 and (F.59), we have

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T |m_t - \widehat{m}_t|^2 &= \frac{1}{T} \left\| \widehat{V}^\top \widehat{\gamma} - V^\top \gamma \right\|^2 = \frac{1}{T} \left\| \sqrt{T} \widehat{\xi} (H_2^\top \gamma + O_P(T^{-1/2})) - V^\top \gamma \right\|^2 \\ &= \frac{1}{T} \left\| \sqrt{T} \widehat{\xi} H_2^\top \gamma - \bar{V}^\top \gamma \right\|^2 + O_P(T^{-1}), \end{aligned} \quad (\text{F.67})$$

where $\widehat{\xi} = (\widehat{\xi}_{(1)}, \dots, \widehat{\xi}_{(p)})$.

Using (F.57), we can write

$$\sqrt{T} \widehat{\xi}_{(k)} h_{k2}^\top = \frac{\widetilde{R}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} h_{k2}^\top = \frac{\bar{V}^\top \widetilde{\beta}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} h_{k2}^\top + \frac{\widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} h_{k2}^\top. \quad (\text{F.68})$$

Using Lemma 7(i), Lemma 9(i) and $\widehat{\lambda}_{(k)} \asymp_P |I_k|$, $|I_k| = qN$, we can derive from (F.68) that

$$\sqrt{T} \widehat{\xi}_{(k)} h_{k2}^\top = \bar{V}^\top h_{k2} h_{k2}^\top + O_P\left(q^{-1/2}N^{-1/2}T^{1/2} + T^{-1/2}\right).$$

That is,

$$\sqrt{T} \widehat{\xi} H_2^\top = \bar{V}^\top H_2 H_2^\top + O_P\left(q^{-1/2}N^{-1/2}T^{1/2} + T^{-1/2}\right). \quad (\text{F.69})$$

Therefore, using (F.69), (F.62) and the assumptions on q , we have

$$\begin{aligned} T^{-1/2} \left\| \sqrt{T} \hat{\xi} H_2^\top \gamma - \bar{V}^\top \gamma \right\| &\lesssim_P T^{-1/2} \left\| \bar{V}^\top H_2 H_2^\top - \bar{V}^\top \right\| \|\gamma\| + q^{-1/2} N^{-1/2} + T^{-1} \\ &\lesssim_P T^{-1/2} \left\| \bar{V} \right\| \|H_2 H_2^\top - \mathbb{I}_p\| + q^{-1/2} N^{-1/2} + T^{-1} \\ &\lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}. \end{aligned}$$

Therefore, it follows from (F.67) that

$$\frac{1}{T} \sum_{t=1}^T |m_t - \hat{m}_t|^2 = \frac{1}{T} \left\| \hat{V}^\top \hat{\gamma} - V^\top \gamma \right\|^2 \lesssim_P \frac{1}{T} + \frac{1}{qN}.$$

In light of the assumptions on q , we can choose q such that $qN \gtrsim N_0 / \log N_0$, which leads to

$$\frac{1}{T} \sum_{t=1}^T |m_t - \hat{m}_t|^2 \lesssim_P \frac{1}{T} + \frac{\log N_0}{N_0}.$$

□

F.9 Proof of Proposition 3

Proof. Write $\tilde{\beta} = \Sigma_u^{-1/2} \beta \Sigma_v^{1/2}$, then by definition $\tilde{m}_t$ can be written as

$$\tilde{m}_t = 1 - \gamma^\top \beta^\top \Sigma_r^{-1} (\beta v_t + u_t) = 1 - \gamma^\top \Sigma_v^{-1/2} \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} (\tilde{\beta} \Sigma_v^{-1/2} v_t + \Sigma_u^{-1/2} u_t), \quad (\text{F.70})$$

or in matrix form

$$\tilde{M} = 1 - \gamma^\top \beta^\top \Sigma_r^{-1} (\beta V + U) = 1 - \gamma^\top \Sigma_v^{-1/2} \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} (\tilde{\beta} \Sigma_v^{-1/2} V + \Sigma_u^{-1/2} U), \quad (\text{F.71})$$

where $\tilde{M} = (\tilde{m}_1, \dots, \tilde{m}_T)$, $V = (v_1, \dots, v_T)$ and $U = (u_1, \dots, u_T)$. Suppose that the SVD of $\tilde{\beta}$ can be written as $\tilde{\beta} = B \Lambda^{1/2} \Gamma$, where $B \in \mathbb{R}^{N \times p}$ and $\Gamma \in \mathbb{R}^{p \times p}$ are matrices of left and right singular vectors, $\Lambda^{1/2} = \text{diag}(\tilde{\lambda}_1^{1/2}, \dots, \tilde{\lambda}_p^{1/2})$ is a diagonal matrix and $\tilde{\lambda}_i$ is the i th eigenvalue of $\tilde{\beta}^\top \tilde{\beta}$. Write $B = (b_1, \dots, b_p)$, then $b_i^\top b_j = 0$ for $i \neq j$. Using the SVD of $\tilde{\beta}$, we have

$$\tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} = \Gamma^\top \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} B^\top.$$

Hence, we have

$$\left\| \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \tilde{\beta} - \mathbb{I}_p \right\| = \left\| \Gamma^\top \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} \Lambda^{1/2} \Gamma - \mathbb{I}_p \right\| = \left\| \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} \Lambda^{1/2} - \mathbb{I}_p \right\| \lesssim_P \tilde{\lambda}_p^{-1}, \quad (\text{F.72})$$

and

$$\left\| \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \Sigma_u^{-1/2} U \right\| = \left\| \Gamma^\top \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} B^\top \Sigma_u^{-1/2} U \right\| \lesssim_P \tilde{\lambda}_p^{-1/2} \left\| B^\top \Sigma_u^{-1/2} U \right\|. \quad (\text{F.73})$$

Since $\text{Cov}(B^\top \Sigma_u^{-1/2} u_t) = \mathbb{I}_p$, we have $\mathbb{E} \left(\left\| B^\top \Sigma_u^{-1/2} U \right\|_F^2 \right) = pT$, which leads to

$$\left\| B^\top \Sigma_u^{-1/2} U \right\| \leq \left\| B^\top \Sigma_u^{-1/2} U \right\|_F \lesssim_P T^{1/2}. \quad (\text{F.74})$$

For the same reason, we have $\left\| \Sigma_v^{-1/2} V \right\| \lesssim_P T^{1/2}$. Then, with Assumption B.10, (F.71), (F.72), (F.73), and (F.74), we have

$$\begin{aligned} \sqrt{\sum_{t=1}^T |m_t - \tilde{m}_t|^2} &\leq \left\| \gamma^\top \Sigma_v^{-1/2} \left(\tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \tilde{\beta} - \mathbb{I}_p \right) \Sigma_v^{-1/2} V \right\| + \left\| \gamma^\top \Sigma_v^{-1} \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \Sigma_u^{-1/2} U \right\| \\ &\lesssim \left\| \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \tilde{\beta} - \mathbb{I}_p \right\| \left\| \Sigma_v^{-1/2} V \right\| + \left\| \tilde{\beta}^\top \left(\tilde{\beta} \tilde{\beta}^\top + \mathbb{I}_N \right)^{-1} \Sigma_u^{-1/2} U \right\| \\ &\lesssim_P T^{1/2} \tilde{\lambda}_p^{-1/2}, \end{aligned}$$

which in turn leads to

$$\frac{1}{T} \sum_{t=1}^T |m_t - \tilde{m}_t|^2 \lesssim_P \tilde{\lambda}_p^{-1},$$

where

$$\tilde{\lambda}_p = \lambda_p \left(\Sigma_v^{1/2} \beta^\top \Sigma_u^{-1} \beta \Sigma_v^{1/2} \right) \geq \lambda_p(\beta \Sigma_v \beta^\top) \lambda_{\min}(\Sigma_u^{-1}) \asymp_P \lambda_p(\beta^\top \beta) \lambda_{\max}^{-1}(\Sigma_u) \gtrsim \lambda_p(\beta^\top \beta),$$

which concludes the proof. $\square$

F.10 Proof of Theorem 4(a)

Proof. For Ridge SDF estimator $\hat{m}_t$, we have

$$\frac{1}{T} \sum_{t=1}^T |m_t - \hat{m}_t|^2 = \frac{1}{T} \left\| \bar{R}^\top (\hat{\Sigma} + \mu \mathbb{I}_N)^{-1} \bar{r} - V^\top \gamma \right\|^2. \quad (\text{F.75})$$

Recall that in the proof of Proposition A.2, we have a condensed form of SVD on $\bar{R}$:

$$\bar{R} = \sqrt{T} \hat{\zeta} \hat{\Lambda}^{1/2} \hat{\xi}^\top + \sqrt{T} \hat{\zeta}_* \hat{\Lambda}_*^{1/2} \hat{\xi}_*^\top,$$

where $\widehat{\Lambda}^{1/2}$ is the diagonal matrix of the first p singular values of $T^{-1/2}\bar{R}$ and $\widehat{\Lambda} = \text{diag}\{\widehat{\lambda}_1, \dots, \widehat{\lambda}_p\}$, $\widehat{\varsigma}$, $\widehat{\xi}$ are the corresponding left and right singular vectors, and $\widehat{\varsigma}_* \in \mathbb{R}^{N \times K}$, $\widehat{\xi}_* \in \mathbb{R}^{T \times K}$ are the singular vectors corresponding to the remaining K nonzero singular values in $\widehat{\Lambda}_*^{1/2} \in \mathbb{R}^{K \times K}$, where $K = \min\{N, T-1\} - p$. Using this representation, (F.75) becomes

$$\begin{aligned} \sqrt{\sum_{t=1}^T |m_t - \widehat{m}_t|^2} &= \left\| (\bar{V}^\top \beta^\top + \bar{U}^\top) \widehat{\varsigma} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\varsigma}^\top \bar{r} - V^\top \gamma + (\bar{V}^\top \beta^\top + \bar{U}^\top) \widehat{\varsigma}_* (\widehat{\Lambda}_* + \mu I)^{-1} \widehat{\varsigma}_*^\top \bar{r} \right\| \\ &\leq \left\| \bar{V}^\top \beta^\top \widehat{\varsigma} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\varsigma}^\top \beta \gamma - \bar{V}^\top \gamma \right\| + \left\| \bar{V}^\top \beta^\top \widehat{\varsigma} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\varsigma}^\top (\beta \bar{v} + \bar{u}) \right\| \\ &\quad + \left\| \bar{U}^\top \widehat{\varsigma} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\varsigma}^\top \bar{r} \right\| + \left\| \bar{V}^\top \beta^\top \widehat{\varsigma}_* (\widehat{\Lambda}_* + \mu I)^{-1} \widehat{\varsigma}_*^\top \bar{r} \right\| \\ &\quad + \left\| \bar{U}^\top \widehat{\varsigma}_* (\widehat{\Lambda}_* + \mu I)^{-1} \widehat{\varsigma}_*^\top \bar{r} \right\| + \left\| V^\top \gamma - \bar{V}^\top \gamma \right\| \end{aligned} \quad (\text{F.76})$$

We analyze these terms one-by-one. Firstly, we consider the properties of $\widehat{\varsigma}$ and $\widehat{\xi}$. Let $\widehat{\varsigma}_k$ and $\widehat{\xi}_k$ denote the k th columns of $\widehat{\varsigma}$ and $\widehat{\xi}$, respectively. Note that $\widehat{\varsigma}_k$ and $\widehat{\xi}_k$ can be regarded as the $\widehat{\varsigma}_{(k)}$ and $\widehat{\xi}_{(k)}$ in our SPCA procedure with $I_k = \langle N \rangle$, where $\widehat{\varsigma}_k$ and $\widehat{\xi}_k$ are the singular vectors at the k th stage. This means we can reuse some of the proofs without repeating essentially the same arguments therein.

Similar to (F.55), we define

$$\tilde{h}_{k1} = T^{-1/2} \bar{V} \widehat{\xi}_k, \quad \tilde{h}_{k2} = \widehat{\lambda}_k^{-1/2} \beta^\top \widehat{\xi}_k, \quad (\text{F.77})$$

and $\tilde{H}_1 = (\tilde{h}_{11}, \dots, \tilde{h}_{p1})$, $\tilde{H}_2 = (\tilde{h}_{12}, \dots, \tilde{h}_{p2})$. Using Lemma 14, we can obtain

$$\left\| \tilde{H}_1 \tilde{H}_2^\top - \mathbb{I}_p \right\| \lesssim_P T^{-1} + \lambda_p^{-1} (T^{-1} N + 1), \quad \left\| \tilde{H}_1 - \tilde{H}_2 \right\| \lesssim_P T^{-1/2} + \lambda_p^{-1} (T^{-1} N + 1). \quad (\text{F.78})$$

Using (F.78) and Lemma 14(i), we have $\left\| \tilde{H}_2 \tilde{H}_2^\top - \mathbb{I}_p \right\| \leq \left\| \tilde{H}_1 \tilde{H}_2^\top - \mathbb{I}_p \right\| + \left\| \tilde{H}_1 - \tilde{H}_2 \right\| \left\| \tilde{H}_2 \right\| \lesssim_P T^{-1/2} + \lambda_p^{-1} (T^{-1} N + 1)$, which, by (F.77), is equivalent to

$$\left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1} \widehat{\varsigma}^\top \beta - \mathbb{I}_p \right\| \lesssim_P \frac{1}{\sqrt{T}} + \frac{N+T}{T\lambda_p}. \quad (\text{F.79})$$

Consequently, with Lemma 11 and $\left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1/2} \right\| = \left\| \tilde{H}_2 \right\| \lesssim_P 1$, we have

$$\begin{aligned} \left\| \beta^\top \widehat{\varsigma} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\varsigma}^\top \beta - \mathbb{I}_p \right\| &\leq \left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1/2} \left(\widehat{\Lambda}^{1/2} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\Lambda}^{1/2} - \mathbb{I}_p \right) \widehat{\Lambda}^{-1/2} \widehat{\varsigma}^\top \beta \right\| + \left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1} \widehat{\varsigma}^\top \beta - \mathbb{I}_p \right\| \\ &\leq \left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1/2} \right\|^2 \left\| \widehat{\Lambda}^{1/2} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\Lambda}^{1/2} - \mathbb{I}_p \right\| + \left\| \beta^\top \widehat{\varsigma} \widehat{\Lambda}^{-1} \widehat{\varsigma}^\top \beta - \mathbb{I}_p \right\| \\ &\lesssim_P \frac{1}{\sqrt{T}} + \frac{N+T}{T\lambda_p} + \frac{\mu}{\lambda_p}, \end{aligned} \quad (\text{F.80})$$

where we use $\left\| \widehat{\Lambda}^{1/2} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\Lambda}^{1/2} - \mathbb{I}_p \right\| = \max_{j \leq p} (\widehat{\lambda}_j + \mu)^{-1} \mu \lesssim_P \lambda_p^{-1} \mu$ in the last step.

With $\|\bar{V}\| \lesssim_P T^{1/2}$ from Lemma 1, it implies from (F.80) that the first term in (F.76) can be bounded:

$$\left\| \bar{V}^\top \beta^\top \widehat{\zeta} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\zeta}^\top \beta \gamma - \bar{V}^\top \gamma \right\| \lesssim_P 1 + \frac{N+T}{\sqrt{T} \lambda_p} + \frac{\mu \sqrt{T}}{\lambda_p}.$$

For the second term in (F.76), using Lemma 11, we have

$$\left\| \bar{V}^\top \beta^\top \widehat{\zeta} (\widehat{\Lambda} + \mu I)^{-1} \widehat{\zeta}^\top (\beta \bar{v} + \bar{u}) \right\| \leq \|\bar{V}\| \left\| \beta^\top \widehat{\zeta} \widehat{\Lambda}^{-1/2} \right\| \left\| \widehat{\Lambda}^{1/2} (\widehat{\Lambda} + \mu I)^{-1} \right\| \|\beta \bar{v} + \bar{u}\| \lesssim_P \sqrt{\frac{N}{\lambda_p}}. \quad (\text{F.81})$$

Next, recall that $\widehat{\zeta}_*$ and $\widehat{\xi}_*$ are singular vectors of $\bar{R}$, we have

$$\bar{V}^\top \beta^\top \widehat{\zeta}_* + \bar{U}^\top \widehat{\zeta}_* = \bar{R}^\top \widehat{\zeta}_* = \sqrt{T} \widehat{\xi}_* \widehat{\Lambda}_*^{1/2}. \quad (\text{F.82})$$

By Weyl's theorem and Assumption B.4, we have

$$|\sigma_j(T^{-1/2} \bar{R}) - \sigma_j(T^{-1/2} \beta \bar{V})| \leq T^{-1/2} \|\bar{R} - \beta \bar{V}\| = T^{-1/2} \|\bar{U}\| \lesssim_P \sqrt{\frac{N}{T}} + 1, \quad (\text{F.83})$$

for $j \leq \min\{N, T\}$. Since $\text{Rank}(T^{-1/2} \beta \bar{V}) \leq p$, we have $\sigma_j(T^{-1/2} \beta \bar{V}) = 0$ for $j > p$ and thus

$$\left\| \widehat{\Lambda}_*^{1/2} \right\| = \sigma_{p+1}(T^{-1/2} \bar{R}) \lesssim_P \sqrt{\frac{N}{T}} + 1. \quad (\text{F.84})$$

Left multiplying (F.82) by $\bar{V}$, we obtain

$$\bar{V} \bar{V}^\top \beta^\top \widehat{\zeta}_* = \sqrt{T} \bar{V} \widehat{\xi}_* \widehat{\Lambda}_*^{1/2} - \bar{V} \bar{U}^\top \widehat{\zeta}_*. \quad (\text{F.85})$$

Together with (F.84) and Assumption B.6, we have

$$\|\beta^\top \widehat{\zeta}_*\| \leq \left\| (\bar{V} \bar{V}^\top)^{-1} \right\| \left(\sqrt{T} \|\bar{V}\| \left\| \widehat{\Lambda}_*^{1/2} \right\| + \|\bar{V} \bar{U}^\top\| \right) \lesssim_P \sqrt{\frac{N}{T}} + 1, \quad (\text{F.86})$$

and consequently,

$$\|\widehat{\zeta}_*^\top \bar{r}\| \leq \|\widehat{\zeta}_*^\top \beta\| \|\gamma + \bar{v}\| + \|\widehat{\zeta}_*^\top \bar{u}\| \lesssim_P \sqrt{\frac{N}{T}} + 1. \quad (\text{F.87})$$

Using (F.86), (F.87), Lemma 13(iv) and $\|\bar{U}\| \lesssim_P N^{1/2} + T^{1/2}$, we have

$$\left\| \beta^\top \widehat{\zeta}_* (\widehat{\Lambda}_* + \mu I)^{-1} \widehat{\zeta}_*^\top \bar{r} \right\| \leq \|\beta^\top \widehat{\zeta}_*\| \left\| (\widehat{\Lambda}_* + \mu I)^{-1} \right\| \|\widehat{\zeta}_*^\top \bar{r}\| \lesssim_P \frac{N+T}{\mu T}, \quad (\text{F.88})$$

and

$$\left\| \bar{U}^\top \hat{\varsigma}_* (\hat{\Lambda}_* + \mu I)^{-1} \hat{\varsigma}_*^\top \bar{r} \right\| \leq \|\bar{U}\| \left\| (\hat{\Lambda}_* + \mu I)^{-1} \right\| \|\hat{\varsigma}_*^\top \bar{r}\| \lesssim_P \frac{N+T}{\mu\sqrt{T}}. \quad (\text{F.89})$$

Using Lemma 13(iii), we have

$$\left\| \hat{\Lambda}^{-1/2} \hat{\varsigma}^\top \bar{r} \right\| \lesssim_P \left\| \hat{\Lambda}^{-1/2} \hat{\varsigma}^\top \beta \right\| + \left\| \hat{\Lambda}^{-1/2} \hat{\varsigma}^\top \bar{u} \right\| \lesssim_P 1 + \frac{N+T}{T\lambda_p} \lesssim_P 1,$$

where we use $\left\| \hat{\Lambda}^{-1/2} \hat{\varsigma}^\top \beta \right\| = \left\| \tilde{H}_2 \right\| \lesssim_P 1$. Then, with Lemma 13(iv), we have

$$\left\| \bar{U}^\top \hat{\varsigma} (\hat{\Lambda} + \mu I)^{-1} \hat{\varsigma}^\top \bar{r} \right\| \leq \|\bar{U}^\top \hat{\varsigma}\| \left\| (\hat{\Lambda} + \mu I)^{-1} \hat{\Lambda}^{1/2} \right\| \left\| \hat{\Lambda}^{-1/2} \hat{\varsigma}^\top \bar{r} \right\| \lesssim_P \sqrt{\frac{T}{\lambda_p}} + \frac{N+T}{\sqrt{T}\lambda_p}. \quad (\text{F.90})$$

Plugging (F.80), (F.81), (F.88), (F.89) and (F.90) into (F.76) and using $\|\bar{V} - V\| \lesssim_P 1$, we obtain

$$\frac{1}{T} \sum_{t=1}^T |m_t - \hat{m}_t|^2 \lesssim_P \frac{\mu^2}{\lambda_p^2} + \frac{1}{T} + \frac{N+T}{T\lambda_p} + \frac{N^2+T^2}{\mu^2 T^2}.$$

With $\mu^2 \asymp T^{-1}\lambda_p(N+T)$, we achieve the best rate from the above bound:

$$\frac{1}{T} \sum_{t=1}^T |m_t - \hat{m}_t|^2 \lesssim_P \frac{1}{T} + \frac{N+T}{T\lambda_p}.$$

□

F.11 Proof of Theorem 4(b)

Proof. i. (Slow rate) Note that (15) is equivalent to a constrained optimization problem:

$$\hat{b} = \arg \min_b \left\| \hat{\Sigma}^{-1/2} \bar{r} - \hat{\Sigma}^{1/2} b \right\|^2, \quad \text{subject to } \|b\|_1 \leq \mu,$$

for some tuning parameter μ . This implies that the vector of the true SDF loadings, b , satisfies that

$$\left\| \hat{\Sigma}^{-1/2} \bar{r} - \hat{\Sigma}^{1/2} \hat{b} \right\|^2 \leq \left\| \hat{\Sigma}^{-1/2} \bar{r} - \hat{\Sigma}^{1/2} b \right\|^2 \quad \text{and} \quad \left\| \hat{b} \right\|_1 \leq \mu, \quad \text{for some } \mu \geq s.$$

Equivalently, expanding the left- and right-hand sides of the above we have

$$\hat{b}^\top \hat{\Sigma} \hat{b} - b^\top \hat{\Sigma} b \leq 2(\hat{b} - b)^\top \bar{r},$$

which leads to

$$(\hat{b} - b)^\top \hat{\Sigma}(\hat{b} - b) \leq 2(\hat{b} - b)^\top (\bar{r} - \hat{\Sigma}b) \leq 2 \left\| \hat{b} - b \right\|_1 \left\| \bar{r} - \hat{\Sigma}b \right\|_\infty.$$

With a tuning parameter $\mu \asymp s$, we have

$$(\hat{b} - b)^\top \hat{\Sigma}(\hat{b} - b) \lesssim s \left\| \bar{r} - \hat{\Sigma}b \right\|_\infty. \quad (\text{F.91})$$

With Lemma 15, we have

$$\left\| \hat{\Sigma}^{1/2}(\hat{b} - b) \right\|^2 \lesssim_P s \sqrt{\frac{\log N}{T}}. \quad (\text{F.92})$$

Therefore, we have

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \left\| \hat{m}_t - \tilde{m}_t \right\|^2 &= \frac{1}{T} \sum_{t=1}^T \left\| \hat{b}^\top (r_t - \bar{r}) - b^\top (r_t - \mathbb{E}(r_t)) \right\|^2 \\ &\leq \frac{2}{T} \sum_{t=1}^T \left\| (\hat{b} - b)^\top (r_t - \bar{r}) \right\|^2 + \frac{2}{T} \sum_{t=1}^T \left\| b^\top (\bar{r} - \mathbb{E}(r_t)) \right\|^2 \\ &\leq 2 \left\| \hat{\Sigma}^{1/2}(\hat{b} - b) \right\|^2 + 2 \left\| b \right\|_1^2 \left\| \bar{r} - \mathbb{E}(r_t) \right\|_\infty^2 \lesssim_P s \sqrt{\frac{\log N}{T}} + s^2 \frac{\log N}{T}. \end{aligned}$$

Since $s \asymp \mu \gtrsim \|b\|_1$, plugging in the optimal rate choice $s \asymp \|b\|_1$, we complete the proof.

ii. (Fast rate) Since $\hat{b}$ is the optimal solution of the minimization problem, it implies that

$$b^\top \hat{\Sigma}b - 2b^\top \bar{r} + b^\top \hat{\Sigma}b + \mu \|b\|_1 \geq \hat{b}^\top \hat{\Sigma}\hat{b} - 2\hat{b}^\top \bar{r} + \hat{b}^\top \hat{\Sigma}\hat{b} + \mu \|\hat{b}\|_1. \quad (\text{F.93})$$

Rewrite (F.93) as

$$(\hat{b} - b)^\top \hat{\Sigma}(\hat{b} - b) \leq 2(\hat{b} - b)^\top (\bar{r} - \hat{\Sigma}b) + \mu(\|b\|_1 - \|\hat{b}\|_1). \quad (\text{F.94})$$

If $\mu \geq 4 \left\| \bar{r} - \hat{\Sigma}b \right\|_\infty$, (F.94) becomes

$$\begin{aligned} \left\| \hat{\Sigma}^{1/2}(\hat{b} - b) \right\|^2 &\leq 2 \left\| \hat{b} - b \right\|_1 \left\| \bar{r} - \hat{\Sigma}b \right\|_\infty + \mu(\|b\|_1 - \|\hat{b}\|_1) \\ &\leq \frac{1}{2}\mu \left\| \hat{b} - b \right\|_1 + \mu(\|b\|_1 - \|\hat{b}\|_1). \end{aligned} \quad (\text{F.95})$$

Let J denote the support of $\hat{b}$, then (F.95) can be written as

$$\begin{aligned} \left\| \hat{\Sigma}^{1/2}(\hat{b} - b) \right\|^2 &\leq \frac{1}{2}\mu \left\| \hat{b}_J - b_J \right\|_1 + \frac{1}{2}\mu \left\| \hat{b}_{J^c} \right\|_1 + \mu \left\| \hat{b}_J - b_J \right\|_1 - \mu \left\| \hat{b}_{J^c} \right\|_1 \\ &= \frac{3}{2}\mu \left\| \hat{b}_J - b_J \right\|_1 - \frac{1}{2}\mu \left\| \hat{b}_{J^c} \right\|_1. \end{aligned} \quad (\text{F.96})$$

Define $b^* = \widehat{b} - b$, then (F.96) implies that $3 \|b_J^*\|_1 \geq \|b_{J^c}^*\|_1$, and we have

$$\frac{b^{*\top}(\Sigma - \widehat{\Sigma})b^*}{\|b^*\|^2} \leq \left\| \Sigma - \widehat{\Sigma} \right\|_{MAX} \frac{\|b^*\|_1^2}{\|b^*\|^2} \lesssim_P \sqrt{\frac{\log N}{T}} \left(\frac{4 \|b_J^*\|_1}{\|b_J^*\|} \right)^2 \lesssim_P |J| \sqrt{\frac{\log N}{T}}.$$

Consequently, with the assumption $|J| \sqrt{\log N/T} \rightarrow 0$ and $\lambda_{\min}(\Sigma) \gtrsim 1$, we have

$$\frac{b^{*\top} \widehat{\Sigma} b^*}{\|b^*\|^2} = \frac{b^{*\top} \Sigma b^*}{\|b^*\|^2} + \frac{b^{*\top}(\Sigma - \widehat{\Sigma})b^*}{\|b^*\|^2} \gtrsim_P 1.$$

Therefore, we have

$$\left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|^2 = b^{*\top} \widehat{\Sigma} b^* \gtrsim_P \|b^*\|^2 \geq \|b_J^*\|^2 \geq |J|^{-1} \|b_J^*\|_1^2 = |J|^{-1} \left\| \widehat{b}_J - b_J \right\|_1^2. \quad (\text{F.97})$$

Plugging (F.97) into (F.96), we have

$$\left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|^2 \leq \frac{3}{2} \mu \left\| \widehat{b}_J - b_J \right\|_1 \lesssim_P \mu |J|^{1/2} \left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|.$$

Thus,

$$\left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|^2 \lesssim_P \mu^2 |J|. \quad (\text{F.98})$$

Choosing $\mu = 4 \left\| \bar{r} - \widehat{\Sigma} b \right\|_\infty$ and by Lemma 15, we obtain

$$\left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|^2 \lesssim_P |J| \frac{\log N}{T}. \quad (\text{F.99})$$

Similar to the slow rate case, we have

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \|\widehat{m}_t - \widetilde{m}_t\|^2 &= \frac{1}{T} \sum_{t=1}^T \left\| \widehat{b}^\top(r_t - \bar{r}) - b^\top(r_t - \mathbb{E}(r_t)) \right\|^2 \\ &\leq \frac{2}{T} \sum_{t=1}^T \left\| (\widehat{b} - b)^\top(r_t - \bar{r}) \right\|^2 + \frac{2}{T} \sum_{t=1}^T \|b^\top(\bar{r} - \mathbb{E}(r_t))\|^2 \\ &\leq 2 \left\| \widehat{\Sigma}^{1/2}(\widehat{b} - b) \right\|^2 + 2 \|b^\top(\bar{r} - \mathbb{E}(r_t))\|^2 \lesssim_P \|b\|_0 \frac{\log N}{T}. \end{aligned}$$

□

F.12 Proof of Theorem 5

Proof. To simplify the notation, we assume $\Sigma_v = \mathbb{I}_p$ without loss of generality and define a function $\text{sr}(\cdot)$:

$$\text{sr}(x_t) = \arg \max_{0 \neq b \in \mathbb{R}^N} \frac{\mathbb{E}(b^\top x_t)}{\sqrt{\text{Var}(b^\top x_t)}}.$$

In other words, $\text{sr}(x_t)$ is the optimal Sharpe ratio we can get from x_t . It is well known that $\text{sr}(x_t) = \sqrt{\mathbb{E}(x_t)^\top \text{Cov}(x_t)^{-1} \mathbb{E}(x_t)}$ and the optimal value can be achieved by $b = \text{Cov}(x_t)^{-1} \mathbb{E}(x_t)$, where $\text{Cov}(x_t)^{-1}$ can be replaced by the Moore–Penrose inverse if it is singular.

Recall that our estimated factors are $\hat{F}_{(k)} = \hat{\zeta}_{(k)}^\top \tilde{R}_{(k)} / \sqrt{\hat{\lambda}_{(k)}} = \hat{\zeta}_{(k)}^\top D_{(k)} \bar{R} / \sqrt{\hat{\lambda}_{(k)}}$. Define $B = (b_1, \dots, b_{\hat{p}})^\top \in \mathbb{R}^{\hat{p} \times N}$, where $b_i = D_{(k)}^\top \hat{\zeta}_{(k)} / \sqrt{\hat{\lambda}_{(k)}}$. From Lemma 7(i), we have $\|B\| = O_P(1)$. With the notation of B , the estimated factors can be written as $\hat{F}_{(k)} = B \bar{R}$ and the SDF loading we get from SPCA is $\hat{\gamma}^\top B$. Therefore, the out-of-sample Sharpe ratio for SPCA estimator is $\text{sr}(\hat{\gamma}^\top B r_t)$. To prove that $\lim_{N,T \rightarrow \infty} \text{sr}(\hat{\gamma}^\top B r_t) \geq \sqrt{\gamma^\top \mathbb{P} \eta^\top \gamma}$, we first show that $\lim_{N,T \rightarrow \infty} \text{sr}(B r_t) \geq \text{sr}(\eta(v_t + \gamma)) = \sqrt{\gamma^\top \mathbb{P} \eta^\top \gamma}$.

The definition of $\text{sr}(B r_t)$ implies that

$$\text{sr}(B r_t) \geq \frac{\mathbb{E}(b^\top \hat{\eta} B r_t)}{\sqrt{\text{Var}(b^\top \hat{\eta} B r_t)}},$$

for any $b \in \mathbb{R}^d$. Therefore, $\text{sr}(B r_t) \geq \text{sr}(\hat{\eta} B r_t)$. Note that (F.50) can be written as $\|\hat{\eta} B \beta - \eta\| = o_P(1)$ and

$$\hat{\eta} B r_t - \eta(v_t + \gamma) = (\hat{\eta} B \beta - \eta)(v_t + \gamma) + \hat{\eta} B u_t,$$

with $\|B\| = o_P(1)$, we have $\|\hat{\eta} B r_t - \eta(v_t + \gamma)\| = o_P(1)$. Consequently, for any $b \in \mathbb{R}^d$, $\mathbb{E}(b^\top \hat{\eta} B r_t) \rightarrow \mathbb{E}(b^\top \eta(v_t + \gamma))$ and $\text{Cov}(b^\top \hat{\eta} B r_t) \rightarrow \text{Cov}(b^\top \eta(v_t + \gamma))$. Therefore, $\text{sr}(\hat{\eta} B r_t) \xrightarrow{P} \text{sr}(\eta(v_t + \gamma))$.

$\text{sr}(\eta(v_t + \gamma))$ can be calculated by $\mathbb{E}(\eta(v_t + \gamma)) = \eta \gamma$ and $\text{Cov}(\eta(v_t + \gamma)) = \eta \eta^\top$ we have

$$\text{sr}(\eta(v_t + \gamma)) = \sqrt{\mathbb{E}(\eta(v_t + \gamma))^\top \text{Cov}(\eta(v_t + \gamma))^{-1} \mathbb{E}(\eta(v_t + \gamma))} = \sqrt{\gamma^\top \eta^\top (\eta \eta^\top)^{-1} \eta \gamma} = \sqrt{\gamma^\top \mathbb{P} \eta^\top \gamma}$$

Again, $(\eta \eta^\top)^{-1}$ here will be the Moore–Penrose inverse if it is singular. To sum up, we have $\lim_{N,T \rightarrow \infty} \text{sr}(B r_t) \geq \text{sr}(\eta(v_t + \gamma)) = \sqrt{\gamma^\top \mathbb{P} \eta^\top \gamma}$. Then, we will show that the optimal Sharpe ratio from $B r_t$ can be achieved approximately by the portfolio $\hat{\gamma}^\top B r_t$.

Note that $B \beta = H_2^\top$ from the definition of H_2 in (F.55), we have $\mathbb{E}(B r_t) = B \beta \gamma = H_2^\top \gamma$. With (F.59), it leads to $\|\hat{\gamma} - H_2^\top \gamma\| = o_P(1)$. For the covariance matrix, write

$$\|\text{Cov}(B r_t) - \mathbb{I}_{\hat{p}}\| \leq \|B \beta \beta^\top B^\top - \mathbb{I}_{\hat{p}}\| + \|B \Sigma_u B^\top\| \leq \|H_2^\top H_2 - \mathbb{I}_{\hat{p}}\| + \|B\|^2 \|\Sigma_u\| = o_P(1).$$

where we use Lemma 14(ii),(iii), $\|B\| = o_P(1)$ and the assumption $\|\Sigma_u\| \lesssim 1$ in the last equation. Consequently, $\hat{\gamma} \xrightarrow{P} \text{Cov}(Br_t)^{-1}E(Br_t)$. With this equation, we have

$$\text{sr}(\hat{\gamma}^\top Br_t) = \frac{E(\hat{\gamma}^\top Br_t)}{\sqrt{\text{Cov}(\hat{\gamma}^\top Br_t)}} \xrightarrow{P} \sqrt{E(Br_t)^\top \text{Cov}(Br_t)^{-1}E(Br_t)} = \text{sr}(Br_t).$$

The proof of the lower bound $\lim_{N,T \rightarrow \infty} \text{sr}(\hat{\gamma}^\top Br_t) \geq \sqrt{\gamma^\top \mathbb{P}_{\eta^\top} \gamma}$ is completed. For the upper bound, for the same reason as $\text{sr}(Br_t) \geq \text{sr}(\hat{\eta}^\top Br_t)$, it is straightforward to obtain $\text{sr}(\hat{\gamma}^\top Br_t) \leq \text{sr}(r_t) \leq \sqrt{\gamma^\top \gamma}$. In the general case that $\Sigma_v \neq \mathbb{I}_p$, replace η, γ by $\eta \Sigma_v^{1/2}, \Sigma_v^{-1/2} \gamma$ to obtain the results. $\square$

F.13 Proof of Proposition C.4

Proof. Using (1) and (4), we have

$$\text{Cov}(g_t, r_{t,[I_0]}) \text{Cov}(r_{t,[I_0]})^{-1} E(r_{t,[I_0]}) = \eta \Sigma_v \beta_{[I_0]}^\top (\beta_{[I_0]} \Sigma_v \beta_{[I_0]}^\top + \Sigma_{u,[I_0]})^{-1} \beta_{[I_0]} \gamma.$$

Therefore, it is sufficient to show that

$$\left\| \Sigma_v \beta_{[I_0]}^\top (\beta_{[I_0]} \Sigma_v \beta_{[I_0]}^\top + \Sigma_{u,[I_0]})^{-1} \beta_{[I_0]} - \mathbb{I}_{N_0} \right\| = O\left(1/\lambda_{\min}(\beta_{[I_0]}^\top \beta_{[I_0]})\right) \quad (\text{F.100})$$

with $N_0 = |I_0|$. Write $\tilde{\beta}_{[I_0]} = \Sigma_{u,[I_0]}^{-1/2} \beta_{[I_0]} \Sigma_v^{1/2}$, then (F.100) becomes

$$\left\| \Sigma_v^{1/2} \tilde{\beta}_{[I_0]}^\top (\tilde{\beta}_{[I_0]} \tilde{\beta}_{[I_0]}^\top + \mathbb{I}_{N_0})^{-1} \tilde{\beta}_{[I_0]} \Sigma_v^{-1/2} - \mathbb{I}_{N_0} \right\| = O\left(1/\lambda_{\min}(\beta_{[I_0]}^\top \beta_{[I_0]})\right) \quad (\text{F.101})$$

Suppose that the SVD of $\tilde{\beta}_{[I_0]}$ can be written as $\tilde{\beta}_{[I_0]} = B \Lambda^{1/2} \Gamma$, where $B \in \mathbb{R}^{N_0 \times p}$ and $\Gamma \in \mathbb{R}^{p \times p}$ are matrices of left and right singular vectors, $\Lambda^{1/2} = \text{diag}(\tilde{\lambda}_1^{1/2}, \dots, \tilde{\lambda}_p^{1/2})$ is a diagonal matrix and $\tilde{\lambda}_i$ is the i th eigenvalue of $\tilde{\beta}_{[I_0]}^\top \tilde{\beta}_{[I_0]}$. Using the SVD of $\tilde{\beta}_{[I_0]}$, we have

$$\tilde{\beta}_{[I_0]}^\top \left(\tilde{\beta}_{[I_0]} \tilde{\beta}_{[I_0]}^\top + \mathbb{I}_N \right)^{-1} \tilde{\beta}_{[I_0]} = \Gamma^\top \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} \Lambda^{1/2} \Gamma.$$

Consequently, with $\lambda_{\max}(\Sigma_v) \lesssim 1$ and $\lambda_{\min}(\Sigma_v) \gtrsim 1$, the left hand side of (F.101) becomes

$$\begin{aligned} \left\| \Sigma_v^{1/2} \tilde{\beta}_{[I_0]}^\top (\tilde{\beta}_{[I_0]} \tilde{\beta}_{[I_0]}^\top + \mathbb{I}_{N_0})^{-1} \tilde{\beta}_{[I_0]} \Sigma_v^{-1/2} - \mathbb{I}_{N_0} \right\| &= \left\| \Sigma_v^{1/2} \Gamma^\top \left(\Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} \Lambda^{1/2} - \mathbb{I}_p \right) \Gamma \Sigma_v^{-1/2} \right\| \\ &\lesssim \left\| \Lambda^{1/2} (\Lambda + \mathbb{I}_p)^{-1} \Lambda^{1/2} - \mathbb{I}_p \right\| = \frac{1}{1 + \tilde{\lambda}_p}. \end{aligned}$$

Note that

$$\begin{aligned} \tilde{\lambda}_p &= \lambda_p \left(\Sigma_v^{1/2} \beta_{[I_0]}^\top \Sigma_u^{-1} \beta_{[I_0]} \Sigma_v^{1/2} \right) \geq \lambda_p (\beta_{[I_0]} \Sigma_v \beta_{[I_0]}^\top) \lambda_{\min}(\Sigma_u^{-1}) \asymp_P \lambda_p (\beta_{[I_0]}^\top \beta_{[I_0]}) \lambda_{\max}^{-1}(\Sigma_u) \\ &\gtrsim \lambda_p (\beta_{[I_0]}^\top \beta_{[I_0]}) = \lambda_{\min}(\beta_{[I_0]}^\top \beta_{[I_0]}), \end{aligned}$$

we have obtained (F.101) and this concludes the proof. $\square$

F.14 Proof of Proposition C.5

Proof. We consider the case $d = 1$ first. Recall that $(\check{p}, \check{q})$ denotes an arbitrary pair of tuning parameter values and that $\check{w}$ is a short-hand notation for $w(\check{p}, \check{q})$. By definition, maximizing $R^2(\check{p}, \check{q})$ is equivalent to minimizing

$$\text{MSE}(\check{w}) = T_{\text{oos}}^{-1} \|\bar{G}_{\text{oos}} - \check{w}\bar{R}_{\text{oos}}\|^2 = \|(\check{w}\beta - \eta)\bar{V}_{\text{oos}} + \check{w}\bar{U}_{\text{oos}} - \bar{Z}_{\text{oos}}\|^2. \quad (\text{F.102})$$

As shown in the proof of Theorem 1, the estimated factor at the k th step is given by $\hat{V}_{(k)} = \sqrt{T}\hat{\xi}_{(k)}^\top$ and the loading of G on $\hat{V}_{(k)}$ is $\hat{\eta}_{(k)} = T^{-1/2}\bar{G}\hat{\xi}_{(k)}^\top$. Using (F.41) and (F.43), we have

$$\hat{V}_{(k)} = \lambda_{(k)}^{-1/2}\hat{\xi}_{(k)}^\top \tilde{R}_{(k)} = \lambda_{(k)}^{-1/2}\hat{\xi}_{(k)}^\top D_{(k)} R_{(k)}. \quad (\text{F.103})$$

Thus, the mimicking portfolio of g_t is given by $\sum_{i=1}^{\check{p}} \hat{\eta}_{(i)} \hat{V}_{(i)} =: \check{w}\bar{R}$, where

$$\check{w} = \sum_{i=1}^{\check{p}} \bar{G}\hat{\xi}_{(i)}^\top \frac{\hat{\xi}_{(i)}^\top D_{(i)}}{\sqrt{T\hat{\lambda}_{(i)}}}. \quad (\text{F.104})$$

Next, we claim that there exists a pair of (c_0, q_0) that satisfy $q_0 \in \mathcal{Q}$ and (11) in Theorem 1. This holds because we can set $q_0 = N^{-\alpha_{nq}}$, and the existence of c_0 is guaranteed under the assumptions that $N^{1-\alpha_{nq}}/N_0 \rightarrow 0$ and $\log T/N^{1-\alpha_{nq}} \rightarrow 0$. Given c_0 and q_0 , let p_0 denote the number of factors extracted. As shown in the proof of Theorem 1, we have $P(p_0 = \tilde{p}) \rightarrow 1$ and $\tilde{p} \leq p$. Therefore, since $p_{\max} \geq p$, it implies that $(p_0, q_0) \in \langle p_{\max} \rangle \times \mathcal{Q}$ with probability approaching 1, and (p_0, q_0) corresponds to (c_0, q_0) mentioned above.

Denote $w_0 = w(p_0, q_0)$. Using (F.50) and the definition $\tilde{\beta}_{(k)} = D_{(k)}\beta$, we have

$$\|w_0\beta - \eta\| = o_P(1). \quad (\text{F.105})$$

Also, with $\|D_{(i)}\| \lesssim_P 1$ from Lemma 7(i), we have

$$\|w_0\| \leq \sum_{i=1}^{p_0} T^{-1/2}\hat{\lambda}_{(i)}^{-1/2} \|\bar{G}\| \|D_{(i)}\| = o_P(1). \quad (\text{F.106})$$

Next, we write (F.102) as

$$\begin{aligned} \text{MSE}(w_0) &= (w_0\beta - \eta) \left(\frac{\bar{V}_{\text{oos}}\bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) (w_0\beta - \eta)^\top + \frac{w_0\bar{U}_{\text{oos}}(w_0\bar{U}_{\text{oos}})^\top}{T_{\text{oos}}} + \frac{\bar{Z}_{\text{oos}}\bar{Z}_{\text{oos}}^\top}{T_{\text{oos}}} \\ &\quad + 2\frac{w_0\bar{U}_{\text{oos}}\bar{V}_{\text{oos}}^\top}{T_{\text{oos}}}(w_0\beta - \eta)^\top - 2(w_0\beta - \eta) \frac{\bar{V}_{\text{oos}}\bar{Z}_{\text{oos}}^\top}{T_{\text{oos}}} - 2\frac{w_0\bar{U}_{\text{oos}}\bar{Z}_{\text{oos}}^\top}{T_{\text{oos}}}, \end{aligned} \quad (\text{F.107})$$

and analyze these terms one by one. Under Assumptions C.13 and C.14, rewriting the proof of Lemma 1 leads to

$$\left\| \frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} - \Sigma_v \right\| \lesssim_P T_{\text{oos}}^{-1/2}, \quad \left\| \frac{\bar{Z}_{\text{oos}} \bar{Z}_{\text{oos}}^\top}{T_{\text{oos}}} - \Sigma_z \right\| \lesssim_P T_{\text{oos}}^{-1/2}, \quad \|w_0 \bar{U}_{\text{oos}}\| \lesssim_P \|w_0\| T_{\text{oos}}^{1/2}, \quad (\text{F.108})$$

and

$$\|w_0 \bar{U}_{\text{oos}} \bar{V}_{\text{oos}}^\top\| \lesssim_P \|w_0\| T_{\text{oos}}^{1/2}, \quad \|\bar{V}_{\text{oos}} \bar{Z}_{\text{oos}}^\top\| \lesssim_P T_{\text{oos}}^{1/2}, \quad \|w_0 \bar{U}_{\text{oos}} \bar{Z}_{\text{oos}}^\top\| \lesssim_P \|w_0\| T_{\text{oos}}^{1/2}. \quad (\text{F.109})$$

Therefore, together with (F.105) and (F.106), (F.107) implies that

$$\begin{aligned} \|\text{MSE}(w_0) - \Sigma_z\| &\lesssim_P \|w_0 \beta - \eta\|^2 \lambda_{\min}(\Sigma_v) + \|w_0\|^2 + T_{\text{oos}}^{-1/2} \\ &\quad + \|w_0 \beta - \eta\| \|w_0\| T_{\text{oos}}^{-1/2} + \|w_0 \beta - \eta\| T_{\text{oos}}^{-1/2} + \|w_0\| T_{\text{oos}}^{-1/2} \\ &= o_P(1). \end{aligned} \quad (\text{F.110})$$

In other words, there exists $(p_0, q_0) \in \langle p_{\max} \rangle \times \mathcal{Q}$ such that $\text{MSE}(w(p_0, q_0)) \xrightarrow{P} \Sigma_z$.

Given that (p^*, q^*) minimize $\text{MSE}(w(\check{p}, \check{q}))$, we have $\text{MSE}(w^*) \leq \text{MSE}(w_0) = \Sigma_z + o_P(1)$. Let δ_1 and δ_2 denote $\|w^* \beta - \eta\|$ and $\|w^*\|$, we will show that $\delta_1 = o_P(1)$ and $\delta_2 = o_P(1)$. To see this, with (F.108) and the assumption $\lambda_{\min}(\Sigma_v) \gtrsim 1$, we have

$$(w^* \beta - \eta) \left(\frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) (w^* \beta - \eta)^\top \geq \delta_1^2 \lambda_{\min} \left(\frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) \gtrsim_P \delta_1^2, \quad (\text{F.111})$$

and

$$\frac{\|w^* \bar{U}\|}{\delta_2 T_{\text{oos}}^{1/2}} = \frac{\|w^* \bar{U}\|}{\|w^*\| T_{\text{oos}}^{1/2}} \geq \min_{\check{p} \leq p_{\max}, \check{q} \in \mathcal{Q}} \frac{\|w(\check{p}, \check{q}) \bar{U}\|}{\|w(\check{p}, \check{q})\| T_{\text{oos}}^{1/2}} \gtrsim_P 1, \quad (\text{F.112})$$

where we use the assumption that $p_{\max}$ and $|\mathcal{Q}| = n_q$ are finite and Assumption C.14 to construct the uniform bound. Combining (F.111) and (F.112), we have

$$\begin{aligned} &(w^* \beta - \eta) \left(\frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) (w^* \beta - \eta)^\top + \frac{w^* \bar{U}_{\text{oos}} (w^* \bar{U}_{\text{oos}})^\top}{T_{\text{oos}}} \\ &\geq \max \left\{ (w^* \beta - \eta) \left(\frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) (w^* \beta - \eta)^\top, \frac{w^* \bar{U}_{\text{oos}} (w^* \bar{U}_{\text{oos}})^\top}{T_{\text{oos}}} \right\} \gtrsim_P \max\{\delta_1^2, \delta_2^2\} \gtrsim (\delta_1 + \delta_2)^2, \end{aligned} \quad (\text{F.113})$$

where the first inequality stems from the fact that the two quadratic forms on the left-hand-side are positive numbers. On the other hand, we have

$$\frac{\|w^* \bar{U}_{\text{oos}} \bar{A}_{\text{oos}}^\top\|}{\|w^*\|} \leq \max_{\check{p} \leq p_{\max}, \check{q} \in \mathcal{Q}} \frac{\|w(\check{p}, \check{q}) \bar{U}_{\text{oos}} \bar{A}_{\text{oos}}^\top\|}{\|w(\check{p}, \check{q})\|} \lesssim_P T_{\text{oos}}^{1/2}, \quad (\text{F.114})$$

for $A = V$ and Z . The decomposition of $\text{MSE}(w^*)$ also has the form (F.107) by replacing w_0 by w^* . With the decomposition, (F.109) and (F.114), we have

$$\begin{aligned} (\delta_1 + \delta_2)^2 &\lesssim_{\text{P}} (w^* \beta - \eta) \left(\frac{\bar{V}_{\text{oos}} \bar{V}_{\text{oos}}^\top}{T_{\text{oos}}} \right) (w^* \beta - \eta)^\top + \frac{w^* \bar{U}_{\text{oos}} (w^* \bar{U}_{\text{oos}})^\top}{T_{\text{oos}}} \\ &\leq \text{MSE}(w^*) - \Sigma_z + O_{\text{P}}(T_{\text{oos}}^{-1/2} \delta_1) + O_{\text{P}}(T_{\text{oos}}^{-1/2} \delta_1 \delta_2) + O_{\text{P}}(T_{\text{oos}}^{-1/2} \delta_2) + o_{\text{P}}(1) \\ &\leq o_{\text{P}}(1 + (\delta_1 + \delta_2) + \delta_1 \delta_2) = o_{\text{P}}(1 + (\delta_1 + \delta_2)^2), \end{aligned} \quad (\text{F.115})$$

where we use $\delta_1 + \delta_2 \leq (1 + (\delta_1 + \delta_2)^2)/2$ and $\delta_1 \delta_2 \leq (\delta_1 + \delta_2)^2/4$ in the last equation. This leads to $\delta_1 = o_{\text{P}}(1)$ and $\delta_2 = o_{\text{P}}(1)$ as δ_1 and δ_2 are non-negative. Plugging them into the second inequality of (F.115), we obtain $\text{MSE}(w^*) - \Sigma_z \geq o_{\text{P}}(1)$. With $\text{MSE}(w^*) \leq \text{MSE}(w_0)$ and (F.110), we have

$$\text{MSE}(w^*) - \Sigma_z = o_{\text{P}}(1). \quad (\text{F.116})$$

In addition, in out-of-sample data, the expected return of g_t 's mimicking portfolio based on w^* satisfies

$$\begin{aligned} \|w^* \bar{r}_{\text{oos}} - \eta \gamma\| &= \|(w^* \beta - \eta)(\gamma + \bar{v}_{\text{oos}}) + \eta \bar{v}_{\text{oos}} - w^* \bar{u}_{\text{oos}}\| \\ &\lesssim_{\text{P}} \|w^* \beta - \eta\| \|\gamma + \bar{v}_{\text{oos}}\| + T_{\text{oos}}^{-1/2} \|\eta\| + T_{\text{oos}}^{-1} \|w^* U_{\text{oos}}\| \|\iota_{T_{\text{oos}}}\| \\ &\lesssim_{\text{P}} \delta_1 + T_{\text{oos}}^{-1/2} + \delta_2 = o_{\text{P}}(1), \end{aligned} \quad (\text{F.117})$$

where we use $\|w^*\|^{-1} \|w^* U_{\text{oos}}\| \leq \max_{\check{p} \leq p_{\text{max}}, \check{q} \in \mathcal{Q}} \|w(\check{p}, \check{q})\|^{-1} \|w(\check{p}, \check{q}) U_{\text{oos}}\| \lesssim_{\text{P}} T_{\text{oos}}^{1/2}$ and $\|\iota_{T_{\text{oos}}}\| = \sqrt{T_{\text{oos}}}$ in the second last inequality. This concludes the proof of the first part of the theorem.

For in-sample data, as we only use $q^* N$ assets at each step, $\|w^*\|_0$ is no larger than $p_{\text{max}} q^* N$. Let S_{w^*} be the set $\{i | (w(p_i^*, q_i^*))_{[i]} \neq 0\}$, we have

$$\|w^* \bar{u}\| = \left| \sum_{i \in S_{w^*}} (w(p_i^*, q_i^*))_{[i]} \bar{u}_i \right| \lesssim_{\text{P}} \|w^*\| \|\bar{u}_{[S_{w^*}]}\| \leq \|w^*\| \sqrt{|S_{w^*}|} \|\bar{u}\|_{\text{MAX}} = o_{\text{P}} \left(\sqrt{\frac{p_{\text{max}} q^* N \log N}{T}} \right). \quad (\text{F.118})$$

With the assumption that $q^* N \log N = O(T)$ and p_{max} is finite, we have $\|w^* \bar{u}\| = o_{\text{P}}(1)$ and thus

$$\begin{aligned} \|\hat{\gamma}_g - \eta \gamma\| &= \|w^* \bar{r} - \eta \gamma\| = \|(w^* \beta - \eta)(\gamma + \bar{v}) + \eta \bar{v} - w^* \bar{u}\| \\ &\lesssim_{\text{P}} \|w^* \beta - \eta\| \|\gamma + \bar{v}\| + T^{-1/2} \|\eta\| + \|w^* \bar{u}\| = o_{\text{P}}(1). \end{aligned} \quad (\text{F.119})$$

Finally, we consider the general case $d \geq 1$. Note that in this case, $w(\check{p}, \check{q}) \in \mathbb{R}^{d \times N}$. We use $(w(\check{p}, \check{q}))_{[i]}$ to denote the i th row of $w(\check{p}, \check{q})$. Suppose that R_i^2 's are maximized separately for each i and denote

$$(p_i^*, q_i^*) = \arg \min_{\check{p} \leq p_{\text{max}}, \check{q} \in \mathcal{Q}} \text{MSE}_i(w(\check{p}, \check{q})),$$

where $\text{MSE}_i(w) = T_{\text{os}}^{-1} \|(\bar{G}_{\text{os}})_{[i]} - w_{[i]} \bar{R}_{\text{os}}\|^2$. Replacing w_0 and w^* by $(w(p_0, q_0))_{[i]}$ and $(w(p_i^*, q_i^*))_{[i]}$ in the above proof, (F.110) and (F.116) become

$$(\Sigma_z)_{ii} + o_P(1) = \text{MSE}_i(w(p_i^*, q_i^*)) \leq \text{MSE}_i(w(p_0, q_0)) = (\Sigma_z)_{ii} + o_P(1). \quad (\text{F.120})$$

Recall that $w^* = w(p^*, q^*)$ is obtained by maximizing the sum of R^2 and

$$(p^*, q^*) = \arg \max_{\check{p} \leq p_{\max}, \check{q} \in \mathcal{Q}} \sum_{i=1}^d R_i^2(\check{p}, \check{q}), \quad (\text{F.121})$$

we then show that $\text{MSE}_i(w^*) = (\Sigma_z)_{ii} + o_P(1)$ also holds. To see this, using the definition of (p_i^*, q_i^*) and (F.120), we have

$$\text{MSE}_i(w^*) = \text{MSE}_i(w(p^*, q^*)) \geq \text{MSE}_i(w(p_i^*, q_i^*)) = (\Sigma_z)_{ii} + o_P(1) = \text{MSE}_i(w_0) + o_P(1). \quad (\text{F.122})$$

Using the definition of (p^*, q^*) in (F.121), we have

$$0 \leq \sum_{i=1}^d R_i^2(p^*, q^*) - \sum_{i=1}^d R_i^2(p_0, q_0) = \sum_{i=1}^d \frac{\text{MSE}_i(w_0) - \text{MSE}_i(w^*)}{T_{\text{os}}^{-1} \|\bar{G}_{[i]}\|^2}. \quad (\text{F.123})$$

Together with $T_{\text{os}}^{-1} \|\bar{G}_{[i]}\|^2 \asymp_P 1$ and (F.122), we have

$$\text{MSE}_i(w_0) - \text{MSE}_i(w^*) \geq \sum_{j \neq i} \frac{\|\bar{G}_{[i]}\|^2}{\|\bar{G}_{[j]}\|^2} (\text{MSE}_j(w^*) - \text{MSE}_j(w_0)) \geq o_P(1). \quad (\text{F.124})$$

Combining (F.122) and (F.124), we have

$$\text{MSE}_i(w^*) = \text{MSE}_i(w_0) + o_P(1) = (\Sigma_z)_{ii} + o_P(1). \quad (\text{F.125})$$

With this equation, replacing w^* , Σ_z by $w_{[i]}^*$ and $(\Sigma_z)_{ii}$ in (F.115) leads to $\|w_{[i]}^* \eta - \beta\| = o_P(1)$ and $\|w_{[i]}^*\| = o_P(1)$. Consequently, we have $\|w_{[i]}^* \bar{r}_{\text{os}} - \eta_{[i]} \gamma\| = o_P(1)$ and $\|w_{[i]}^* \bar{r} - \eta_{[i]} \gamma\| = o_P(1)$, which concludes the proof. $\square$

F.15 Technical Lemmas and Their Proofs

Without loss of generality, we assume that $\Sigma_v = \mathbb{I}_p$ in the following lemmas. Also, except for Lemma 4, we assume that $\hat{p} = \tilde{p}$ and $\hat{I}_k = I_k$ for $k = 1, \dots, \tilde{p}$, which hold with probability approaching one as we will show in Lemma 4.

Lemma 1. *Under Assumptions B.1-B.7, for any $I \subset \langle N \rangle$, we have the following results:*

$$(i) \quad \|T^{-1} \bar{V} \bar{V}^\top - \Sigma_v\| \lesssim_P T^{-1/2}, \quad \|T^{-1} \bar{Z} \bar{Z}^\top - \Sigma_z\| \lesssim_P T^{-1/2}.$$

- (ii) $\left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \right\| \lesssim_P T^{1/2}.$
- (iii) $\left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \bar{V}^\top \right\| \lesssim_P T^{1/2}, \quad \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \bar{Z}^\top \right\| \lesssim_P T^{1/2}.$
- (iv) $\|\bar{U}\|_{\text{MAX}} \lesssim_P (\log NT)^{1/2}, \quad \|\bar{U} \bar{V}^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{1/2}, \quad \|\bar{U} \bar{Z}^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{1/2}.$
- (v) $\|\bar{U}_{[I]}\| \lesssim_P |I|^{1/2} + T^{1/2}, \quad \|\bar{U}_{[I]} \bar{V}^\top\| \lesssim_P |I|^{1/2} T^{1/2}, \quad \|\bar{U}_{[I]} \bar{Z}^\top\| \lesssim_P |I|^{1/2} T^{1/2}.$
- (vi) $\|\bar{V}\| \lesssim_P T^{1/2}, \quad \|\bar{Z}\| \lesssim_P T^{1/2}, \quad \|\bar{V} \bar{Z}^\top\| \lesssim_P T^{1/2}, \quad \|\bar{V} \bar{Z}^\top - V Z^\top\| \lesssim_P 1$

Proof. (i) Using Assumption B.1, we have

$$\left\| \frac{\bar{V} \bar{V}^\top}{T} - \Sigma_v \right\| \leq \left\| \frac{V V^\top}{T} - \Sigma_v \right\| + \left\| \frac{V \iota_T \iota_T^\top V^\top}{T^2} \right\| = \left\| \frac{V V^\top}{T} - \mathbb{I}_p \right\| + \|\bar{v}\|^2 \lesssim_P T^{-1/2}.$$

Replacing $\bar{V}$ by $\bar{Z}$, we also have the second inequality.

(ii) Using Assumption B.5, we have

$$\left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \right\| \leq \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} \right\| + T^{-1} \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} \iota_T \iota_T^\top \right\| \lesssim_P T^{1/2}.$$

(iii) By Assumptions B.1, B.5 and B.6, we have

$$\begin{aligned} \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \bar{V}^\top \right\| &\leq \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} V^\top \right\| + T^{-1} \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} \iota_T \iota_T^\top V \right\| \\ &\leq \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} V^\top \right\| + \left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top U_{[I]} \iota_T \right\| \|\bar{v}\| \lesssim_P T^{1/2}. \end{aligned}$$

Replacing $\bar{V}$ by $\bar{Z}$ in the above proof, with Assumptions B.2, B.5 and B.7, we also have

$$\left\| \left(\beta_{[I]}^\top \beta_{[I]} \right)^{-\frac{1}{2}} \beta_{[I]}^\top \bar{U}_{[I]} \bar{Z}^\top \right\| \lesssim_P T^{1/2}.$$

(iv) Using Assumption B.4, we have

$$\|\bar{U}\|_{\text{MAX}} \leq \|U\|_{\text{MAX}} + T^{-1} \|U \iota_T \iota_T^\top\|_{\text{MAX}} \leq \|U\|_{\text{MAX}} + \|\bar{u}\|_{\text{MAX}} \|\iota_T\| \lesssim_P (\log N)^{1/2} + (\log T)^{1/2}.$$

Using Assumptions B.1, B.4, B.6, we have

$$\|\bar{U} \bar{V}^\top\|_{\text{MAX}} \leq \|U V^\top\|_{\text{MAX}} + T^{-1} \|U \iota_T \iota_T^\top V^\top\|_{\text{MAX}} \leq \|U V^\top\|_{\text{MAX}} + T \|\bar{u}\|_{\text{MAX}} \|\bar{v}\| \lesssim_P (\log N)^{1/2} T^{1/2}.$$

Replacing $\bar{V}$ by $\bar{Z}$ in the above proof, with Assumptions B.2, B.4 and B.7, we also have

$$\|\bar{U} \bar{Z}^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{1/2}.$$

(v) Using Assumption B.4 , we have

$$\|\bar{U}_{[I]}\| \leq \|U_{[I]}\| + T^{-1} \|U_{[I]}\iota_T \iota_T^\top\| \leq \|U_{[I]}\| + \|\bar{u}_{[I]}\| \|\iota_T\| \lesssim_P |I|^{1/2} + T^{1/2}.$$

Using Assumptions B.1, B.4, B.6, we have

$$\|\bar{U}_{[I]}\bar{V}^\top\| \leq \|U_{[I]}V^\top\| + T^{-1} \|U_{[I]}\iota_T \iota_T^\top V^\top\| \leq \|U_{[I]}V^\top\| + T \|\bar{u}_{[I]}\| \|\bar{v}\| \lesssim_P |I|^{1/2} T^{1/2}.$$

Replacing $\bar{V}$ by $\bar{Z}$ in the above proof, with Assumptions B.2, B.4 and B.7, we also have

$$\|\bar{U}_{[I]}\bar{Z}^\top\| \lesssim_P |I|^{1/2} T^{1/2}.$$

(vi) Using Assumption B.1, we have

$$\|\bar{V}\| \leq \|V\| + T^{-1} \|V\iota_T \iota_T^\top\| \leq \|V\| + \|\bar{v}\| \|\iota_T\| \lesssim_P T^{1/2}.$$

Using Assumption B.2, we have

$$\|\bar{Z}\| \leq \|Z\| + T^{-1} \|Z\iota_T \iota_T^\top\| \leq \|Z\| + \|\bar{z}\| \|\iota_T\| \lesssim_P T^{1/2}.$$

Using Assumptions B.1 and B.2, we have

$$\|\bar{V}\bar{Z}^\top\| \leq \|VZ\| + T^{-1} \|V\iota_T \iota_T^\top Z\| \leq \|V\| + T \|\bar{v}\| \|\bar{z}\| \lesssim_P T^{1/2},$$

and

$$\|\bar{V}\bar{Z}^\top - VZ^\top\| = \|T^{-1}V\iota_T \iota_T^\top Z\| = T \|\bar{v}\| \|\bar{z}\| \lesssim_P 1.$$

□

Lemma 2. *The singular vectors $\hat{\xi}_{(k)}$ s we obtain from Algorithm 3 satisfy $\hat{\xi}_{(j)}^\top \hat{\xi}_{(k)} = \delta_{jk}$ for $j, k \leq \hat{p}$.*

Proof. If $j = k$, this result holds from the definition of $\hat{\xi}_{(k)}$. If $j < k$, recall that $\tilde{R}_{(k)}$ is defined in (F.42) and $\hat{\xi}_{(k)}$ is the first right singular vector of $\tilde{R}_{(k)}$, we have

$$\tilde{R}_{(k)} = \bar{R}_{[I_k]} \prod_{i < k} (\mathbb{I}_T - \hat{\xi}_{(i)} \hat{\xi}_{(i)}^\top) \quad \text{and} \quad \hat{\xi}_{(k)} = \arg \max_{\alpha} \frac{\|\tilde{R}_{(k)} \alpha\|}{\|\alpha\|}.$$

If $\hat{\xi}_{(k)}^\top \hat{\xi}_{(j)} = c_0 \neq 0$ for some $j < k$, then

$$\|\tilde{R}_{(k)}(\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)})\| = \|\tilde{R}_{(k)} \hat{\xi}_{(k)} - c_0 \tilde{R}_{(k)} \hat{\xi}_{(j)}\| = \|\tilde{R}_{(k)} \hat{\xi}_{(k)}\|, \quad (\text{F.126})$$

since the definition of $\tilde{R}_{(k)}$ implies that $\tilde{R}_{(k)}\hat{\xi}_{(j)} = 0$ for $j < k$.

On the other hand, since $\hat{\xi}_{(k)}^\top \hat{\xi}_{(j)} = c_0 \neq 0$, we have $(\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)})^\top \hat{\xi}_{(j)} = 0$, and consequently,

$$\|\hat{\xi}_{(k)}\|^2 = \|\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)}\|^2 + \|c_0 \hat{\xi}_{(j)}\|^2 > \|\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)}\|^2. \quad (\text{F.127})$$

Apparently, if $\|\tilde{R}_{(k)}\| = 0$, the process will stop so we have $\|\tilde{R}_{(k)}\| > 0$ for $k \leq \hat{p}$. Together with (F.126) and (F.127), we have

$$\|\tilde{R}_{(k)}\| = \frac{\|\tilde{R}_{(k)}\hat{\xi}_{(k)}\|}{\|\hat{\xi}_{(k)}\|} \leq \frac{\|\tilde{R}_{(k)}(\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)})\|}{\|\hat{\xi}_{(k)} - c_0 \hat{\xi}_{(j)}\|},$$

which contradicts with the definition of $\hat{\xi}_{(k)}$. Therefore, $\hat{\xi}_{(k)}^\top \hat{\xi}_{(j)} = 0$ for $j < k$. This completes the proof. $\square$

Lemma 3. Under Assumption B.3, if $c \rightarrow 0$, $qN/N_0 \rightarrow 0$ then b_k , $\beta_{(k)}$ and $\tilde{p}$ defined in Section B satisfy

- (i) $\langle b_j, b_k \rangle = \delta_{jk}$ for $j \leq k \leq \tilde{p}$.
- (ii) $\|\beta_{(k)}\| \asymp q^{1/2} N^{1/2}$.
- (iii) $\tilde{p} \leq p$.
- (iv) $\tilde{p} = p$, if we further have $\lambda_p(\eta^\top \eta) \gtrsim 1$.

Proof. (i) Recall that b_k is the first right singular vector of $\beta_{(k)}$ and $\beta_{(k)} = \beta_{[I_k]} \prod_{j < k} \mathbb{M}_{b_j}$. Using the same arguments as in the proof of Lemma 2, we have $\langle b_j, b_k \rangle = \delta_{jk}$ for $j \leq k \leq \tilde{p}$.

(ii) The selection rule at k th step implies that

$$\frac{1}{|I_k|} \sum_{i \in I_k} \left\| \beta_{[i]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}}^2 \geq \frac{1}{N_0} \sum_{i \in I_0} \left\| \beta_{[i]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}}^2. \quad (\text{F.128})$$

For any matrix $A \in \mathbb{R}^{N \times d}$ and set $I \subset \langle N \rangle$, we have

$$\sum_{i \in I} \|A_{[i]}\|_{\text{MAX}}^2 \leq \|A\|_{\text{F}}^2 \leq d \sum_{i \in I} \|A_{[i]}\|_{\text{MAX}}^2,$$

and

$$\|A\|^2 \leq \|A\|_{\text{F}}^2 \leq d \|A\|^2,$$

we thereby have

$$\|A\|^2 \asymp \sum_{i \in I} \|A_{[i]}\|_{\text{MAX}}^2. \quad (\text{F.129})$$

Using this result, (F.128) becomes

$$\frac{1}{|I_k|} \left\| \beta_{[I_k]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\|^2 \gtrsim \frac{1}{N_0} \left\| \beta_{[I_0]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\|^2.$$

Then, we have

$$\frac{1}{\sqrt{|I_k|}} \|\beta_{(k)}\| \left\| \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\| \geq \frac{1}{\sqrt{|I_k|}} \left\| \beta_{[I_k]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\| \gtrsim \frac{1}{\sqrt{N_0}} \left\| \beta_{[I_0]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\| \geq \frac{1}{\sqrt{N_0}} \sigma_p(\beta_{[I_0]}) \left\| \prod_{j < k} \mathbb{M}_{b_j} \eta^\top \right\|, \quad (\text{F.130})$$

where we use $\beta_{[I_k]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top = \beta_{[I_k]} (\prod_{j < k} \mathbb{M}_{b_j})^2 \eta^\top = \beta_{(k)} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top$ in the first inequality. With $\sigma_p(\beta_{[I_0]}) \gtrsim \sqrt{N_0}$ from Assumption B.3, (F.130) leads to $\|\beta_{(k)}\| \gtrsim |I_k|^{1/2}$. In addition, $\|\beta\|_{\text{MAX}} \lesssim 1$ from Assumption B.3 leads to $\|\beta_{(k)}\| \lesssim |I_k|^{1/2}$. Therefore, we have $\|\beta_{(k)}\| \asymp |I_k|^{1/2} \asymp q^{1/2} N^{1/2}$.

(iii) From (i), we have shown that b_k 's are pairwise orthogonal for $k \leq \tilde{p}$. It is impossible to have more than p pairwise orthogonal p dimensional vectors. Thus, $\tilde{p} \leq p$.

(iv) Recall that $\tilde{p}$ is defined in Section B. Since the procedure in its definition stops at $\tilde{p} + 1$, we have at most $qN - 1$ rows of β satisfying $\left\| \beta_{[i]} \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}} \geq c$, which implies

$$\left\| \beta_{[I_0]} \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \eta^\top \right\|^2 \lesssim qN + (N_0 - qN)c^2 = o(N_0),$$

where we use (F.129) and the assumptions $c \rightarrow 0$, $qN/N_0 \rightarrow 0$. With $\sigma_p(\beta_{[I_0]}) \gtrsim \sqrt{N_0}$ from Assumption B.3, we have

$$\left\| \eta \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \right\| \leq \sigma_p(\beta_{[I_0]})^{-1} \left\| \beta_{[I_0]} \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \eta^\top \right\| = o(1). \quad (\text{F.131})$$

If $\tilde{p} \leq p - 1$, using (i), we have

$$\eta \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} = \eta - \eta \sum_{j \leq \tilde{p}} b_j b_j^\top,$$

which implies that

$$\sigma_p(\eta) \leq \sigma_1 \left(\eta \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \right) + \sigma_p \left(\eta \sum_{j \leq \tilde{p}} b_j b_j^\top \right). \quad (\text{F.132})$$

Since

$$\text{Rank} \left(\eta \sum_{j \leq \tilde{p}} b_j b_j^\top \right) \leq \tilde{p} \leq p-1, \quad (\text{F.133})$$

we have $\sigma_p \left(\eta \sum_{j \leq \tilde{p}} b_j b_j^\top \right) \leq 0$ and thus (F.132) and (F.131) lead to $\sigma_p(\eta) \lesssim \sigma_1 \left(\eta \prod_{j \leq \tilde{p}} \mathbb{M}_{b_j} \right) \rightarrow 0$. This contradicts with the assumption that $\lambda_p(\eta^\top \eta) \gtrsim 1$. Therefore, we have $\tilde{p} \geq p$. Together with the result in (iii), we have $\tilde{p} = p$. $\square$

Lemma 4. Suppose Assumptions B.1-B.8 hold. If $c^{-1} \log(NT)^{1/2} (q^{-1/2} N^{-1/2} + T^{-1/2}) \rightarrow 0$ and $c \rightarrow 0$, then for $k \leq \tilde{p}$ and for I_k , $\tilde{p}$ and $\beta_{(k)}$ defined in Section B, we have

$$(i) \ P(\hat{I}_k = I_k) \rightarrow 1.$$

$$(ii) \ \left\| \tilde{R}_{(k)} - \beta_{(k)} \bar{V} \right\| \lesssim_P q^{1/2} N^{1/2} + T^{1/2}.$$

$$(iii) \ |\hat{\lambda}_{(k)}^{1/2} / \|\beta_{(k)}\| - 1| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}.$$

$$(iv) \ \left\| \mathbb{P}_{\hat{V}_{(k)}^\top} - T^{-1} \bar{V}^\top \mathbb{P}_{b_k} \bar{V} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}.$$

$$(v) \ P(\hat{p} = \tilde{p}) \rightarrow 1.$$

Proof. We prove (i)-(iv) by induction. First, we show that (i)-(iv) hold when $k = 1$:

(i) Recall that $\hat{I}_1$ is selected based on $T^{-1} \bar{R} \bar{G}^\top$ and I_1 based on $\beta \eta^\top$. With simple algebra, we have

$$T^{-1} \bar{R} \bar{G}^\top - \beta \eta^\top = \beta (T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p) \eta^\top + T^{-1} \bar{U} \bar{V}^\top \eta^\top + T^{-1} \beta \bar{V} \bar{Z}^\top + T^{-1} \bar{U} \bar{Z}^\top.$$

With Assumptions B.1, B.2, B.3, B.6 B.7, we have

$$\begin{aligned} \|T^{-1} \bar{R} \bar{G}^\top - \beta \eta^\top\|_{\text{MAX}} &\lesssim \|\beta\|_{\text{MAX}} \|T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p\| \|\eta\| + T^{-1} \|\bar{U} \bar{V}^\top\|_{\text{MAX}} \|\eta\| \\ &\quad + T^{-1} \|\beta\|_{\text{MAX}} \|\bar{V} \bar{Z}^\top\| + T^{-1} \|\bar{U} \bar{Z}^\top\|_{\text{MAX}} \lesssim_P (\log N)^{1/2} T^{-1/2}. \end{aligned}$$

From Assumption B.8, we have $c_{qN}^{(1)} - c_{qN+1}^{(1)} \gtrsim c_{qN}^{(1)}$ and the definition of $\tilde{p}$ implies that $c_{qN}^{(k)} \geq c$ for $k \leq \tilde{p}$. Thus, we have $c_{qN}^{(1)} - c_{qN+1}^{(1)} \gtrsim c$. Define the events

$$A_1 := \left\{ \|T^{-1} \bar{R}_{[i]} \bar{G}^\top\|_{\text{MAX}} > (c_{qN}^{(1)} + c_{qN+1}^{(1)})/2 \text{ for all } i \in I_1 \right\},$$

$$\begin{aligned} A_2 &:= \left\{ \|T^{-1}\bar{R}_{[i]}\bar{G}^\top\|_{\text{MAX}} < (c_{qN}^{(1)} + c_{qN+1}^{(1)})/2 \text{ for all } i \in I_1^c \right\}, \\ A_3 &:= \left\{ \|T^{-1}\bar{R}_{[i]}\bar{G}^\top - \beta_{[i]}\eta^\top\|_{\text{MAX}} \geq (c_{qN}^{(1)} - c_{qN+1}^{(1)})/2 \text{ for some } i \in \langle N \rangle \right\}. \end{aligned} \quad (\text{F.134})$$

It is easy to observe that $\{\hat{I}_1 = I_1\} \supset A_1 \cap A_2$. In addition, from the definition of I_1 , we have $\|\beta_{[i]}\eta^\top\|_{\text{MAX}} \geq c_{qN}^{(1)}$ for all $i \in I_1$ and $\|\beta_{[i]}\eta^\top\|_{\text{MAX}} \leq c_{qN+1}^{(1)}$ for all $i \in I_1^c$. Therefore, if A_1^c occurs, we have

$$\|T^{-1}\bar{R}_{[i]}\bar{G}^\top - \beta_{[i]}\eta^\top\|_{\text{MAX}} \geq (c_{qN}^{(1)} - c_{qN+1}^{(1)})/2,$$

for some $i \in I_1$, which implies $A_1^c \subset A_3$. Similarly, we have $A_2^c \subset A_3$. Using $\{\hat{I}_1 = I_1\} \supset A_1 \cap A_2$ and $A_1^c \cup A_2^c \subset A_3$, we have

$$\mathbb{P}(\hat{I}_1 = I_1) \geq \mathbb{P}(A_1 \cap A_2) = 1 - \mathbb{P}(A_1^c \cup A_2^c) \geq 1 - \mathbb{P}(A_3). \quad (\text{F.135})$$

Using $c^{-1}(\log N)^{1/2}T^{-1/2} \rightarrow 0$ and $c_{qN}^{(1)} - c_{qN+1}^{(1)} \gtrsim c$, we have $\mathbb{P}(A_3) \rightarrow 0$ and consequently, $\mathbb{P}(\hat{I}_1 = I_1) \rightarrow 1$.

(ii) Since $\hat{I}_1 = I_1$ with high probability, we impose $\hat{I}_1 = I_1$ below. Then, we have $\tilde{R}_{(1)} = \bar{R}_{[I_1]}$ and Assumption B.12 gives $\|\tilde{R}_{(1)} - \beta_{(1)}\bar{V}\| = \|\bar{U}_{[I_1]}\| \lesssim_P q^{1/2}N^{1/2} + T^{1/2}$.

(iii) From Lemma 10, we have $\sigma_j(\beta_{(1)}\bar{V})/\sigma_j(\beta_{(1)}) = T^{1/2} + O_P(1)$. The result in (ii) implies that

$$\left| \|\tilde{R}_{(1)}\| - \|\beta_{(1)}\bar{V}\| \right| \leq \|\tilde{R}_{(1)} - \beta_{(1)}\bar{V}\| \lesssim_P q^{1/2}N^{1/2} + T^{1/2}.$$

Together with $\|\beta_{(1)}\| \asymp qN$ from Lemma 3, we have

$$\left| \frac{\hat{\lambda}_{(1)}^{1/2}}{\|\beta_{(k)}\|} - 1 \right| = \left| \frac{\|\tilde{R}_{(1)}\|}{T^{1/2}\|\beta_{(1)}\|} - 1 \right| \leq \frac{\left| \|\tilde{R}_{(1)}\| - \|\beta_{(1)}\bar{V}\| \right|}{T^{1/2}\|\beta_{(1)}\|} + \frac{\left| \|\beta_{(1)}\bar{V}\| - T^{1/2}\|\beta_{(1)}\| \right|}{T^{1/2}\|\beta_{(1)}\|} \lesssim_P q^{-1/2}N^{-1/2} + T^{-1/2}.$$

(iv) Let $\tilde{\xi}_{(1)} \in \mathbb{R}^{T \times 1}$ denote the first right singular vector of $\beta_{(1)}\bar{V}$. From Lemma 10, we have

$$\left\| \mathbb{P}_{\tilde{\xi}_{(1)}} - T^{-1}\bar{V}^\top \mathbb{P}_{b_k} \bar{V} \right\| \lesssim_P T^{-1/2} \quad (\text{F.136})$$

and $\sigma_j(\beta_{(1)}\bar{V})/\sigma_j(\beta_{(1)}) = T^{1/2} + O_P(1)$ for $j \leq p$, which leads to

$$\sigma_1(\beta_{(1)}\bar{V}) - \sigma_2(\beta_{(1)}\bar{V}) = T^{1/2}(\sigma_1(\beta_{(1)}) - \sigma_2(\beta_{(1)})) + O_P(\sigma_1(\beta_{(1)})) \asymp_P T^{1/2}\sigma_1(\beta_{(1)}), \quad (\text{F.137})$$

where we use the assumption that $\sigma_2(\beta_{(1)}) \leq (1 + \delta)^{-1}\sigma_1(\beta_{(1)})$ in the last equation.

Using $\|\tilde{R}_{(1)} - \beta_{(1)}\bar{V}\| \lesssim_P q^{1/2}N^{1/2} + T^{1/2}$ as proved in (ii), (F.137), Lemma 3 and Wedin's sin-theta theorem for singular vectors in Wedin (1972), we have

$$\left\| \mathbb{P}_{\hat{V}_{(k)}} - \mathbb{P}_{\tilde{\xi}_{(1)}} \right\| \lesssim_P \frac{q^{1/2}N^{1/2} + T^{1/2}}{\sigma_1(\beta_{(1)}\bar{V}) - \sigma_2(\beta_{(1)}\bar{V})} \lesssim_P q^{-1/2}N^{-1/2} + T^{-1/2}, \quad (\text{F.138})$$

In light of (F.136) and (F.138), we have that (iv) holds for $k = 1$.

So far, we have proved that (i)-(iv) hold for $k = 1$. Now, assuming that (i)-(iv) hold for $j \leq k-1$, we will show that (i)-(iv) continue to hold for $j = k$.

(i) Again, we show the difference between the sample covariances and their population counterparts introduced in the SPCA procedure are tiny. At the k th step, the difference can be written as

$$\begin{aligned}
& \left\| \beta \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \eta^\top - T^{-1}(\beta \bar{V} + \bar{U}) \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} (\eta \bar{V} + \bar{Z})^\top \right\|_{\text{MAX}} \\
& \leq \left\| \beta \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \eta^\top - T^{-1} \beta \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top \eta^\top \right\|_{\text{MAX}} + T^{-1} \left\| \beta \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{Z}^\top \right\|_{\text{MAX}} \\
& \quad + T^{-1} \left\| \bar{U} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top \eta^\top \right\|_{\text{MAX}} + T^{-1} \left\| \bar{U} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{Z}^\top \right\|_{\text{MAX}} \tag{F.139}
\end{aligned}$$

Since (iv) holds for $j \leq k-1$, we have

$$\left\| \sum_{j=1}^{k-1} \mathbb{P}_{\hat{V}_{(j)}^\top} - T^{-1} \bar{V}^\top \sum_{j=1}^{k-1} \mathbb{P}_{b_j} \bar{V} \right\| = \left\| \sum_{j=1}^{k-1} \left(\mathbb{P}_{\hat{V}_{(j)}^\top} - T^{-1} \bar{V}^\top \mathbb{P}_{b_j} \bar{V} \right) \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}. \tag{F.140}$$

Using Lemma 2 and Lemma 3(i), we have

$$\prod_{j=1}^{k-1} \mathbb{M}_{b_j} = \mathbb{I}_p - \sum_{j=1}^{k-1} \mathbb{P}_{b_j}, \quad \text{and} \quad \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} = \mathbb{I}_T - \sum_{j=1}^{k-1} \mathbb{P}_{\hat{V}_{(j)}}.$$

Using the above equations, (F.140), and $\|T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p\| \lesssim_P T^{-1/2}$, we have

$$T^{-1/2} \left\| \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \bar{V} \right\| = T^{-1/2} \left\| \bar{V} \sum_{j=1}^{k-1} \mathbb{P}_{\hat{V}_{(j)}^\top} - \sum_{j=1}^{k-1} \mathbb{P}_{b_j} \bar{V} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}. \tag{F.141}$$

Similarly, right multiplying $\bar{V}^\top$ to the term inside the $\|\cdot\|$ of (F.141), we have

$$\left\| T^{-1} \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}. \tag{F.142}$$

Then, we analyze these four terms in (F.139) one by one. For the first term, using (F.142) and

Assumption B.3, we have

$$\begin{aligned} \left\| \beta \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \eta^\top - T^{-1} \beta \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top \eta^\top \right\|_{\text{MAX}} &\lesssim \|\beta\|_{\text{MAX}} \left\| \prod_{j=1}^{k-1} \mathbb{M}_{b_j} - T^{-1} \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top \right\| \|\eta\| \\ &\lesssim_{\text{P}} q^{-1/2} N^{-1/2} + T^{-1/2}. \end{aligned}$$

For the second term, using (F.141), Lemma 1 and Assumptions B.3 and B.2, we have

$$\begin{aligned} T^{-1} \left\| \beta \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{Z}^\top \right\|_{\text{MAX}} &\lesssim T^{-1} \|\beta\|_{\text{MAX}} \left\| \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \right\| \|\bar{V} \bar{Z}^\top\| + T^{-1} \|\beta\|_{\text{MAX}} \left\| \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \bar{V} \right\| \|\bar{Z}\| \\ &\lesssim_{\text{P}} q^{-1/2} N^{-1/2} + T^{-1/2}. \end{aligned}$$

For the third term, using (F.141) and Lemma 1, we have

$$\begin{aligned} T^{-1} \left\| \bar{U} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{V}^\top \eta^\top \right\|_{\text{MAX}} &\lesssim T^{-1} \|\bar{U} \bar{V}^\top\|_{\text{MAX}} \left\| \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \right\| \|\eta\| \\ &\quad + T^{-1} \|\bar{U}\|_{\text{MAX}} T^{1/2} \left\| \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \bar{V} \right\| \|\eta\| \\ &\lesssim_{\text{P}} (\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right). \end{aligned}$$

For the forth term, using (F.140) and Lemma 1, we have

$$\begin{aligned} T^{-1} \left\| \bar{U} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{Z}^\top \right\|_{\text{MAX}} &\lesssim T^{-1} \|\bar{U} \bar{Z}^\top\|_{\text{MAX}} + T^{-2} \|\bar{U} \bar{V}^\top\|_{\text{MAX}} \left\| \sum_{j=1}^{k-1} \mathbb{P}_{b_j} \right\| \|\bar{V} \bar{Z}^\top\| \\ &\quad + T^{-1/2} \|\bar{U}\|_{\text{MAX}} \left\| T^{-1} \bar{V}^\top \sum_{j=1}^{k-1} \mathbb{P}_{b_j} \bar{V} - \sum_{j=1}^{k-1} \mathbb{P}_{\hat{V}_{(j)}^\top} \right\| \|\bar{Z}\| \\ &\lesssim_{\text{P}} (\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right). \end{aligned}$$

Hence, we have

$$\left\| T^{-1} \bar{R} \prod_{j=1}^{k-1} \mathbb{M}_{\hat{V}_{(j)}^\top} \bar{G}^\top - \beta \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}} \lesssim_{\text{P}} (\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right). \quad (\text{F.143})$$

As in the case of $k = 1$, from Assumption B.8, we have $c_{qN}^{(k)} - c_{qN+1}^{(k)} \gtrsim c_{qN}^{(k)}$. In addition, since the stopping rule for the procedure in Section B is $c_{qN}^{(\tilde{p}+1)} < c$, we have $c_{qN}^{(k)} \geq c$ for $k \leq \tilde{p}$. With the

assumption that

$$c^{-1}(\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right) \rightarrow 0,$$

we can reuse the arguments for (F.134) and (F.135) in the case of $k = 1$ and obtain $P(\widehat{I}_k = I_k) \rightarrow 1$.

(ii) We impose $\widehat{I}_k = I_k$ below. Then, we have $\widetilde{R}_{(k)} = \bar{R}_{[I_k]} \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top}$ and thus

$$\widetilde{R}_{(k)} - \beta_{(k)} \bar{V} = \bar{R}_{[I_k]} \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top} - \beta_{(k)} \bar{V} = \bar{\beta}_{[I_k]} \left(\bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top} - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \bar{V} \right) + \bar{U}_{[I_k]} \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top}.$$

Hence, using Assumptions B.3, Lemma 1, and (F.141), we have

$$\left\| \widetilde{R}_{(k)} - \beta_{(k)} \bar{V} \right\| \leq \left\| \beta_{[I_k]} \right\| \left\| \bar{V} \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top} - \prod_{j=1}^{k-1} \mathbb{M}_{b_j} \bar{V} \right\| + \left\| \bar{U}_{[I_k]} \right\| \left\| \prod_{j=1}^{k-1} \mathbb{M}_{\widehat{V}_{(j)}^\top} \right\| \lesssim_P q^{1/2} N^{1/2} + T^{1/2}.$$

(iii) The proof of (iii) is analogous to the case $k = 1$. Rewrite the proof of the case $k = 1$ by replacing $\widetilde{R}_{(1)}$ and $\beta_{(1)}$ by $\widetilde{R}_{(k)}$ and $\beta_{(k)}$. We have $|\widehat{\lambda}_{(k)}^{1/2} / \|\beta_{(k)}\| - 1| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}$.

(iv) The proof of (iv) is analogous to the case $k = 1$. Let $\tilde{\xi}_{(k)}$ denote the first right singular vector of $\beta_{(k)} \bar{V}$, then we have $\left\| \mathbb{M}_{\tilde{\xi}_{(k)}} - T^{-1} \bar{V}^\top \mathbb{M}_{b_k} \bar{V} \right\| \lesssim_P T^{-1/2}$ from Lemma 10. Since we have $\left\| \widetilde{R}_{(k)} - \beta_{(k)} \bar{V} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}$ from (ii), using the same proof as in the case $k = 1$, we have

$$\left\| \mathbb{M}_{\widehat{V}_{(k)}^\top} - \mathbb{M}_{\tilde{\xi}_{(k)}} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2},$$

by Wedin's sin-theta theorem. Combining these two inequalities completes the proof.

To sum up, by induction, we have shown that (i)-(iv) hold for $k \leq \tilde{p}$.

(v) Recall that $\tilde{p}$ is determined by $\beta_{[i]} \prod_{j < k} \mathbb{M}_{b_j} \eta^\top$ whereas $\widehat{p}$ is determined by $T^{-1} \bar{R}_{[i]} \prod_{j < k} \mathbb{M}_{\widehat{V}_{(j)}^\top} \bar{G}^\top$. Since (iv) holds for $j \leq \tilde{p}$ as shown above, using the same proof for (F.143), we have

$$\left\| T^{-1} \bar{R} \prod_{j=1}^{\tilde{p}} \mathbb{M}_{\widehat{V}_{(j)}^\top} \bar{G}^\top - \beta \prod_{j=1}^{\tilde{p}} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}} \lesssim_P (\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right). \quad (\text{F.144})$$

The assumption $c_{qN}^{(\tilde{p}+1)} \leq (1 + \delta)^{-1} c$ in Assumption B.8 implies that $c - c_{qN}^{(\tilde{p}+1)} \asymp c$. Together with

$$c^{-1}(\log NT)^{1/2} \left(q^{-1/2} N^{-1/2} + T^{-1/2} \right) \rightarrow 0,$$

we can reuse the arguments for (F.134) and (F.135) with events

$$\begin{aligned} B_1 &:= \left\{ \left\| T^{-1} \bar{R}_{[i]} \prod_{j=1}^{\tilde{p}} \mathbb{M}_{\widehat{V}_{(j)}^\top} \bar{G}^\top \right\|_{\text{MAX}} > (c + c_{qN}^{(\tilde{p}+1)})/2 \text{ for at most } qN - 1 \text{ rows } i \in \langle N \rangle \right\}, \\ B_2 &:= \left\{ \left\| T^{-1} \bar{R}_{[i]} \prod_{j=1}^{\tilde{p}} \mathbb{M}_{\widehat{V}_{(j)}^\top} \bar{G}^\top - \beta_{[i]} \prod_{j=1}^{\tilde{p}} \mathbb{M}_{b_j} \eta^\top \right\|_{\text{MAX}} \geq (c - c_{qN}^{(\tilde{p}+1)})/2 \text{ for some } i \in \langle N \rangle \right\}, \end{aligned} \quad (\text{F.145})$$

to obtain $\mathbb{P}(\widehat{p} = \tilde{p}) \geq \mathbb{P}(B_1) = 1 - \mathbb{P}(B_1^c) \geq 1 - \mathbb{P}(B_2) \rightarrow 1$. $\square$

Lemma 5. Suppose that $\Gamma_{(k)} \in \mathbb{R}^{|I_k| \times |I_k|}$ is an orthogonal matrix with the first p rows equals to $(\beta_{[I_k]}^\top \beta_{[I_k]})^{-\frac{1}{2}} \beta_{[I_k]}^\top$ and we define

$$\begin{pmatrix} s_{(k)}^1 \\ s_{(k)}^2 \end{pmatrix} := \Gamma_{(k)} \widehat{\varsigma}_{(k)} \quad \text{and} \quad \begin{pmatrix} \tilde{U}_{(k)}^1 \\ \tilde{U}_{(k)}^2 \end{pmatrix} := \Gamma_{(k)} \bar{U}_{[I_k]},$$

where $s_{(k)}^1 \in \mathbb{R}^{p \times 1}$ and $\tilde{U}_{(k)}^1 \in \mathbb{R}^{p \times T}$ are the first p rows of $\Gamma_{(k)} \widehat{\varsigma}_{(k)}$ and $\Gamma_{(k)} \bar{U}_{[I_k]}$, respectively. Then, under Assumptions B.1-B.8, we have

$$\begin{aligned} (i) \quad & \left\| s_{(k)}^2 \right\| \lesssim_P T^{-1/2} \widehat{\lambda}_{(k)}^{-1/2} (|I_k|^{1/2} + T^{1/2}). \\ (ii) \quad & \left\| \tilde{U}_{(k)}^1 \right\| \lesssim_P T^{1/2}, \quad \left\| \tilde{U}_{(k)}^1 \bar{V}^\top \right\| \lesssim_P T^{1/2}, \quad \left\| \tilde{U}_{(k)}^1 \bar{Z}^\top \right\| \lesssim_P T^{1/2}. \end{aligned}$$

Proof. (i) The assumption $\widehat{I}_k = I_k$ and the definition (F.42) of $\tilde{R}_{(k)}$ together lead to

$$\tilde{R}_{(k)} = \bar{R}_{[I_k]} \prod_{i < k} \left(\mathbb{I}_T - \widehat{\xi}_{(i)} \widehat{\xi}_{(i)}^\top \right).$$

Then, with (F.57) and Lemma 2, we have $\widehat{\varsigma}_{(k)} = \bar{R}_{[I_k]} \widehat{\xi}_{(k)} / \sqrt{T \widehat{\lambda}_{(k)}}$. From the construction of $\Gamma_{(k)}$, we have

$$\Gamma_{(k)} \bar{R}_{(k)} = \begin{pmatrix} \left(\beta_{[I_k]}^\top \beta_{[I_k]} \right)^{\frac{1}{2}} \bar{V} + \tilde{U}_{(k)}^1 \\ \tilde{U}_{(k)}^2 \end{pmatrix},$$

which in turn gives

$$\begin{pmatrix} s_{(k)}^1 \\ s_{(k)}^2 \end{pmatrix} = \Gamma_{(k)} \widehat{\varsigma}_{(k)} = \frac{1}{\sqrt{T \widehat{\lambda}_{(k)}}} \begin{pmatrix} \left(\beta_{[I_k]}^\top \beta_{[I_k]} \right)^{\frac{1}{2}} \bar{V} + \tilde{U}_{(k)}^1 \\ \tilde{U}_{(k)}^2 \end{pmatrix} \widehat{\xi}_{(k)}.$$

With Lemma 1(v), we have

$$\left\| s_{(k)}^2 \right\| = \left\| \frac{\tilde{U}_{(k)}^2}{\sqrt{T\hat{\lambda}_{(k)}}} \right\| \leq \left\| \frac{\bar{U}_{[I_k]}}{\sqrt{T\hat{\lambda}_{(k)}}} \right\| \lesssim_P T^{-1/2} \hat{\lambda}_{(k)}^{-1/2} (|I_k|^{1/2} + T^{1/2}).$$

(ii) With Lemma 1(ii)(iii) and the definition of $\Gamma_{(k)}$, these results follow immediately. $\square$

Lemma 6. Under Assumptions B.1-B.8, if $\hat{\lambda}_{(k)} \asymp_P |I_k|$ and $|I_k| \asymp qN$ for $k \leq \tilde{p}$, then we have

$$\begin{aligned} (i) \quad & \left\| \frac{\bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1}. \\ (ii) \quad & \left\| \frac{\bar{V} \bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P q^{-1} N^{-1} + T^{-1}, \quad \left\| \frac{\bar{Z} \bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P q^{-1} N^{-1} + T^{-1}, \quad \left| \frac{\hat{\varsigma}_{(k)}^\top \bar{u}_{[I_k]}}{\sqrt{\hat{\lambda}_{(k)}}} \right| \lesssim_P q^{-1} N^{-1} + T^{-1}. \end{aligned}$$

Proof. (i) Using the equation $\hat{\varsigma}_{(k)}^\top \bar{U}_{[I_k]} = (s_{(k)}^1)^\top \tilde{U}_{(k)}^1 + (s_{(k)}^2)^\top \tilde{U}_{(k)}^2$ and Lemma 5, we have

$$\begin{aligned} \left\| \hat{\varsigma}_{(k)}^\top \bar{U}_{[I_k]} \right\| & \leq \left\| s_{(k)}^1 \right\| \left\| \tilde{U}_{(k)}^1 \right\| + \left\| s_{(k)}^2 \right\| \left\| \tilde{U}_{(k)}^2 \right\| \leq \left\| s_{(k)}^1 \right\| \left\| \tilde{U}_{(k)}^1 \right\| + \left\| s_{(k)}^2 \right\| \left\| \bar{U}_{[I_k]} \right\| \\ & \lesssim_P \sqrt{T} + \frac{|I_k| + T}{\sqrt{T\hat{\lambda}_{(k)}}}, \end{aligned} \tag{F.146}$$

which leads to

$$\left\| \frac{\bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}} \right\| \lesssim_P \frac{1}{\sqrt{\hat{\lambda}_{(k)}}} + \frac{|I_k| + T}{T\hat{\lambda}_{(k)}} \lesssim_P q^{-1/2} N^{-1/2} + T^{-1}.$$

(ii) From Lemmas 1 and 5, we have

$$\begin{aligned} \left\| \bar{V} \bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)} \right\| & \leq \left\| \bar{V} \left(\tilde{U}_{(k)}^1 \right)^\top s_{(k)}^1 \right\| + \left\| \bar{V} \left(\tilde{U}_{(k)}^2 \right)^\top s_{(k)}^2 \right\| \leq \left\| \bar{V} \left(\tilde{U}_{(k)}^1 \right)^\top \right\| + \left\| \bar{V} \bar{U}_{[I_k]}^\top \right\| \left\| s_{(k)}^2 \right\| \\ & \lesssim_P \sqrt{T} + \frac{|I_k| + T}{\sqrt{\hat{\lambda}_{(k)}}}, \end{aligned}$$

which leads to

$$\left\| \frac{\bar{V} \bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P \frac{1}{\sqrt{T\hat{\lambda}_{(k)}}} + \frac{|I_k| + T}{T\hat{\lambda}_{(k)}} \lesssim_P q^{-1} N^{-1} + T^{-1}.$$

Replacing $\bar{V}$ by $\bar{Z}$ and $\iota_T^\top$ in the above proof and using Lemmas 1 and 5, we have similar results:

$$\left\| \frac{\bar{Z} \bar{U}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| \lesssim_P q^{-1} N^{-1} + T^{-1}, \quad \text{and} \quad \left| \frac{\bar{u}_{[I_k]}^\top \hat{\varsigma}_{(k)}}{\sqrt{\hat{\lambda}_{(k)}}} \right| \lesssim_P q^{-1} N^{-1} + T^{-1}. \tag{F.147}$$

□

Lemma 7. Under Assumptions B.1-B.8, if $\widehat{\lambda}_{(j)} \asymp_{\mathbb{P}} |I_j|$ and $|I_j| \asymp qN$ for $j \leq \tilde{p}$, then for $k, l \leq \tilde{p}$, we have

$$\begin{aligned}
 (i) \quad & \left\| \frac{\tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1}, \quad \left\| \frac{\tilde{U}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1/2}, \quad \|D_{(k)}\| \lesssim_{\mathbb{P}} 1. \\
 (ii) \quad & \left\| \frac{\bar{V}\tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{T\sqrt{\widehat{\lambda}_{(k)}}} \right\| \lesssim_{\mathbb{P}} q^{-1}N^{-1} + T^{-1}, \quad \left\| \frac{\bar{Z}\tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{T\sqrt{\widehat{\lambda}_{(k)}}} \right\| \lesssim_{\mathbb{P}} q^{-1}N^{-1} + T^{-1}, \quad \left| \frac{\widehat{\varsigma}_{(k)}^\top \tilde{u}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} \right| \lesssim_{\mathbb{P}} q^{-1}N^{-1} + T^{-1}. \\
 (iii) \quad & \left| \frac{\widehat{\xi}_{(l)}^\top \tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right| \lesssim_{\mathbb{P}} q^{-1}N^{-1} + T^{-1}.
 \end{aligned}$$

Proof. (i) Recall that the definition of $U_{(k)}$ is

$$\tilde{U}_{(k)} = \bar{U}_{[I_k]} - \sum_{i=1}^{k-1} \frac{\bar{R}_{[I_k]} \widehat{\xi}_{(i)}}{\sqrt{T}} \frac{\widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{\widehat{\lambda}_{(i)}}}. \quad (\text{F.148})$$

Then, a direct multiplication of $\widehat{\varsigma}_{(k)}^\top / \sqrt{T\widehat{\lambda}_{(k)}}$ from the left side of (F.148) leads to

$$\frac{\widehat{\varsigma}_{(k)}^\top \tilde{U}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} = \frac{\widehat{\varsigma}_{(k)}^\top \bar{U}_{[I_k]}}{\sqrt{T\widehat{\lambda}_{(k)}}} - \sum_{i=1}^{k-1} \frac{\widehat{\varsigma}_{(k)}^\top \bar{R}_{[I_k]} \widehat{\xi}_{(i)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \frac{\widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}}.$$

Consequently, with Lemma 6(i) we have

$$\begin{aligned}
 \left\| \frac{\widehat{\varsigma}_{(k)}^\top \tilde{U}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| & \leq \left\| \frac{\widehat{\varsigma}_{(k)}^\top \bar{U}_{[I_k]}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| + \sum_{i=1}^{k-1} \left\| \frac{\bar{R}_{[I_k]}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| \left\| \frac{\widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}} \right\| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{\widehat{\lambda}_{(k)}}} + \frac{|I_k| + T}{T\widehat{\lambda}_{(k)}} + \sqrt{\frac{|I_k|}{\widehat{\lambda}_{(k)}}} \sum_{i=1}^{k-1} \left\| \frac{\widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}} \right\| \\
 & \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1} + \sum_{i=1}^{k-1} \left\| \frac{\widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}} \right\|. \quad (\text{F.149})
 \end{aligned}$$

If $\left\| T^{-1/2} \widehat{\lambda}_{(i)}^{-1/2} \widehat{\varsigma}_{(i)}^\top \tilde{U}_{(i)} \right\| \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1}$ holds for $i \leq k-1$, then (F.149) implies that this inequality also holds for k . In addition, when $k=1$, $\tilde{U}_{(1)} = \bar{U}_{[I_1]}$ and this equation is implied from Lemma 6(i). Therefore, we have $\left\| T^{-1/2} \widehat{\lambda}_{(k)}^{-1/2} \widehat{\varsigma}_{(k)}^\top \tilde{U}_{(k)} \right\| \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1}$ for $k \leq \tilde{p}$ by induction.

Using (F.148) again, with Assumption B.4, we have

$$\left\| \frac{\tilde{U}_{(k)}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| \leq \left\| \frac{\bar{U}_{[I_k]}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| + \sum_{i=1}^{k-1} \left\| \frac{\bar{R}_{[I_k]}}{\sqrt{T\widehat{\lambda}_{(k)}}} \right\| \left\| \frac{\tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}} \right\| \lesssim_{\mathbb{P}} q^{-1/2}N^{-1/2} + T^{-1/2} + \sum_{i=1}^{k-1} \left\| \frac{\tilde{U}_{(i)}}{\sqrt{T\widehat{\lambda}_{(i)}}} \right\|. \quad (\text{F.150})$$

When $k = 1$, Assumption B.4 implies that $\left\| T^{-1/2} \widehat{\lambda}_{(k)}^{-1/2} \widetilde{U}_{(k)} \right\| \lesssim_P q^{-1/2} N^{-1/2} + T^{-1/2}$. Then, using the same induction argument with (F.150), we have this inequality holds for $k \leq \tilde{p}$.

Recall that $D_{(k)}$ is defined by

$$D_{(k)} = \mathbb{I}_{[I_k]} - \sum_{i=1}^{k-1} \bar{R}_{[I_k]} \hat{\xi}_{(i)} \frac{\zeta'_{(i)} D_{(i)}}{\sqrt{T \widehat{\lambda}_{(i)}}}$$

and $D_{(1)} = \mathbb{I}_{[I_1]} \lesssim 1$, we have $\|D_{(k)}\| \lesssim_P 1$ by induction as $\left\| T^{-1/2} \widehat{\lambda}_{(i)}^{-1/2} \bar{R}_{[I_k]} \right\| \lesssim_P 1$.

(ii) Similarly, by simple multiplication of $\bar{V}^\top$ from the right side of (F.148), we have

$$\frac{\zeta_{(k)}^\top \widetilde{U}_{(k)} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(k)}}} = \frac{\zeta_{(k)}^\top \bar{U}_{[I_k]} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(k)}}} - \sum_{i=1}^{k-1} \frac{\zeta_{(k)}^\top \bar{R}_{[I_k]} \hat{\xi}_{(i)}}{\sqrt{T \widehat{\lambda}_{(k)}}} \frac{\zeta_{(i)}^\top \widetilde{U}_{(i)} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(i)}}}.$$

Consequently, we have

$$\begin{aligned} \left\| \frac{\zeta_{(k)}^\top \widetilde{U}_{(k)} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(k)}}} \right\| &\leq \left\| \frac{\zeta_{(k)}^\top \bar{U}_{[I_k]} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(k)}}} \right\| + \sum_{i=1}^{k-1} \left\| \frac{\bar{R}_{[I_k]}}{\sqrt{T \widehat{\lambda}_{(k)}}} \right\| \left\| \frac{\zeta_{(i)}^\top \widetilde{U}_{(i)} \bar{V}^\top}{T \sqrt{\widehat{\lambda}_{(i)}}} \right\| \\ &\lesssim_P q^{-1} N^{-1} + T^{-1} + \sqrt{\frac{|I_k|}{\widehat{\lambda}_{(k)}}} \sum_{i=1}^{k-1} \left\| \frac{\zeta_{(i)}^\top \widetilde{U}_{(i)} \bar{V}^\top}{\sqrt{T \widehat{\lambda}_{(i)}}} \right\| \\ &\lesssim_P q^{-1} N^{-1} + T^{-1} + \sum_{i=1}^{k-1} \left\| \frac{\zeta_{(i)}^\top \widetilde{U}_{(i)} \bar{V}^\top}{\sqrt{T \widehat{\lambda}_{(i)}}} \right\|. \end{aligned} \quad (\text{F.151})$$

When $k = 1$, $\left\| T^{-1} \widehat{\lambda}_{(k)}^{-1/2} \zeta_{(k)}^\top \widetilde{U}_{(k)} \bar{V}^\top \right\| \lesssim_P q^{-1} N^{-1} + T^{-1}$ is a result of Lemma 6(ii). Then, a direct induction argument using (F.151) leads to this inequality for $k \leq \tilde{p}$.

Replacing $\bar{V}$ by $\bar{Z}$ and $\iota_T^\top$ in the above proof, and using Lemma 6(ii), we have the following results:

$$\left\| \frac{\bar{Z} \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{T \sqrt{\widehat{\lambda}_{(k)}}} \right\| \lesssim_P q^{-1} N^{-1} + T^{-1} \quad \text{and} \quad \left| \frac{\widetilde{u}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} \right| \lesssim_P q^{-1} N^{-1} + T^{-1}.$$

(iii) Recall that $\widetilde{R}_{(k)} = \widetilde{\beta}_{(k)} \bar{V} + \widetilde{U}_{(k)}$ as defined in (F.42), we have

$$|\zeta_{(l)}^\top \widetilde{R}_{(l)} \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}| \leq |\zeta_{(l)}^\top \widetilde{\beta}_{(l)} \bar{V} \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}| + |\zeta_{(l)}^\top \widetilde{U}_{(l)} \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}| \leq \left\| \zeta_{(l)}^\top \widetilde{\beta}_{(l)} \right\| \left\| \bar{V} \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)} \right\| + \left\| \zeta_{(l)}^\top \widetilde{U}_{(l)} \right\| \left\| \widetilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)} \right\|.$$

Using (F.57), we have

$$\left| \frac{\hat{\xi}_{(k)}^\top \tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}} \right| = \left| \frac{\hat{\varsigma}_{(l)}^\top \tilde{R}_{(l)} \tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}\hat{\lambda}_{(l)}}} \right| \leq \left\| \frac{\hat{\varsigma}_{(l)}^\top \tilde{\beta}_{(l)}}{\sqrt{\hat{\lambda}_{(l)}}} \right\| \left\| \frac{\tilde{V} \tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}} \right\| + \left\| \frac{\tilde{U}_{(k)}^\top \hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}} \right\| \left\| \frac{\tilde{U}_{(l)}^\top \hat{\varsigma}_{(l)}}{\sqrt{T\hat{\lambda}_{(l)}}} \right\|. \quad (\text{F.152})$$

With Lemma 1 and (i), we have

$$T^{1/2} \left\| \tilde{\beta}_{(k)} \right\| \lesssim_P \sigma_p(\tilde{V}) \left\| \tilde{\beta}_{(k)} \right\| \leq \left\| \tilde{\beta}_{(k)} \tilde{V} \right\| \leq \left\| \tilde{U}_{(k)} \right\| + \left\| \tilde{R}_{(k)} \right\| \leq \left\| \tilde{U}_{(k)} \right\| + \left\| \tilde{R}_{[I_k]} \right\| \lesssim_P T^{1/2} q^{1/2} N^{1/2}, \quad (\text{F.153})$$

which leads to $\left\| \hat{\lambda}_{(k)}^{-1/2} \hat{\varsigma}_{(k)}^\top \tilde{\beta}_{(k)} \right\| \lesssim_P q^{-1/2} N^{-1/2} \left\| \tilde{\beta}_{(k)} \right\| \lesssim_P 1$. Using this inequality and results of (i) and (ii) in (F.152) completes the proof. $\square$

Lemma 8. Under Assumptions B.1-B.8, if $\hat{\lambda}_{(j)} \asymp_P |I_j|$ and $|I_j| \asymp qN$ for $j \leq \tilde{p}$, then for $k \leq \tilde{p} + 1$, we have

$$(i) \quad \left\| \tilde{Z}_{(k)} \tilde{V}^\top \right\| \lesssim_P T^{1/2} + Tq^{-1}N^{-1}.$$

$$(ii) \quad \left\| \tilde{Z}_{(k)} \tilde{U}_{[I_0]}^\top \right\| \lesssim_P N_0^{1/2} T^{1/2} + Tq^{-1/2} N^{-1/2}.$$

Proof. (i) From the definition (F.45) of $\tilde{Z}_{(k)}$, we have

$$\tilde{Z}_{(k)} \tilde{V}^\top = \bar{Z} \tilde{V}^\top - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{U}_{(i)} \tilde{V}^\top}{\sqrt{T\hat{\lambda}_{(i)}}}.$$

Then, with Lemma 7(ii), we have

$$\left\| \tilde{Z}_{(k)} \tilde{V}^\top \right\| \leq \left\| \bar{Z} \tilde{V}^\top \right\| + \sum_{i=1}^{k-1} \left\| \bar{G} \hat{\xi}_{(i)} \right\| \left\| \frac{\hat{\varsigma}_{(i)}^\top \tilde{U}_{(i)} \tilde{V}^\top}{\sqrt{T\hat{\lambda}_{(i)}}} \right\| \lesssim_P T^{1/2} + T(q^{-1}N^{-1} + T^{-1}) \lesssim_P T^{1/2} + Tq^{-1}N^{-1}.$$

(ii) With (F.45) again, we have

$$\tilde{Z}_{(k)} \tilde{U}_{[I_0]}^\top = \bar{Z} \tilde{U}_{[I_0]}^\top - \sum_{i=1}^{k-1} \bar{G} \hat{\xi}_{(i)} \frac{\hat{\varsigma}_{(i)}^\top \tilde{U}_{(i)} \tilde{U}_{[I_0]}^\top}{\sqrt{T\hat{\lambda}_{(i)}}},$$

which, along with Lemma 7(i) and the assumptions on q , lead to

$$\begin{aligned} \left\| \tilde{Z}_{(k)} \tilde{U}_{[I_0]}^\top \right\| &\leq \left\| \bar{Z} \tilde{U}_{[I_0]}^\top \right\| + \sum_{i=1}^{k-1} \left\| \bar{G} \hat{\xi}_{(i)} \right\| \left\| \frac{\hat{\varsigma}_{(i)}^\top \tilde{U}_{(i)}}{\sqrt{T\hat{\lambda}_{(i)}}} \right\| \left\| \tilde{U}_{[I_0]} \right\| \\ &\lesssim_P N_0^{1/2} T^{1/2} + \left(q^{-1/2} N^{-1/2} + T^{-1} \right) \left(N_0^{1/2} T^{1/2} + T \right) \end{aligned}$$

$$\lesssim_P N_0^{1/2} T^{1/2} + T q^{-1/2} N^{-1/2}.$$

□

Lemma 9. Suppose that Assumptions **B.1-B.8** hold. If $\widehat{\lambda}_{(j)} \asymp_P |I_j|$ and $|I_j| \asymp qN$ for $j \leq \tilde{p}$, then H_1, H_2 defined by (F.55) satisfy

$$(i) \quad \|H_1\| \lesssim_P 1, \quad \|H_2\| \lesssim_P 1.$$

$$(ii) \quad \|H_1^\top H_2 - \mathbb{I}_{\tilde{p}}\| \lesssim_P T^{-1} + q^{-1} N^{-1}.$$

$$(iii) \quad \|H_1 - H_2\| \lesssim_P T^{-1/2} + q^{-1} N^{-1}.$$

Proof. (i) Using the definition (F.55) of H_1 and Lemma 1, we have

$$\|h_{k1}\| = \left\| \frac{\bar{V} \widehat{\xi}_{(k)}}{\sqrt{T}} \right\| \leq T^{-1/2} \|\bar{V}\| \lesssim_P 1,$$

which leads to $\|H_1\| \lesssim_P 1$.

Using the definition (F.55) of H_2 , we have

$$\|h_{k2}\| = \left\| \frac{\tilde{\beta}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{\widehat{\lambda}_{(k)}}} \right\| \leq q^{-1/2} N^{-1/2} \|\tilde{\beta}_{(k)}\|. \quad (\text{F.154})$$

With Lemma 1 and Lemma 7(i), we have

$$T^{1/2} \|\tilde{\beta}_{(k)}\| \lesssim_P \sigma_p(\bar{V}) \|\tilde{\beta}_{(k)}\| \leq \|\tilde{\beta}_{(k)} \bar{V}\| \leq \|\tilde{U}_{(k)}\| + \|\tilde{R}_{(k)}\| \leq \|\tilde{U}_{(k)}\| + \|\bar{R}_{[I_k]}\| \lesssim_P T^{1/2} q^{1/2} N^{1/2}. \quad (\text{F.155})$$

Combining (F.154) and (F.155), we have $\|h_{k2}\| \lesssim_P 1$ and thus $\|H_2\| \lesssim_P 1$.

(ii) By (F.57) and Lemma 2, we have

$$\delta_{lk} = \widehat{\xi}_{(l)}^\top \widehat{\xi}_{(k)} = \frac{\widehat{\xi}_{(l)}^\top \bar{V}^\top \tilde{\beta}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{T \widehat{\lambda}_{(k)}}} + \frac{\widehat{\xi}_{(l)}^\top \tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{T \widehat{\lambda}_{(k)}}} = h_{l1}^\top h_{k2} + \frac{\widehat{\xi}_{(l)}^\top \tilde{U}_{(k)}^\top \widehat{\varsigma}_{(k)}}{\sqrt{T \widehat{\lambda}_{(k)}}}.$$

By Lemma 7(iii), we have

$$|h_{l1}^\top h_{k2} - \delta_{lk}| \lesssim_P q^{-1} N^{-1} + T^{-1},$$

and thus $\|H_1^\top H_2 - \mathbb{I}_{\tilde{p}}\| \lesssim_P q^{-1} N^{-1} + T^{-1}$.

(iii) Using (F.57), we have

$$\bar{V}\hat{\xi}_{(k)} = \frac{\bar{V}\bar{V}^\top\tilde{\beta}_{(k)}^\top}{\sqrt{T\hat{\lambda}_{(k)}}}\hat{\varsigma}_{(k)} + \frac{\bar{V}\tilde{U}_{(k)}^\top\hat{\varsigma}_{(k)}}{\sqrt{T\hat{\lambda}_{(k)}}}.$$

With the definition of h_{k1} and h_{k2} , it becomes

$$h_{k1} = \frac{\bar{V}\bar{V}^\top}{T}h_{k2} + \frac{\bar{V}\tilde{U}_{(k)}^\top\hat{\varsigma}_{(k)}}{T\sqrt{\hat{\lambda}_{(k)}}}. \quad (\text{F.156})$$

With $\|h_{k2}\| \lesssim_P 1$, Lemma 1 and Lemma 7(ii), (F.156) leads to

$$h_{k1} - h_{k2} \lesssim_P T^{-1/2} + q^{-1}N^{-1}.$$

This completes the proof. $\square$

Lemma 10. For any $N \times p$ matrix β , if $\|T^{-1}\bar{V}\bar{V}^\top - \mathbb{I}_p\| \lesssim_P T^{-1/2}$, we have

(i) $\sigma_j(\beta\bar{V})/\sigma_j(\beta) = T^{1/2} + O_P(1)$ for $j \leq p$.

(ii) If $\sigma_1(\beta) - \sigma_2(\beta) \asymp \sigma_1(\beta)$, then $\left\|\mathbb{P}_{\tilde{\xi}} - T^{-1}\bar{V}^\top\mathbb{P}_b\bar{V}\right\| \lesssim_P T^{-1/2}$, where b is the first right singular vector of β and $\tilde{\xi}$ is the first right singular vector of $\beta\bar{V}$.

Proof. (i) For $j \leq p$, $\sigma_j(\beta\bar{V})^2 = \lambda_j(\beta\bar{V}\bar{V}^\top\beta^\top) = \lambda_j(\beta^\top\beta\bar{V}\bar{V}^\top)$ which implies

$$\lambda_j(\beta^\top\beta)\lambda_p(\bar{V}\bar{V}^\top) \leq \sigma_j(\beta\bar{V})^2 \leq \lambda_j(\beta^\top\beta)\lambda_1(\bar{V}\bar{V}^\top).$$

With the assumption $\|T^{-1}\bar{V}\bar{V}^\top - \mathbb{I}_p\| \lesssim_P T^{-1/2}$, we have $T^{-1/2}\sigma_j(\beta\bar{V})/\sigma_j(\beta) = 1 + O_P(T^{-1/2})$ by Weyl's theorem.

(ii) Let $\hat{\varsigma}$ and $\tilde{\varsigma}$ be the first left singular vectors of β and $\beta\bar{V}$, respectively. Equivalently, $\hat{\varsigma}$ and $\tilde{\varsigma}$ are the eigenvectors of $\beta\beta^\top$ and $T^{-1}\beta\bar{V}\bar{V}^\top\beta^\top$. Since $\|\beta\beta^\top - T^{-1}\beta\bar{V}\bar{V}^\top\beta^\top\| \leq \|\beta\|^2\|T^{-1}\bar{V}\bar{V}^\top - \mathbb{I}_p\| \lesssim_P \sigma_1(\beta)^2 T^{-1/2}$ and $\sigma_1(\beta) - \sigma_2(\beta) \asymp \sigma_1(\beta)$, by sin-theta theorem we have

$$\|\hat{\varsigma}\hat{\varsigma}^\top - \tilde{\varsigma}\tilde{\varsigma}^\top\| \lesssim \frac{\|\beta\beta^\top - T^{-1}\beta\bar{V}\bar{V}^\top\beta^\top\|}{\sigma_1(\beta)^2 - \sigma_2(\beta)^2 - O(\|\beta\beta^\top - T^{-1}\beta\bar{V}\bar{V}^\top\beta^\top\|)} \lesssim_P T^{-1/2}.$$

Using the relationship between left and right singular vectors, we have

$$b^\top = \frac{\hat{\varsigma}^\top\beta}{\sigma_1(\beta)}, \quad \tilde{\xi}^\top = \frac{\tilde{\varsigma}^\top\beta\bar{V}}{\|\beta\bar{V}\|}.$$

Therefore,

$$\left\| \mathbb{P}_{\tilde{\xi}} - \frac{\sigma_1(\beta)^2}{\|\beta\bar{V}\|^2} \bar{V}^\top \mathbb{P}_b \bar{V} \right\| = \left\| \tilde{\xi} \tilde{\xi}^\top - \frac{\bar{V}^\top \beta^\top \tilde{\xi} \tilde{\xi}^\top \beta \bar{V}}{\|\beta\bar{V}\|^2} \right\| = \left\| \frac{\bar{V}^\top \beta^\top \tilde{\xi} \tilde{\xi}^\top \beta \bar{V}}{\|\beta\bar{V}\|^2} - \frac{\bar{V}^\top \beta^\top \tilde{\xi} \tilde{\xi}^\top \beta \bar{V}}{\|\beta\bar{V}\|^2} \right\| \lesssim_P T^{-1/2}. \quad (\text{F.157})$$

By Weyl's inequality, we have $T^{-1} \|\beta\bar{V}\|^2 = \lambda_1(T^{-1} \beta\bar{V}\bar{V}^\top \beta^\top) = \lambda_1(\beta\beta^\top) + O_P(\sigma_1(\beta)^2 T^{-1/2}) = \sigma_1(\beta)^2 + O_P(\sigma_1(\beta)^2 T^{-1/2})$. Plugging this result into (F.157), we have $\left\| \mathbb{P}_{\tilde{\xi}} - T^{-1} \bar{V}^\top \mathbb{P}_b \bar{V} \right\| \lesssim_P T^{-1/2}$. $\square$

Lemmas 11-13 below are concerned with the singular values and singular vectors of $T^{-1/2} \bar{R}$. We use $\hat{\xi}_j$, $\hat{\xi}_j$ and $\hat{\lambda}_j^{1/2}$, $j \leq p$ to denote them throughout Lemmas 11-13.

Lemma 11. *Under the assumptions of Theorem 4(a), we have*

$$\frac{\hat{\lambda}_j}{\lambda_j} - 1 \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) + T^{-1/2},$$

where $\lambda_j = \lambda_j(\beta^\top \beta)$ and $\hat{\lambda}_j = \lambda_j(T^{-1} \bar{R} \bar{R}^\top)$.

Proof. Since $\lambda_j(\beta\bar{V}\bar{V}^\top \beta^\top) = \lambda_j(\beta^\top \beta \bar{V}\bar{V}^\top)$, we have

$$\lambda_j(\beta^\top \beta) \lambda_p \left(\frac{\bar{V}\bar{V}^\top}{T} \right) \leq \frac{\lambda_j(\beta^\top \beta \bar{V}\bar{V}^\top)}{T} \leq \lambda_j(\beta^\top \beta) \lambda_1 \left(\frac{\bar{V}\bar{V}^\top}{T} \right). \quad (\text{F.158})$$

By Lemma 1(i) and Weyl's inequality, we have $\lambda_j(T^{-1} \bar{V}\bar{V}^\top) - 1 \lesssim_P T^{-1/2}$ for $j \leq p$. Then, (F.158) becomes

$$\frac{\lambda_j(\beta\bar{V}\bar{V}^\top \beta^\top)}{T \lambda_j(\beta^\top \beta)} - 1 \lesssim_P T^{-1/2},$$

which is equivalent to

$$\frac{\sigma_j(\beta\bar{V})}{\sqrt{T} \sigma_j(\beta)} - 1 \lesssim_P T^{-1/2}. \quad (\text{F.159})$$

Using Weyl's inequality again, we have $|\sigma_j(\bar{R}) - \sigma_j(\beta\bar{V})| \leq \|\bar{U}\| \lesssim_P N^{1/2} + T^{1/2}$, which is equivalent to

$$\frac{\hat{\lambda}_j^{1/2}}{\lambda_j^{1/2}} - \frac{\sigma_j(\beta\bar{V})}{\sqrt{T} \sigma_j(\beta)} \lesssim_P \frac{1}{\sqrt{T}} + \frac{\sqrt{N} + \sqrt{T}}{\sqrt{T} \lambda_j}. \quad (\text{F.160})$$

Combine (F.159) and (F.160), we complete the proof. $\square$

Lemma 12. Suppose that the SVD of β is given by:

$$\beta = \Gamma^\top \begin{pmatrix} \Lambda^{\frac{1}{2}} \\ 0 \end{pmatrix} H, \quad (\text{F.161})$$

where $\Gamma \in \mathbb{R}^{N \times N}$, $H \in \mathbb{R}^{p \times p}$ are orthogonal matrices, and Λ is a diagonal matrix of the eigenvalues of $\beta^\top \beta$. If we write $\Gamma \hat{\zeta}_j = (s_{j1}^\top, s_{j2}^\top)^\top$, where $s_{j1} \in \mathbb{R}^p$, $s_{j2} \in \mathbb{R}^{N-p}$. Then under the assumptions of Theorem 4(a), we have

- (i) $\left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1)$, where e_{j1} is a $p \times 1$ unit vector with the i th entry being equal to 1.
- (ii) $\|s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1)$.
- (iii) $\left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\| \lesssim_P 1$.
- (iv) $\|s_{j2}\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1)$.

Proof. With the orthogonal matrix Γ defined above, we can write

$$\tilde{U} = \Gamma \bar{U} = \begin{pmatrix} \tilde{U}_{1_{p \times T}} \\ \tilde{U}_{2_{(N-p) \times T}} \end{pmatrix}, \quad (\text{F.162})$$

so that

$$\Gamma \bar{R} = \begin{pmatrix} \Lambda^{\frac{1}{2}} \\ 0 \end{pmatrix} \bar{V} + \tilde{U} = \begin{pmatrix} \Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1 \\ \tilde{U}_2 \end{pmatrix}.$$

The relationship between singular vectors $\hat{\zeta}_j$ and $\hat{\xi}_j$ can be written as

$$\Gamma \hat{\zeta}_j = \frac{(\Gamma \bar{R}) \hat{\xi}_j}{\sqrt{T \hat{\lambda}_j}}, \quad \hat{\xi}_j = \frac{(\Gamma \bar{R})^\top (\Gamma \hat{\zeta}_j)}{\sqrt{T \hat{\lambda}_j}}. \quad (\text{F.163})$$

Specifically, we have

$$s_{j1} = \frac{(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1) \hat{\xi}_j}{\sqrt{T \hat{\lambda}_j}}, \quad s_{j2} = \frac{\tilde{U}_2 \hat{\xi}_j}{\sqrt{T \hat{\lambda}_j}}, \quad \hat{\xi}_j = \frac{(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1)^\top s_{j1} + \tilde{U}_2^\top s_{j2}}{\sqrt{T \hat{\lambda}_j}}. \quad (\text{F.164})$$

From (F.164), we have

$$(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1) (\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1)^\top s_{j1} + (\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1) \tilde{U}_2^\top s_{j2} = T \hat{\lambda}_j s_{j1}. \quad (\text{F.165})$$

We can rewrite (F.165) as

$$\begin{aligned} \left(\mathbb{I}_p - \frac{\Lambda}{\lambda_j}\right) s_{j1} &= \frac{1}{T\lambda_j} \left(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1\right) \tilde{U}_2^\top s_{j2} + \frac{1}{\lambda_j} \Lambda^{\frac{1}{2}} \left(\frac{\bar{V}\bar{V}^\top}{T} - I\right) \Lambda^{\frac{1}{2}} s_{j1} + \frac{\Lambda^{\frac{1}{2}} \bar{V} \tilde{U}_1^\top}{T\lambda_j} s_{j1} \\ &\quad + \frac{\tilde{U}_1 \bar{V}^\top \Lambda^{\frac{1}{2}}}{T\lambda_j} s_{j1} + \frac{\tilde{U}_1 \tilde{U}_1^\top}{T\lambda_j} s_{j1} - \left(\frac{\hat{\lambda}_j}{\lambda_j} - 1\right) s_{j1}. \end{aligned} \quad (\text{F.166})$$

Define $L = \text{diag}(l_1, \dots, l_p)$, where l_i is equal to $\lambda_j/(\lambda_j - \lambda_i)$ if $i \neq j$ and 0 otherwise.

By left multiplying L to both sides of (F.166), we have

$$\begin{aligned} s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1} &= \frac{1}{T\lambda_j} L \Lambda^{\frac{1}{2}} \bar{V} \frac{\tilde{U}_2^\top \tilde{U}_2}{\sqrt{T\hat{\lambda}_j}} \hat{\xi}_j + \frac{1}{T\lambda_j} L \tilde{U}_1 \frac{\tilde{U}_2^\top \tilde{U}_2}{\sqrt{T\hat{\lambda}_j}} \hat{\xi}_j + \frac{1}{\lambda_j} L \Lambda^{\frac{1}{2}} \left(\frac{\bar{V}\bar{V}^\top}{T} - \mathbb{I}_p\right) \Lambda^{\frac{1}{2}} s_{j1} \\ &\quad + \frac{L \Lambda^{\frac{1}{2}} \bar{V} \tilde{U}_1^\top}{T\lambda_j} s_{j1} + L \frac{\tilde{U}_1 \bar{V}^\top \Lambda^{\frac{1}{2}}}{T\lambda_j} s_{j1} + L \frac{\tilde{U}_1 \tilde{U}_1^\top}{T\lambda_j} s_{j1} - \left(\frac{\hat{\lambda}_j}{\lambda_j} - 1\right) L s_{j1}. \end{aligned} \quad (\text{F.167})$$

Now left multiplying $\left(\frac{\Lambda}{\lambda_j}\right)^{\frac{1}{2}}$ again, we have

$$\begin{aligned} \left(\frac{\Lambda}{\lambda_j}\right)^{\frac{1}{2}} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) &= \frac{1}{T\lambda_j^{3/2}} \Lambda^{\frac{1}{2}} L \Lambda^{\frac{1}{2}} \bar{V} \frac{\tilde{U}_2^\top \tilde{U}_2}{\sqrt{T\hat{\lambda}_j}} \hat{\xi}_j + \frac{1}{T\lambda_j^{3/2}} \Lambda^{\frac{1}{2}} L \tilde{U}_1 \frac{\tilde{U}_2^\top \tilde{U}_2}{\sqrt{T\hat{\lambda}_j}} \hat{\xi}_j \\ &\quad + \frac{1}{\lambda_j} \Lambda^{\frac{1}{2}} L \Lambda^{\frac{1}{2}} \left(\frac{\bar{V}\bar{V}^\top}{T} - \mathbb{I}_p\right) \left(\frac{\Lambda}{\lambda_j}\right)^{\frac{1}{2}} s_{j1} + \Lambda^{\frac{1}{2}} L \Lambda^{\frac{1}{2}} \frac{\bar{V} \tilde{U}_1^\top}{T\lambda_j^{3/2}} s_{j1} \\ &\quad + \Lambda^{\frac{1}{2}} L \frac{\tilde{U}_1 \bar{V}^\top}{T\lambda_j} \left(\frac{\Lambda}{\lambda_j}\right)^{\frac{1}{2}} s_{j1} + \Lambda^{\frac{1}{2}} L \frac{\tilde{U}_1 \tilde{U}_1^\top}{T\lambda_j^{3/2}} s_{j1} - \left(\frac{\hat{\lambda}_j}{\lambda_j} - 1\right) \left(\frac{\Lambda}{\lambda_j}\right)^{\frac{1}{2}} L s_{j1} \\ &= K_1 + K_2 + K_3 + K_4 + K_5 + K_6 + K_7. \end{aligned} \quad (\text{F.168})$$

Before we analyze these seven terms in (F.168), we first analyze $\|L\|$, $\|L\Lambda^{1/2}\|$ and $\|L\Lambda\|$. Since L and Λ are diagonal matrices, by Assumption B.12 we can easily show that

$$\|L\| \lesssim 1, \quad \|L\Lambda^{1/2}\| \lesssim \lambda_j^{1/2}, \quad \|L\Lambda\| \lesssim \lambda_j. \quad (\text{F.169})$$

In addition, Lemma 1(ii)(iii)(v) imply that

$$\|\tilde{U}_1\| = \|(\beta^\top \beta)^{-1/2} \beta^\top \bar{U}\| \lesssim_P T^{1/2}, \quad \|\tilde{U}_1 \bar{V}^\top\| = \|(\beta^\top \beta)^{-1/2} \beta^\top \bar{U} \bar{V}^\top\| \lesssim_P T^{1/2}, \quad \|\tilde{U}_2\| \leq \|\bar{U}\| \lesssim_P N^{1/2} + T^{1/2}. \quad (\text{F.170})$$

Using Lemma 1(i)(vi), Lemma 11, (F.169) and (F.170), we analyze these seven terms in (F.168) one

by one. For the first term, we have

$$\|K_1\| \leq T^{-3/2} \lambda_j^{-3/2} \widehat{\lambda}_j^{-1/2} \|L\Lambda\| \|\bar{V}\| \|\tilde{U}_2^\top \tilde{U}_2\| \|\widehat{\xi}_j\| \lesssim_P \lambda_j^{-1} (T^{-1}N + 1),$$

where we also use $\|\tilde{U}_2^\top \tilde{U}_2\| \leq \|\bar{U}^\top \bar{U}\| \lesssim_P N + T$ in the last equation. For the second term, we have

$$\|K_2\| \leq T^{-3/2} \lambda_j^{-3/2} \widehat{\lambda}_j^{-1/2} \|\Lambda^{1/2} L\| \|\tilde{U}_1\| \|\tilde{U}_2^\top \tilde{U}_2\| \|\widehat{\xi}_j\| \lesssim_P \lambda_j^{-3/2} (T^{-1}N + 1).$$

For the third term, we have

$$\|K_3\| \leq \lambda_j^{-1} \|L\Lambda\| \|T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p\| \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\| \lesssim_P T^{-1/2} \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\|.$$

For the forth term, we have

$$\|K_4\| \leq T^{-1} \lambda_j^{-3/2} \|L\Lambda\| \|\bar{V} \tilde{U}_1^\top\| \lesssim_P \lambda_j^{-1/2} T^{-1/2},$$

where we use $\|\bar{V} \tilde{U}_1^\top\| \lesssim_P T^{1/2}$ from Lemma 1. For the fifth term, we have

$$\|K_5\| \leq T^{-1} \lambda_j^{-1} \|L\Lambda^{1/2}\| \|\tilde{U}_1 \bar{V}^\top\| \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\| \lesssim_P \lambda_j^{-1/2} T^{-1/2} \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\|.$$

For the sixth term, we have

$$\|K_6\| \leq T^{-1} \lambda_j^{-3/2} \|L\Lambda^{1/2}\| \|\tilde{U}_1 \tilde{U}_1^\top\| \lesssim_P \lambda_j^{-1},$$

where we use $\|\tilde{U}_1 \tilde{U}_1^\top\| \lesssim_P T$ as shown in Lemma 1. For the last term, we have

$$\|K_7\| \leq \lambda_j^{-2} |\widehat{\lambda}_j - \lambda_j| \|L\Lambda^{1/2}\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) + T^{-1/2}.$$

To sum up, (F.168) gives

$$\left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) + T^{-1/2} + T^{-1/2} \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\|. \quad (\text{F.171})$$

Note that

$$\begin{aligned} \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\| &\leq \left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| + \left\| (\Lambda/\lambda_j)^{1/2} \langle s_{j1}, e_{j1} \rangle e_{j1} \right\| \\ &\leq \left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| + |\langle s_{j1}, e_{j1} \rangle| \sqrt{\lambda_j^{-1} e_{j1}^\top \Lambda e_{j1}} \\ &= \left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| + O_P(1). \end{aligned}$$

Plugging this into (F.171), we have

$$\left\| (\Lambda/\lambda_j)^{1/2} (s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}) \right\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) + T^{-1/2}, \quad (\text{F.172})$$

which in turn leads to $\|(\Lambda/\lambda_j)^{1/2} s_{j1}\| \lesssim_P 1$ as by assumption $\lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) \rightarrow 0$. Similarly, we can analyze corresponding terms in (F.167), and obtain

$$\|s_{j1} - \langle s_{j1}, e_{j1} \rangle e_{j1}\| \lesssim_P T^{-1/2} \left\| (\Lambda/\lambda_j)^{1/2} s_{j1} \right\| + \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1) + T^{-1/2}.$$

From (F.164), we have

$$\|s_{j2}\| \leq \left\| \frac{\tilde{U}_2}{\sqrt{T\lambda_j}} \right\| \left\| \left(\frac{\lambda_j}{\hat{\lambda}_j} \right)^{\frac{1}{2}} \right\| \left\| \hat{\xi}_j \right\| \lesssim_P \lambda_j^{-1/2} (T^{-1/2} N^{1/2} + 1). \quad (\text{F.173})$$

This concludes the proof. $\square$

Lemma 13. *Under the assumptions of Theorem 4(a), we have*

$$\begin{aligned} (i) \quad & \left\| \frac{\hat{\xi}_i^\top \bar{U}^\top \hat{\zeta}_j}{\sqrt{T\hat{\lambda}_j}} \right\| \lesssim_P \frac{1}{T} + \frac{N+T}{T\lambda_i} + \frac{N+T}{T\lambda_j}. \\ (ii) \quad & \left\| \frac{\bar{V} \bar{U}^\top \hat{\zeta}_i}{T\sqrt{\hat{\lambda}_i}} \right\| \lesssim_P \frac{1}{T} + \frac{N+T}{T\lambda_i}, \quad \left| \frac{\hat{\zeta}_i^\top \bar{u}}{\sqrt{\hat{\lambda}_i}} \right| \lesssim_P \frac{1}{T} + \frac{N+T}{T\lambda_i}. \\ (iv) \quad & \left\| \frac{\hat{\zeta}_i^\top \bar{U}}{\sqrt{T\hat{\lambda}_i}} \right\| \lesssim_P \frac{1}{\sqrt{\lambda_i}} + \frac{N+T}{T\lambda_i}. \end{aligned}$$

Proof. (i) From (F.163), we have

$$\frac{\hat{\xi}_i^\top \bar{U}^\top \hat{\zeta}_j}{\sqrt{T\hat{\lambda}_j}} = \frac{\hat{\zeta}_i^\top \bar{R} \bar{U}^\top \hat{\zeta}_j}{T\sqrt{\hat{\lambda}_i \hat{\lambda}_j}}.$$

Using the orthogonal matrix Γ and the notations in Lemma 11 and Lemma 12, we have

$$\begin{aligned} \hat{\zeta}_i^\top \bar{R} \bar{U}^\top \hat{\zeta}_j &= s_i^\top \left(\Gamma \beta \bar{V} + \tilde{U} \right) \tilde{U}^\top s_j = s_{i1}^\top \left(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1 \right) \tilde{U}_1^\top s_{j1} + s_{i2}^\top \tilde{U}_2 \tilde{U}_1^\top s_{j1} \\ &\quad + s_{i1}^\top \left(\Lambda^{\frac{1}{2}} \bar{V} + \tilde{U}_1 \right) \tilde{U}_2^\top s_{j2} + s_{i2}^\top \tilde{U}_2 \tilde{U}_2^\top s_{j2} \\ &= K_1 + K_2 + K_3 + K_4. \end{aligned}$$

Recall that from Lemma 12, we have $\|(\Lambda/\lambda_j)^{1/2} s_{j1}\| \lesssim_P 1$. Using this result and Lemma 1, we analyze these four terms one by one. For the first term, we have

$$\|K_1\| \leq \left\| s_{i1}^\top \Lambda^{\frac{1}{2}} \right\| \left\| \bar{V} \tilde{U}_1^\top \right\| \|s_{j1}\| + \|s_{i1}\| \left\| \tilde{U}_1 \tilde{U}_1^\top \right\| \|s_{j1}\| \lesssim_P \sqrt{\lambda_i T} + T.$$

For the second term, we have

$$\|K_2\| \leq \|s_{i2}\| \|\tilde{U}_2\| \|\tilde{U}_1\| \lesssim_P \sqrt{\frac{N+T}{T\lambda_i}} (\sqrt{N} + \sqrt{T}) \sqrt{T} \lesssim_P \lambda_i^{-1/2} (N+T).$$

For the third term, we have

$$\|K_3\| \leq \left(\|s_{i1}^\top \Lambda^{\frac{1}{2}}\| \|\bar{V}\| + \|\tilde{U}_1\| \right) \|\tilde{U}_2\| \|s_{j2}\| \lesssim_P \sqrt{\lambda_i T} (\sqrt{N} + \sqrt{T}) \sqrt{\frac{N+T}{T\lambda_j}} = \lambda_j^{-1/2} \lambda_i^{1/2} (N+T).$$

For the last term, we have

$$\|K_4\| \leq \|\tilde{U}_2 \tilde{U}_2^\top\| \|s_{i2}\| \|s_{j2}\| \lesssim_P \lambda_i^{-1/2} \lambda_j^{-1/2} T^{-1} (N+T)^2.$$

Using above equations and Lemma 11, we get

$$\left\| \frac{\hat{\xi}_i^\top \bar{U}^\top \hat{\varsigma}_j}{\sqrt{T\hat{\lambda}_j}} \right\| = \left\| \frac{\hat{\xi}_i^\top \bar{R} \bar{U}^\top \hat{\varsigma}_j}{T\sqrt{\hat{\lambda}_i \hat{\lambda}_j}} \right\| \lesssim_P \frac{1}{T} + \frac{N+T}{T\lambda_i} + \frac{N+T}{T\lambda_j}.$$

(ii) Using $\bar{U}^\top \hat{\varsigma}_i = \tilde{U}_1^\top s_{i1} + \tilde{U}_2^\top s_{i2}$ and (F.170), we have

$$\|\bar{V} \bar{U}^\top \hat{\varsigma}_i\| \leq \|\bar{V} \tilde{U}_1^\top s_{i1}\| + \|\bar{V} \tilde{U}_2^\top s_{i2}\| \leq \|\bar{V} \tilde{U}_1^\top\| + \|\bar{V}\| \|\bar{U}\| \|s_{i2}\| \lesssim_P \sqrt{T} + \frac{N+T}{\sqrt{\lambda_i}}.$$

Then, with Lemma 11, we have $\|T^{-1} \hat{\lambda}_i^{-1/2} \bar{V} \bar{U}^\top \hat{\varsigma}_i\| \lesssim_P T^{-1} + \lambda_i^{-1} (T^{-1}N + 1).$

Replace $\bar{V}$ in the above proof by $\iota_T^\top$, we can get $\|\hat{\lambda}_i^{-1/2} \bar{u}^\top \hat{\varsigma}_i\| \lesssim_P T^{-1} + \lambda_i^{-1} (T^{-1}N + 1).$

(iii) Using $\bar{U}^\top \hat{\varsigma}_i = \tilde{U}_1^\top s_{i1} + \tilde{U}_2^\top s_{i2}$ and (F.170), we have

$$\|\hat{\varsigma}_i^\top \bar{U}\| \leq \|s_{i1}^\top \tilde{U}_1\| + \|s_{i2}^\top \tilde{U}_2\| \leq \|\tilde{U}_1\| + \|\bar{U}\| \lesssim_P \sqrt{T} + \frac{N+T}{\sqrt{T\lambda_i}}.$$

Applying Lemma 11 again completes the proof. □

Lemma 14. Under the assumptions of Theorem 4(a), $\tilde{H}_1, \tilde{H}_2$ defined by (F.77) satisfy

$$(i) \quad \|\tilde{H}_1\| \lesssim_P 1, \quad \|\tilde{H}_2\| \lesssim_P 1.$$

$$(ii) \quad \|\tilde{H}_1^\top \tilde{H}_2 - \mathbb{I}_{\tilde{p}}\| \lesssim_P T^{-1} + \lambda_p^{-1} (T^{-1}N + 1).$$

$$(iii) \quad \|\tilde{H}_1 - \tilde{H}_2\| \lesssim_P T^{-1/2} + \lambda_p^{-1} (T^{-1}N + 1).$$

Proof. (i) Using the definition of $\tilde{H}_1$ in (F.77) and Lemma 1, we have

$$\|\tilde{h}_{k1}\| = \left\| \frac{\bar{V} \hat{\xi}_k}{\sqrt{T}} \right\| \leq T^{-1/2} \|\bar{V}\| \lesssim_P 1,$$

which leads to $\|\tilde{H}_1\| \lesssim_P 1$.

Using $\Gamma\hat{\zeta}_k = (s_{k1}^\top, s_{k2}^\top)^\top$, the SVD of β in (F.161), the definition of $\tilde{H}_2$ in (F.77), Lemma 11 and Lemma 12(iii), we have

$$\|\tilde{h}_{k2}\| = \left\| \frac{\beta^\top \hat{\zeta}_k}{\sqrt{\hat{\lambda}_k}} \right\| = \left\| \frac{\Lambda^{1/2} s_{k1}}{\sqrt{\hat{\lambda}_k}} \right\| \lesssim_P 1, \quad (\text{F.174})$$

which leads to $\|\tilde{H}_2\| \lesssim_P 1$.

(ii) By (F.163) and Lemma 2, for $l, k \leq p$, we have

$$\delta_{lk} = \hat{\xi}_l^\top \hat{\xi}_k = \frac{\hat{\xi}_l^\top \bar{V}^\top \beta^\top \hat{\zeta}_k}{\sqrt{T \hat{\lambda}_k}} + \frac{\hat{\xi}_l^\top \bar{U}^\top \hat{\zeta}_k}{\sqrt{T \hat{\lambda}_k}} = \tilde{h}_{l1}^\top \tilde{h}_{k2} + \frac{\hat{\xi}_l^\top \bar{U}^\top \hat{\zeta}_k}{\sqrt{T \hat{\lambda}_k}}.$$

By Lemma 13(i), we have

$$|\tilde{h}_{l1}^\top \tilde{h}_{k2} - \delta_{lk}| \lesssim_P \frac{1}{T} + \frac{N+T}{T \min\{\lambda_l, \lambda_k\}} \leq \frac{1}{T} + \frac{N+T}{T \lambda_p},$$

and thus $\|\tilde{H}_1^\top \tilde{H}_2 - \mathbb{I}_p\| \lesssim_P T^{-1} + \lambda_p^{-1}(T^{-1}N + 1)$.

(iii) Using (F.163), we have

$$\bar{V} \hat{\xi}_k = \frac{\bar{V} \bar{V}^\top \beta^\top}{\sqrt{T \hat{\lambda}_k}} \hat{\zeta}_k + \frac{\bar{V} \bar{U}^\top \hat{\zeta}_k}{\sqrt{T \hat{\lambda}_k}}.$$

With the definition of h_{k1} and h_{k2} , it becomes

$$\tilde{h}_{k1} = \frac{\bar{V} \bar{V}^\top}{T} \tilde{h}_{k2} + \frac{\bar{V} \bar{U}^\top \hat{\zeta}_k}{T \sqrt{\hat{\lambda}_k}}. \quad (\text{F.175})$$

With $\|\tilde{h}_{k2}\| \lesssim_P 1$, Lemma 1 and Lemma 13(ii), (F.175) leads to

$$\|\tilde{h}_{k1} - \tilde{h}_{k2}\| \leq \|T^{-1} \bar{V} \bar{V}^\top - \mathbb{I}_p\| \|\tilde{h}_{k2}\| + \left\| \frac{\bar{V} \bar{U}^\top \hat{\zeta}_k}{T \sqrt{\hat{\lambda}_k}} \right\| \lesssim_P T^{-1/2} + \lambda_p^{-1}(T^{-1}N + 1),$$

which concludes the proof of (iii). $\square$

Lemma 15. *Under Assumption B.12, we have*

$$\|\bar{r} - \hat{\Sigma} b\|_\infty \lesssim_P \sqrt{\frac{\log N}{T}}, \quad \|b^\top (\bar{r} - \mathbb{E}(r_t))\| \lesssim_P \frac{1}{\sqrt{T}}.$$

Proof. For the first inequality, we have

$$\left\| \bar{r} - \hat{\Sigma} b \right\|_{\infty} \leq \left\| \bar{r} - E(r) \right\|_{\infty} + \left\| \Sigma b - \hat{\Sigma} b \right\|_{\infty} \lesssim_P \sqrt{\frac{\log N}{T}},$$

where we use large deviation inequalities in Assumption B.11:

$$\left\| \bar{r} - E(r_t) \right\|_{\infty} \lesssim_P \sqrt{\frac{\log N}{T}}, \quad \text{and} \quad \left\| \Sigma b - \hat{\Sigma} b \right\|_{\infty} = \left\| \frac{1}{T} \bar{R} \bar{R}^T b - \text{Cov}(r_t, r_t^T b) \right\|_{\infty} \lesssim_P \sqrt{\frac{\log N}{T}}.$$

The second inequality follows immediately from Assumption B.11:

$$\|b^T(\bar{r} - E(r_t))\| = \left| \frac{1}{T} \sum_{t=1}^T m_t - E(m_t) \right| \lesssim_P \frac{1}{\sqrt{T}}.$$

□

References

- Bai, Z. and J. W. Silverstein (2009). *Spectral Analysis of Large Dimensional Random Matrices*. Springer.
- Davis, C. and W. M. Kahan (1970). The rotation of eigenvectors by a perturbation. iii. *SIAM Journal on Numerical Analysis* 7(1), 1–46.
- Fan, J., Y. Liao, and M. Mincheva (2011). High-dimensional covariance matrix estimation in approximate factor models. *Annals of Statistics* 39(6), 3320–3356.
- Giglio, S. W. and D. Xiu (2021). Asset pricing with omitted factors. *Journal of Political Economy* 129(7), 1947–1990.
- Kan, R. and C. Zhang (1999). Two-Pass Tests of Asset Pricing Models with Useless Factors. *The Journal of Finance* 54(1), 203–235.
- Kelly, B. and S. Pruitt (2013). Market expectations in the cross-section of present values. *The Journal of Finance* 68(5), 1721–1756.
- Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics* 149(2), 149–173.
- Lettau, M. and M. Pelger (2020). Estimating latent asset-pricing factors. *Journal of Econometrics* 218, 1–31.
- Wang, W. and J. Fan (2017, 06). Asymptotics of empirical eigenstructure for high dimensional spiked covariance. *Ann. Statist.* 45(3), 1342–1374.
- Wedin, P.-Å. (1972). Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics* 12(1), 99–111.